in region 2:

$$a_1^2 = \frac{\gamma+1}{2}a^{*2} - \frac{\gamma-1}{2}u_1^2 \tag{8.53}$$

and

$$a_2^2 = \frac{\gamma+1}{2}a^{*2} - \frac{\gamma-1}{2}u_2^2 \tag{8.54}$$

In Equations (8.53) and (8.54), a^* is the same constant value because the flow across the shock wave is adiabatic (see Sections 7.5 and 8.5). Substituting Equations (8.53) and (8.54) into Equation (8.52), we have

$$\frac{\gamma+1}{2}\frac{a^{*2}}{\gamma u_1} - \frac{\gamma-1}{2\gamma}u_1 - \frac{\gamma+1}{2}\frac{a^{*2}}{\gamma u_2} + \frac{\gamma-1}{2\gamma}u_2 = u_2 - u_1$$

or

$$\frac{\gamma+1}{2\gamma u_1 u_2}(u_2-u_1)a^{*2} + \frac{\gamma-1}{2\gamma}(u_2-u_1) = u_2 - u_1$$

Dividing by $u_2 - u_1$, we obtain

$$\frac{\gamma+1}{2\gamma u_1 u_2}a^{*2} + \frac{\gamma-1}{2\gamma} = 1$$

Solving for a^*, we obtain

$$\boxed{a^{*2} = u_1 u_2} \tag{8.55}$$

Equation (8.55) is called the *Prandtl relation* and is a useful intermediate relation for normal shock waves. For example, from Equation (8.55),

$$1 = \frac{u_1}{a^*}\frac{u_2}{a^*} \tag{8.56}$$

Recall the definition of characteristic Mach number, $M^* = u/a^*$, given in Section 8.4. Hence, Equation (8.56) becomes

$$1 = M_1^* M_2^*$$

or

$$M_2^* = \frac{1}{M_1^*} \tag{8.57}$$

Substituting Equation (8.48) into (8.57), we have

$$\frac{(\gamma+1)M_2^2}{2+(\gamma-1)M_2^2} = \left[\frac{(\gamma+1)M_1^2}{2+(\gamma-1)M_1^2}\right]^{-1} \tag{8.58}$$

Solving Equation (8.58) for M_2^2, we obtain

$$\boxed{M_2^2 = \frac{1+[(\gamma-1)/2]M_1^2}{\gamma M_1^2 - (\gamma-1)/2}} \tag{8.59}$$

Equation (8.59) is our first major result for a normal shock wave. Examine Equation (8.59) closely; it states that the Mach number behind the wave M_2 is a

function only of the Mach number ahead of the wave M_1. Moreover, if $M_1 = 1$, then $M_2 = 1$. This is the case of an infinitely weak normal shock wave, defined as a *Mach wave*. Furthermore, if $M_1 > 1$, then $M_2 < 1$; that is, the Mach number behind the normal shock wave is *subsonic*. As M_1 increases above 1, the normal shock wave becomes stronger, and M_2 becomes progressively less than 1. However, in the limit as $M_1 \to \infty$, M_2 approaches a finite minimum value, $M_2 \to \sqrt{(\gamma - 1)/2\gamma}$, which for air is 0.378.

Let us now obtain the ratios of the thermodynamic properties ρ_2/ρ_1, p_2/p_1, and T_2/T_1 across a normal shock wave. Rearranging Equation (8.2) and using Equation (8.55), we have

$$\frac{\rho_2}{\rho_1} = \frac{u_1}{u_2} = \frac{u_1^2}{u_2 u_1} = \frac{u^2}{a^{*2}} = M_1^{*2} \tag{8.60}$$

Substituting Equation (8.48) into (8.60), we obtain

$$\boxed{\frac{\rho_2}{\rho_1} = \frac{u_1}{u_2} = \frac{(\gamma + 1)M_1^2}{2 + (\gamma - 1)M_1^2}} \tag{8.61}$$

To obtain the pressure ratio, return to the momentum equation, Equation (8.6), combined with the continuity equation, Equation (8.2):

$$p_2 - p_1 = \rho_1 u_1^2 - \rho_2 u_2^2 = \rho_1 u_1 (u_1 - u_2) = \rho_1 u_1^2 \left(1 - \frac{u_2}{u_1}\right) \tag{8.62}$$

Dividing Equation (8.62) by p_1, and recalling that $a_1^2 = \gamma p_1/\rho_1$, we obtain

$$\frac{p_2 - p_1}{p_1} = \frac{\gamma \rho_1 u_1^2}{\gamma p_1}\left(1 - \frac{u_2}{u_1}\right) = \frac{\gamma u_1^2}{a_1^2}\left(1 - \frac{u_2}{u_1}\right) = \gamma M_1^2 \left(1 - \frac{u_2}{u_1}\right) \tag{8.63}$$

For u_2/u_1 in Equation (8.63), substitute Equation (8.61):

$$\frac{p_2 - p_1}{p_1} = \gamma M_1^2 \left[1 - \frac{2 + (\gamma - 1)M_1^2}{(\gamma + 1)M_1^2}\right] \tag{8.64}$$

Equation (8.64) simplifies to

$$\boxed{\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1}(M_1^2 - 1)} \tag{8.65}$$

To obtain the temperature ratio, recall the equation of state $p = \rho RT$. Hence,

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)\left(\frac{\rho_1}{\rho_2}\right) \tag{8.66}$$

Substituting Equations (8.61) and (8.65) into (8.66), and recalling that $h = c_p T$, we obtain

$$\boxed{\frac{T_2}{T_1} = \frac{h_2}{h_1} = \left[1 + \frac{2\gamma}{\gamma + 1}(M_1^2 - 1)\right]\frac{2 + (\gamma - 1)M_1^2}{(\gamma + 1)M_1^2}} \tag{8.67}$$

Equations (8.61), (8.65), and (8.67) are important. Examine them closely. Note that ρ_2/ρ_1, p_2/p_1, and T_2/T_1 are *functions of the upstream Mach number* M_1 *only*. Therefore, in conjunction with Equation (8.59) for M_2, we see that the upstream Mach number M_1 is *the* determining parameter for changes across a normal shock wave in a calorically perfect gas. This is a dramatic example of the power of the Mach number as a governing parameter in compressible flows. In the above equations, if $M_1 = 1$, then $p_2/p_1 = \rho_2/\rho_1 = T_2/T_1 = 1$; that is, we have the case of a normal shock wave of vanishing strength—a Mach wave. As M_1 increases above 1, p_2/p_1, ρ_2/ρ_1, and T_2/T_1 progressively increase above 1. In the limiting case of $M_1 \to \infty$ in Equations (8.59), (8.61), (8.65), and (8.67), we find, for $\gamma = 1.4$,

$$\lim_{M_1\to\infty} M_2 = \sqrt{\frac{\gamma-1}{2\gamma}} = 0.378 \qquad \text{(as discussed previously)}$$

$$\lim_{M_1\to\infty} \frac{\rho_2}{\rho_1} = \frac{\gamma+1}{\gamma-1} = 6$$

$$\lim_{M_1\to\infty} \frac{p_2}{p_1} = \infty \qquad \lim_{M_1\to\infty} \frac{T_2}{T_1} = \infty$$

Note that, as the upstream Mach number increases toward infinity, the pressure and temperature increase without bound, whereas the density approaches a rather moderate finite limit.

We have stated earlier that shock waves occur in supersonic flows; a stationary normal shock such as shown in Figure 8.3 does not occur in subsonic flow. That is, in Equations (8.59), (8.61), (8.65), and (8.67), the upstream Mach number is supersonic $M_1 \geq 1$. However, on a *mathematical basis,* these equations also allow solutions for $M_1 \leq 1$. These equations embody the continuity, momentum, and energy equations, which in principle do not care whether the value of M_1 is subsonic or supersonic. Here is an ambiguity which can only be resolved by appealing to the second law of thermodynamics (see Section 7.2). Recall that the second law of thermodynamics determines the *direction* which a given process can take. Let us apply the second law to the flow across a normal shock wave, and examine what it tells us about allowable values of M_1.

First, consider the entropy change across the normal shock wave. From Equation (7.25),

$$s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} \tag{7.25}$$

with Equations (8.65) and (8.67), we have

$$s_2 - s_1 = c_p \ln \left\{ \left[1 + \frac{2\gamma}{\gamma+1}(M_1^2 - 1) \right] \frac{2 + (\gamma-1)M_1^2}{(\gamma+1)M_1^2} \right\} - R \ln \left[1 + \frac{2\gamma}{\gamma+1}(M_1^2 - 1) \right] \tag{8.68}$$

From Equation (8.68), we see that the entropy change $s_2 - s_1$ across the shock is a function of M_1 only. The second law dictates that

$$s_2 - s_1 \geq 0$$

In Equation (8.68), if $M_1 = 1$, $s_2 = s_1$, and if $M_1 > 1$, then $s_2 - s_1 > 0$, both of which obey the second law. However, if $M_1 < 1$, then Equation (8.68) gives $s_2 - s_1 < 0$, which is *not* allowed by the second law. Consequently, in nature, only cases involving $M_1 \geq 1$ are valid; that is, normal shock waves can occur only in supersonic flow.

Why does the entropy increase across the shock wave? The second law tells us that it must, but what mechanism does nature use to accomplish this increase? To answer these questions, recall that a shock wave is a very thin region (on the order of 10^{-5} cm) across which some large changes occur almost discontinuously. Therefore, within the shock wave itself, large gradients in velocity and temperature occur; that is, the mechanisms of friction and thermal conduction are strong. These are dissipative, irreversible mechanisms that always increase the entropy. Therefore, the precise entropy increase predicted by Equation (8.68) for a given supersonic M_1 is appropriately provided by nature in the form of friction and thermal conduction within the interior of the shock wave itself.

In Section 7.5, we defined the total temperature T_0 and total pressure p_0. What happens to these total conditions across a shock wave? To help answer this question, consider Figure 8.7, which illustrates the definition of total conditions ahead of and behind the shock. In region 1 ahead of the shock, a fluid element has the actual conditions of M_1, p_1, T_1, and s_1. Now imagine that we bring this fluid element to rest isentropically, creating the "imaginary" state 1a ahead of the shock. In state 1a, the fluid element at rest would have a pressure and temperature $p_{0,1}$ and $T_{0,1}$, respectively, that is, the total pressure and total temperature, respectively,

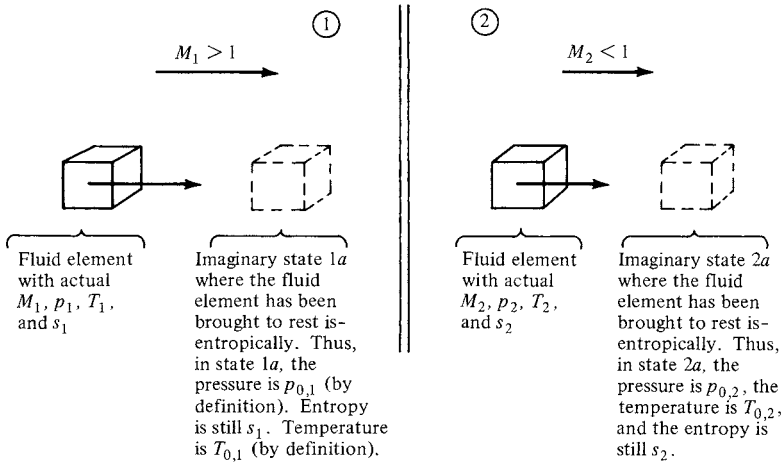


Figure 8.7 Total conditions ahead of and behind a normal shock wave.

in region 1. The entropy in state $1a$ would still be s_1 because the fluid element is brought to rest isentropically; $s_{1a} = s_1$. Now consider region 2 behind the shock. Again consider a fluid element with the actual conditions of M_2, p_2, T_2, and s_2, as sketched in Figure 8.7. And again let us imagine that we bring this fluid element to rest isentropically, creating the "imaginary" state $2a$ behind the shock. In state $2a$, the fluid element at rest would have pressure and temperature $p_{0,2}$ and $T_{0,2}$, respectively, that is, the total pressure and total temperature, respectively, in region 2. The entropy in state $2a$ would still be s_2 because the fluid element is brought to rest isentropically; $s_{2a} = s_2$. The questions are now asked: How does $T_{0,2}$ compare with $T_{0,1}$, and how does $p_{0,2}$ compare with $p_{0,1}$?

To answer the first of these questions, consider Equation (8.30):

$$c_p T_1 + \frac{u_1^2}{2} = c_p T_2 + \frac{u_2^2}{2} \tag{8.30}$$

From Equation (8.38), the total temperature is given by

$$c_p T_0 = c_p T + \frac{u^2}{2} \tag{8.38}$$

Combining Equations (8.30) and (8.38), we have

$$c_p T_{0,1} = c_p T_{0,2}$$

or

$$\boxed{T_{0,1} = T_{0,2}} \tag{8.69}$$

Equation (8.69) states that *total temperature is constant across a stationary normal shock wave*. This should come as no surprise; the flow across a shock wave is adiabatic, and in Section 7.5 we demonstrated that in a steady, adiabatic, inviscid flow of a calorically perfect gas, the total temperature is constant.

To examine the variation of total pressure across a normal shock wave, write Equation (7.25) between the imaginary states $1a$ and $2a$:

$$s_{2a} - s_{1a} = c_p \ln \frac{T_{2a}}{T_{1a}} - R \ln \frac{p_{2a}}{p_{1a}} \tag{8.70}$$

However, from the above discussion, as well as the sketch in Figure 8.7, we have $s_{2a} = s_2$, $s_{1a} = s_1$, $T_{2a} = T_{0,2}$, $T_{1a} = T_{0,1}$, $p_{2a} = p_{0,2}$, and $p_{1a} = p_{0,1}$. Thus, Equation (8.70) becomes

$$s_2 - s_1 = c_p \ln \frac{T_{0,2}}{T_{0,1}} - R \ln \frac{p_{0,2}}{p_{0,1}} \tag{8.71}$$

We have already shown that $T_{0,2} = T_{0,1}$; hence, Equation (8.71) yields

$$\boxed{s_2 - s_1 = -R \ln \frac{p_{0,2}}{p_{0,1}}} \tag{8.72}$$

or

$$\boxed{\frac{p_{0,2}}{p_{0,1}} = e^{-(s_2 - s_1)/R}} \tag{8.73}$$

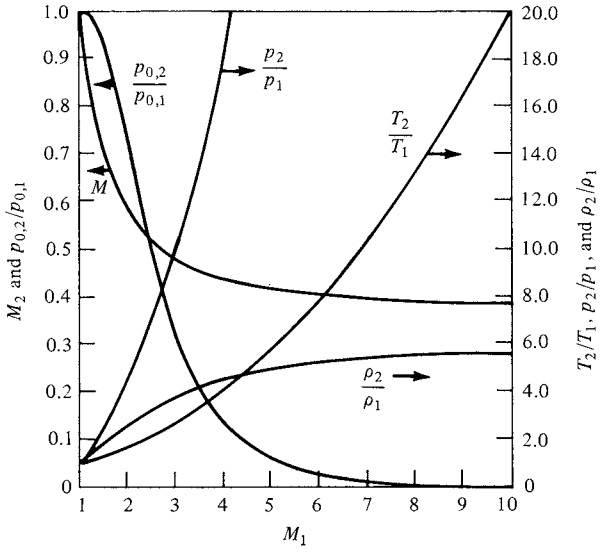


Figure 8.8 The variation of properties across a normal shock wave as a function of upstream Mach number: $\gamma = 1.4$.

From Equation (8.68), we know that $s_2 - s_1 > 0$ for a normal shock wave. Hence, Equation (8.73) states that $p_{0,2} < p_{0,1}$. *The total pressure decreases across a shock wave.* Moreover, since $s_2 - s_1$ is a function of M_1 only [from Equation (8.68)], then Equation (8.73) clearly states that the total pressure ratio $p_{0,2}/p_{0,1}$ across a normal shock wave is a function of M_1 only.

In summary, we have now verified the qualitative changes across a normal shock wave as sketched in Figure 7.4*b* and as originally discussed in Section 7.6. Moreover, we have obtained closed-form analytic expressions for these changes in the case of a calorically perfect gas. We have seen that p_2/p_1, ρ_2/ρ_1, T_2/T_1, M_2, and $p_{0,2}/p_{0,1}$ are functions of the upstream Mach number M_1 only. To help you obtain a stronger physical feeling of normal shock-wave properties, these variables are plotted in Figure 8.8 as a function of M_1. Note that (as stated earlier) these curves show how, as M_1 becomes very large, T_2/T_1 and p_2/p_1 also become very large, whereas ρ_2/ρ_1 and M_2 approach finite limits. Examine Figure 8.8 carefully, and become comfortable with the trends shown.

The results given by Equations (8.59), (8.61), (8.65), (8.67), and (8.73) are so important that they are tabulated as a function of M_1 in Appendix B for $\gamma = 1.4$.

EXAMPLE 8.11

Consider a normal shock wave in air where the upstream flow properties are $u_1 = 680$ m/s, $T_1 = 288$ K, and $p_1 = 1$ atm. Calculate the velocity, temperature, and pressure downstream of the shock.

■ Solution

$$a_1 = \sqrt{\gamma RT_1} = \sqrt{1.4(287)(288)} = 340 \text{ m/s}$$

$$M_1 = \frac{u_1}{a_1} = \frac{680}{340} = 2$$

From Appendix B, since $p_2/p_1 = 4.5$, $T_2/T_1 = 1.687$, $M_2 = 0.5774$, then

$$p_2 = \frac{p_2}{p_1} p_1 = 4.5(1 \text{ atm}) = \boxed{4.5 \text{ atm}}$$

$$T_2 = \frac{T_2}{T_1} T_1 = 1.687(288) = \boxed{486 \text{ K}}$$

$$a_2 = \sqrt{\gamma RT_2} = \sqrt{1.4(287)(486)} = 442 \text{ m/s}$$

$$u_2 = M_2 a_2 = 0.5774(442) = \boxed{255 \text{ m/s}}$$

EXAMPLE 8.12

Consider a normal shock wave in a supersonic airstream where the pressure upstream of the shock is 1 atm. Calculate the loss of total pressure across the shock wave when the upstream Mach number is (*a*) $M_1 = 2$, and (*b*) $M_1 = 4$. Compare these two results and comment on their implication.

■ Solution

(*a*) The upstream total pressure is obtained from

$$p_{0,1} = \left(\frac{p_{0,1}}{p_1} \right) p_1$$

where from Appendix A for $M_1 = 2$, $p_{0,1}/p_1 = 7.824$. Hence,

$$p_{0,1} = (7.824)(1 \text{ atm}) = 7.824 \text{ atm}$$

The total pressure behind the normal shock is obtained from

$$p_{0,2} = \left(\frac{p_{0,2}}{p_{0,1}} \right) p_{0,1}$$

where from Appendix B, for $M_1 = 2$, $p_{0,2}/p_{0,1} = 0.7209$. Hence,

$$p_{0,2} = (0.7209)(7.824) = 5.64 \text{ atm}$$

The *loss* of total pressure is

$$p_{0,1} - p_{0,2} = 7.824 - 5.64 = \boxed{2.184 \text{ atm}}$$

(*b*) For $M_1 = 4$, from Appendix A,

$$p_{0,1} = \left(\frac{p_{0,1}}{p_1} \right) p_1 = (151.8)(1 \text{ atm}) = 151.8 \text{ atm}$$

The total pressure behind the normal shock is obtained from Appendix B, for $M_1 = 4$, as

$$p_{0,2} = \left(\frac{p_{0,2}}{p_{0,1}}\right) p_{0,1} = (0.1388)(151.8) = 21.07 \text{ atm}$$

The *loss* of total pressure is

$$p_{0,1} - p_{0,2} = 151.8 - 21.07 = \boxed{130.7 \text{ atm}}$$

Note: In any flow, total pressure is a precious commodity. Any loss of total pressure reduces the flow's ability to do useful work. Losses of total pressure reduce the performance of any flow device, and cost money. We will see this time-and-time-again in subsequent chapters. In this example, we see that for a normal shock at Mach 2, the loss of total pressure was 2.184 atm, whereas simply by doubling the Mach number to 4, the loss of total pressure was a whopping 130.7 atm. The moral to this story is that, if you are going to suffer a normal shock wave in a flow, everything else being equal, you want the normal shock to occur at the lowest possible upstream Mach number.

EXAMPLE 8.13

A ramjet engine is an air-breathing propulsion device with essentially no rotating machinery (no rotating compressor blades, turbine, etc.). The basic generic parts of a conventional ramjet are sketched in Figure 8.9. The flow, moving from left to right, enters the inlet, where it is compressed and slowed down. The compressed air then enters the combustor at very low subsonic speed, where it is mixed and burned with a fuel. The hot gas then expands through a nozzle. The net result is the production of thrust toward the left in Figure 8.9. In this figure the ramjet is shown in a supersonic freestream with a detached shock wave ahead of the inlet. The portion of the shock just to the left of point 1 is a normal shock. (A detached normal shock wave in front of the inlet of a ramjet in a supersonic flow is not the ideal operating condition; rather, it is desirable that the flow pass through one or more *oblique* shock waves before entering the inlet. Oblique shock waves are discussed

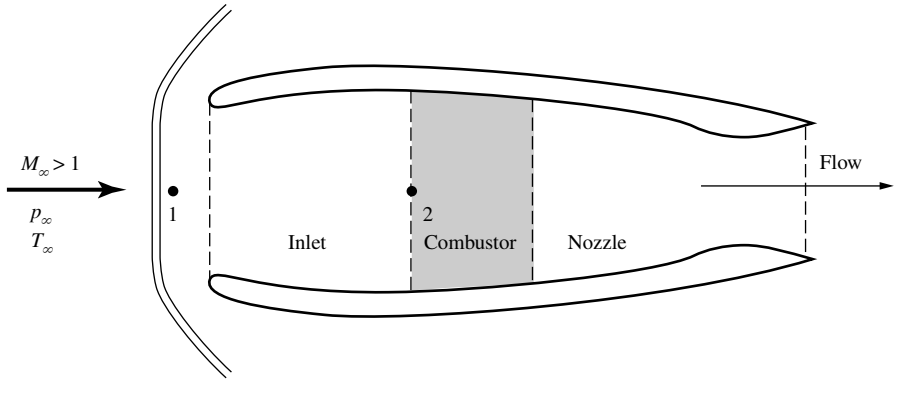


Figure 8.9 Schematic of a conventional subsonic-combustion ramjet engine.

in Chapter 9.) After passing through the shock wave, the flow from point 1 to point 2, located at the entrance to the combustor, is isentropic. The ramjet is flying at Mach 2 at a standard altitude of 10 km, where the air pressure and temperature are 2.65×10^4 N/m^2 and 223.3 K, respectively. Calculate the air temperature and pressure at point 2 when the Mach number at that point is 0.2.

■ Solution

The total pressure and total temperature of the freestream at $M_\infty = 2$ can be obtained from Appendix A.

$$p_{0,\infty} = \left(\frac{p_{0,\infty}}{p_\infty}\right) p_\infty = (7.824)(2.65 \times 10^4) = 2.07 \times 10^5 \text{ N/m}^2$$

$$T_{0,\infty} = \left(\frac{T_{0,\infty}}{T_\infty}\right) T_\infty = (1.8)(223.3) = 401.9 \text{ K}$$

At point 1 behind the normal shock, the total pressure is, from Appendix B, for $M_\infty = 2$

$$p_{0,1} = \left(\frac{p_{0,1}}{p_{0,\infty}}\right) p_{0,\infty} = (0.7209)(2.07 \times 10^5) = 1.49 \times 10^5 \text{ N/m}^2$$

The total temperature is constant across the shock, hence

$$T_{0,1} = T_{0,\infty} = 401.9 \text{ K}$$

The flow is isentropic between points 1 and 2, hence p_0 and T_0 are constant between these points. Therefore, $p_{0,2} = 1.49 \times 10^5$ N/m^2 and $T_{0,2} = 401.9$ K. At point 2, where $M_2 = 0.2$, the ratios of the total-to-static pressure and total-to-static temperature, from Appendix A, are $p_{0,2}/p_2 = 1.028$ and $T_{0,2}/T_2 = 1.008$. Hence,

$$p_2 = \left(\frac{p_2}{p_{0,2}}\right)(p_{02}) = \frac{1.49 \times 10^5}{1.028} = \boxed{1.45 \times 10^5 \text{ N/m}^2}$$

$$T_2 = \left(\frac{T_2}{T_{0,2}}\right)(T_{02}) = \frac{401.9}{1.008} = \boxed{399 \text{ K}}$$

Recall that 1 atm $= 1.02 \times 10^5$ N/m^2. Hence, p_2 in atmospheres is

$$p_2 = \frac{1.45 \times 10^5}{1.02 \times 10^5} = \boxed{1.42 \text{ atm}}$$

Note: Air pressures and temperatures on the order of 1.42 atm and 399 K entering the combustor are very tolerable conditions for low-speed subsonic combustion. In Example 8.11, we will draw a comparison between this result and the conditions that would exist at a much higher freestream Mach number.

EXAMPLE 8.14

Repeat Example 8.13, except for a freestream Mach number $M_\infty = 10$. Assume that the ramjet has been redesigned so that the Mach number at point 2 remains equal to 0.2.

■ Solution

From Appendix A, for $M = 10$, we have

$$p_{0,\infty} = \left(\frac{p_{0,\infty}}{p_\infty}\right) p_\infty = (0.4244 \times 10^5)(2.65 \times 10^4) = 1.125 \times 10^9 \text{ N/m}^2$$

$$T_{0,\infty} = \left(\frac{T_{0,\infty}}{T_\infty}\right) T_\infty = (21)(223.3) = 4690 \text{ K}$$

At point 1, from Appendix B for $M_\infty = 10$, we have

$$p_{0,1} = \left(\frac{p_{0,1}}{p_{0,\infty}}\right) (p_{0,\infty}) = (0.3045 \times 10^{-2})(1.125 \times 10^9) = 3.43 \times 10^6 \text{ N/m}^2$$

and

$$T_{0,1} = T_{0,\infty} = 4690 \text{ K}$$

At point 2, where $M_2 = 0.2$, we have from Example 8.8, $p_{0,2}/p_2 = 1.028$ and $T_{0,2}/T_2 = 1.008$. Also at point 2, since the flow is isentropic between points 1 and 2,

$$p_{0,2} = p_{0,1} = 3.43 \times 10^6 \text{ N/m}^2$$

$$T_{0,2} = T_{0,1} = 4690 \text{ K}$$

Hence,

$$p_2 = \left(\frac{p_2}{p_{0,2}}\right) (p_{0,2}) = \frac{3.43 \times 10^6}{1.028} = \boxed{3.34 \times 10^6 \text{ N/m}^2}$$

$$T_2 = \left(\frac{T_2}{T_{0,2}}\right) (T_{0,2}) = \frac{4690}{1.008} = \boxed{4653 \text{ K}}$$

In atmospheres,

$$p_2 = \frac{3.34 \times 10^6}{1.02 \times 10^5} = \boxed{32.7 \text{ atm}}$$

Compared to the rather benign conditions at point 2 existing for the case treated in Example 8.13, in the present example the air entering the combustor is at a pressure and temperature of 32.7 atm and 4653 K—both extremely severe conditions. The temperature is so hot that the fuel injected into the combustor will decompose rather than burn, with little or no thrust being produced. Moreover, the pressure is so high that the structural design of the combustor would have to be extremely heavy, assuming in the first place that some special heat-resistant material could be found that could handle the high temperature. In short, a conventional ramjet, where the flow is slowed down to a low subsonic Mach number before entering the combustor, *will not work at high, hypersonic Mach numbers*. The solution to this problem is not to slow the flow inside the engine to low subsonic speeds, but rather to slow it only to a lower but still supersonic speed. In this manner, the temperature and pressure increase inside the engine will be smaller and can be made tolerable. In such a ramjet, the entire flowpath through the engine remains at supersonic speed, including inside the combustor. This necessitates the injection and mixing of the fuel in a supersonic stream—a challenging technical problem. This type of ramjet, where the flow is supersonic throughout, is called a supersonic combustion ramjet—SCRAMjet

for short. SCRAMjets are a current area of intense research and advanced development. In November 2005, for the first time in history, a SCRAMjet engine successfully powered a hypersonic flight vehicle, the experimental X-43 shown in Figure 9.31, achieving a Mach number of almost 10. SCRAMjet engines are the only viable airbreathing power plants for hypersonic cruise vehicles. Aspects of SCRAMjet engine design will be discussed in Chapter 9.

EXAMPLE 8.15

The pressure ratio across a normal shock wave in air is 4.5. What are the Mach numbers in front of and behind the wave? What are the density and temperature ratios across the wave?

■ Solution

From Appendix B, for $p_2/p_1 = 4.5$,

$$M_1 = \boxed{2} \quad \text{and} \quad M_2 = \boxed{0.5774}$$

Also, from the same table,

$$\frac{\rho_2}{\rho_1} = \boxed{2.667} \quad \text{and} \quad \frac{T_2}{T_1} = \boxed{1.687}$$

Note: For a normal shock, the specification of the pressure *ratio* across the wave uniquely determines the Mach number in front of the wave, the Mach number behind the wave, and the ratio of all other thermodynamic properties across the wave.

EXAMPLE 8.16

The temperature ratio across a normal shock wave in air is 5.8. What are the Mach numbers in front of and behind the wave? What are the density and pressure ratios across the wave?

■ Solution

From Appendix B, for $T_2/T_1 = 5.8$,

$$M_1 = \boxed{5} \quad \text{and} \quad M_2 = \boxed{0.4152}$$

Also, from the table,

$$\frac{\rho_2}{\rho_1} = \boxed{5} \quad \text{and} \quad \frac{p_2}{p_1} = \boxed{29}$$

Note: For a normal shock wave, the specification of the temperature *ratio* uniquely determines the Mach number in front of the wave, the Mach number behind the wave, and the ratio of all other thermodynamic properties across the wave.

EXAMPLE 8.17

The Mach number behind a normal shock wave is 0.4752. What is the Mach number in front of the wave? What are the density, pressure, and temperature ratios across the shock?

■ Solution

From Appendix B, for $M_2 = 0.4752$,

$$M_1 = \boxed{3} \qquad \frac{\rho_2}{\rho_1} = \boxed{3.857} \qquad \frac{p_2}{p_1} = \boxed{10.33} \qquad \frac{T_2}{T_1} = \boxed{2.679}$$

Note: For a normal shock wave, the specification of the Mach number behind the shock uniquely determines the Mach number in front of the wave and the ratios of all thermodynamic properties across the shock.

EXAMPLE 8.18

The velocity and temperature of the flow ahead of a normal shock wave are 1215 m/s and 300 K, respectively. Calculate the velocity of the flow behind the shock.

■ Solution

$$a_1 = \sqrt{\gamma R T_1} = \sqrt{(1.4)(287)(300)} = 347.2 \text{ m/s}$$

$$M_1 = \frac{u_1}{a_1} = \frac{1215}{347.2} = 3.5$$

From Appendix B, for $M_1 = 3.5$, $M_2 = 0.4512$ and $T_2/T_1 = 3.315$,

$$T_2 = \left(\frac{T_2}{T_1}\right) T_1 = (3.315)(300) = 994.5 \text{ K}$$

$$a_2 = \sqrt{\gamma R T_2} = \sqrt{(1.4)(287)(994.5)} = 632.1 \text{ m/s}$$

$$u_2 = M_2\, a_2 = (0.4512)(632.1) = \boxed{285.2 \text{ m/s}}$$

Note: Unlike the previous three examples where only *one dimensionless* quantity (M_1 or M_2, or p_2/p_1, etc.) uniquely specified the shock wave, in this example two quantities are needed to specify the shock wave. This is because we were given actual dimensional quantities such as velocity in meters per second and temperature in kelvins. Just the velocity by itself will not define a specific normal shock, nor will the temperature by itself. We needed both quantities to define the specific shock. Of course, to solve this example, the first thing we did was to calculate the Mach number from the given u_1 and T_1. Emphasis is again made that Mach number, not velocity, is the powerful single quantity that specifies a particular normal shock.

EXAMPLE 8.19

The velocity and temperature behind a normal shock wave are 329 m/s and 1500 K, respectively. Calculate the velocity in front of the shock wave.

■ Solution

$$a_2 = \sqrt{\gamma R T_2} = \sqrt{(1.4)(287)(1500)} = 776.3 \text{ m/s}$$

$$M_2 = \frac{u_2}{a_2} = \frac{329}{776.3} = 0.4238$$

Examining Appendix B, we see there is no precise entry for $M_2 = 0.4238$; rather, this number lies between 0.4236 at $M_1 = 4.5$ and 0.4245 at $M_1 = 4.45$. By interpolation,

$$M_1 = 4.45 + \frac{(0.4245 - 0.4238)}{(0.4245 - 0.4236)}(4.5 - 4.45)$$

$$M_1 = 4.45 + 0.0389 = 4.4898$$

From Appendix B we note that $T_2/T_1 = 4.875$ at $M_2 = 0.4236$ and $T_2/T_1 = 4.788$ at $M_2 = 0.4245$. Interpolating to find T_2/T_1 at $M_2 = 0.4238$, we find

$$\frac{T_2}{T_1} = 4.788 + \frac{0.4245 - 0.4238}{0.4245 - 0.4236}(4.875 - 4.788)$$

$$\frac{T_2}{T_1} = 4.788 + 0.068 = 4.856$$

Thus,

$$T_1 = \frac{T_2}{T_2/T_1} = \frac{1500}{4.856} = 308.9$$

$$a_1 = \sqrt{\gamma R T_1} = \sqrt{(1.4)(287)(308.9)} = 352.3 \text{ m/s}$$

$$u_1 = M_1 \, a_1 = (4.489)(352.3) = \boxed{158 \text{ m/s}}$$

Note: Once again we see that a single velocity does not specify a normal shock wave. However, a velocity in combination with temperature does specify the normal shock. In contrast to Example 8.18, where u_1 and T_1 ahead of the shock specified the shock, we see that u_2 and T_2 behind the shock also are sufficient to specify the shock.

EXAMPLE 8.20

Repeat Example 8.19, but use the "nearest entry" in tables rather than interpolating between entries. Using the nearest entry is a less accurate calculation than interpolation, but it is simpler and quicker. Compare this less accurate result with the more accurate result from Example 8.19.

■ Solution

From Example 8.19, $M_2 = 0.4238$. The nearest entry in Appendix B is $M_2 = 0.4236$, which corresponds to $M_1 = 4.5$ and $T_2/T_1 = 4.875$. Using the nearest entry, we have

$$T_1 = \frac{T_2}{T_2/T_1} = \frac{1500}{4.875} = 307.7 \text{ K}$$

$$a_1 = \sqrt{\gamma R T_1} = \sqrt{(1.4)(287)(307.7)} = 351.6 \text{ m/s}$$

$$u_1 = M_1\, a_1 = 4.5(351.6) = \boxed{1582 \text{ m/s}}$$

Comparing this result with that from Example 8.19, we have:

$$u_1 = 1581 \text{ m/s} \qquad \text{(interpolation)}$$
$$u_1 = 1582 \text{ m/s} \qquad \text{(nearest entry)}$$

We conclude, at least in this case, that using the nearest entry caused only a 0.06 percent error, not enough to worry about in the context of a worked example.

EXAMPLE 8.21

Consider a normal shock with an upstream Mach number of 3.53. Obtain the downstream Mach number by:

(a) Using the nearest entry in the tables.

(b) Interpolating the tabulated values.

(c) Exact analytical calculations.

Compare the accuracy of the results.

■ Solution

(a) The nearest entry in Appendix B is for $M_1 = 3.55$. For this entry in the tables,

$$M_2 = \boxed{0.4492}$$

(b) $M_1 = 3.53$ lies between the entries for $M_1 = 3.5$, where $M_2 = 0.4512$, and $M_1 = 3.55$, where $M_2 = 0.4492$. Interpolating to obtain M_2 corresponding to $M_1 = 3.53$, we have

$$M_2 = 0.4492 + \frac{(3.55 - 3.53)}{(3.55 - 3.5)}(0.4512 - 0.4492)$$

$$M_2 = 0.4492 + 0.0008 = \boxed{0.45}$$

(c) From Equation (8.59),

$$M_2^2 = \frac{1 + [\gamma - 1)/2]M_1^2}{\gamma M_1^2 - (\gamma - 1)/2} = \frac{1 + 0.2(3.53)^2}{(1.4)(3.53)^2 - 0.2} = 0.2025$$

Thus,

$$M_2 = (0.2025)^{1/2} = \boxed{0.45}$$

Compare the results:

(a) $M_2 = 0.4492$ (nearest entry)

(b) $M_2 = 0.45$ (interpolation)

(c) $M_2 = 0.45$ (exact)

Conclusion: For all practical purposes, all three approaches yield almost identical results.

8.6.1 Comment on the Use of Tables to Solve Compressible Flow Problems

Appendices A, B, and C provide tables for the convenient calculation of certain problems in compressible flow. Many of the previous worked examples in this chapter illustrate the usefulness of these tables. Even when you are dealing with conditions that do not correspond *exactly* to a direct entry in the tables, and in practice this is usually the case, simple linear interpolation between lines in the table gave quite accurate numbers for answers, as the results of Examples 8.19, 8.20, and 8.21 demonstrated. This accuracy is verified by calculations made with exact analytical formulas as demonstrated in Example 8.21. Various compressible flow tables have been in existence since the 1940s, and their purpose was, as it is now, to provide a quick and convenient tool for the solution of various compressible flow problems. They are particularly convenient when we adopt the method of using the nearest entry in the tables, rather than take the time to interpolate between entries. Examples 8.20 and 8.21 demonstrate that little accuracy is lost by using tables that contain many more closely spaced entries than presented in Appendices A, B, and C in this book (limited in length because of space constraints). A classic example is the compressible flow tables contained in NACA TR-1135 (Reference 119), a "bible" on the desk of most aerodynamicists working in high-speed flow.

The modern alternative to these tables is, of course, the digital computer into which the analytical equations, such as Equations (8.40), (8.42), (8.43), (8.59), (8.61), and (8.65), can easily be programmed, and the numbers in Appendices A, B, and C can be reproduced on your hand calculator. This is particularly straightforward if you have an explicit calculation, such as calculating M_2 behind a normal shock explicitly from a known value of M_1 ahead of the shock using Equation (8.59). But return to Equation (8.59) for a moment. What if you are given M_2, and you want to find M_1? This is a not-so-convenient implicit calculation in Equation (8.59), whereas, using the tables, you can immediately go to Appendix B,

scan down the column for M_2, find the given value of M_2, and then find the corresponding value of M_1 by reading directly across the page.

For our purposes, for the remainder of our discussions on compressible flow in this book, we will frequently use the tables, and we will for simplicity adopt the "nearest entry" method.

8.7 MEASUREMENT OF VELOCITY IN A COMPRESSIBLE FLOW

The use of a Pitot tube for measuring the velocity of a low-speed, incompressible flow was discussed in Section 3.4. Before progressing further, return to Section 3.4, and review the principal aspects of a Pitot tube, as well as the formulas used to obtain the flow velocity from the Pitot pressure, assuming incompressible flow.

For low-speed, incompressible flow, we saw in Section 3.4 that the velocity can be obtained from a knowledge of both the total pressure and the static pressure at a point. The total pressure is measured by a Pitot tube, and the static pressure is obtained from a static pressure orifice or by some independent means. The important aspect of Section 3.4 is that the pressure sensed by a Pitot tube, along with the static pressure, is all that is necessary to extract the flow velocity for an incompressible flow. In the present section, we see that the same is true for a compressible flow, both subsonic and supersonic, if we consider the Mach number rather than the velocity. In both subsonic and supersonic compressible flows, a knowledge of the Pitot pressure and the static pressure is sufficient to calculate Mach number, although the formulas are different for each Mach-number regime. Let us examine this matter further.

8.7.1 Subsonic Compressible Flow

Consider a Pitot tube in a subsonic, compressible flow, as sketched in Figure 8.10*a*. As usual, the mouth of the Pitot tube (point *b*) is a stagnation region. Hence, a fluid element moving along streamline *ab* is brought to rest isentropically at point *b*. In turn, the pressure sensed at point *b* is the total pressure of the freestream, $p_{0,1}$. This is the Pitot pressure read at the end of the tube. If, in addition, we know the freestream static pressure p_1, then the Mach number in region 1 can be obtained from Equation (8.42),

$$\frac{p_{0,1}}{p_1} = \left(1 + \frac{\gamma - 1}{2} M_1^2\right)^{\gamma/(\gamma-1)} \tag{8.42}$$

or solving for M_1^2,

$$\boxed{M_1^2 = \frac{2}{\gamma - 1}\left[\left(\frac{p_{0,1}}{p_1}\right)^{(\gamma-1)/\gamma} - 1\right]} \tag{8.74}$$

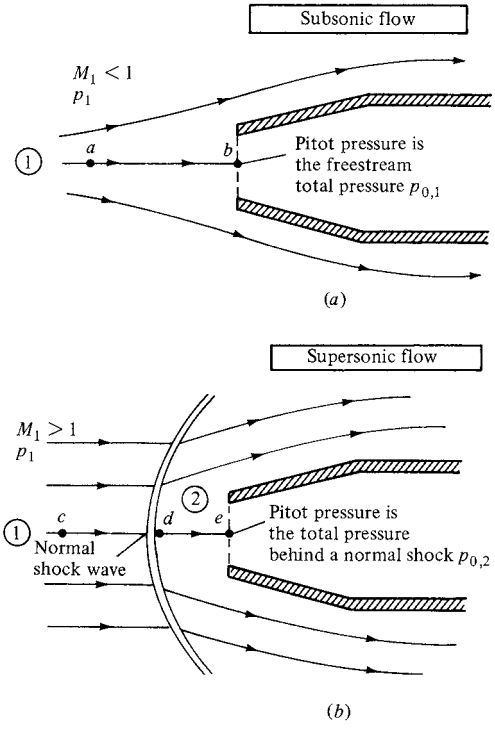


Figure 8.10 A Pitot tube in (*a*) subsonic flow and (*b*) supersonic flow.

Clearly, from Equation (8.74), the Pitot pressure $p_{0,1}$ and the static pressure p_1 allow the direct calculation of Mach number.

The flow velocity can be obtained from Equation (8.74) by recalling that $M_1 = u_1/a_1$. Hence,

$$u_1^2 = \frac{2a_1^2}{\gamma - 1}\left[\left(\frac{p_{0,1}}{p_1}\right)^{(\gamma-1)/\gamma} - 1\right] \tag{8.75}$$

From Equation (8.75), we see that, unlike incompressible flow, a knowledge of $p_{0,1}$ and p_1 is not sufficient to obtain u_1; we also need the freestream speed of sound, a_1.

8.7.2 Supersonic Flow

Consider a Pitot tube in a supersonic freestream, as sketched in Figure 8.10*b*. As usual, the mouth of the Pitot tube (point e) is a stagnation region. Hence, a fluid element moving along streamline cde is brought to rest at point e. However, because the freestream is supersonic and the Pitot tube presents an obstruction to the flow, there is a strong bow shock wave in front of the tube, much like the picture shown at the left of Figure 8.1 for supersonic flow over a blunt body. Hence,

streamline *cde* crosses the normal portion of the bow shock. A fluid element moving along streamline *cde* will first be decelerated *nonisentropically* to a subsonic velocity at point *d* just behind the shock. Then it is isentropically compressed to zero velocity at point *e*. As a result, the pressure at point *e* is *not* the total pressure of the freestream but rather the total pressure *behind a normal shock wave,* $p_{0,2}$. This is the Pitot pressure read at the end of the tube. Keep in mind that because of the entropy increase across the shock, there is a loss in total pressure across the shock, $p_{0,2} < p_{0,1}$. However, knowing $p_{0,2}$ and the freestream static pressure p_1 is still sufficient to calculate the freestream Mach number M_1, as follows:

$$\frac{p_{0,2}}{p_1} = \frac{p_{0,2}}{p_2}\frac{p_2}{p_1} \tag{8.76}$$

Here, $p_{0,2}/p_2$ is the ratio of total pressure to static pressure in region 2 immediately behind the normal shock, and p_2/p_1 is the static pressure ratio across the shock. From Equation (8.42),

$$\frac{p_{0,2}}{p_2} = \left(1 + \frac{\gamma - 1}{2}M_2^2\right)^{\gamma/(\gamma-1)} \tag{8.77}$$

where, from Equation (8.59),

$$M_2^2 = \frac{1 + [(\gamma - 1)/2]M_1^2}{\gamma M_1^2 - (\gamma - 1)/2} \tag{8.78}$$

Also, from Equation (8.65),

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1}(M_1^2 - 1) \tag{8.79}$$

Substituting Equation (8.78) into (8.77), and substituting the result as well as Equation (8.79) into Equation (8.76), we obtain, after some algebraic simplification (see Problem 8.14),

$$\boxed{\frac{p_{0,2}}{p_1} = \left(\frac{(\gamma + 1)^2 M_1^2}{4\gamma M_1^2 - 2(\gamma - 1)}\right)^{\gamma/(\gamma-1)} \frac{1 - \gamma + 2\gamma M_1^2}{\gamma + 1}} \tag{8.80}$$

Equation (8.80) is called the *Rayleigh Pitot tube formula*. It relates the Pitot pressure $p_{0,2}$ and the freestream static pressure p_1 to the freestream Mach number M_1. Equation (8.80) gives M_1 as an implicit function of $p_{0,2}/p_1$ and allows the calculation of M_1 from a known $p_{0,2}/p_1$. For convenience in making calculations, the ratio $p_{0,2}/p_1$ is tabulated versus M_1 in Appendix B.

EXAMPLE 8.22

A Pitot tube is inserted into an airflow where the static pressure is 1 atm. Calculate the flow Mach number when the Pitot tube measures (*a*) 1.276 atm, (*b*) 2.714 atm, (*c*) 12.06 atm.

■ Solution

First, we must assess whether the flow is subsonic or supersonic. At Mach 1, the Pitot tube would measure $p_0 = p/0.528 = 1.893p$. Hence, when $p_0 < 1.893$ atm, the flow is subsonic, and when $p_0 > 1.893$ atm, the flow is supersonic.

(a) Pitot tube measurement = 1.276 atm. The flow is subsonic. Hence, the Pitot tube is directly sensing the total pressure of the flow. From Appendix A, for $p_0/p = 1.276$,

$$\boxed{M = 0.6}$$

(b) Pitot tube measurement = 2.714 atm. The flow is supersonic. Hence, the Pitot tube is sensing the total pressure behind a normal shock wave. From Appendix B, for $p_{0,2}/p_1 = 2.714$,

$$\boxed{M_1 = 1.3}$$

(c) Pitot tube measurement = 12.06 atm. The flow is supersonic. From Appendix B, for $p_{0,2}/p_1 = 12.06$,

$$\boxed{M_1 = 3.0}$$

EXAMPLE 8.23

Consider a hypersonic missile flying at Mach 8 at an altitude of 20,000 ft, where the pressure is 973.3 lb/ft^2. The nose of the missile is blunt and is shaped like that shown at the left of Figure 8.1. Calculate the pressure at the stagnation point on the nose.

■ Solution

Examining the blunt body shown in Figure 8.1, the streamline that impinges at the stagnation point has traversed the normal portion of the bow shock wave. By definition, $V = 0$ at the stagnation point. Since the flow is isentropic between the shock and the body, the pressure at the stagnation point on the body is the total pressure behind a normal shock with an upstream Mach number of 8. Let us denote the pressure at the stagnation point by p_s. Since $p_{0,2}$ is the total pressure behind the normal shock, then $p_s = p_{0,2}$. From Appendix B, for Mach 8, $p_{0,2}/p_1 = 82.87$. Hence,

$$p_s = p_{0,2} = \left(\frac{p_{0,2}}{p_1}\right)(p_1) = 82.87(973.3) = \boxed{8.07 \times 10^4 \text{ lb/ft}^2}$$

Since 1 atm = 2116 lb/ft^2,

$$p_s = \frac{8.07 \times 10^4}{2116} = \boxed{38.1 \text{ atm}}$$

Note that the pressure at the nose of the missile is quite high—38.1 atm. This is typical of hypersonic flight at low altitude.

Check on the calculation This problem can also be solved by first calculating the upstream total pressure from Appendix A, and then using the total pressure ratio across the normal shock from Appendix B. From Appendix A for Mach 8, $p_{0,1}/p_1 = 0.9763 \times 10^{-4}$.

Hence,

$$p_{0,1} = \left(\frac{p_{0,1}}{p_1}\right) p_1 = (0.9763 \times 10^4)973.3 = 9.502 \times 10^6$$

From Appendix B for Mach 8, $p_{0,2}/p_{0,1} = 8.8488 \times 10^{-2}$. Hence,

$$p_s = p_{0,2} = \left(\frac{p_{0,2}}{p_{0,1}}\right) p_{0,1} = (0.8488 \times 10^{-2})(9.502 \times 10^6) = \boxed{8.07 \times 10^4 \text{ lb/ft}^2}$$

This is the same result as obtained earlier.

EXAMPLE 8.24

Consider the Lockheed SR-71 Blackbird shown in Figure 8.11 flying at a standard altitude of 25 km. The pressure measured by a Pitot tube on this airplane is 3.88×10^4 N/m^2. Calculate the velocity of the airplane.

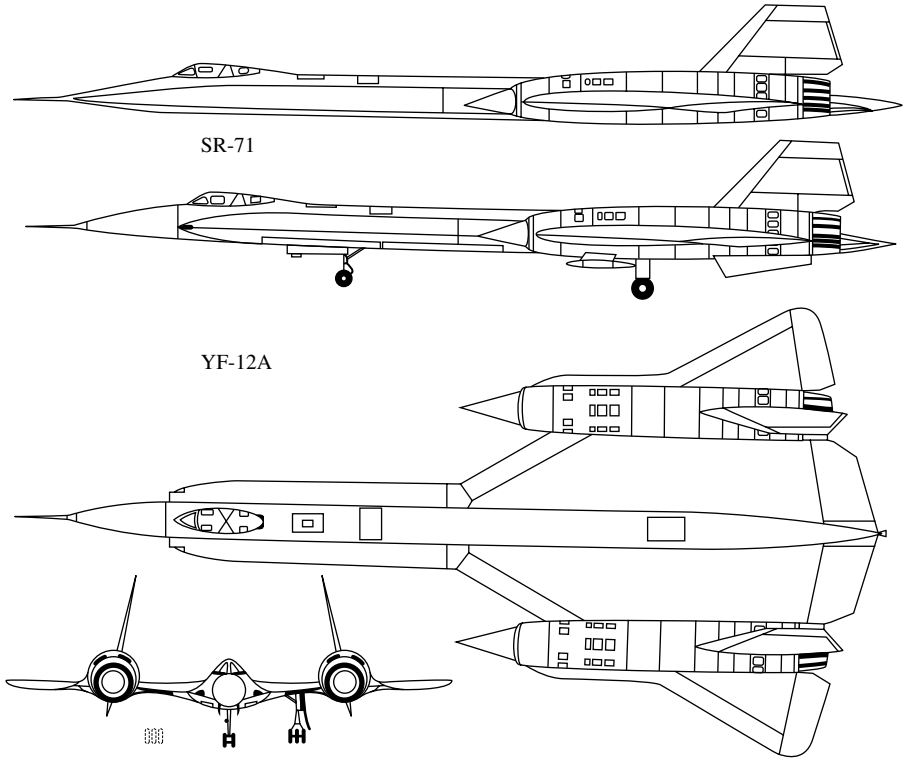


Figure 8.11 The Lockheed SR-71/YF-12A Blackbird.

■ Solution

From Appendix D, at an altitude of 25 km, $p = 2.5273 \times 10^3$ N/m^2 and $T = 216.66$ K. Hence,

$$\frac{p_{0,1}}{p_1} = \frac{3.88 \times 10^4}{2.5273 \times 10^3} = 15.35$$

From Appendix B, for $p_{0,1}/p_1 = 15.35$, $M_1 = 3.4$:

$$a_1 = \sqrt{\gamma RT} = \sqrt{(1.4)(287)(216.66)} = 295 \text{ m/s}.$$

Thus, the velocity of the airplane is

$$V_1 = M_1 a_1 = (3.4)(295) = \boxed{1003 \text{ m/s}}$$

8.8 SUMMARY

Return to the road map given in Figure 8.2, and make certain that you are comfortable with the areas we have covered in this chapter. A brief summary of the more important relations follows.

The speed of sound in a gas is given by

$$a = \sqrt{\left(\frac{\partial p}{\partial \rho}\right)_s} \tag{8.18}$$

For a calorically perfect gas,

$$a = \sqrt{\frac{\gamma p}{\rho}} \tag{8.23}$$

or

$$a = \sqrt{\gamma RT} \tag{8.25}$$

The speed of sound depends only on the gas temperature.

For a steady, adiabatic, inviscid flow, the energy equation can be expressed as

$$h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2} \tag{8.29}$$

$$c_p T_1 + \frac{u_1^2}{2} = c_p T_2 + \frac{u_2^2}{2} \tag{8.30}$$

(*continued*)

$$\frac{a_1^2}{\gamma-1}+\frac{u_1^2}{2}=\frac{a_2^2}{\gamma-1}+\frac{u_2^2}{2} \tag{8.32}$$

$$\frac{a^2}{\gamma-1}+\frac{u^2}{2}=\frac{a_0^2}{\gamma-1} \tag{8.33}$$

$$\frac{a^2}{\gamma-1}+\frac{u^2}{2}=\frac{\gamma+1}{2(\gamma-1)}a^{*2} \tag{8.35}$$

Total conditions in a flow are related to static conditions via

$$c_pT+\frac{u^2}{2}=c_pT_0 \tag{8.38}$$

$$\frac{T_0}{T}=1+\frac{\gamma-1}{2}M^2 \tag{8.40}$$

$$\frac{p_0}{p}=\left(1+\frac{\gamma-1}{2}M^2\right)^{\gamma/(\gamma-1)} \tag{8.42}$$

$$\frac{\rho_0}{\rho}=\left(1+\frac{\gamma-1}{2}M^2\right)^{1/(\gamma-1)} \tag{8.43}$$

Note that the ratios of total to static properties are a function of local Mach number only. These functions are tabulated in Appendix A.

The basic normal shock equations are

Continuity: $$\rho_1u_1=\rho_2u_2 \tag{8.2}$$

Momentum: $$p_1+\rho_1u_1^2=p_2+\rho_2u_2^2 \tag{8.6}$$

Energy: $$h_1+\frac{u_1^2}{2}=h_2+\frac{u_2^2}{2} \tag{8.10}$$

These equations lead to relations for changes across a normal shock as a function of upstream Mach number M_1 only:

$$M_2^2=\frac{1+[(\gamma-1)/2]M_1^2}{\gamma M_1^2-(\gamma-1)/2} \tag{8.59}$$

(*continued*)

$$\frac{\rho_2}{\rho_1} = \frac{u_1}{u_2} = \frac{(\gamma+1)M_1^2}{2+(\gamma-1)M_1^2} \tag{8.61}$$

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma+1}(M_1^2 - 1) \tag{8.65}$$

$$\frac{T_2}{T_1} = \frac{h_2}{h_1} = \left[1 + \frac{2\gamma}{\gamma+1}(M_1^2-1)\right]\frac{2+(\gamma-1)M_1^2}{(\gamma+1)M_1^2} \tag{8.67}$$

$$s_2 - s_1 = c_p \ln\left\{\left[1 + \frac{2\gamma}{\gamma+1}(M_1^2-1)\right]\frac{2+(\gamma-1)M_1^2}{(\gamma+1)M_1^2}\right\} - R\ln\left[1 + \frac{2\gamma}{\gamma+1}(M_1^2-1)\right] \tag{8.68}$$

$$\frac{p_{0,2}}{p_{0,1}} = e^{-(s_2-s_1)/R} \tag{8.73}$$

The normal shock properties are tabulated versus M_1 in Appendix B.

For a calorically perfect gas, the total temperature is constant across a normal shock wave:

$$T_{0,2} = T_{0,1}$$

However, there is a loss in total pressure across the wave:

$$p_{0,2} < p_{0,1}$$

For subsonic and supersonic compressible flow, the freestream Mach number is determined by the ratio of Pitot pressure to freestream static pressure. However, the equations are different:

Subsonic flow:
$$M_1^2 = \frac{2}{\gamma-1}\left[\left(\frac{p_{0,1}}{p_1}\right)^{(\gamma-1)/\gamma} - 1\right] \tag{8.74}$$

Supersonic flow:
$$\frac{p_{0,2}}{p_1} = \left[\frac{(\gamma+1)^2 M_1^2}{4\gamma M_1^2 - 2(\gamma-1)}\right]^{\gamma/(\gamma-1)} \frac{1-\gamma+2\gamma M_1^2}{\gamma+1} \tag{8.80}$$

8.9 PROBLEMS

8.1 Consider air at a temperature of 230 K. Calculate the speed of sound.

8.2 The temperature in the reservoir of a supersonic wind tunnel is 519°R. In the test section, the flow velocity is 1385 ft/s. Calculate the test-section Mach number. Assume the tunnel flow is adiabatic.

8.3 At a given point in a flow, $T = 300$ K, $p = 1.2$ atm, and $V = 250$ m/s. At this point, calculate the corresponding values of p_0, T_0, p^*, T^*, and M^*.

8.4 At a given point in a flow, $T = 700$°R, $p = 1.6$ atm, and $V = 2983$ ft/s. At this point, calculate the corresponding values of p_0, T_0, p^*, T^*, and M^*.

8.5 Consider the isentropic flow through a supersonic nozzle. If the test-section conditions are given by $p = 1$ atm, $T = 230$ K, and $M = 2$, calculate the reservoir pressure and temperature.

8.6 Consider the isentropic flow over an airfoil. The freestream conditions correspond to a standard altitude of 10,000 ft and $M_\infty = 0.82$. At a given point on the airfoil, $M = 1.0$. Calculate p and T at this point. (*Note:* You will have to use the standard atmosphere table in Appendix E for this problem.)

8.7 The flow just upstream of a normal shock wave is given by $p_1 = 1$ atm, $T_1 = 288$ K, and $M_1 = 2.6$. Calculate the following properties just downstream of the shock: p_2, T_2, ρ_2, M_2, $p_{0,2}$, $T_{0,2}$, and the change in entropy across the shock.

8.8 The pressure upstream of a normal shock wave is 1 atm. The pressure and temperature downstream of the wave are 10.33 atm and 1390°R, respectively. Calculate the Mach number and temperature upstream of the wave and the total temperature and total pressure downstream of the wave.

8.9 The entropy increase across a normal shock wave is 199.5 J/(kg · K). What is the upstream Mach number?

8.10 The flow just upstream of a normal shock wave is given by $p_1 = 1800$ lb/ft^2, $T_1 = 480$°R, and $M_1 = 3.1$. Calculate the velocity and M^* behind the shock.

8.11 Consider a flow with a pressure and temperature of 1 atm and 288 K. A Pitot tube is inserted into this flow and measures a pressure of 1.555 atm. What is the velocity of the flow?

8.12 Consider a flow with a pressure and temperature of 2116 lb/ft^2 and 519°R, respectively. A Pitot tube is inserted into this flow and measures a pressure of 7712.8 lb/ft^2. What is the velocity of this flow?

8.13 Repeat Problems 8.11 and 8.12 using (incorrectly) Bernoulli's equation for incompressible flow. Calculate the percent error induced by using Bernoulli's equation.

8.14 Derive the Rayleigh Pitot tube formula, Equation (8.80).

8.15 On March 16, 1990, an Air Force SR-71 set a new continental speed record, averaging a velocity of 2112 mi/h at an altitude of 80,000 ft.

Calculate the temperature (in degrees Fahrenheit) at a stagnation point on the vehicle.

8.16 In the test section of a supersonic wind tunnel, a Pitot tube in the flow reads a pressure of 1.13 atm. A static pressure measurement (from a pressure tap on the sidewall of the test section) yields 0.1 atm. Calculate the Mach number of the flow in the test section.

8.17 When the Apollo command module returned to earth from the moon, it entered the earth's atmosphere at a Mach number of 36. Using the results from the present chapter for a calorically perfect gas with the ratio of specific heats equal to 1.4, predict the gas temperature at the stagnation point of the Apollo at Mach 36 at an altitude where the freestream temperature is 300 K. Comment on the validity of your answer.

8.18 The stagnation temperature on the Apollo vehicle at Mach 36 as it entered the atmosphere was 11,000 K, a much different value than predicted in Problem 8.17 for the case of a calorically perfect gas with a ratio of specific heats equal to 1.4. The difference is due to chemical reactions that occur in air at these high temperatures—dissociation and ionization. The analyses in this book assuming a calorically perfect gas with constant specific heats are not valid for such chemically reacting flows. However, as an engineering approximation, the calorically perfect gas results are sometimes applied with a lower value of the ratio of specific heats, a so-called "effective gamma," in order to try to simulate the effects of high temperature chemically reacting flows. For the condition stated in this problem, calculate the value of the effective gamma necessary to yield a temperature of 11,000 K at the stagnation point. Assume the freestream temperature is 300 K.

8.19 Prove that the total pressure is constant throughout an isentropic flow.

CHAPTER 9

Oblique Shock and Expansion Waves

In the case of air (and the same is true for all gases) the shock wave is extremely thin so that calculations based on one-dimensional flow are still applicable for determining the changes in velocity and density on passing through it, even when the rest of the flow system is not limited to one dimension, provided that only the velocity component normal to the wave is considered.

G. I. Taylor and J. W. Maccoll, 1934

PREVIEW BOX

Take a look at Figure 9.1. What you see is the computed wave pattern—both shock and expansion waves—generated by a generic supersonic transport configuration flying at Mach 1.7 at an altitude of 15 km. All these waves are *oblique* to the flow, in contrast to the normal shock waves discussed in Chapter 8. The present chapter is all about *oblique* shock and expansion waves.

The material in this chapter is vital to a fundamental understanding of supersonic flow. Moreover, it is vital to you if you are interested in designing an economically feasible and environmentally acceptable supersonic transport. The shock waves in Figure 9.1 create the major source of drag (wave drag) on the airplane, and the waves, when they propagate to the ground, cause the much discussed "sonic boom." The material in this chapter is also vital to you if you are interested in designing SCRAMjet engines for hypersonic airplanes. The performance of such engines depends in part on the nature of the oblique wave patterns both upstream of and inside the engine. The material in this chapter is vital to you if you are interested in designing supersonic and hypersonic wind tunnels, where the oblique wave patterns created by models in the tunnel and in the diffuser downstream of the models affect the performance of the tunnel. In fact, the material in this chapter is vital to a whole host of applications in supersonic flow.

By now you get the message—the material in this chapter is simply vital to your study of supersonic flows. It is the bread and butter of such flows. So lay out the bread, spread it with butter, and consume this chapter. And on top of everything else, learning about oblique shock and expansion waves is exciting. I predict that you are going to enjoy this.

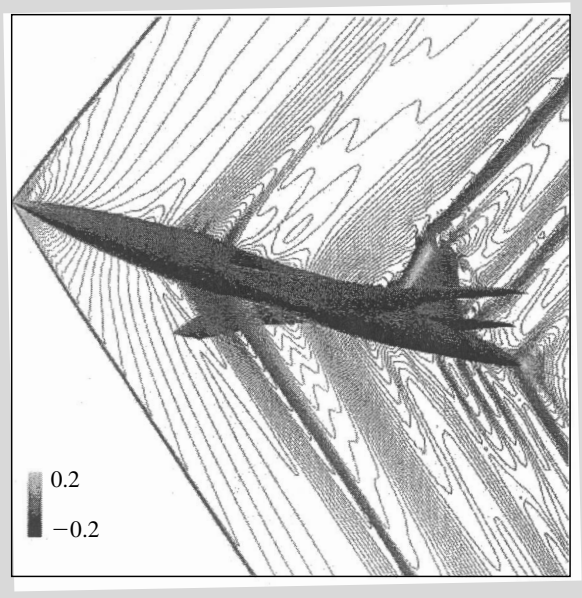
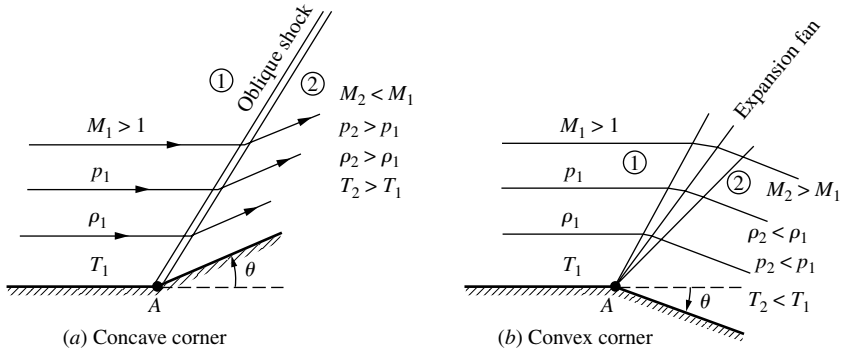


Figure 9.1 Wave pattern on a supersonic transport configuration (without nacelles) at Mach 1.7. Computational fluid dynamic calculations by Y. Makino, et al., "Nonaxisymmetrical Fuselage Shape Modification for Drag Reduction of Low-Sonic-Boom Airplane," *AIAA Journal,* vol. 41, no. 8, August 2003, p. 1415.

9.1 INTRODUCTION

In Chapter 8, we discussed normal shock waves, that is, shock waves that make an angle of 90° with the upstream flow. The behavior of normal shock waves is important; moreover, the study of normal shock waves provides a relatively straightforward introduction to shock-wave phenomena. However, examining Figure 7.5 and the photographs shown in Figure 7.6, we see that, in general, a shock wave will make an oblique angle with respect to the upstream flow. These are called *oblique shock waves* and are the subject of part of this chapter. A normal shock wave is simply a special case of the general family of oblique shocks, namely, the case where the wave angle is 90°.

In addition to oblique shock waves, where the pressure increases discontinuously across the wave, supersonic flows are also characterized by oblique expansion waves, where the pressure *decreases continuously* across the wave. Let us examine these two types of waves further. Consider a supersonic flow over a wall with a corner at point A, as sketched in Figure 9.2. In Figure 9.2*a*, the



(*a*) Concave corner (*b*) Convex corner

Figure 9.2 Supersonic flow over a corner.

wall is turned upward at the corner through the deflection angle θ; that is, the corner is concave. The flow at the wall must be tangent to the wall; hence, the streamline at the wall is also deflected upward through the angle θ. The bulk of the gas is above the wall, and in Figure 9.2*a*, the streamlines are turned upward, *into* the main bulk of the flow. Whenever a supersonic flow is "turned into itself" as shown in Figure 9.2*a*, an oblique shock wave will occur. The originally horizontal streamlines ahead of the wave are uniformly deflected in crossing the wave, such that the streamlines behind the wave are parallel to each other and inclined upward at the deflection angle θ. Across the wave, the Mach number discontinuously decreases, and the pressure, density, and temperature discontinuously increase. In contrast, Figure 9.2*b* shows the case where the wall is turned downward at the corner through the deflection angle θ; that is, the corner is convex. Again, the flow at the wall must be tangent to the wall; hence, the streamline at the wall is deflected downward through the angle θ. The bulk of the gas is above the wall, and in Figure 9.2*b*, the streamlines are turned downward, *away from* the main bulk of the flow. Whenever a supersonic flow is "turned away from itself" as shown in Figure 9.2*b*, an expansion wave will occur. This expansion wave is in the shape of a fan centered at the corner. The fan continuously opens in the direction away from the corner, as shown in Figure 9.2*b*. The originally horizontal streamlines ahead of the expansion wave are deflected smoothly and continuously through the expansion fan such that the streamlines behind the wave are parallel to each other and inclined downward at the deflection angle θ. Across the expansion wave, the Mach number increases, and the pressure, temperature, and density decrease. Hence, an expansion wave is the direct antithesis of a shock wave.

Oblique shock and expansion waves are prevalent in two- and three-dimensional supersonic flows. These waves are inherently two-dimensional in nature, in contrast to the one-dimensional normal shock waves discussed in Chapter 8. That is, in Figure 9.2*a* and *b*, the flow-field properties are a function of x and y. The purpose of the present chapter is to determine and study the properties of these oblique waves.

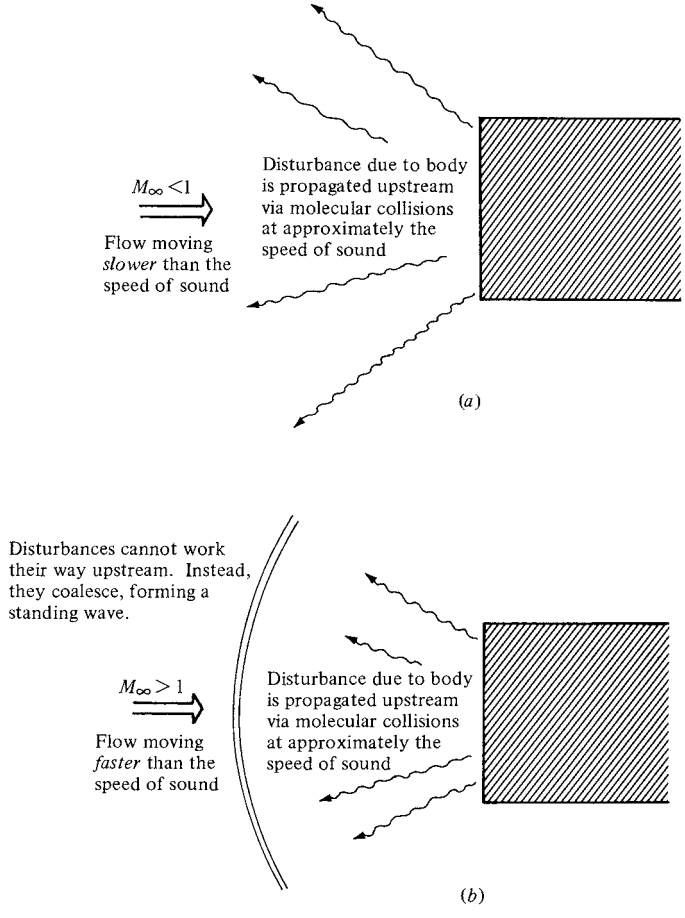


Figure 9.3 Propagation of disturbances. (*a*) Subsonic flow. (*b*) Supersonic flow.

What is the physical mechanism that creates waves in a supersonic flow? To address this question, recall our picture of the propagation of a sound wave via molecular collisions, as portrayed in Section 8.3. If a slight disturbance takes place at some point in a gas, information is transmitted to other points in the gas by sound waves which propagate in all directions away from the source of the disturbance. Now consider a body in a flow, as sketched in Figure 9.3. The gas molecules which impact the body surface experience a change in momentum. In turn, this change is transmitted to neighboring molecules by random molecular collisions. In this fashion, information about the presence of the body attempts to be transmitted to the surrounding flow via molecular collisions; that is, the information is propagated upstream at approximately the local speed of sound. If the upstream flow is subsonic, as shown in Figure 9.3*a*, the disturbances have no

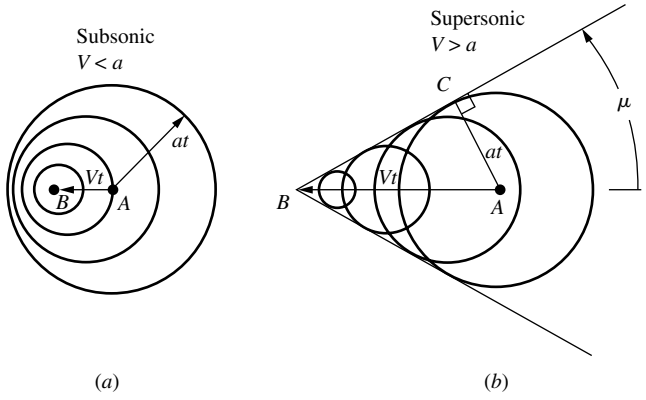


Figure 9.4 Another way of visualizing the propagation of disturbances in (*a*) subsonic and (*b*) supersonic flow.

problem working their way far upstream, thus giving the incoming flow plenty of time to move out of the way of the body. On the other hand, if the upstream flow is supersonic, as shown in Figure 9.3*b*, the disturbances cannot work their way upstream; rather, at some finite distance from the body, the disturbance waves pile up and coalesce, forming a standing wave in front of the body. Hence, the physical generation of waves in a supersonic flow—both shock and expansion waves—is due to the propagation of information via molecular collisions and due to the fact that such propagation cannot work its way into certain regions of the supersonic flow.

Why are most waves oblique rather than normal to the upstream flow? To answer this question, consider a small source of disturbance moving through a stagnant gas. For lack of anything better, let us call this disturbance source a "beeper," which periodically emits sound. First, consider the beeper moving at *subsonic* speed through the gas, as shown in Figure 9.4*a*. The speed of the beeper is V, where $V < a$. At time $t = 0$, the beeper is located at point A; at this point, it emits a sound wave that propagates in all directions at the speed of sound, a. At a later time t this sound wave has propagated a distance at from point A and is represented by the circle of radius at shown in Figure 9.4*a*. During the same time, the beeper has moved a distance Vt and is now at point B in Figure 9.4*a*. Moreover, during its transit from A to B, the beeper has emitted several other sound waves, which at time t are represented by the smaller circles in Figure 9.4*a*. Note that the beeper always stays *inside* the family of circular sound waves and that the waves continuously move ahead of the beeper. This is because the beeper is traveling at a subsonic speed $V < a$. In contrast, consider the beeper moving at a *supersonic* speed $V > a$ through the gas, as shown in Figure 9.4*b*. At time $t = 0$, the beeper is located at point A, where it emits a sound wave. At a later time t this sound wave has propagated a distance at from point A and is represented by the circle of radius at shown in Figure 9.4*b*. During the same time, the beeper

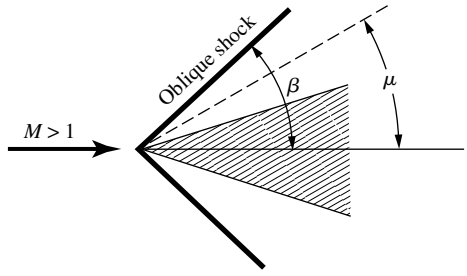


Figure 9.5 Relation between the oblique shock-wave angle and the Mach angle.

has moved a distance Vt to point B. Moreover, during its transit from A to B, the beeper has emitted several other sound waves, which at time t are represented by the smaller circles in Figure 9.4*b*. However, in contrast to the subsonic case, the beeper is now constantly *outside* the family of circular sound waves; that is, it is moving ahead of the wave fronts because $V > a$. Moreover, something new is happening; these wave fronts form a disturbance envelope given by the straight line BC, which is tangent to the family of circles. This line of disturbances is defined as a *Mach wave*. In addition, the angle ABC that the Mach wave makes with respect to the direction of motion of the beeper is defined as the *Mach angle* μ. From the geometry of Figure 9.4*b*, we readily find that

$$\sin \mu = \frac{at}{Vt} = \frac{a}{V} = \frac{1}{M}$$

Thus, the Mach angle is simply determined by the local Mach number as

$$\boxed{\mu = \sin^{-1} \frac{1}{M}} \tag{9.1}$$

Examining Figure 9.4*b*, the Mach wave, that is, the envelope of disturbances in the supersonic flow, is clearly *oblique* to the direction of motion. If the disturbances are stronger than a simple sound wave, then the wave front becomes stronger than a Mach wave, creating an oblique shock wave at an angle β to the freestream, where $\beta > \mu$. This comparison is shown in Figure 9.5. However, the physical mechanism creating the oblique shock is essentially the same as that described above for the Mach wave. Indeed, a Mach wave is a limiting case for oblique shock (i.e., it is an infinitely weak oblique shock).

This finishes our discussion of the physical source of oblique waves in a supersonic flow. Let us now proceed to develop the equations that allow us to calculate the change in properties across these oblique waves, first for oblique shock waves, and then for expansion waves. In the process, we follow the road map given in Figure 9.6.

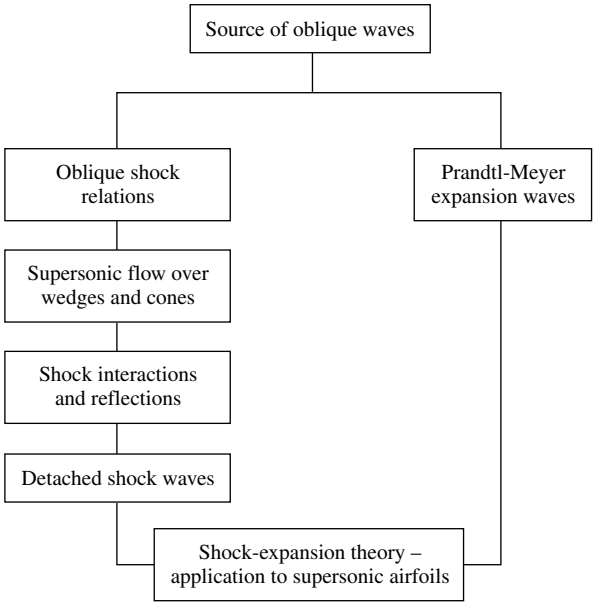


Figure 9.6 Road map for Chapter 9.

EXAMPLE 9.1

A supersonic airplane is flying at Mach 2 at an altitude of 16 km. Assume the shock wave pattern from the airplane (see Figure 9.1) quickly coalesces into a Mach wave that intersects the ground behind the airplane, causing a "sonic boom" to be heard by a bystander on the ground. At the instant the sonic boom is heard, how far ahead of the bystander is the airplane?

■ Solution

Examine Figure 9.7, which shows the airplane at an altitude of 16 km with a Mach wave trailing behind it. The Mach wave intersects the ground at a ground distance d from the airplane. From Equation (9.1),

$$\mu = \sin^{-1}\left(\frac{1}{M}\right) = \sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$$

From Figure 9.7,

$$\tan\mu = \frac{16 \text{ km}}{d}$$

or,

$$d = \frac{16 \text{ km}}{\tan\mu} = \frac{16}{0.577} = \boxed{27.7 \text{ km}}$$

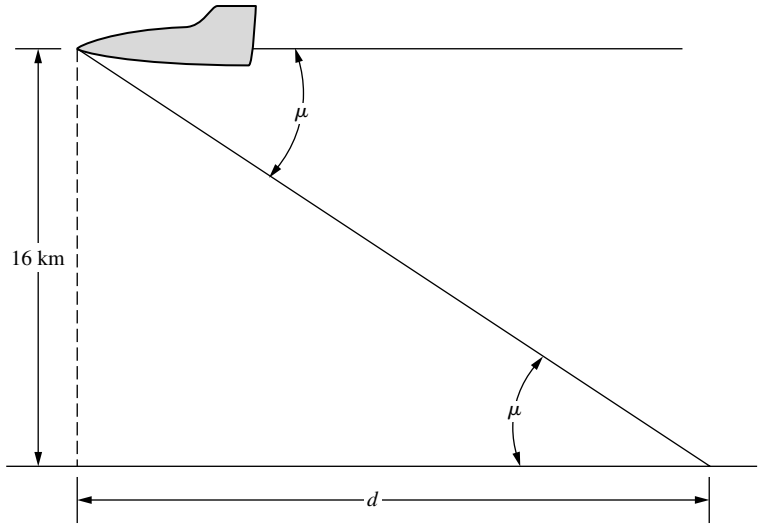


Figure 9.7 Mach wave from a supersonic vehicle, and its impingement on the ground.

9.2 OBLIQUE SHOCK RELATIONS

Consider the oblique shock wave sketched in Figure 9.8. The angle between the shock wave and the upstream flow direction is defined as the *wave angle,* denoted by β. The upstream flow (region 1) is horizontal, with a velocity V_1 and Mach number M_1. The downstream flow (region 2) is inclined upward through the deflection angle θ and has velocity V_2 and Mach number M_2. The upstream velocity V_1 is split into components tangential and normal to the shock wave, w_1 and u_1, respectively, with the associated tangential and normal Mach numbers $M_{t,1}$ and $M_{n,1}$, respectively. Similarly, the downstream velocity is split into tangential and normal components w_2 and u_2, respectively, with the associated Mach numbers $M_{t,2}$ and $M_{n,2}$.

Consider the control volume shown by the dashed lines in the upper part of Figure 9.8. Sides a and d are parallel to the shock wave. Segments b and c follow the upper streamline, and segments e and f follow the lower streamline. Let us apply the integral form of the conservation equations to this control volume, keeping in mind that we are dealing with a steady, inviscid, adiabatic flow with no body forces. For these assumptions, the continuity equation, Equation (2.48), becomes

$$\oiint_S \rho \mathbf{V} \cdot \mathbf{dS} = 0$$

This surface integral evaluated over faces a and d yields $-\rho_1 u_1 A_1 + \rho_2 u_2 A_2$, where $A_1 = A_2 =$ area of faces a and d. The faces b, c, e, and f are parallel to

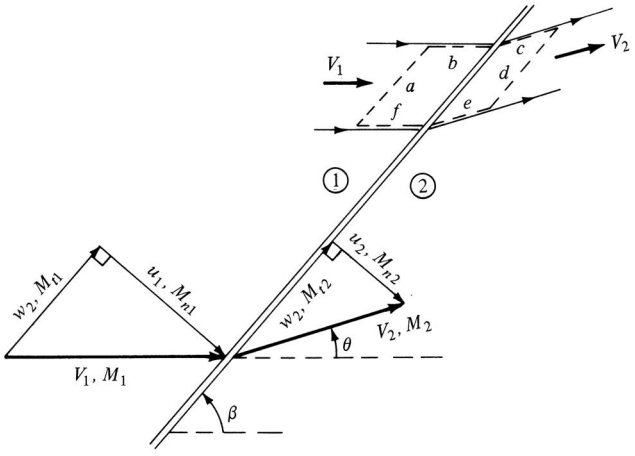


Figure 9.8 Oblique shock geometry.

the velocity, and hence contribute nothing to the surface integral (i.e., $\mathbf{V} \cdot \mathbf{dS} = 0$ for these faces). Thus, the continuity equation for an oblique shock wave is

$$-\rho_1 u_1 A_1 + \rho_2 u_2 A_2 = 0$$

or

$$\boxed{\rho_1 u_1 = \rho_2 u_2} \tag{9.2}$$

Keep in mind that u_1 and u_2 in Equation (9.2) are *normal* to the shock wave.

The integral form of the momentum equation, Equation (2.64), is a vector equation. Hence, it can be resolved into two components, tangential and normal to the shock wave. First, consider the tangential component, keeping in mind the type of flow we are considering:

$$\oiint_S (\rho \mathbf{V} \cdot \mathbf{dS}) w = -\oiint_S (p\, dS)_{\text{tangential}} \tag{9.3}$$

In Equation (9.3), w is the component of velocity tangential to the wave. Since **dS** is perpendicular to the control surface, then $(p\, dS)_{\text{tangential}}$ over faces a and d is zero. Also, since the vectors $p\,\mathbf{dS}$ on faces b and f are equal and opposite, the pressure integral in Equation (9.3) involves two tangential forces that cancel each other over faces b and f. The same is true for faces c and e. Hence, Equation (9.3) becomes

$$-(\rho_1 u_1 A_1) w_1 + (\rho_2 u_2 A_2) w_2 = 0 \tag{9.4}$$

Dividing Equation (9.4) by Equation (9.2), we have

$$\boxed{w_1 = w_2} \tag{9.5}$$

Equation (9.5) is an important result; it states that the *tangential component of the flow velocity is constant across an oblique shock.*

The normal component of the integral momentum equation is, from Equation (2.64),

$$\oiint_S (\rho \mathbf{V} \cdot \mathbf{dS})u = -\oiint_S (p\, dS)_{\text{normal}} \tag{9.6}$$

Here, the pressure integral evaluated over faces a and d yields the net sum $-p_1A_1 + p_2A_2$. Once again, the equal and opposite pressure forces on b and f cancel, as do those on c and e. Hence, Equation (9.6) becomes, for the control volume shown in Figure 9.8,

$$-(\rho_1 u_1 A_1)u_1 + (\rho_2 u_2 A_2)u_2 = -(-p_1A_1 + p_2A_2)$$

Since $A_1 = A_2$, this becomes

$$\boxed{p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2} \tag{9.7}$$

Again, note that the only velocities appearing in Equation (9.7) are the components *normal* to the shock.

Finally, consider the integral form of the energy equation, Equation (2.95). For our present case, this can be written as

$$\oiint_S \rho \left(e + \frac{V^2}{2}\right) \mathbf{V} \cdot \mathbf{dS} = -\oiint_S p\mathbf{V} \cdot \mathbf{dS} \tag{9.8}$$

Again noting that the flow is tangent to faces b, c, f, and e, and hence $\mathbf{V} \cdot \mathbf{dS} = 0$ on these faces, Equation (9.8) becomes, for the control volume in Figure 9.6,

$$-\rho_1 \left(e_1 + \frac{V_1^2}{2}\right) u_1 A_1 + \rho_2 \left(e_2 + \frac{V_2^2}{2}\right) u_2 A_2 = -(-p_1 u_1 A_1 + p_2 u_2 A_2) \tag{9.9}$$

Collecting terms in Equation (9.9), we have

$$-\rho_1 u_1 \left(e_1 + \frac{p_1}{\rho_1} + \frac{V_1^2}{2}\right) + \rho_2 u_2 \left(e_2 + \frac{p_2}{\rho_2} + \frac{V_2^2}{2}\right) = 0$$

or

$$\rho_1 u_1 \left(h_1 + \frac{V_1^2}{2}\right) = \rho_2 u_2 \left(h_2 + \frac{V_2^2}{2}\right) \tag{9.10}$$

Dividing Equation (9.10) by (9.2), we have

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \tag{9.11}$$

Since $h + V^2/2 = h_0$, we have again the familiar result that the *total enthalpy is constant across the shock wave*. Moreover, for a calorically perfect gas, $h_0 = c_p T_0$; hence, the *total temperature is constant across the shock wave*. Carrying Equation (9.11) a bit further, note from Figure 9.8 that $V^2 = u^2 + w^2$. Also, from Equation (9.5), we know that $w_1 = w_2$. Hence,

$$V_1^2 - V_2^2 = (u_1^2 + w_1^2) - (u_2^2 + w_2^2) = u_1^2 - u_2^2$$

Thus, Equation (9.11) becomes

$$\boxed{h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2}} \tag{9.12}$$

Let us now gather our results. Look carefully at Equations (9.2), (9.7), and (9.12). They are the continuity, normal momentum, and energy equations, respectively, for an oblique shock wave. Note that they involve the *normal components only* of velocity u_1 and u_2; the tangential component w does not appear in these equations. Hence, we deduce that *changes across an oblique shock wave are governed only by the component of velocity normal to the wave*.

Again, look hard at Equations (9.2), (9.7), and (9.12). *They are precisely the governing equations for a normal shock wave,* as given by Equations (8.2), (8.6), and (8.10). Hence, precisely the same algebra as applied to the normal shock equations in Section 8.6, when applied to Equations (9.2), (9.7), and (9.12), will lead to identical expressions for changes across an oblique shock in terms of the normal component of the upstream Mach number $M_{n,1}$. Note that

$$M_{n,1} = M_1 \sin \beta \tag{9.13}$$

Hence, for an oblique shock wave, with $M_{n,1}$ given by Equation (9.13), we have, from Equations (8.59), (8.61), and (8.65),

$$M_{n,2}^2 = \frac{1 + [(\gamma - 1)/2]M_{n,1}^2}{\gamma M_{n,1}^2 - (\gamma - 1)/2} \tag{9.14}$$

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1)M_{n,1}^2}{2 + (\gamma - 1)M_{n,1}^2} \tag{9.15}$$

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1}(M_{n,1}^2 - 1) \tag{9.16}$$

The temperature ratio T_2/T_1 follows from the equation of state:

$$\frac{T_2}{T_1} = \frac{p_2}{p_1}\frac{\rho_1}{\rho_2} \tag{9.17}$$

Note that $M_{n,2}$ is the *normal* Mach number behind the shock wave. The downstream Mach number itself, M_2, can be found from $M_{n,2}$ and the geometry of Figure 9.8 as

$$M_2 = \frac{M_{n,2}}{\sin(\beta - \theta)} \tag{9.18}$$

Examine Equations (9.14) to (9.17). They state that oblique shock-wave properties in a calorically perfect gas depend only on the normal component of the upstream Mach number $M_{n,1}$. However, note from Equation (9.13) that $M_{n,1}$ depends on both M_1 and β. Recall from Section 8.6 that changes across a normal shock wave depend on one parameter only—the upstream Mach number M_1.

In contrast, we now see that changes across an oblique shock wave depend on two parameters—say, M_1 and β. However, this distinction is slightly moot because in reality a normal shock wave is a special case of oblique shocks where $\beta = \pi/2$.

Equation (9.18) introduces the deflection angle θ into our oblique shock analysis; we need θ to be able to calculate M_2. However, θ is not an independent, third parameter; rather, θ is a function of M_1 and β, as derived below. From the geometry of Figure 9.8,

$$\tan \beta = \frac{u_1}{w_1} \tag{9.19}$$

and

$$\tan(\beta - \theta) = \frac{u_2}{w_2} \tag{9.20}$$

Dividing Equation (9.20) by (9.19), recalling that $w_1 = w_2$, and invoking the continuity equation, Equation (9.2), we obtain

$$\frac{\tan(\beta - \theta)}{\tan \beta} = \frac{u_2}{u_1} = \frac{\rho_1}{\rho_2} \tag{9.21}$$

Combining Equation (9.21) with Equations (9.13) and (9.15), we obtain

$$\frac{\tan(\beta - \theta)}{\tan \beta} = \frac{2 + (\gamma - 1)M_1^2 \sin^2 \beta}{(\gamma + 1)M_1^2 \sin^2 \beta} \tag{9.22}$$

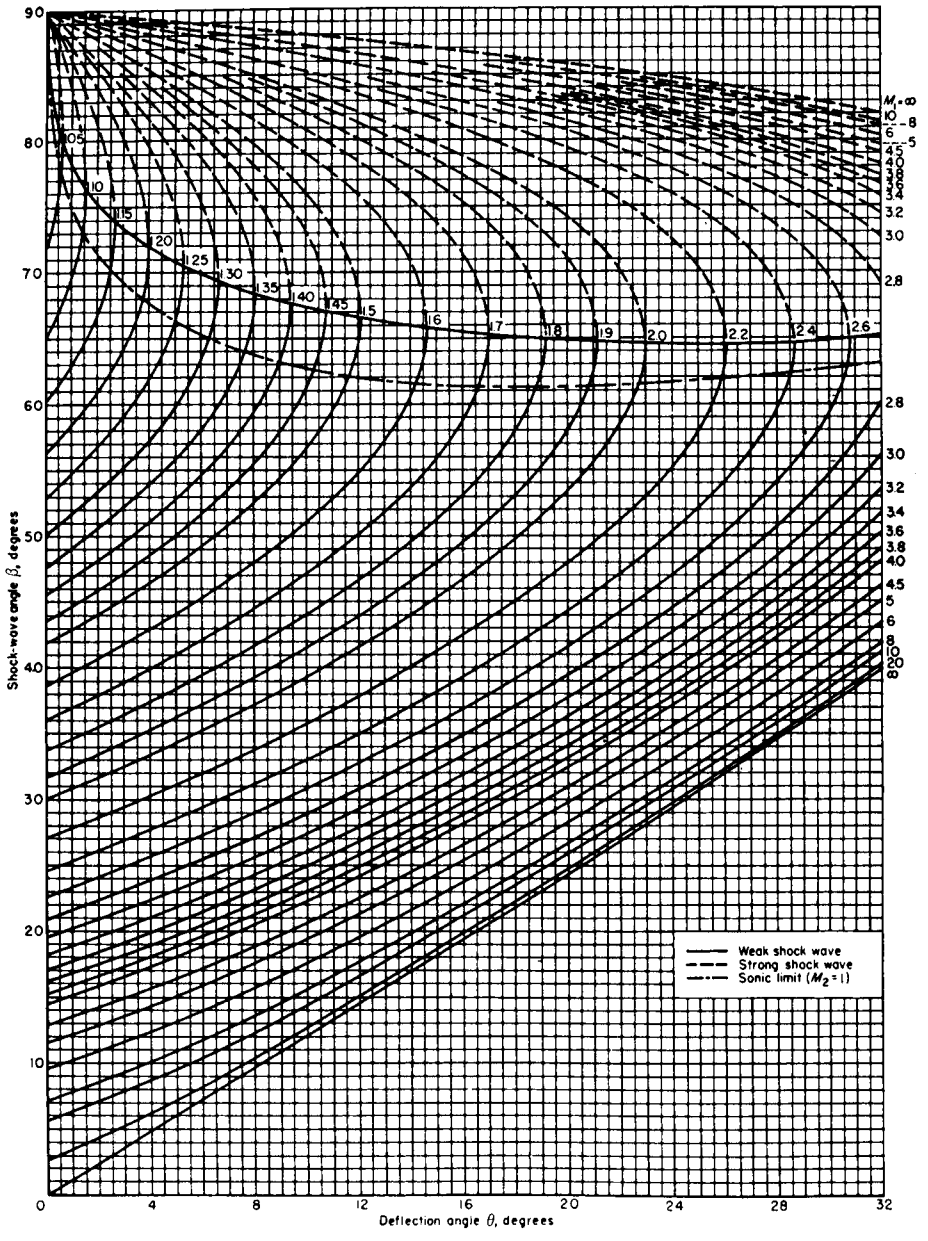
which gives θ as an implicit function of M_1 and β. After some trigonometric substitutions and rearrangement, Equation (9.22) can be cast explicitly for θ as

$$\boxed{\tan \theta = 2 \cot \beta \frac{M_1^2 \sin^2 \beta - 1}{M_1^2(\gamma + \cos 2\beta) + 2}} \tag{9.23}$$

Equation (9.23) is an important equation. It is called the θ-β-M relation, and it specifies θ as a unique function of M_1 and β. This relation is vital to the analysis of oblique shock waves, and results from it are plotted in Figure 9.9 for $\gamma = 1.4$. Examine this figure closely. It is a plot of wave angle versus deflection angle, with the Mach number as a parameter. The results given in Figure 9.9 are plotted in some detail—this is a chart which you will need to use for solving oblique shock problems.

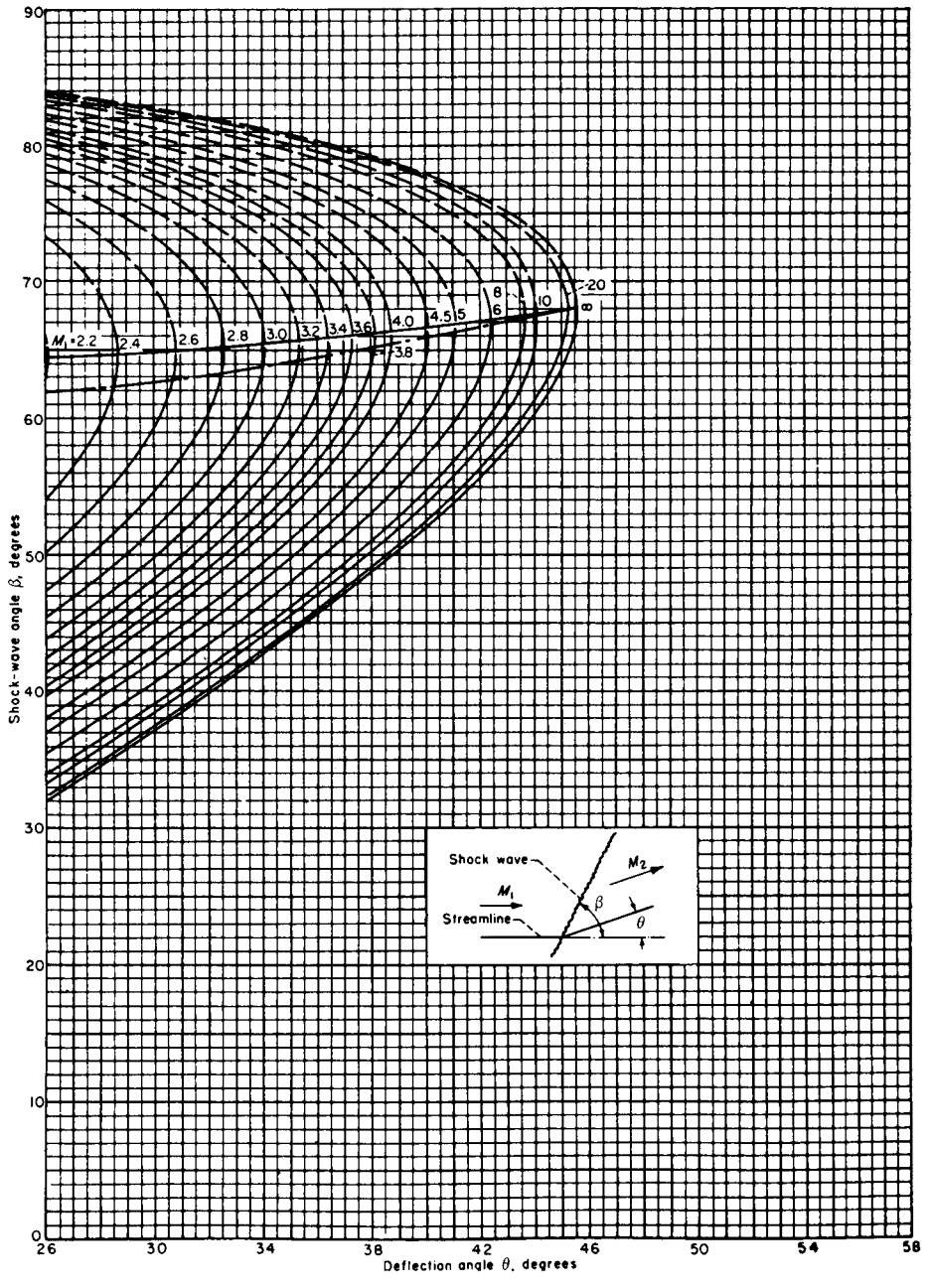
Figure 9.9 illustrates a wealth of physical phenomena associated with oblique shock waves. For example:

1. For any given upstream Mach number M_1, there is a maximum deflection angle θ_{max}. If the physical geometry is such that $\theta > \theta_{max}$, then no solution exists for a *straight* oblique shock wave. Instead, nature establishes a curved shock wave, detached from the corner or the nose of a body. This is illustrated in Figure 9.10. Here, the left side of the figure illustrates flow over a wedge and a concave corner where the deflection angle is less than θ_{max} for the given upstream Mach number. Therefore, we see a straight oblique shock wave attached to the nose of the wedge and to the corner. The right side of Figure 9.10 gives the case where the deflection angle is greater



(*a*)

Figure 9.9 Oblique shock properties: $\gamma = 1.4$. The θ-β-M diagram. (*Source: NACA Report 1135, Ames Research Staff, "Equations, Tables and Charts for Compressible Flow," 1953.*)



(*b*)

Figure 9.9 (continued) (*Source: NACA Report 1135, Ames Research Staff, "Equations, Tables and Charts for Compressible Flow," 1953.*)

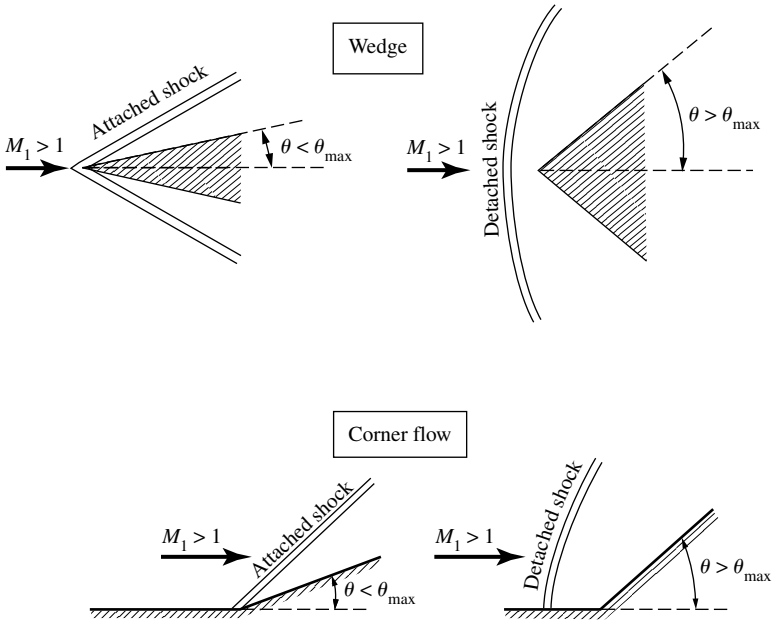


Figure 9.10 Attached and detached shocks.

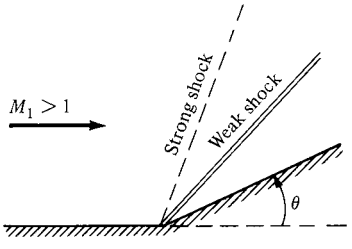


Figure 9.11 The weak and strong shock cases.

than θ_{max}; hence, there is no allowable straight oblique shock solution from the theory developed earlier in this section. Instead, we have a curved shock wave detached from the nose of the wedge or from the corner. Return to Figure 9.9, and note that the value of θ_{max} increases with increasing M_1. Hence, at higher Mach numbers, the straight oblique shock solution can exist at higher deflection angles. However, there is a limit; as M_1 approaches infinity, θ_{max} approaches 45.5° (for $\gamma = 1.4$).

2. For any given θ *less than* θ_{max}, there are *two* straight oblique shock solutions for a given upstream Mach number. For example, if $M_1 = 2.0$ and $\theta = 15°$, then from Figure 9.9, β can equal either 45.3 or 79.8°. The smaller value of β is called the *weak* shock solution, and the larger value of β is the *strong* shock solution. These two cases are illustrated in Figure 9.11.

The classifications "weak" and "strong" derive from the fact that for a given M_1, the larger the wave angle, the larger the normal component of upstream Mach number $M_{n,1}$, and from Equation (9.16) the larger the pressure ratio p_2/p_1. Thus, in Figure 9.11, the higher-angle shock wave will compress the gas more than the lower-angle shock wave, hence the terms "strong" and "weak" solutions. In nature, the weak shock solution usually prevails. Whenever you see straight, attached oblique shock waves, such as sketched at the left of Figure 9.10, they are almost always the weak shock solution. It is safe to make this assumption, unless you have specific information to the contrary. Note in Figure 9.9 that the locus of points connecting all the values of θ_{max} (the curve that sweeps approximately horizontally across the middle of Figure 9.9) divides the weak and strong shock solutions. Above this curve, the strong shock solution prevails (as further indicated by the θ-β-M curves being dashed); below this curve, the weak shock solution prevails (where the θ-β-M curves are shown as solid lines). Note that slightly below this curve is another curve which also sweeps approximately horizontally across Figure 9.9. This curve is the dividing line above which $M_2 < 1$ and below which $M_2 > 1$. For the strong shock solution, the downstream Mach number is always subsonic $M_2 < 1$. For the weak shock solution very near θ_{max}, the downstream Mach number is also subsonic, but barely so. For the vast majority of cases involving the weak shock solution, the downstream Mach number is supersonic $M_2 > 1$. Since the weak shock solution is almost always the case encountered in nature, we can readily state that the Mach number downstream of a straight, attached oblique shock is almost always supersonic.

3. If $\theta = 0$, then β equals either 90° or μ. The case of $\beta = 90^\circ$ corresponds to a normal shock wave (i.e., the normal shocks discussed in Chapter 8 belong to the family of strong shock solutions). The case of $\beta = \mu$ corresponds to the Mach wave illustrated in Figure 9.4*b*. In both cases, the flow streamlines experience no deflection across the wave.

4. (In all of the following discussions, we consider the weak shock solution exclusively, unless otherwise noted.) Consider an experiment where we have supersonic flow over a wedge of given semiangle θ, as sketched in Figure 9.12. Now assume that we increase the freestream Mach number M_1. As M_1 increases, we observe that β decreases. For example, consider $\theta = 20^\circ$ and $M_1 = 2.0$, as shown on the left of Figure 9.12. From Figure 9.9, we find that $\beta = 53.3^\circ$. Now assume M_1 is increased to 5, keeping θ constant at 20°, as sketched on the right of Figure 9.12. Here, we find that $\beta = 29.9^\circ$. Interestingly enough, although this shock is at a lower wave angle, it is a stronger shock than the one on the left. This is because $M_{n,1}$ is larger for the case on the right. Although β is smaller, which decreases $M_{n,1}$, the upstream Mach number M_1 is larger, which increases $M_{n,1}$ by an amount which more than compensates for the decreased β. For example, note the values of $M_{n,1}$ and p_2/p_1 given in Figure 9.12. Clearly,

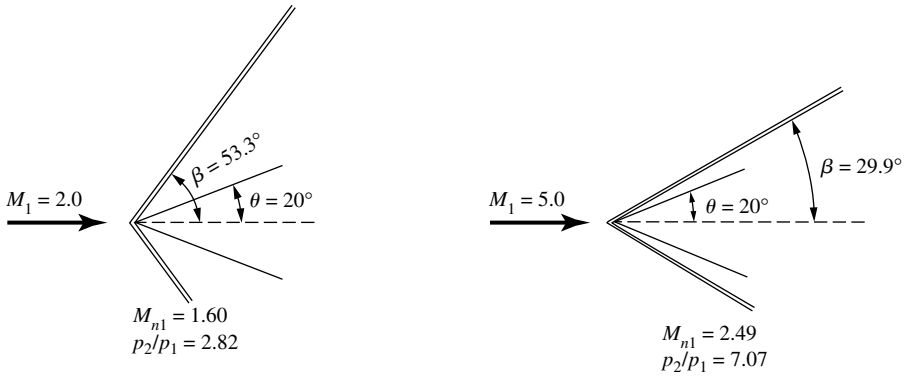


Figure 9.12 Effects of increasing the upstream Mach number.

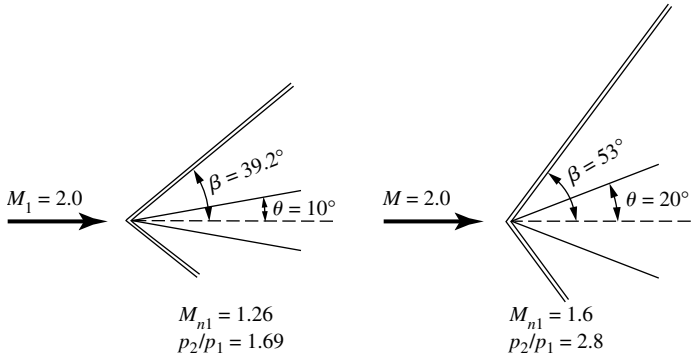


Figure 9.13 Effect of increasing the deflection angle.

the Mach 5 case on the right yields the stronger shock wave. Hence, in general for attached shocks with a fixed deflection angle, as the upstream Mach number M_1 increases, the wave angle β decreases, and the shock wave becomes stronger. Going in the other direction, as M_1 decreases, the wave angle increases, and the shock becomes weaker. Finally, if M_1 is decreased enough, the shock wave will become detached. For the case of $\theta = 20^\circ$ shown in Figure 9.12, the shock will be detached for $M_1 < 1.84$.

5. Consider another experiment. Here, let us keep M_1 fixed and increase the deflection angle. For example, consider the supersonic flow over a wedge shown in Figure 9.13. Assume that we have $M_1 = 2.0$ and $\theta = 10^\circ$, as sketched at the left of Figure 9.13. The wave angle will be 39.2° (from Figure 9.9). Now assume that the wedge is hinged so that we can increase its deflection angle, keeping M_1 constant. In such a case, the wave angle will increase, as shown on the right of Figure 9.13. Also, $M_{n,1}$ will increase,

and hence the shock will become stronger. Therefore, in general for attached shocks with a fixed upstream Mach number, as the deflection angle increases, the wave angle β increases, and the shock becomes stronger. However, once θ exceeds θ_{max}, the shock wave will become detached. For the case of $M_1 = 2.0$ in Figure 9.13, this will occur when $\theta > 23°$.

The physical properties of oblique shocks just discussed are very important. Before proceeding further, make certain to go over this discussion several times until you feel perfectly comfortable with these physical variations.

EXAMPLE 9.2

Consider a supersonic flow with $M = 2$, $p = 1$ atm, and $T = 288$ K. This flow is deflected at a compression corner through 20°. Calculate M, p, T, p_0, and T_0 behind the resulting oblique shock wave.

■ Solution

From Figure 9.9, for $M_1 = 2$ and $\theta = 20°$, $\beta = 53.4°$. Hence, $M_{n,1} = M_1 \sin\beta = 2 \sin 53.4° = 1.606$. From Appendix B, for $M_{n,1} = 1.60$ (rounded to the nearest table entry),

$$M_{n,2} = 0.6684 \qquad \frac{p_2}{p_1} = 2.82 \qquad \frac{T_2}{T_1} = 1.388 \qquad \frac{p_{0,2}}{p_{0,1}} = 0.8952$$

Hence,

$$M_2 = \frac{M_{n,2}}{\sin(\beta - \theta)} = \frac{0.6684}{\sin(53.4 - 20)} = \boxed{1.21}$$

$$p_2 = \frac{p_2}{p_1} p_1 = 2.82(1 \text{ atm}) = \boxed{2.82 \text{ atm}}$$

$$T_2 = \frac{T_2}{T_1} T_1 = 1.388(288) = \boxed{399.7 \text{ K}}$$

For $M_1 = 2$, from Appendix A, $p_{0,1}/p_1 = 7.824$ and $T_{0,1}/T_1 = 1.8$; thus,

$$p_{0,2} = \frac{p_{0,2}}{p_{0,1}} \frac{p_{0,1}}{p_1} p_1 = 0.8952(7.824)(1 \text{ atm}) = \boxed{7.00 \text{ atm}}$$

The total temperature is constant across the shock. Hence,

$$T_{0,2} = T_{0,1} = \frac{T_{0,1}}{T_1} T_1 = 1.8(288) = \boxed{518.4 \text{ K}}$$

Note: For oblique shocks, the entry for $p_{0,2}/p_1$ in Appendix B *cannot* be used to obtain $p_{0,2}$; this entry in Appendix B is for normal shocks only and is obtained directly from Equation (8.80). In turn, Equation (8.80) is derived using (8.77), where M_2 is the *actual* flow Mach number, not the normal component. Only in the case of a normal shock is this also the Mach number normal to the wave. Hence, Equation (8.80) holds only for normal shocks; it *cannot* be used for oblique shocks with M_1 replaced by $M_{n,1}$. For example, an *incorrect* calculation would be to use

$p_{0,2}/p_1 = 3.805$ for $M_{n,1} = 1.60$. This gives $p_{0,2} = 3.805$ atm, a totally incorrect result compared with the correct value of 7.00 atm obtained above.

EXAMPLE 9.3

Consider an oblique shock wave with a wave angle of 30°. The upstream flow Mach number is 2.4. Calculate the deflection angle of the flow, the pressure and temperature ratios across the shock wave, and the Mach number behind the wave.

■ Solution

From Figure 9.9, for $M_1 = 2.4$ and $\beta = 30°$, we have $\boxed{\theta = 6.5°}$. Also,

$$M_{n,1} = M_1 \sin\beta = 2.4 \sin 30° = 1.2$$

From Appendix B,

$$\frac{p_2}{p_1} = \boxed{1.513}$$

$$\frac{T_2}{T_1} = \boxed{1.128}$$

$$M_{n,2} = 0.8422$$

Thus,

$$M_2 = \frac{M_{n,2}}{\sin(\beta - \theta)} = \frac{0.8422}{\sin(30 - 6.5)} = \boxed{2.11}$$

Note: Two aspects are illustrated by this example:

1. This is a fairly weak shock wave—only a 51 percent increase in pressure across the wave. Indeed, examining Figure 9.9, we find that this case is close to that of a Mach wave, where $\mu = \sin^{-1}(1/M) = \sin^{-1}(\frac{1}{2.4}) = 24.6°$. The shock-wave angle of 30° is not much larger than μ; the deflection angle of 6.5° is also small—consistent with the relative weakness of the shock wave.
2. Only two properties need to be specified in order to define uniquely a given oblique shock wave. In this example, M_1 and β were those two properties. In Example 9.2, the specified M_1 and θ were the two properties. Once any two properties about the oblique shock are specified, the shock is uniquely defined. This is analogous to the case of a normal shock wave studied in Chapter 8. There, we proved that all the changes across a normal shock wave were uniquely defined by specifying only *one* property, such as M_1. However, implicit in all of Chapter 8 was an additional property, namely, the wave angle of a normal shock wave is 90°. Of course, a normal shock is simply one example of the whole spectrum of oblique shocks, namely, a shock with $\beta = 90°$. An examination of Figure 9.9 shows that the normal shock belongs to the family of strong shock solutions, as discussed earlier.

EXAMPLE 9.4

Consider an oblique shock wave with $\beta = 35^\circ$ and a pressure ratio $p_2/p_1 = 3$. Calculate the upstream Mach number.

■ Solution

From Appendix B, for $p_2/p_1 = 3$, $M_{n,1} = 1.64$ (nearest entry). Since

$$M_{n,1} = M_1 \sin\beta$$

then

$$M_1 = \frac{M_{n,1}}{\sin\beta} = \frac{1.66}{\sin 35^\circ} = \boxed{2.86}$$

Note: Once again, the oblique shock is uniquely defined by two properties, in this case β and p_2/p_1.

EXAMPLE 9.5

Consider a Mach 3 flow. It is desired to slow this flow to a subsonic speed. Consider two separate ways of achieving this: (1) the Mach 3 flow is slowed by passing directly through a normal shock wave; (2) the Mach 3 flow first passes through an oblique shock with a 40° wave angle, and then subsequently through a normal shock. These two cases are sketched in Figure 9.14. Calculate the ratio of the final total pressure values for the two cases, that is, the total pressure behind the normal shock for case 2 divided by the total pressure behind the normal shock for case 1. Comment on the significance of the result.

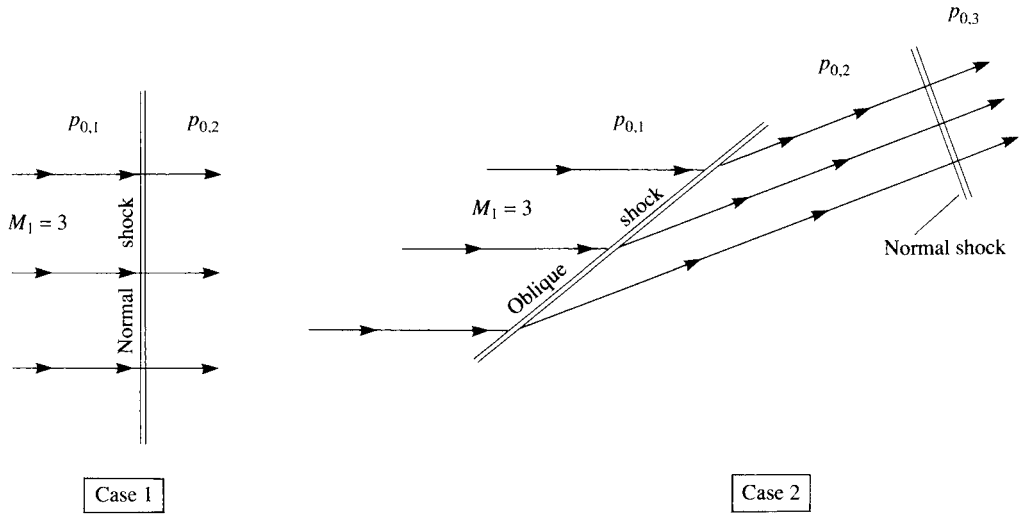


Figure 9.14 Illustration for Example 9.4.

■ Solution

For case 1, at $M = 3$, we have, from Appendix B,

$$\left(\frac{p_{0_2}}{p_{0_1}}\right)_{\text{case 1}} = 0.3283$$

For case 2, we have $M_{n,1} = M_1 \sin\beta = 3\sin 40^\circ = 1.93$. From Appendix B,

$$\frac{p_{0_2}}{p_{0_1}} = 0.7535 \quad \text{and} \quad M_{n,2} = 0.588$$

From Figure 9.9, for $M_1 = 3$ and $\beta = 40^\circ$, we have the deflection angle $\theta = 22^\circ$. Hence,

$$M_2 = \frac{M_{n,2}}{\sin(\beta - \theta)} = \frac{0.588}{\sin(40 - 22)} = 1.90$$

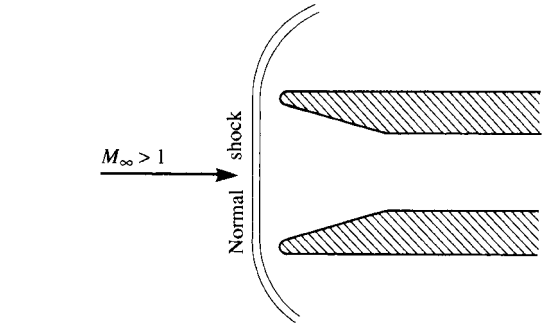
From Appendix B, for a normal shock with an upstream Mach number of 1.9, we have $p_{0_3}/p_{0_2} = 0.7674$. Thus, for case 2,

$$\left(\frac{p_{0_3}}{p_{0_1}}\right)_{\text{case 2}} = \left(\frac{p_{0_2}}{p_{0_1}}\right)\left(\frac{p_{0_3}}{p_{0_2}}\right) = (0.7535)(0.7674) = 0.578$$

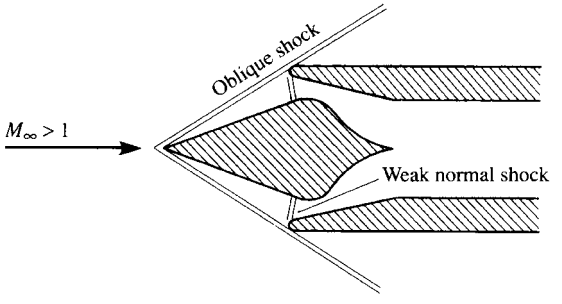
Hence,

$$\left(\frac{p_{0_3}}{p_{0_1}}\right)_{\text{case 2}} \Bigg/ \left(\frac{p_{0_2}}{p_{0_1}}\right)_{\text{case 1}} = \frac{0.578}{0.3283} = \boxed{1.76}$$

The result of Example 9.5 shows that the final total pressure is 76 percent higher for the case of the multiple shock system (case 2) in comparison to the single normal shock (case 1). In principle, the total pressure is an indicator of how much useful work can be done by the gas; this is described later in Section 10.4. Everything else being equal, the higher the total pressure, the more useful is the flow. Indeed, losses of total pressure are an index of the *efficiency* of a fluid flow—the lower the total pressure loss, the more efficient is the flow process. In this example, case 2 is more efficient in slowing the flow to subsonic speeds than case 1 because the loss in total pressure across the multiple shock system of case 2 is actually less than that for case 1 with a single, strong, normal shock wave. The physical reason for this is straightforward. The loss in total pressure across a normal shock wave becomes particularly severe as the upstream Mach number increases; a glance at the $p_{0,2}/p_{0,1}$ column in Appendix B attests to this. If the Mach number of a flow can be reduced *before* passing through a normal shock, the loss in total pressure is much less because the normal shock is weaker. This is the function of the oblique shock in case 2, namely, to reduce the Mach number of the flow before passing through the normal shock. Although there is a total pressure loss across the oblique shock also, it is much less than across a normal shock at the same upstream Mach number. The net effect of the oblique shock reducing the flow Mach number before passing through the normal shock more than makes up for the total pressure loss across the oblique shock, with the beneficial result that the multiple shock system in case 2 produces a *smaller* loss in total pressure than a single normal shock at the same freestream Mach number.



(*a*) Normal shock inlet



(*b*) Oblique shock inlet

Figure 9.15 Illustration of (*a*) normal shock inlet and (*b*) oblique shock inlet.

A practical application of these results is in the design of supersonic inlets for jet engines. A normal shock inlet is sketched in Figure 9.15*a*. Here, a normal shock forms ahead of the inlet, with an attendant large loss in total pressure. In contrast, an oblique shock inlet is sketched in Figure 9.15*b*. Here, a central cone creates an oblique shock wave, and the flow subsequently passes through a relatively weak normal shock at the lip of the inlet. For the same flight conditions (Mach number and altitude), the total pressure loss for the oblique shock inlet is less than for a normal shock inlet. Hence, everything else being equal, the resulting engine thrust will be higher for the oblique shock inlet. This, of course, is why most modern supersonic aircraft have oblique shock inlets.

9.3 SUPERSONIC FLOW OVER WEDGES AND CONES

For the supersonic flow over wedges, as shown in Figures 9.12 and 9.13, the oblique shock theory developed in Section 9.2 is an *exact* solution of the flow field; no simplifying assumptions have been made. Supersonic flow over a wedge

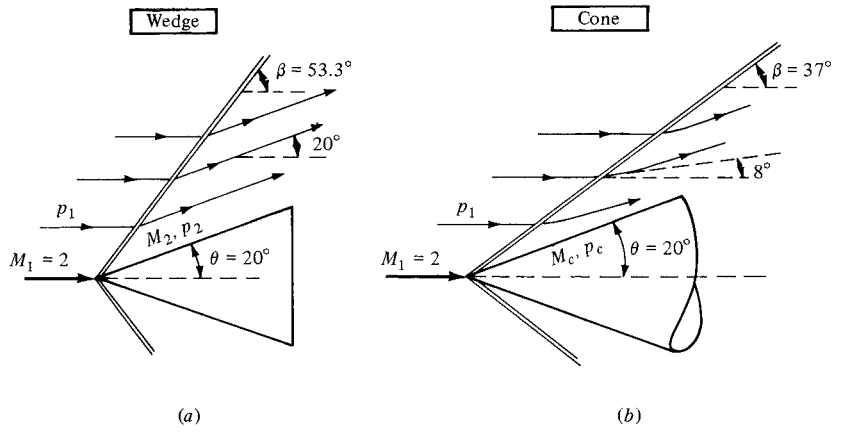


Figure 9.16 Relation between wedge and cone flow; illustration of the three-dimensional relieving effect.

is characterized by an attached, straight oblique shock wave from the nose, a uniform flow downstream of the shock with streamlines parallel to the wedge surface, and a surface pressure equal to the static pressure behind the oblique shock p_2. These properties are summarized in Figure 9.16*a*. Note that the wedge is a two-dimensional profile; in Figure 9.16*a*, it is a section of a body that stretches to plus or minus infinity in the direction perpendicular to the page. Hence, wedge flow is, by definition, two-dimensional flow, and our two-dimensional oblique shock theory fits this case nicely.

In contrast, consider the supersonic flow over a cone, as sketched in Figure 9.16*b*. There is a straight oblique shock which emanates from the tip, just as in the case of a wedge, but the similarity stops there. Recall from Chapter 6 that flow over a three-dimensional body experiences a "three-dimensional relieving effect." That is, in comparing the wedge and cone in Figure 9.16, both with the same 20° angle, the flow over the cone has an extra dimension in which to move, and hence it more easily adjusts to the presence of the conical body in comparison to the two-dimensional wedge. One consequence of this three-dimensional relieving effect is that the shock wave on the cone is weaker than on the wedge; that is, it has a smaller wave angle, as compared in Figure 9.16. Specifically, the wave angles for the wedge and cone are 53.3 and 37°, respectively, for the same body angle of 20° and the same upstream Mach number of 2.0. In the case of the wedge (Figure 9.16*a*), the streamlines are deflected by exactly 20° through the shock wave, and hence downstream of the shock the flow is exactly parallel to the wedge surface. In contrast, because of the weaker shock on the cone, the streamlines are deflected by only 8° through the shock, as shown in Figure 9.16*b*. Therefore, between the shock wave and the cone surface, the streamlines must gradually curve upward in order to accommodate the 20° cone. Also, as a consequence of the three-dimensional relieving effect, the pressure on the surface of the cone, p_c, is less than the wedge surface pressure p_2, and the cone surface

Mach number M_c is greater than that on the wedge surface M_2. In short, the main differences between the supersonic flow over a cone and wedge, both with the same body angle, are that (1) the shock wave on the cone is weaker, (2) the cone surface pressure is less, and (3) the streamlines above the cone surface are curved rather than straight.

The analysis of the supersonic flow over a cone is more sophisticated than the oblique shock theory given in this chapter. The calculation of the supersonic flow over a cone is discussed in Section 13.6. For details concerning supersonic conical flow analysis, see Chapter 10 of Reference 21. However, it is important for you to recognize that conical flows are inherently different from wedge flows and to recognize in what manner they differ. This has been the purpose of the present section.

EXAMPLE 9.6

Consider a wedge with a 15° half angle in a Mach 5 flow, as sketched in Figure 9.17. Calculate the drag coefficient for this wedge. (Assume that the pressure over the base is equal to freestream static pressure, as shown in Figure 9.17.)

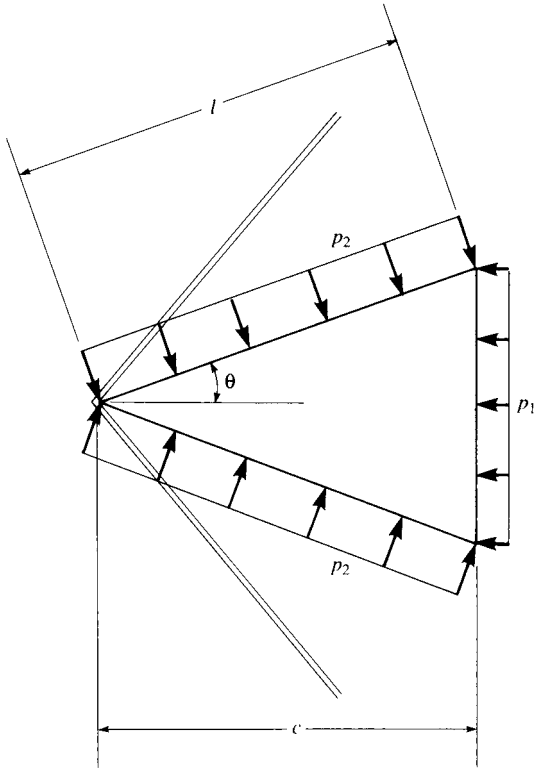


Figure 9.17 Illustration for Example 9.6.

■ Solution

Consider the drag on a unit span of the wedge D'. Hence,

$$c_d = \frac{D'}{q_1 S} = \frac{D'}{q_1 c(1)} = \frac{D'}{q_1 c}$$

From Figure 9.17,

$$D' = 2p_2 l \sin\theta - 2p_1 l \sin\theta = (2l \sin\theta)(p_2 - p_1)$$

However,

$$l = \frac{c}{\cos\theta}$$

Thus,

$$D' = (2c \tan\theta)(p_2 - p_1)$$

and

$$c_d = (2 \tan\theta)\left(\frac{p_2 - p_1}{q_1}\right)$$

Note that

$$q_1 \equiv \frac{1}{2}\rho_1 V_1^2 = \frac{1}{2}\rho_1 \frac{\gamma p_1}{\gamma p_1} V_1^2 = \frac{\gamma p_1}{2a_1^2} V_1^2 = \frac{\gamma}{2} p_1 M_1^2$$

Thus,

$$c_d = (2 \tan\theta)\left(\frac{p_2 - p_1}{(\gamma/2) p_1 M_1^2}\right) = \frac{4 \tan\theta}{\gamma M_1^2}\left(\frac{p_2}{p_1} - 1\right)$$

From Figure 9.9, for $M_1 = 5$ and $\theta = 15^\circ$, $\beta = 24.2^\circ$. Hence,

$$M_{n,1} = M_1 \sin\beta = 5 \sin(24.2^\circ) = 2.05$$

From Appendix B, for $M_{n,1} = 2.05$, we have

$$\frac{p_2}{p_1} = 4.736$$

Hence,

$$c_d = \frac{4 \tan\theta}{\gamma M_1^2}\left(\frac{p_2}{p_1} - 1\right) = \frac{4 \tan 15^\circ}{(1.4)(5)^2}(4.736 - 1) = \boxed{0.114}$$

(*Note:* The drag is *finite* for this case. In a supersonic or hypersonic inviscid flow over a two-dimensional body, the drag is always finite. D'Alembert's paradox does *not* hold for freestream Mach numbers such that shock waves appear in the flow. The fundamental reason for the generation of drag here is the presence of shock waves. Shocks are always a dissipative, drag-producing mechanism. For this reason, the drag in this case is called *wave drag,* and c_d is the wave-drag coefficient, more properly denoted as $c_{d,w}$.)

9.3.1 A Comment on Supersonic Lift and Drag Coefficients

The result obtained in Example 9.6 is a stunning verification of the validity of the dimensional analysis discussed in Section 1.7. There we proved that, for a body of a given shape at a given angle of attack, the aerodynamic coefficients are simply a function of Mach number and Reynolds number [see Equations (1.42), (1.43)

and (1.44)]. Consider the 15° half-angle wedge at zero angle of attack shown in Figure 9.17. This is a body of a given shape at a given angle of attack. Moreover, in Example 9.6 we are given *only* the freestream Mach number, and are asked to calculate the drag coefficient for the wedge. Since the flow is inviscid, the Reynolds number does not play a role. At first glance, one might intuitively think that we need to be given at least a freestream pressure and velocity to obtain the drag coefficient. After all, the physical source of the drag is the pressure distribution integrated all over the surface of the body, as emphasized in Section 1.5. And the surface pressure distribution is shown schematically in Figure 9.17. Why is it, then, that we are not given some information about the pressure level and the freestream velocity?

The answer, clearly demonstrated by the dimensional analysis in Section 1.7, is that the drag *coefficient* depends just on Mach number. In Example 9.6 the freestream Mach number is given as Mach 5. The solution progresses by treating the surface pressure distribution that is responsible for the drag, but the solution ultimately requires only pressure ratios, not the actual value of the pressure itself. At the end of the calculation in Example 9.6, the drag coefficient is finally obtained, and all we needed for the calculation was the freestream Mach number. What a nice verification of the validity of the dimensional analysis discussed in Section 1.7 and the concept of flow similarity given in Section 1.8! Moreover, we have verified these concepts for a supersonic flow. Of course, the concepts in Sections 1.7 and 1.8 are fundamental; they hold no matter what is the flow regime—subsonic, supersonic, hypersonic, etc.

Finally, if we had been asked for the drag *force* in Example 9.6, additional information would have been required, such as the size of the wedge, and the pressure of the freestream. But one of the beauties of dealing with the aerodynamic *coefficients* rather than the forces or moments themselves is that the *coefficients* for an inviscid flow depend on Mach number, and Mach number only.

9.4 SHOCK INTERACTIONS AND REFLECTIONS

Return to the oblique shock wave illustrated in Figure 9.2*a*. In this picture, we can imagine the shock wave extending unchanged above the corner to infinity. However, in real life this does not happen. In reality, the oblique shock in Figure 9.2*a* will impinge somewhere on another solid surface and/or will intersect other waves, either shock or expansion waves. Such wave intersections and interactions are important in the practical design and analysis of supersonic airplanes, missiles, wind tunnels, rocket engines, etc. A perfect historical example of this, as well as the consequences that can be caused by not paying suitable attention to wave interactions, is a ramjet flight-test program conducted in the early 1960s. During this period, a ramjet engine was mounted underneath the X-15 hypersonic airplane for a series of flight tests at high Mach numbers, in the range from 4 to 7. (The X-15, shown in Figure 9.18, was an experimental, rocket-powered airplane designed to probe the lower end of hypersonic manned flight.) During the first high-speed tests, the shock wave from the engine cowling impinged on the

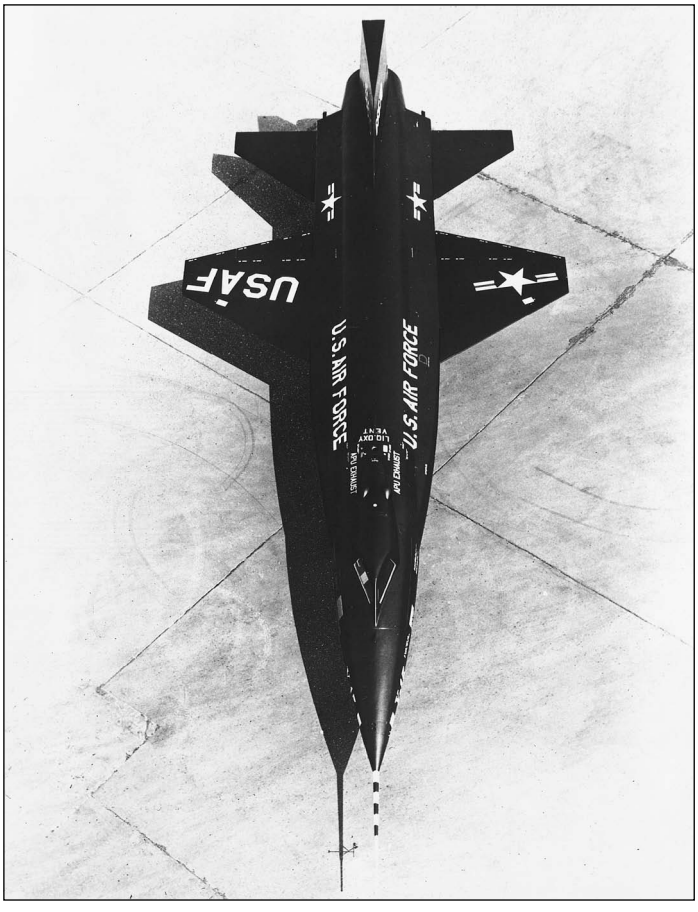


Figure 9.18 The X-15 hypersonic research vehicle. Designed and built during the late 1950s, it served as a test vehicle for the U.S. Air Force and NASA. (*Courtesy the John Anderson Collection.*)

bottom surface of the X-15, and because of locally high aerodynamic heating in the impingement region, a hole was burned in the X-15 fuselage. Although this problem was later fixed, it is a graphic example of what shock-wave interactions can do to a practical configuration.

The purpose of this section is to present a mainly qualitative discussion of shock-wave interactions. For more details, see Chapter 4 of Reference 21.

First, consider an oblique shock wave generated by a concave corner, as shown in Figure 9.19. The deflection angle at the corner is θ, thus generating an oblique shock at point A with a wave angle β_1. Assume that a straight, horizontal wall is present above the corner, as also shown in Figure 9.19. The shock wave generated at point A, called the *incident shock wave,* impinges on the upper wall

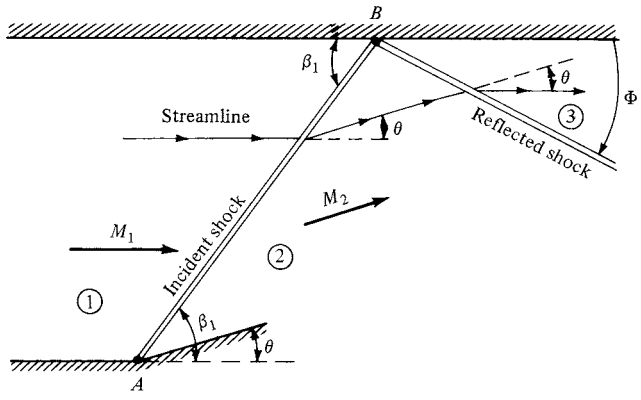


Figure 9.19 Regular reflection of a shock wave from a solid boundary.

at point B. *Question:* Does the shock wave simply disappear at point B? If not, what happens to it? To answer this question, we appeal to our knowledge of shock-wave properties. Examining Figure 9.19, we see that the flow in region 2 behind the incident shock is inclined upward at the deflection angle θ. However, the flow must be tangent everywhere along the upper wall; if the flow in region 2 were to continue unchanged, it would run into the wall and have no place to go. Hence, the flow in region 2 must eventually be bent downward through the angle θ in order to maintain a flow tangent to the upper wall. Nature accomplishes this downward deflection via a second shock wave originating at the impingement point B in Figure 9.19. This second shock is called the *reflected shock wave*. The purpose of the reflected shock is to deflect the flow in region 2 so that it is parallel to the upper wall in region 3, thus preserving the wall boundary condition.

The strength of the reflected shock wave is weaker than the incident shock. This is because $M_2 < M_1$, and M_2 represents the upstream Mach number for the reflected shock wave. Since the deflection angles are the same, whereas the reflected shock sees a lower upstream Mach number, we know from Section 9.2 that the reflected wave must be weaker. For this reason, the angle the reflected shock makes with the upper wall Φ is not equal to β_1 (i.e., the wave reflection is not specular). The properties of the reflected shock are uniquely defined by M_2 and θ; since M_2 is in turn uniquely defined by M_1 and θ, then the properties in region 3 behind the reflected shock as well as the angle Φ are easily determined from the given conditions of M_1 and θ by using the results of Section 9.2 as follows:

1. Calculate the properties in region 2 from the given M_1 and θ. In particular, this gives us M_2.
2. Calculate the properties in region 3 from the value of M_2 calculated above and the known deflection angle θ.

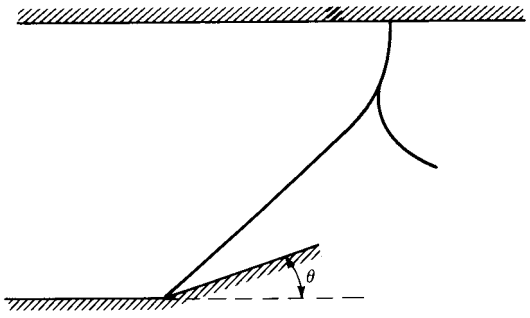


Figure 9.20 Mach reflection.

An interesting situation can arise as follows. Assume that M_1 is only slightly above the minimum Mach number necessary for a straight, attached shock wave at the given deflection angle θ. For this case, the oblique shock theory from Section 9.2 allows a solution for a straight, attached incident shock. However, we know that the Mach number decreases across a shock (i.e., $M_2 < M_1$). This decrease may be enough such that M_2 is *not* above the minimum Mach number for the required deflection θ through the reflected shock. In such a case, our oblique shock theory does not allow a solution for a straight reflected shock wave. The regular reflection as shown in Figure 9.19 is not possible. Nature handles this situation by creating the wave pattern shown in Figure 9.20. Here, the originally straight incident shock becomes curved as it nears the upper wall and becomes a normal shock wave at the upper wall. This allows the streamline at the wall to continue parallel to the wall behind the shock intersection. In addition, a curved reflected shock branches from the normal shock and propagates downstream. This wave pattern, shown in Figure 9.20, is called a *Mach reflection*. The calculation of the wave pattern and general properties for a Mach reflection requires numerical techniques such as those to be discussed in Chapter 13.

Another type of shock interaction is shown in Figure 9.21. Here, a shock wave is generated by the concave corner at point G and propagates upward. Denote this wave as shock A. Shock A is a *left-running wave*, so-called because if you stand on top of the wave and look downstream, you see the shock wave running in front of you toward the left. Another shock wave is generated by the concave corner at point H, and propagates downward. Denote this wave as shock B. Shock B is a *right-running wave,* so-called because if you stand on top of the wave and look downstream, you see the shock running in front of you toward the right. The picture shown in Figure 9.21 is the intersection of right- and left-running shock waves. The intersection occurs at point E. At the intersection, wave A is refracted and continues as wave D. Similarly, wave B is refracted and continues as wave C. The flow behind the refracted shock D is denoted by region 4; the flow behind the refracted shock C is denoted by region 4′. These two regions are divided by a slip line EF. Across the slip line, the pressures are constant (i.e., $p_4 = p_{4'}$), and the direction (but not necessarily the

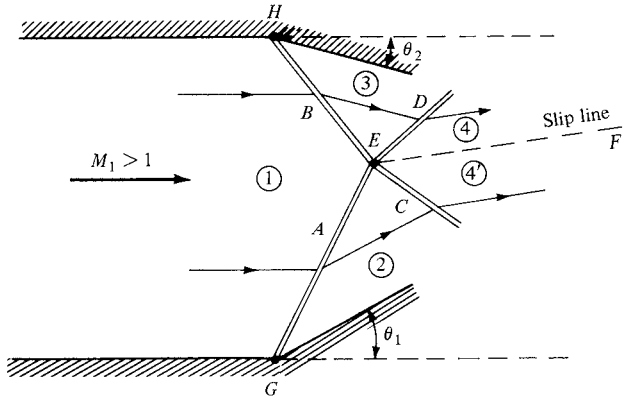


Figure 9.21 Intersection of right- and left-running shock waves.

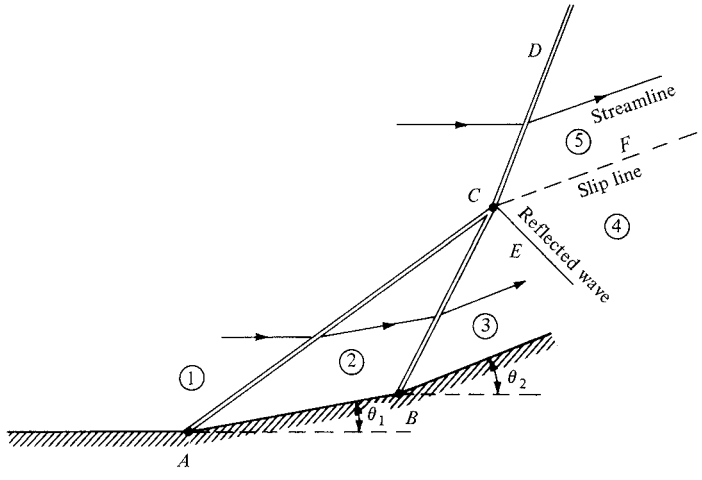


Figure 9.22 Intersection of two left-running shock waves.

magnitude) of velocity is the same, namely, parallel to the slip line. All other properties in regions 4 and 4′ are different, most notably the entropy ($s_4 \neq s_{4'}$). The conditions which must hold across the slip line, along with the known M_1, θ_1, and θ_2, uniquely determine the shock-wave interaction shown in Figure 9.21. (See Chapter 4 of Reference 21 for details concerning the calculation of this interaction.)

Figure 9.22 illustrates the intersection of two left-running shocks generated at corners A and B. The intersection occurs at point C, at which the two shocks merge and propagate as the stronger shock CD, usually along with a weak reflected wave CE. This reflected wave is necessary to adjust the flow so that the velocities in

regions 4 and 5 are in the same direction. Again, a slip line *CF* trails downstream of the intersection point.

The above cases are by no means all the possible wave interactions in a supersonic flow. However, they represent some of the more common situations encountered frequently in practice.

EXAMPLE 9.7

Consider an oblique shock wave generated by a compression corner with a 10° deflection angle. The Mach number of the flow ahead of the corner is 3.6; the flow pressure and temperature are standard sea level conditions. The oblique shock wave subsequently impinges on a straight wall opposite the compression corner. The geometry for this flow is given in Figure 9.19. Calculate the angle of the reflected shock wave Φ relative to the straight wall. Also, obtain the pressure, temperature, and Mach number behind the reflected wave.

■ Solution

From the θ-β-M diagram, Figure 9.9, for $M_1 = 3.6$ and $\theta = 10^\circ$, $\beta_1 = 24^\circ$. Hence,

$$M_{n,1} = M_1 \sin\beta_1 = 3.6 \sin 24^\circ = 1.464$$

From Appendix B,

$$M_{n,2} = 0.7157 \qquad \frac{p_2}{p_1} = 2.32 \qquad \frac{T_2}{T_1} = 1.294$$

Also,

$$M_2 = \frac{M_{n,2}}{\sin(\beta - \theta)} = \frac{0.7157}{\sin(24 - 10)} = 2.96$$

These are the conditions behind the incident shock wave. They constitute the upstream flow properties for the reflected shock wave. We know that the flow must be deflected again by $\theta = 10^\circ$ in passing through the reflected shock. Thus, from the θ-β-M diagram, for $M_2 = 2.96$ and $\theta = 10^\circ$, we have the wave angle for the reflected shock, $\beta_2 = 27.3^\circ$. Note that β_2 is *not* the angle the reflected shock makes with respect to the upper wall; rather, by definition of the wave angle, β_2 is the angle between the reflected shock and the direction of the flow in region 2. The shock angle relative to the wall is, from the geometry shown in Figure 9.19,

$$\Phi = \beta_2 - \theta = 27.3 - 10 = \boxed{17.3^\circ}$$

Also, the normal component of the upstream Mach number relative to the reflected shock is $M_2 \sin\beta_2 = (2.96) \sin 27.3^\circ = 1.358$. From Appendix B,

$$\frac{p_3}{p_2} = 1.991 \qquad \frac{T_3}{T_2} = 1.229 \qquad M_{n,3} = 0.7572$$

Hence,

$$M_3 = \frac{M_{n,3}}{\sin(\beta_2 - \theta)} = \frac{0.7572}{\sin(27.3 - 10)} = \boxed{2.55}$$

For standard sea level conditions, $p_1 = 2116$ lb/ft^3 and $T_1 = 519°$R. Thus,

$$p_3 = \frac{p_3}{p_2}\frac{p_2}{p_1}p_1 = (1.991)(2.32)(2116) = \boxed{9774 \text{ lb/ft}^3}$$

$$T_3 = \frac{T_3}{T_2}\frac{T_2}{T_1}T_1 = (1.229)(1.294)(519) = \boxed{825°\text{R}}$$

Note that the reflected shock is weaker than the incident shock, as indicated by the smaller pressure ratio for the reflected shock, $p_3/p_2 = 1.991$ as compared to $p_2/p_1 = 2.32$ for the incident shock.

9.5 DETACHED SHOCK WAVE IN FRONT OF A BLUNT BODY

The curved bow shock which stands in front of a blunt body in a supersonic flow is sketched in Figure 8.1. We are now in a position to better understand the properties of this bow shock, as follows.

The flow in Figure 8.1 is sketched in more detail in Figure 9.23. Here, the shock wave stands a distance δ in front of the nose of the blunt body; δ is defined

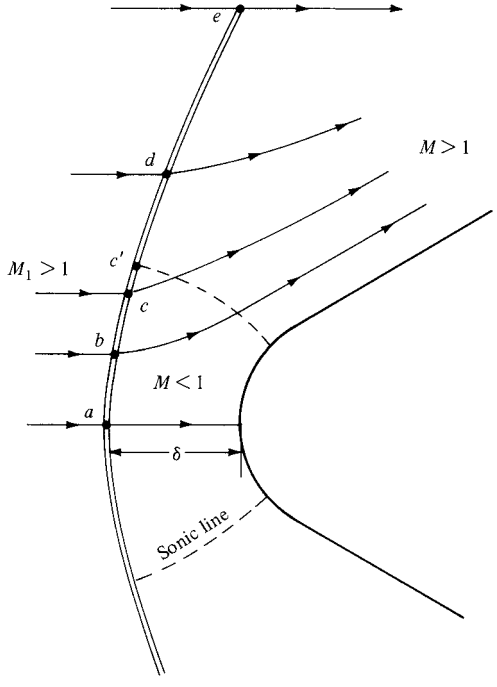


Figure 9.23 Flow over a supersonic blunt body.

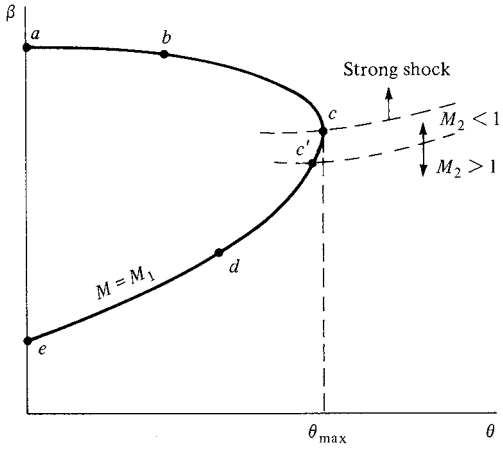


Figure 9.24 θ-β-M diagram for the sketch shown in Figure 9.23.

as the *shock detachment distance*. At point a, the shock wave is normal to the upstream flow; hence, point a corresponds to a normal shock wave. Away from point a, the shock wave gradually becomes curved and weaker, eventually evolving into a Mach wave at large distances from the body (illustrated by point e in Figure 9.23).

A curved bow shock wave is one of the instances in nature when you can observe *all* possible oblique shock solutions at once for a given freestream Mach number M_1. This takes place between points a and e. To see this more clearly, consider the θ-β-M diagram sketched in Figure 9.24 in conjunction with Figure 9.23. In Figure 9.24, point a corresponds to the normal shock, and point e corresponds to the Mach wave. Slightly above the centerline, at point b in Figure 9.23, the shock is oblique but pertains to the strong shock-wave solution in Figure 9.24. The flow is deflected slightly upward behind the shock at point b. As we move further along the shock, the wave angle becomes more oblique, and the flow deflection increases until we encounter point c. Point c on the bow shock corresponds to the maximum deflection angle shown in Figure 9.24. Above point c, from c to e, all points on the shock correspond to the weak shock solution. Slightly above point c, at point c', the flow behind the shock becomes sonic. From a to c', the flow is subsonic behind the bow shock; from c' to e, it is supersonic. Hence, the flow field between the curved bow shock and the blunt body is a mixed region of both subsonic and supersonic flow. The dividing line between the subsonic and supersonic regions is called the *sonic line,* shown as the dashed line in Figure 9.23.

The shape of the detached shock wave, its detachment distance δ, and the complete flow field between the shock and the body depend on M_1 and the size and shape of the body. The solution of this flow field is not trivial. Indeed, the

supersonic blunt-body problem was a major focus for supersonic aerodynamicists during the 1950s and 1960s, spurred by the need to understand the high-speed flow over blunt-nosed missiles and reentry bodies. Indeed, it was not until the late 1960s that truly sufficient numerical techniques became available for satisfactory engineering solutions of supersonic blunt-body flows. These modern techniques are discussed in Chapter 13.

EXAMPLE 9.8

Consider the detached curved bow shock wave in front of the two-dimensional parabolic blunt body drawn in Figure 9.25. The freestream is at Mach 8. Consider the two streamlines passing through the shock at points *a* and *b* shown in Figure 9.25. The wave angle at point *a* is 90°, and that at point *b* is 60°. Calculate and compare the value of entropy (relative to the free stream) for streamlines *a* and *b* in the flow behind the shock.

■ Solution

The oblique shock properties studied in this chapter are derived on the basis of a straight oblique shock wave with a uniform flow field behind the shock wave, such as sketched in Figure 9.2*a*. These solutions do not apply to the nonuniform flow field behind a curved shock wave such as shown in Figure 9.25. Such blunt-body solutions are treated in Section 13.5. The straight oblique shock solutions treated in the present chapter, however, give the shock wave properties at any local point *immediately behind the curved shock* in Figure 9.25 as long as we know the local wave angle at the point. Therefore, immediately behind the shock at point *a*, because the shock is a normal shock at that point, we have

$$M_{n,1} = 8$$

From Appendix B, for $M_{n,1} = 8$, $p_2/p_1 = 74.5$ and $T_2/T_1 = 13.39$. From Equation (7.25), the entropy increase across the shock is

$$s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}$$

Since

$$c_p = \frac{\gamma R}{\gamma - 1} = \frac{(1.4)(287)}{0.4} = 1004.5 \ \frac{\text{J}}{\text{Kg} \cdot \text{K}}$$

Then

$$s_2 - s_1 = (1004.5) \ln 13.39 - (287) \ln 74.5$$

$$s_2 - s_1 = 1370 \ \frac{\text{J}}{\text{Kg} \cdot \text{K}}$$

Downstream of the shock wave, the flow along any given streamline is both adiabatic (no heat transfer) and reversible (no friction etc.), hence the flow along a given streamline behind the shock wave is *isentropic*. Therefore, along streamline *a*, the entropy is constant

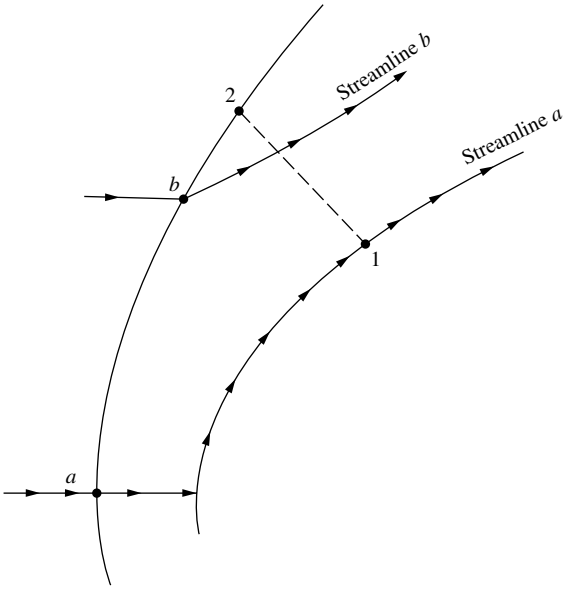


Figure 9.25 Two streamlines crossing a detached bow shockwave in front of a blunt body at Mach 8.

and equal to its value at point a just behind the shock wave. Thus,

$$s_2 - s_1 = \boxed{1370 \frac{\text{J}}{\text{Kg} \cdot \text{K}}} \qquad \text{along streamline } a$$

Immediately behind the curved bow shock at point b, where $\beta = 60^\circ$, we have

$$\begin{aligned} M_{n,1} &= M_1 \sin \beta = 8 \sin 60^\circ \\ &= 8(0.866) = 6.928 \end{aligned}$$

From Appendix B, using the nearest entry of $M_{n,1} = 6.9$, we have $p_2/p_1 = 55.38$ and $T_2/T_1 = 10.2$. Thus, at point b,

$$\begin{aligned} s_2 - s_1 &= c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1} \\ &= (1004.5) \ln 10.2 - (287) \ln 55.38 \\ &= \boxed{1180 \frac{\text{J}}{\text{Kg} \cdot \text{K}}} \qquad \text{along streamline } b \end{aligned}$$

The entropy along streamline b is smaller than that along streamline a because streamline b passes through a weaker part of the bow shock wave.

9.5.1 Comment on the Flow Field behind a Curved Shock Wave: Entropy Gradients and Vorticity

As illustrated by Example 9.8, the entropy is different along different streamlines behind a curved shock wave. For the case treated in Example 9.8, streamline *a* passes through the normal shock at point *a* and then flows downstream, wetting the surface of the body, as shown in Figure 9.25. This is the maximum entropy streamline. All other streamlines have smaller values of entropy; streamline *b* has a smaller entropy than streamline *a* because it passes through a weaker part of the curved shock wave at point *b*. Therefore, if you visualize a line that cuts through the flow field from point 1 to point 2 in Figure 9.25, the entropy decreases along this line from the body to the shock. That is, there exists an entropy gradient, ∇s, in the flow. For blunt-nosed hypersonic bodies the entropy gradient can be quite large, and is the source of the "entropy layer" that interacts with the boundary layer on hypersonic bodies (see, for example, Reference 55).

The presence of entropy gradients in the flow behind a curved shock wave has another consequence—the production of vorticity in the flow. The physical connection between entropy gradients and vorticity is quantified by *Crocco's theorem*, a combination of the momentum equation and the combined first and second laws of thermodynamics:

$$T\nabla s = \nabla h_o - \mathbf{V} \times (\nabla \times \mathbf{V}) \quad \text{Crocco's theorem}$$

In this equation, ∇s is the entropy gradient, ∇h_o is the gradient in the total enthalpy, and $\nabla \times \mathbf{V}$ is the vorticity. For a derivation of Crocco's theorem, see, for example, Section 6.6 of Reference 21. For our discussion, we present Crocco's theorem simply to emphasize an important feature of the flow behind the curved shock shown in Figure 9.25. The flow is adiabatic, hence ∇h_o is zero everywhere in the flow. However, ∇s is finite, and therefore from Crocco's theorem $\nabla \times \mathbf{V}$ must be finite.

Conclusion: The flow field behind a curved shock wave is *rotational*. As a result, a velocity potential with all its analytical advantages discussed earlier in this book can not be defined for the blunt-body flow field. Consequently, the flow field behind a curved shock is computed by means of numerical solutions of the continuity, momentum, and energy equations. Such computational fluid dynamic solutions are discussed in Section 13.5.

9.6 PRANDTL-MEYER EXPANSION WAVES

Oblique shock waves, as discussed in Sections 9.2 to 9.5, occur when a supersonic flow is turned into itself (see again Figure 9.2*a*). In contrast, when a supersonic flow is turned away from itself, an expansion wave is formed, as sketched in Figure 9.2*b*. Examine this figure carefully, and review the surrounding discussion in Section 9.1 before progressing further. The purpose of the present section is to develop a theory which allows us to calculate the changes in flow properties

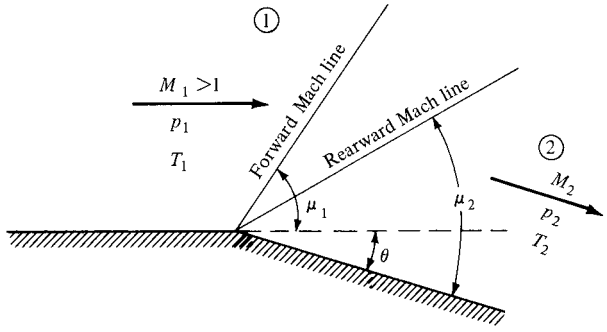


Figure 9.26 Prandtl-Meyer expansion.

across such expansion waves. To this stage in our discussion of oblique waves, we have completed the left-hand branch of the road map in Figure 9.6. In this section, we cover the right-hand branch.

The expansion fan in Figure 9.2*b* is a *continuous* expansion region that can be visualized as an infinite number of Mach waves, each making the Mach angle μ [see Equation (9.1)] with the local flow direction. As sketched in Figure 9.26, the expansion fan is bounded upstream by a Mach wave which makes the angle μ_1 with respect to the upstream flow, where $\mu_1 = \arcsin(1/M_1)$. The expansion fan is bounded downstream by another Mach wave which makes the angle μ_2 with respect to the downstream flow, where $\mu_2 = \arcsin(1/M_2)$. Since the expansion through the wave takes place across a continuous succession of Mach waves, and since $ds = 0$ for each Mach wave, the expansion is *isentropic*. This is in direct contrast to flow across an oblique shock, which always experiences an entropy increase. The fact that the flow through an expansion wave is isentropic is a greatly simplifying aspect, as we will soon appreciate.

An expansion wave emanating from a sharp convex corner as sketched in Figures 9.2*b* and 9.26 is called a *centered* expansion wave. Ludwig Prandtl and his student Theodor Meyer first worked out a theory for centered expansion waves in 1907–1908, and hence such waves are commonly denoted as *Prandtl-Meyer expansion waves*.

The problem of an expansion wave is as follows: Referring to Figure 9.26, given the upstream flow (region 1) and the deflection angle θ, calculate the downstream flow (region 2). Let us proceed.

Consider a very weak wave produced by an infinitesimally small flow deflection $d\theta$ as sketched in Figure 9.27. We consider the limit of this picture as $d\theta \to 0$; hence, the wave is essentially a Mach wave at the angle μ to the upstream flow. The velocity ahead of the wave is V. As the flow is deflected downward through the angle $d\theta$, the velocity is increased by the infinitesimal amount dV, and hence the flow velocity behind the wave is $V + dV$ inclined at the angle $d\theta$. Recall from the treatment of the momentum equation in Section 9.2 that any change in

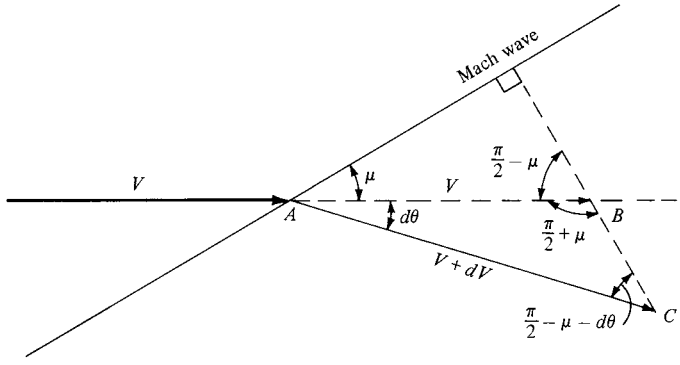


Figure 9.27 Geometrical construction for the infinitesimal changes across an infinitesimally weak wave (in the limit, a Mach wave).

velocity across a wave takes place *normal* to the wave; the tangential component is unchanged across the wave. In Figure 9.27, the horizontal line segment AB with length V is drawn behind the wave. Also, the line segment AC is drawn to represent the new velocity $V + dV$ behind the wave. Then line BC is normal to the wave because it represents the line along which the change in velocity occurs. Examining the geometry in Figure 9.27, from the law of sines applied to triangle ABC, we see that

$$\frac{V + dV}{V} = \frac{\sin(\pi/2 + \mu)}{\sin(\pi/2 - \mu - d\theta)} \tag{9.24}$$

However, from trigonometric identities,

$$\sin\left(\frac{\pi}{2} + \mu\right) = \sin\left(\frac{\pi}{2} - \mu\right) = \cos\mu \tag{9.25}$$

$$\sin\left(\frac{\pi}{2} - \mu - d\theta\right) = \cos(\mu + d\theta) = \cos\mu\cos d\theta - \sin\mu\sin d\theta \tag{9.26}$$

Substituting Equations (9.25) and (9.26) into (9.24), we have

$$1 + \frac{dV}{V} = \frac{\cos\mu}{\cos\mu\cos d\theta - \sin\mu\sin d\theta} \tag{9.27}$$

For small $d\theta$, we can make the small-angle assumptions $\sin d\theta \approx d\theta$ and $\cos d\theta \approx 1$. Then, Equation (9.27) becomes

$$1 + \frac{dV}{V} = \frac{\cos\mu}{\cos\mu - d\theta\sin\mu} = \frac{1}{1 - d\theta\tan\mu} \tag{9.28}$$

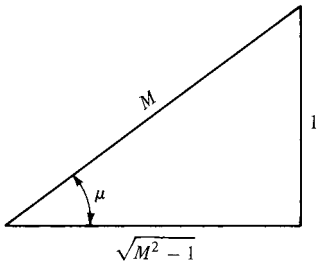


Figure 9.28 Right triangle associated with the Mach angle.

Note that the function $1/(1-x)$ can be expanded in a power series (for $x < 1$) as

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$$

Hence, Equation (9.28) can be expanded as (ignoring terms of second order and higher)

$$1 + \frac{dV}{V} = 1 + d\theta \tan\mu + \cdots \tag{9.29}$$

Thus, from Equation (9.29),

$$d\theta = \frac{dV/V}{\tan\mu} \tag{9.30}$$

From Equation (9.1), we know that $\mu = \arcsin(1/M)$. Hence, the right triangle in Figure 9.28 demonstrates that

$$\tan\mu = \frac{1}{\sqrt{M^2-1}} \tag{9.31}$$

Substituting Equation (9.31) into (9.30), we obtain

$$\boxed{d\theta = \sqrt{M^2-1}\,\frac{dV}{V}} \tag{9.32}$$

Equation (9.32) relates the infinitesimal change in velocity dV to the infinitesimal deflection $d\theta$ across a wave of vanishing strength. In the precise limit of a Mach wave, of course dV and hence $d\theta$ are zero. In this sense, Equation (9.32) is an approximate equation for a finite $d\theta$, but it becomes a true equality as $d\theta \to 0$. Since the expansion fan illustrated in Figures 9.2*b* and 9.26 is a region of an infinite number of Mach waves, Equation (9.32) is a differential equation which precisely describes the flow inside the expansion wave.

Return to Figure 9.26. Let us integrate Equation (9.32) from region 1, where the deflection angle is zero and the Mach number is M_1, to region 2, where the deflection angle is θ and the Mach number is M_2:

$$\int_0^\theta d\theta = \theta = \int_{M_1}^{M_2} \sqrt{M^2 - 1}\,\frac{dV}{V} \tag{9.33}$$

To carry out the integral on the right-hand side of Equation (9.33), dV/V must be obtained in terms of M, as follows. From the definition of Mach number, $M = V/a$, we have $V = Ma$, or

$$\ln V = \ln M + \ln a \tag{9.34}$$

Differentiating Equation (9.34), we obtain

$$\frac{dV}{V} = \frac{dM}{M} + \frac{da}{a} \tag{9.35}$$

From Equations (8.25) and (8.40), we have

$$\left(\frac{a_0}{a}\right)^2 = \frac{T_0}{T} = 1 + \frac{\gamma - 1}{2}M^2 \tag{9.36}$$

Solving Equation (9.36) for a, we obtain

$$a = a_0\left(1 + \frac{\gamma - 1}{2}M^2\right)^{-1/2} \tag{9.37}$$

Differentiating Equation (9.37), we obtain

$$\frac{da}{a} = -\left(\frac{\gamma - 1}{2}\right)M\left(1 + \frac{\gamma - 1}{2}M^2\right)^{-1} dM \tag{9.38}$$

Substituting Equation (9.38) into (9.35), we have

$$\frac{dV}{V} = \frac{1}{1 + [(\gamma - 1)/2]M^2}\frac{dM}{M} \tag{9.39}$$

Equation (9.39) is a relation for dV/V strictly in terms of M—this is precisely what is desired for the integral in Equation (9.33). Hence, substituting Equation (9.39) into (9.33), we have

$$\theta = \int_{M_1}^{M_2} \frac{\sqrt{M^2 - 1}}{1 + [(\gamma - 1)/2]M^2}\frac{dM}{M} \tag{9.40}$$

In Equation (9.40), the integral

$$\nu(M) \equiv \int \frac{\sqrt{M^2 - 1}}{1 + [(\gamma - 1)/2]M^2}\frac{dM}{M} \tag{9.41}$$

is called the *Prandtl-Meyer function,* denoted by ν. Carrying out the integration, Equation (9.41) becomes

$$\boxed{\nu(M) = \sqrt{\frac{\gamma+1}{\gamma-1}}\tan^{-1}\sqrt{\frac{\gamma-1}{\gamma+1}(M^2-1)} - \tan^{-1}\sqrt{M^2-1}} \tag{9.42}$$

The constant of integration that would ordinarily appear in Equation (9.42) is not important, because it drops out when Equation (9.42) is used for the definite integral in Equation (9.40). For convenience, it is chosen as zero, such that $\nu(M) = 0$ when $M = 1$. Finally, we can now write Equation (9.40), combined with (9.41), as

$$\boxed{\theta = \nu(M_2) - \nu(M_1)} \tag{9.43}$$

where $\nu(M)$ is given by Equation (9.42) for a calorically perfect gas. The Prandtl-Meyer function ν is very important; it is the key to the calculation of changes across an expansion wave. Because of its importance, ν is tabulated as a function of M in Appendix C. For convenience, values of μ are also tabulated in Appendix C.

How do the above results solve the problem stated in Figure 9.26; that is how can we obtain the properties in region 2 from the known properties in region 1 and the known deflection angle θ? The answer is straightforward:

1. For the given M_1, obtain $\nu(M_1)$ from Appendix C.
2. Calculate $\nu(M_2)$ from Equation (9.43), using the known θ and the value of $\nu(M_1)$ obtained in step 1.
3. Obtain M_2 from Appendix C corresponding to the value of $\nu(M_2)$ from step 2.
4. The expansion wave is isentropic; hence, p_0 and T_0 are constant through the wave. That is, $T_{0,2} = T_{0,1}$ and $p_{0,2} = p_{0,1}$. From Equation (8.40), we have

$$\frac{T_2}{T_1} = \frac{T_2/T_{0,2}}{T_1/T_{0,1}} = \frac{1+[(\gamma-1)/2]M_1^2}{1+[(\gamma-1)/2]M_2^2} \tag{9.44}$$

From Equation (8.42), we have

$$\frac{p_2}{p_1} = \frac{p_2/p_0}{p_1/p_0} = \left(\frac{1+[(\gamma-1)/2]M_1^2}{1+[(\gamma-1)/2]M_2^2}\right)^{\gamma/(\gamma-1)} \tag{9.45}$$

Since we know both M_1 and M_2, as well as T_1 and p_1, Equations (9.44) and (9.45) allow the calculation of T_2 and p_2 downstream of the expansion wave.

EXAMPLE 9.9

A supersonic flow with $M_1 = 1.5$, $p_1 = 1$ atm, and $T_1 = 288$ K is expanded around a sharp corner (see Figure 9.26) through a deflection angle of 15°. Calculate M_2, p_2, T_2, $p_{0,2}$, $T_{0,2}$, and the angles that the forward and rearward Mach lines make with respect to the upstream flow direction.

■ Solution

From Appendix C, for $M_1 = 1.5$, $\nu_1 = 11.91°$. From Equation (9.43), $\nu_2 = \nu_1 + \theta = 11.91 + 15 = 26.91°$. Thus, $\boxed{M_2 = 2.0}$ (rounding to the nearest entry in the table).

From Appendix A, for $M_1 = 1.5$, $p_{0,1}/p_1 = 3.671$ and $T_{0,1}/T_1 = 1.45$, and for $M_2 = 2.0$, $p_{0,2}/p_2 = 7.824$ and $T_{0,2}/T_2 = 1.8$.

Since the flow is isentropic, $T_{0,2} = T_{0,1}$ and $p_{0,2} = p_{0,1}$. Thus,

$$p_2 = \frac{p_2}{p_{0,2}}\frac{p_{0,2}}{p_{0,1}}\frac{p_{0,1}}{p_1}p_1 = \frac{1}{7.824}(1)(3.671)(1 \text{ atm}) = \boxed{0.469 \text{ atm}}$$

$$T_2 = \frac{T_2}{T_{0,2}}\frac{T_{0,2}}{T_{0,1}}\frac{T_{0,1}}{T_1}T_1 = \frac{1}{1.8}(1)(1.45)(288) = \boxed{232 \text{ K}}$$

$$p_{0,2} = p_{0,1} = \frac{p_{0,1}}{p_1}p_1 = 3.671(1 \text{ atm}) = \boxed{3.671 \text{ atm}}$$

$$T_{0,2} = T_{0,1} = \frac{T_{0,1}}{T_1}T_1 = 1.45(288) = \boxed{417.6 \text{ K}}$$

Returning to Figure 9.26, we have

$$\text{Angle of forward Mach line} = \mu_1 = \boxed{41.81°}$$

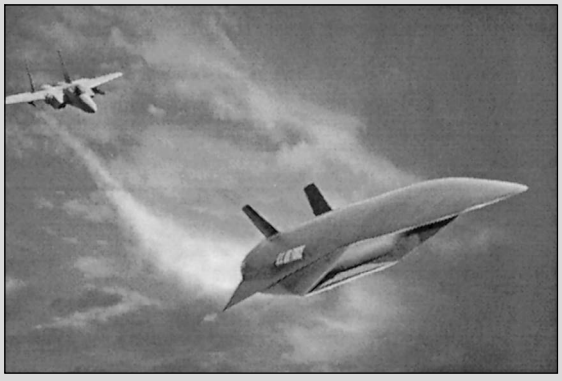
$$\text{Angle of rearward Mach line} = \mu_2 - \theta = 30 - 15 = \boxed{15°}$$

DESIGN BOX

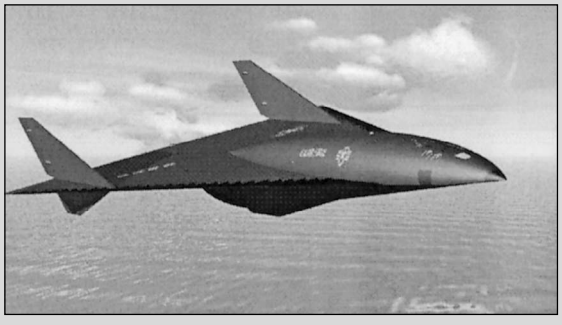
In Example 8.14 we indicated the reasons why an air-breathing power plant for high Mach number, hypersonic vehicles would have to be a supersonic combustion ramjet engine—a SCRAMjet. The design of such an engine depends heavily on the properties of oblique shock waves and expansion waves—the subjects of this chapter. In this design box, we examine some of the basic design features of SCRAMjet engines. Looking ahead to the future of aerodynamics in the twenty-first century, hypersonic flight is essentially the last frontier of our quest to fly faster and higher. Many of the hypersonic vehicles of the future will be powered by SCRAMjet engines. So the material in this design box is much like peering through a window into the future.

Several types of conceptual SCRAMjet-powered vehicles are shown in Figure 9.29, from Reference 71. Figure 9.29*a* illustrates a hypersonic air-to-surface missile, Figure 9.29*b* shows a conceptual strike/reconnaissance airplane, and Figure 9.29*c* shows the first stage of a two-stage-to-orbit vehicle for access to space. All of these futuristic vehicles would fly in the Mach 8–12 range, and all would be powered by SCRAMjet engines.

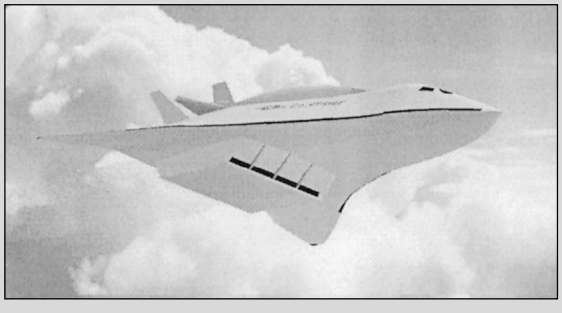
The side view of a generic hypersonic vehicle powered by a SCRAMjet is shown in Figure 9.30. Essentially, the entire bottom surface of the vehicle is an integrated portion of the air-breathing SCRAMjet engine. The forebody shock wave (1) from the nose of the vehicle is the initial part of the compression process for the engine. Air flowing through this shock wave is compressed, and then enters the SCRAMjet engine module (2) where it is further compressed by reflected shock waves inside the engine duct, mixed



(*a*) SCRAMjet-powered air-to-surface-missile concept



(*b*) SCRAMjet-powered strike/reconnaissance vehicle concept



(*c*) SCRAMjet-powered space access vehicle concept

Figure 9.29 Computer-generated images of possible future SCRAMjet-powered hypersonic vehicles. (*Courtesy of James Weber, United States Air Force.*)

with fuel, and then expanded out the back end of the module. The back end of the vehicle is scooped out (3) in order to further enhance the expansion of the exhaust gas. At the design flight condition, the forebody shock wave impinges right at the leading edge of the cowl (4), so that all the flow passing through the shock will enter the engine, rather than some of the air spilling around the external surface.

It is also possible to further compress the air before it enters the engine module by creating an isentropic compression wave downstream of the shock. This is shown in Figure 9.31, patterned after Reference 72. Here, the bottom surface of the vehicle is contoured just right to form an isentropic compression wave that will focus at the leading edge of the cowl, right where the forebody shock wave is impinging as well. An isentropic compression wave is the opposite of the isentropic expansion wave discussed in Section 9.6, but the calculation of its properties is governed by the same Prandtl-Meyer function given in Equation (9.42), except in this case the local Mach number decreases through the wave, and the pressure increases. To create such an isentropic compression wave in reality is quite difficult; the contour of the body surface must be a specific shape for a specific upstream Mach number, and most efforts over the years to produce isentropic compression waves in various supersonic and hypersonic flow devices have usually resulted in the wave prematurely coalescing into several weak shock waves with associated entropy increases and total pressure loss. SCRAMjet-powered vehicles might incorporate such an isentropic compression surface. Other physical phenomena that influence SCRAMjet engine performance and vehicle aerodynamics are also noted in Figure 9.31. The leading edge must be blunted in order to reduce the aerodynamic heating at the nose (to be discussed in Chapter 14). The viscous boundary layer over the surface of the body creates drag and aerodynamic heating, and when a shock wave impinges on the boundary layer, flow separation and local reattachment may occur, creating local regions of high heat transfer (the shock wave/boundary layer interaction problem). There is always the important question as to where transition from laminar to turbulent boundary layer flow occurs along the body, because turbulent boundary layers result in increased aerodynamic heating and skin

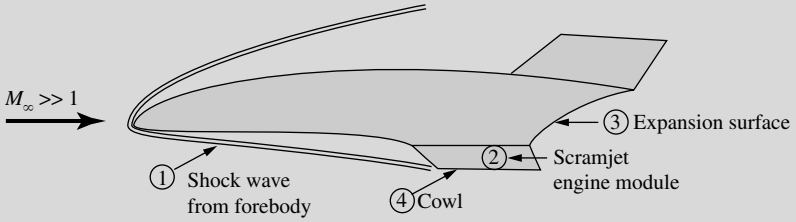


Figure 9.30 Sketch of a generic hypersonic vehicle powered by a SCRAMjet engine.

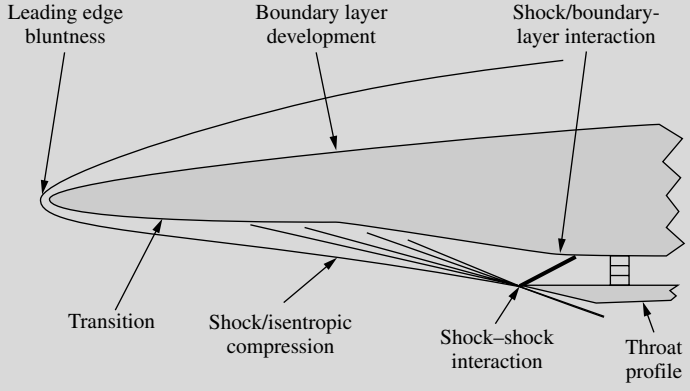


Figure 9.31 Sketch of some of the flow features on the forebody of a SCRAMjet-powered hypersonic vehicle.

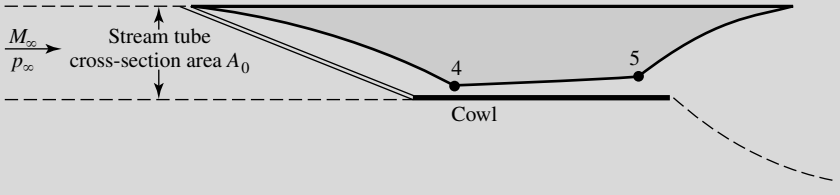


Figure 9.32 Flow path through a SCRAMjet engine.

friction. Finally, when the forebody shock impinges on the leading edge of the cowl, it will interact with the local shock wave created at the blunt leading edge of the cowl, resulting in a shock–shock interaction problem that may create a local region of intense heating at the cowl leading edge. All of these phenomena influence the quality of the flow entering the SCRAMjet module, and they pose challenging problems for the designers of future SCRAMjet engines.

A generic sketch indicating the flow path through the SCRAMjet is shown in Figure 9.32. Here again we see the forebody shock impinging at the leading edge of the cowl, and a contoured compression surface to encourage an isentropic compression behind the shock. The cross-sectional area of the streamtube flowing through the shock wave, as noted in Figure 9.32, is greatly reduced behind the shock and through the compression wave owing to the large

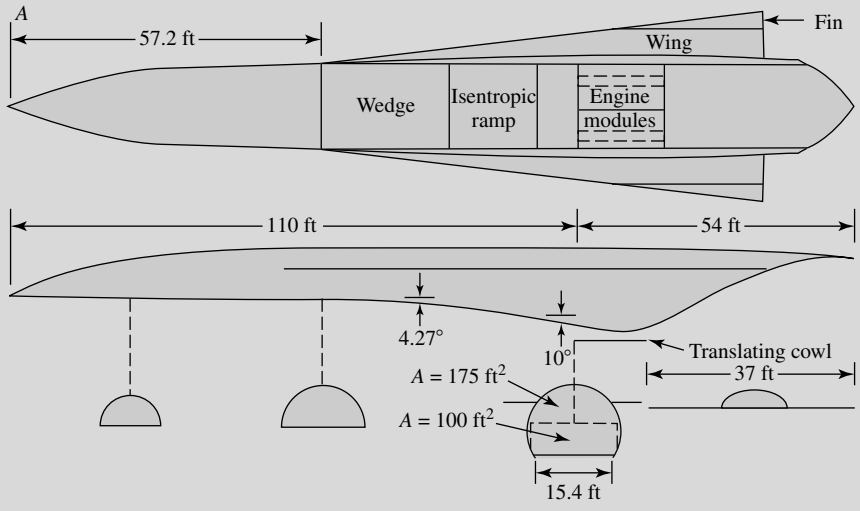


Figure 9.33 Two-sided view of a conceptual vehicle design. (*Source:* Billig, Reference 73.)

M_∞	Altitude (ft)	M_4	$\frac{A_0}{A_4}$	$\frac{p_4}{p_\infty}$	p_4 (lb/in^2)	T_4 (°R)	V_4 (ft/s)
7	80,077	3.143	10.85	47	19.03	1451	5,757
10	95,500	4.143	16.49	89.6	17.78	1958	8,744
15	114,250	5.502	25.23	185.9	15.94	2880	13,908
20	137,760	6.650	33.11	313.6	10.02	4074	19,648

increase of air density through these compression processes. Because of this, the flow path through the combustor has a much smaller cross-sectional area. In Figure 9.32, points 4 and 5 denote the entrance and exit, respectively, of the combustor. Billig (Reference 73) has calculated typical flow conditions entering the combustor (at point 4) as a function of free stream Mach number M_∞ and flight altitude. Some of his results are tabulated above, where A_0 is the cross-sectional area of the streamtube in the freestream (see Figure 9.32), A_4 is its cross-sectional area at location 4, and M_4, p_4, T_4, and V_4 are the local Mach number, pressure, temperature, and flow velocity, respectively, at location 4.

Note from the tabulation that the local Mach number entering the combustor for the given conditions ranges from about 3 to above 6. The combustion takes place in this high Mach number stream—the very essence of a SCRAMjet engine. Also note from Figure 9.32 that the cross-sectional area of the combustor increases from point 4 to point 5 along its length; this is to accommodate the local heat addition to the flow due to burning with the fuel, and still keep the flow moving at supersonic speed. (In Reference 21, among others, it is shown that heat addition to a supersonic flow slows the flow, whereas we will prove in Chapter 10 that increasing the cross-sectional area of a supersonic flow increases its speed. Hence, the combustor area in a SCRAMjet must be increased in the flow direction in order to keep the heat addition process from slowing the flow too much.)

A two-sided (bottom and side) view of a conceptual design for a SCRAMjet powered hypersonic vehicle is given in Figure 9.33, patterned after the design discussed by Billig (Reference 73). Note the slender shape for aerodynamic efficiency (high lift-to-drag ratio), the wedge and isentropic ramp to compress the flow before entering the engine modules, and the translating cowl in order to properly position the impinging shock wave on the cowl leading edge as the free stream Mach number changes. We again

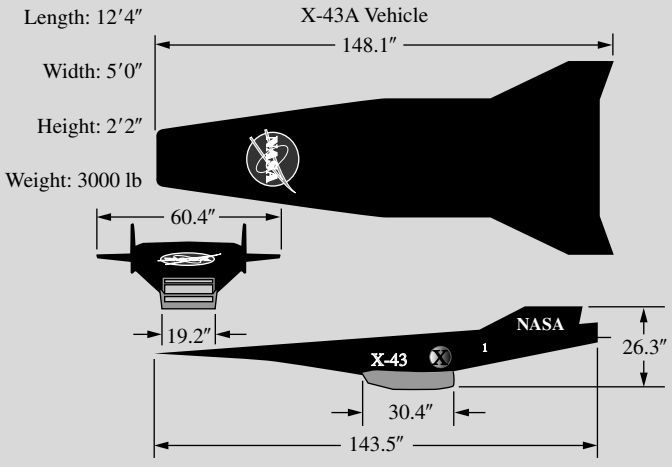


Figure 9.34 The X-43A, Hyper-X, hypersonic research vehicle (NASA).

make note that the whole undersurface of the vehicle is an integral part of the SCRAMjet engine cycle. For air-breathing hypersonic vehicles, the problem of airframe/propulsion integration is paramount; it is a major driving design feature of such aircraft.

America flew its first SCRAMjet-powered flight vehicle, the X-43, also labeled the Hyper-X, in 2004. A three-view of the X-43 is given in Figure 9.34. This small unpiloted test vehicle was launched from a modified Orbital Sciences Pegasus first stage rocket booster, which in turn was launched from a B-52 bomber in flight. The primary purpose of the X-43 is to demonstrate the viability of a SCRAMjet engine under actual flight conditions, as opposed to research results in ground test facilities. In particular, in two successful test flights it successively demonstrated performance at $M_\infty = 7$ and 10. The X-43 is a NASA project; it was the first free-flight of an airframe integrated supersonic combustion ramjet engine.

EXAMPLE 9.10

In the preceding discussion on SCRAMjet engines, an isentropic compression wave was mentioned as one of the possible compression mechanisms. Consider the isentropic compression surface sketched in Figure 9.35*a*. The Mach number and pressure upstream of the wave are $M_1 = 10$ and $p_1 = 1$ atm, respectively. The flow is turned through a total angle of 15°. Calculate the Mach number and pressure in region 2 behind the compression wave.

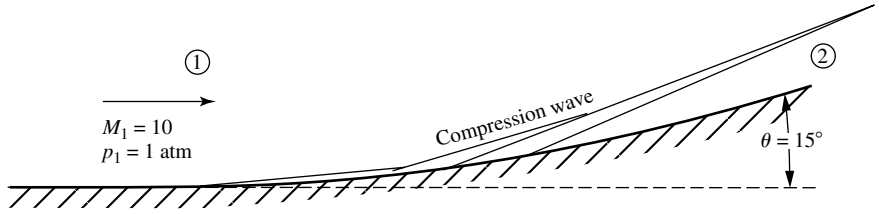
■ Solution

From Appendix C, for $M_1 = 10$, $\nu_1 = 102.3°$. In Region 2,

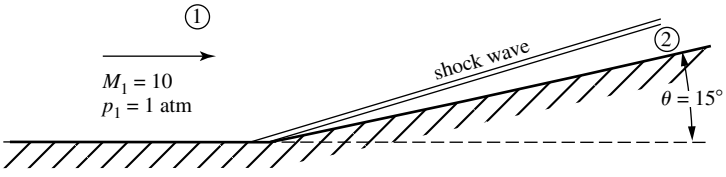
$$\nu_2 = \nu_1 - \theta = 102.3 - 15 = 87.3°$$

From Appendix C for $\nu_2 = 87.3°$, we have (closest entry)

$$\boxed{M_2 = 6.4}$$



(*a*) Isentropic compression corner



(*b*) Shock compression corner

Figure 9.35 Figures for (*a*) Example 9.9 and (*b*) Example 9.10.

From Appendix A, for $M_1 = 10$, $p_{0,1}/p_1 = 0.4244 \times 10^5$ and for $M_2 = 6.4$, $p_{0,2}/p_2 = 0.2355 \times 10^4$. Since the flow is isentropic, $p_{0,2} = p_{0,1}$, and hence

$$p_2 = \left(\frac{p_2}{p_{0,2}}\right)\left(\frac{p_{0,2}}{p_{0,1}}\right)\left(\frac{p_{0,1}}{p_1}\right)p_1 = \left(\frac{1}{0.2355 \times 10^4}\right)(1)(0.4244 \times 10^5)(1)$$

$$= \boxed{18.02 \text{ atm}}$$

EXAMPLE 9.11

Consider the flow over a compression corner with the same upstream conditions of $M_1 = 10$ and $p_1 = 1$ atm as in Example 9.10, and the same turning angle of 15°, except in this case the corner is sharp and the compression takes place through an oblique shock wave as sketched in Figure 9.35*b*. Calculate the downstream Mach number, static pressure, and total pressure in region 2. Compare the results with those obtained in Example 9.10, and comment on the significance of the comparison.

■ Solution

From Figure 9.9 for $M_1 = 10$ and $\theta = 15°$, the wave angle is $\beta = 20°$. The component of the upstream Mach number perpendicular to the wave is

$$M_{n,1} = M_1 \sin\beta = (10)\sin 20° = 34.2$$

From Appendix B for $M_{n,1} = 3.42$, we have (nearest entry), $p_2/p_1 = 13.32$, $p_{0,2}/p_{0,1} = 0.2322$, and $M_{n,2} = 0.4552$. Hence

$$M_2 = \frac{M_{n,2}}{\sin(\beta - \theta)} = \frac{0.4552}{\sin(20 - 15)} = \boxed{5.22}$$

$$p_2 = (p_2/p_1)p_1 = 13.32(1) = \boxed{13.32 \text{ atm}}$$

The total pressure in region 1 can be obtained from Appendix A as follows. For $M_1 = 10$, $p_{0,1}/p_1 = 0.4244 \times 10^5$. Hence, the total pressure in region 2 is

$$p_{0,2} = \left(\frac{p_{0,2}}{p_{0,1}}\right)\left(\frac{p_{0,1}}{p_1}\right)(p_1) = (0.2322)(0.4244 \times 10^5)(1) = \boxed{9.85 \times 10^3 \text{ atm}}$$

As a check, we can calculate $p_{0,2}$ as follows. (This check also alerts us to the error incurred when we round to the nearest entry in the tables.) From Appendix A for $M_2 = 5.22$, $p_{0,2}/p_2 = 0.6661 \times 10^3$ (nearest entry). Hence,

$$p_{0,2} = \left(\frac{p_{0.2}}{p_2}\right)(p_2) = (0.6661 \times 10^3)(13.32) = 8.87 \times 10^3 \text{ atm}$$

Note this answer is 10 percent lower than that obtained above, which is simply due to our rounding to the nearest entry in the tables. The error incurred by taking the nearest entry is exacerbated by the very high Mach numbers in this example. Much better accuracy can be obtained by properly *interpolating* between table entries.

Comparing the results from this example and Example 9.10, we clearly see that the isentropic compression is a more efficient compression process, yielding a downstream Mach number and pressure that are both considerably higher than in the case of the shock wave. The inefficiency of the shock wave is measured by the loss of total pressure across the shock; total pressure drops by about 77 percent across the shock. This emphasizes why designers of supersonic and hypersonic inlets would love to have the compression process carried out via isentropic compression waves. However, as noted in our discussion on SCRAMjets, it is very difficult to achieve such a compression in real life; the contour of the compression surface must be quite precise, and it is a point design for the given upstream Mach number. At off-design Mach numbers, even the best-designed compression contour will result in shocks.

9.7 SHOCK-EXPANSION THEORY: APPLICATIONS TO SUPERSONIC AIRFOILS

Consider a flat plate of length c at an angle of attack α in a supersonic flow, as sketched in Figure 9.36. On the top surface, the flow is turned away from itself; hence, an expansion wave occurs at the leading edge, and the pressure on the top surface p_2 is less than the freestream pressure $p_2 < p_1$. At the trailing edge, the flow must return to approximately (but not precisely) the freestream direction. Here, the flow is turned back into itself, and consequently a shock wave occurs at the trailing edge. On the bottom surface, the flow is turned into itself; an oblique shock wave occurs at the leading edge, and the pressure on the bottom surface p_3 is greater than the freestream pressure $p_3 > p_1$. At the trailing edge, the flow is turned into approximately (but not precisely) the freestream direction by means of an expansion wave. Examining Figure 9.36, note that the top and bottom surfaces of the flat plate experience uniform pressure distribution of p_2 and p_3, respectively,

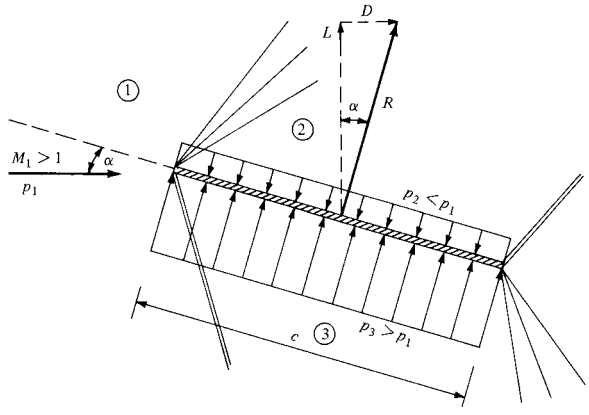


Figure 9.36 Flat plate at an angle of attack in a supersonic flow.

and that $p_3 > p_2$. This creates a net pressure imbalance that generates the resultant aerodynamic force R, shown in Figure 9.36. Indeed, for a unit span, the resultant force and its components, lift and drag, per unit span are

$$R' = (p_3 - p_2)c \tag{9.46}$$

$$L' = (p_3 - p_2)c \cos \alpha \tag{9.47}$$

$$D' = (p_3 - p_2)c \sin \alpha \tag{9.48}$$

In Equations (9.47) and (9.48), p_3 is calculated from oblique shock properties (Section 9.2), and p_2 is calculated from expansion-wave properties (Section 9.6). Moreover, these are *exact* calculations; no approximations have been made. The inviscid, supersonic flow over a flat plate at angle of attack is exactly given by the combination of shock and expansion waves sketched in Figure 9.36.

The flat-plate case given above is the simplest example of a general technique called *shock-expansion theory*. Whenever we have a body made up of straight-line segments and the deflection angles are small enough so that no detached shock waves occur, the flow over the body goes through a series of distinct oblique shock and expansion waves, and the pressure distribution on the surface (hence the lift and drag) can be obtained *exactly* from both the shock- and expansion-wave theories discussed in this chapter.

As another example of the application of shock-expansion theory, consider the diamond-shape airfoil in Figure 9.37. Assume the airfoil is at 0° angle of attack. The supersonic flow over the airfoil is first compressed and deflected through the angle ε by the oblique shock wave at the leading edge. At midchord, the flow is expanded through an angle 2ε, creating an expansion wave. At the trailing edge, the flow is turned back to the freestream direction through another oblique shock. The pressure distributions on the front and back faces of the airfoil are sketched in Figure 9.37; note that the pressures on faces a and c are uniform

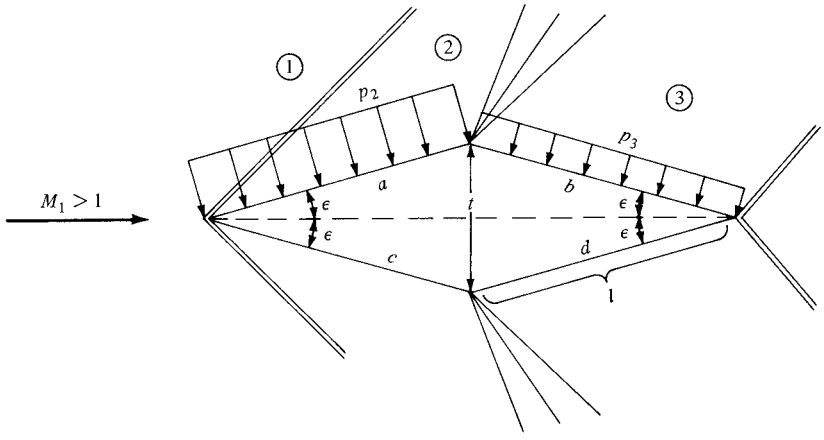


Figure 9.37 Diamond-wedge airfoil at zero angle of attack in a supersonic flow.

and equal to p_2 and that the pressures on faces b and d are also uniform but equal to p_3, where $p_3 < p_2$. In the lift direction, perpendicular to the freestream, the pressure distributions on the top and bottom faces exactly cancel (i.e., $L' = 0$). In contrast, in the drag direction, parallel to the freestream, the pressure on the front faces a and c is larger than on the back faces b and d, and this results in a finite drag. To calculate this drag (per unit span), consider the geometry of the diamond airfoil in Figure 9.37, where l is the length of each face and t is the airfoil thickness. Then,

$$D' = 2(p_2 l \sin \varepsilon - p_3 l \sin \varepsilon) = 2(p_2 - p_3)\frac{t}{2}$$

Hence,

$$D' = (p_2 - p_3)t \tag{9.49}$$

In Equation (9.49), p_2 is calculated from oblique shock theory, and p_3 is obtained from expansion-wave theory. Moreover, these pressures are the *exact* values for supersonic, inviscid flow over the diamond airfoil.

At this stage, it is worthwhile to recall our discussion in Section 1.5 concerning the source of aerodynamic force on a body. In particular, examine Equations (1.1), (1.2), (1.7), and (1.8). These equations give the means to calculate L' and D' from the pressure and shear stress distributions over the surface of a body of general shape. The results of the present section, namely, Equations (9.47) and (9.48) for a flat plate and Equation (9.49) for the diamond airfoil, are simply specialized results from the more general formulas given in Section 1.5. However, rather than formally going through the integration indicated in Equations (1.7) and (1.8), we obtained our results for the simple bodies in Figures 9.36 and 9.37 in a more direct fashion.

The results of this section illustrate a very important aspect of inviscid, supersonic flow. Note that Equation (9.48) for the flat plate and Equation (9.49) for the diamond airfoil predict a *finite drag* for these two-dimensional profiles.

This is in direct contrast to our results for two-dimensional bodies in a low-speed, incompressible flow, as discussed in Chapters 3 and 4, where the drag was theoretically zero. That is, in supersonic flow, d'Alembert's paradox does not occur. In a supersonic, inviscid flow, the drag per unit span on a two-dimensional body is finite. This new source of drag is called *wave drag,* and it represents a serious consideration in the design of all supersonic airfoils. The existence of wave drag is inherently related to the increase in entropy and consequently to the loss of total pressure across the oblique shock waves created by the airfoil.

Finally, the results of this section represent a merger of both the left- and right-hand branches of our road map shown in Figure 9.6. As such, it brings us to a logical conclusion of our discussion of oblique waves in supersonic flows.

EXAMPLE 9.12

Calculate the lift and drag coefficiens for a flat plate at a 5° angle of attack in a Mach 3 flow.

■ Solution

Refer to Figure 9.36. First, calculate p_2/p_1 on the upper surface. From Equation (9.43),

$$\nu_2 = \nu_1 + \theta$$

where $\theta = \alpha$. From Appendix C, for $M_1 = 3$, $\nu_1 = 49.76°$. Thus,

$$\nu_2 = 49.76° + 5° = 54.76°$$

From Appendix C,

$$M_2 = 3.27$$

From Appendix A, for $M_1 = 3$, $p_{0_1}/p_1 = 36.73$; for $M_2 = 3.27$, $p_{0_2}/p_2 = 55$. Since $p_{0_1} = p_{0_2}$,

$$\frac{p_2}{p_1} = \frac{p_{0_1}}{p_1} \Big/ \frac{p_{0_2}}{p_2} = \frac{36.73}{55} = 0.668$$

Next, calculate p_3/p_1 on the bottom surface. From the θ-β-M diagram (Figure 9.9), for $M_1 = 3$ and $\theta = 5°$, $\beta = 23.1°$. Hence,

$$M_{n,1} = M_1 \sin\beta = 3 \sin 23.1° = 1.177$$

From Appendix B, for $M_{n,1} = 1.177$, $p_3/p_1 = 1.458$ (nearest entry).

Returning to Equation (9.47), we have

$$L' = (p_3 - p_2)c \cos\alpha$$

The lift coefficient is obtained from

$$c_l = \frac{L'}{q_1 S} = \frac{L'}{(\gamma/2)p_1 M_1^2 c} = \frac{2}{\gamma M_1^2}\left(\frac{p_3}{p_1} - \frac{p_2}{p_1}\right)\cos\alpha$$

$$= \frac{2}{(1.4)(3)^2}(1.458 - 0.668)\cos 5° = \boxed{0.125}$$

From Equation (9.48),

$$D' = (p_3 - p_2)c \sin\alpha$$

Hence,

$$c_d = \frac{D'}{q_1 S} = \frac{2}{\gamma M_1^2}\left(\frac{p_3}{p_1} - \frac{p_2}{p_1}\right)\sin\alpha$$

$$= \frac{2}{(1.4)(3^2)}(1.458 - 0.668)\sin 5^\circ = \boxed{0.011}$$

A slightly simpler calculation for c_d is to recognize from Equations (9.47) and (9.48), or from the geometry of Figure 9.36, that

$$\frac{c_d}{c_l} = \tan\alpha$$

Hence,

$$c_d = c_l \tan\alpha = 0.125 \tan 5^\circ = 0.011$$

9.8 A COMMENT ON LIFT AND DRAG COEFFICIENTS

Expanding on the comments made in Section 9.3.1, reflect again on the result obtained in Example 9.6, where the drag coefficient was calculated for a 15° half-angle wedge in a Mach 5 flow. Reflect also on the result obtained in Example 9.12, where the lift and drag coefficients were calculated for a flat plate at a 5° angle of attack in a Mach 3 flow. Note that to calculate these coefficients we did not need to know the freestream pressure, density, or velocity. All we needed to know was:

1. The shape of the body
2. The angle of attack
3. The freestream Mach number

These examples are clear-cut illustrations of the results of dimensional analysis discussed in Section 1.7, and are totally consistent with Equations (1.42) and (1.43), which emphasize that lift and drag coefficients for a body of given shape are functions of *only* Reynolds number, Mach number, and angle of attack. For the examples in this chapter, we are dealing with an inviscid flow, so Re is not relevant—only M_∞ and α.

9.9 THE X-15 AND ITS WEDGE TAIL

Examine the photograph of the X-15 hypersonic research vehicle shown in Figure 9.18. The viewpoint in this photograph is looking down at the top of the vehicle. Concentrate on the vertical tail at the rear of the airplane. The cross section of the vertical tail is clearly seen—it is a *wedge* cross section in contrast to the type of thin symmetric airfoil sections usually employed for vertical tails on airplanes. The wedge shape is further emphasized by examining Figure 9.38,

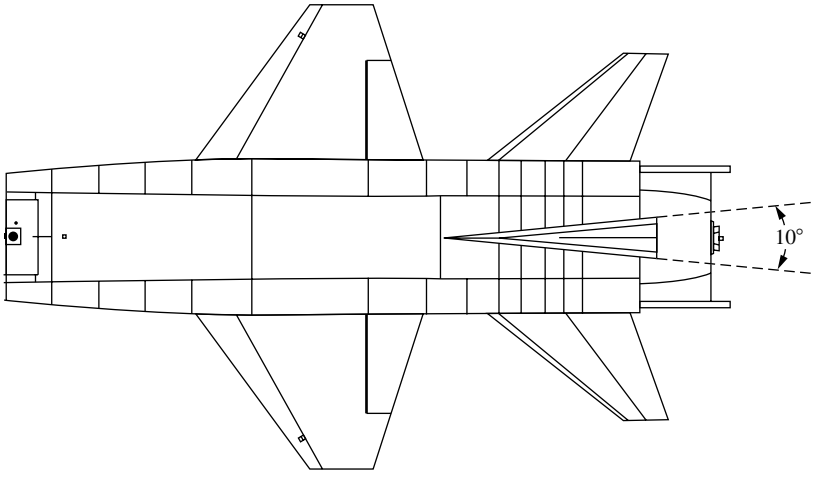


Figure 9.38 Illustration of the wedge-shaped tail on the X-15.

which shows a drawing of the rear top portion of the X-15. The included angle of the wedge cross section is 10°.

The wedge tail is one of the unique design aspects of the X-15. It came out of concern for stability problems at hypersonic speeds. The earlier X-1 and X-2 supersonic research vehicles had encountered such problems at much lower Mach numbers, and one of the major early concerns in the design of the X-15 was to find a solution that would provide stability up to Mach 7. The answer was provided by C. H. McLellan, an NACA engineer at the NACA Langley Memorial Laboratory. McLellan had carried out theoretical calculations of the influence of airfoil shape on normal force at hypersonic speeds. He found that a 10° wedge was more effective than a thin supersonic section. The X-15 designers were aware of McLellan's work, and designed the aircraft with a 10° wedge tail that provided adequate directional stability for the hypersonic vehicle.

Why is a wedge tail more effective than one with a thin section? To help answer this question, consider the following example.

EXAMPLE 9.13

Consider the flat plate shown in Figure 9.39*a* and the 10° included angle wedge shown in Figure 9.39*b*, both at an angle of attack of 10° in a Mach 7 airstream. (*a*) Calculate the lift coefficient of the flat plate. (*b*) Calculate the lift coefficient of the wedge.

■ Solution

(*a*) First, consider the expansion wave over the top of the plate. From Appendix C, for $M_1 = 7$, $\nu_1 = 90.97°$. From Equation (9.43)

$$\nu_2 = \nu_1 + \alpha = 90.97° + 10° = 100.97°$$

Interpolating to obtain M_2 from Appendix C,

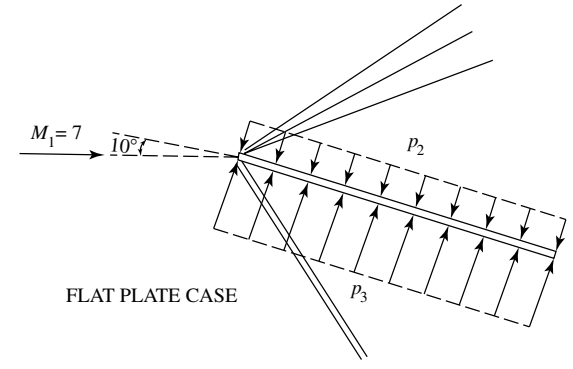
$$M_2 = 9 + \frac{100.97 - 99.32}{102.3 - 99.32}\ (1) = 9.56$$

Going to the isentropic flow tables in Appendix A, and interpolating for p_o/p between entries, we have $p_{o_2}/p_2 = 0.33 \times 10^5$. Also from Appendix A, for $M_1 = 7$, we have $p_{o_1}/p_1 = 0.14 \times 10^4$. Since p_o is constant across the expansion wave, then

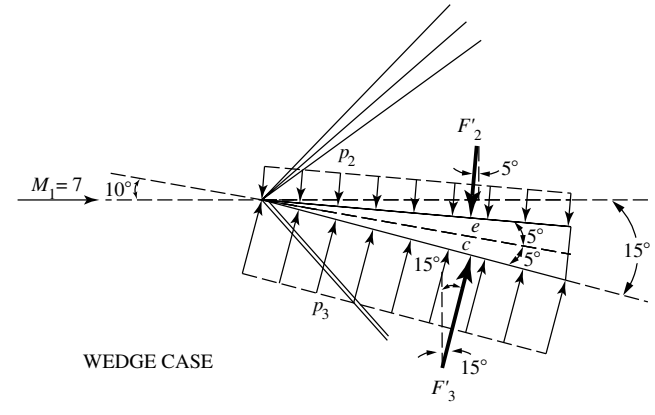
$$\frac{p_2}{p_1} = \frac{p_{o_1}/p_1}{p_{o_2}/p_2} = \frac{0.414 \times 10^4}{0.33 \times 10^5} = 0.1255$$

Now consider the shock under the bottom of the plate in Figure 9.39*a*. From the θ–β–M diagram in Figure 9.9, for $M_1 = 7$ and $\alpha = 10^\circ$, $\beta = 16.5^\circ$,

$$M_{n,1} = M_1 \sin \beta = 7 \sin 16.5^\circ = 1.99$$



(*a*)



(*b*)

Note: Angles not to scale

Figure 9.39 Schematic of hypersonic flow over (*a*) a flat plate, and (*b*) a wedge, both at a 10° angle of attack. Not to scale.

From Appendix B, for $M_{n,1} = 1.99$, interpolating, we have

$$\frac{p_3}{p_1} = 4.407 + (0.093)(0.5) = 4.45$$

The lift coefficient for a supersonic or hypersonic flat plate was derived in Example 9.12 as

$$c_\ell = \frac{2}{\gamma M_1^2}\left(\frac{p_3}{p_1} - \frac{p_2}{p_1}\right)\cos\alpha$$

$$= \frac{2}{(1.4)(7)^2}\,(4.45 - 0.1255) = \boxed{0.126}$$

(b) First consider the expansion wave over the top of the wedge.

$$\nu_2 = \nu_1 + 5^\circ = 90.97^\circ + 5^\circ = 95.97^\circ$$

From Appendix C, interpolating,

$$M_2 = 8 + \frac{95.97 - 96.62}{99.32 - 95.62}\,(1) = 8.1$$

From Appendix A, interpolating,

$$\frac{p_{o2}}{p_2} = 0.9763 \times 10^4 + (0.211 \times 10^5 - 0.9763 \times 10^4)(1) = 1.0897 \times 10^4$$

The relation between the chord length, c and the length of the face of the wedge ℓ is

$$\ell = \frac{c}{\cos 5^\circ} = \frac{c}{0.996} = 1.004c$$

The force per unit span, F_2', acting on the top surface of the wedge, is

$$F_2' = p_2\,\ell = \left(\frac{p_{o_1}/p_1}{p_{o_2}/p_2}\right) p_1\,\ell$$

For $M_1 = 7$, from Appendix A, $p_{o_1}/p_1 = 0.414 \times 10^4$. Thus,

$$F_2' = \left(\frac{0.414 \times 10^4}{1.0897 \times 10^4}\right) p_1\ell = 0.38 p_1 \ell$$

Considering the shock wave under the bottom of the wedge, we have, from the θ–β–M diagram, for $M_1 = 7$ and $\theta = 15^\circ$, $\beta = 23.5^\circ$. Thus,

$$M_{n,1} = M_1 \sin\beta = 7 \sin 23.5^\circ = 2.79$$

From Appendix B, interpolating,

$$\frac{p_3}{p_1} = 8.656 + (8.98 - 8.656)(0.8) = 8.915$$

Thus, the force per unit span, F_3', acting on the bottom surface of the wedge, is

$$F_3' = p_3\,\ell = \left(\frac{p_3}{p_1}\right) p_1\,\ell = 8.915 p_1\,\ell$$

The lift per unit span is the combination of the *components* of F_2' and F_3' perpendicular to the free stream. Examining Figure 9.39*b*, we see that

$$L' = F_3' \cos 15^\circ - F_2' \cos 5^\circ = 0.9659F_3' - 0.9962F_2'$$

$$L' = (0.9659)(8.915)p_1\ \ell - (0.9962)(0.38)p_1\ \ell$$

$$L' = 8.232p_1\ \ell$$

However, $\ell = 1.004c$, Thus,

$$L' = 8.232p_1(1.004c) = 8.265p_1c$$

The lift coefficient is

$$c_\ell = \frac{L'}{q_1 c} = \frac{L'}{(\gamma/2)p_1 M_1^2 c} = \frac{2L'}{\gamma p_1 M_1^2 c}$$

Since $L' = 8.265p_1c$, we have

$$c_\ell = \frac{2(8.265)p_1 c}{(1.4)p_1(7)^2 c} = \boxed{0.241}$$

From the results of Example 9.13, the lift coefficient for the wedge is double that for the flat plate. For a vertical tail surface, the "lift is the *side force* that creates the restoring yaw moment provided by the vertical tail when the airplane experiences a disturbance or displacement in yaw." Clearly, the wedge provides a much stronger restoring moment than a very thin airfoil shape represented by the flat plate in Figure 9.39*a*.

Physically, the wedge at angle of attack, when compared to a flat plate at the same angle of attack, is taking advantage of the nonlinear nature of supersonic shock waves. When the flat plate in Example 9.13*a* is pitched to a 10° angle of attack, the deflection angle of the flow over the bottom surface is also 10°. In contrast, with the wedge in Example 9.13*b*, the deflection angle of the flow over the bottom surface is already at 5° when the wedge is at zero angle of attack, and then is increased to 15° when the wedge is pitched to an angle of attack of 10°. This gives a shock wave angle for the wedge of 23.5°, larger than the 16.5° wave angle for the flat plate. Examining Equations (9.13) and (9.16), we note that the pressure ratio across an oblique shock varies essentially as the *square* of the wave angle. This is why the pressure ratio on the bottom surface of the wedge from Examples 9.13*b* is *twice* as large as that for the flat plate in Example 9.13*a*, $p_3/p_1 = 8.915$ for the wedge as compared to $p_3/p_1 = 4.45$ for the flat plate. The wedge, by already starting with a flow deflection angle at zero angle of attack, gets "more bang for the buck" at angle of attack.

Finally, we note that the wedge tail on the X-15 is a beautiful example of how theoretical aerodynamic research, done to extend the aerodynamic state of the art, was taken off the library shelves later on to solve a show-stopping problem of major importance to the practical design of a pioneering airplane, the

X-15. McLellan's research at the NACA helped to make the X-15 possible. (See Reference 120.)

9.10 VISCOUS FLOW: SHOCK-WAVE/ BOUNDARY-LAYER INTERACTION

Shock waves and boundary layers do not mix; bad things can happen when a shock wave impinges on a boundary layer. Unfortunately, shock-wave/boundary-layer interactions frequently occur in practical supersonic flows, and therefore we pay attention to this interaction in the present section. The fluid dynamics of a shock-wave/boundary-layer interaction is complex (and extremely interesting), and a detailed presentation is beyond the scope of this book. Here we give a brief qualitative discussion—just enough to acquaint you with the basic picture.

Consider a supersonic flow over a surface wherein an oblique shock wave impinges on the surface, such as sketched in Figure 9.19. In this figure the flow is assumed to be inviscid, and the incident shock impinges at point B on the upper wall, giving rise to a reflected shock emanating from the same point. There is a discontinuous pressure increase at point B, a combination of the pressure increases across the incident and reflected shocks. Indeed, point B is a singular point where there is an infinitely large adverse pressure gradient.

Imagine that suddenly we have a boundary layer along the wall in Figure 9.19. At point B, the boundary layer would experience an infinitely large adverse pressure gradient. In Section 4.12, we discussed what happens to a boundary layer when it experiences a large adverse pressure gradient—it separates from the surface. These are the basic elements of the shock-wave/boundary-layer interaction. The incident shock wave imposes a strong adverse pressure gradient on the boundary layer, which in turn separates from the surface, and the resulting flow field in the vicinity of the shock wave impingement becomes one of a mutual interaction between the boundary layer and the shock wave.

This mutual interaction is sketched qualitatively in Figure 9.40. Here, for ease of presentation, we show a shock wave impinging on a lower wall rather than on the upper wall as in Figure 9.19. In Figure 9.40, we see a boundary layer growing along a flat plate. Because the external flow is supersonic, the boundary-layer velocity profile is subsonic near the wall and supersonic near the outer edge. At some downstream location an incident shock impinges on the boundary layer. The large pressure rise across the shock wave acts as a severe adverse pressure gradient imposed on the boundary layer, thus causing the boundary layer to locally separate from the surface. Because the high pressure behind the shock feeds upstream through the subsonic portion of the boundary layer, the separation takes place ahead of the theoretical inviscid flow impingement point of the incident shock wave. In turn, the separated boundary layer deflects the external supersonic flow into itself, thus inducing a second shock wave, identified here as the induced separation shock wave. The separated boundary layer subsequently turns back toward the plate, reattaching to the surface at some downstream location.

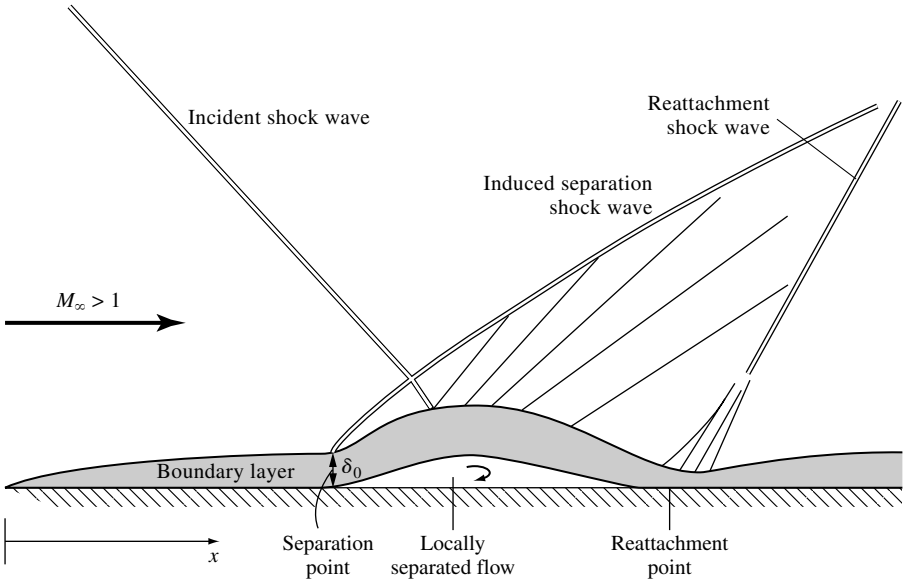


Figure 9.40 Schematic of the shock-wave/boundary-layer interaction.

Here again the supersonic flow is deflected into itself, causing a third shock wave called the *reattachment shock*. Between the separation and reattachment shocks, where the boundary layer is turning back toward the surface, the supersonic flow is turned away from itself, generating expansion waves shown in Figure 9.40. At the point of reattachment, the boundary layer has become relatively thin, the pressure is high, and consequently this becomes a region of high local aerodynamic heating. Further away from the plate, the separation and reattachment shocks merge to form the conventional reflected shock wave that is expected from the inviscid picture, as shown in Figure 9.19. The scale and severity of the interaction shown in Figure 9.40 depends on whether the boundary layer is laminar or turbulent. Since laminar boundary layers separate more readily than turbulent boundary layers (see Section 4.12), the laminar interaction usually takes place more readily with more severe attendant consequences than the turbulent interaction. However, the general qualitative aspects of the interaction shown in Figure 9.40 are the same for both cases.

The shock-wave/boundary-layer interaction has a major effect on the pressure, shear stress, and heat-transfer distributions along the wall. Of particular consequence is the high local heat-transfer rate at the reattachment point, which at hypersonic speeds can peak to an order of magnitude larger than at neighboring locations. An example of the effect on the wall pressure distribution is shown in Figure 9.41*a*, patterned after the work of Baldwin and Lomax at the NASA Ames Research Center (B. S. Baldwin, and H. Lomax, "Thin Layer Approximation and Algebraic Model for Separated Turbulent Flows," AIAA Paper No. 78-257, January 1978). Here, the pressure distribution along the wall

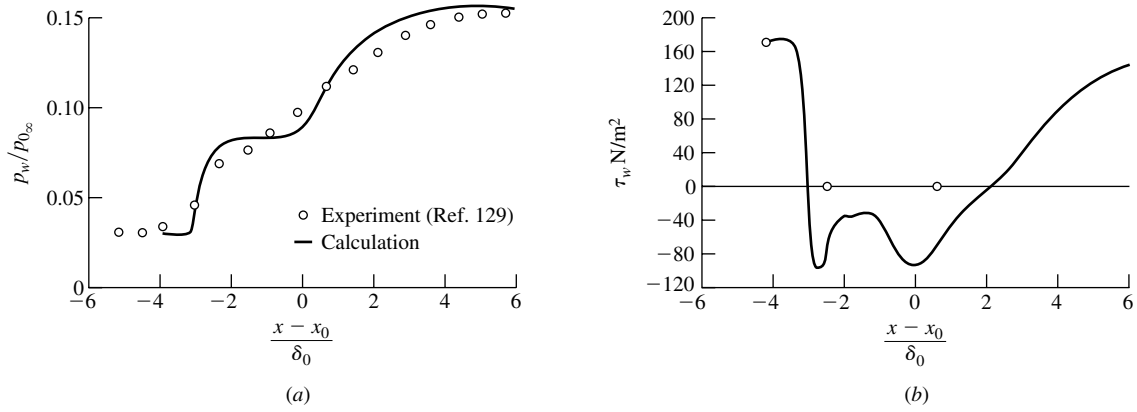


Figure 9.41 Effects of shock-wave/boundary-layer interaction on (*a*) pressure distribution, and (*b*) shear stress for Mach 3 turbulent flow over a flat plate.

in the interaction region is plotted versus distance along the wall, x, where x_0 is the theoretical point of impingement for the incident shock in the inviscid flow case. The pressure distribution shows a steplike increase, with an intermediate plateau; this distribution is typical of the shock-wave/boundary-layer interaction. The external flow is at Mach 3 ahead of the incident shock, and the boundary layer is turbulent. The solid curve is a computational fluid dynamic (CFD) calculation, and the circles are experimental data. Notice that the pressure increase extends a distance equal to about four times the boundary-layer thickness ahead of the theoretical inviscid impingement point. The shear stress distribution is shown in Figure 9.41*b*. In the pocket of separated flow, τ_w becomes small and reverses its direction (negative values of τ_w) due to the low energy recirculating flow.

Because of the consequent creation of separated flow, increased loss of total pressure, and high peak heat-transfer rates, shock-wave/boundary-layer interactions usually should be avoided as much as possible in the design of supersonic aircraft and flow devices. However, this is easier said than done. Shock-wave/boundary-layer interactions are a fact of life in the practical world of supersonic flow, and that is why we have discussed their basic nature in this section. On the other hand, modern creative ideas have led to the beneficial use of the separated flow from a shock-wave/boundary-layer interaction to actually enhance the off-design performance of jet-engine exhaust nozzles and for certain types of flow control. So the picture is not entirely black.

9.11 HISTORICAL NOTE: ERNST MACH—A BIOGRAPHICAL SKETCH

The Mach number is named in honor of Ernst Mach, an Austrian physicist and philosopher who was an illustrious and controversial figure in late nineteenth-century physics. Mach conducted the first meaningful experiments in supersonic flight, and his results triggered a similar interest in Ludwig Prandtl 20 years later.

Who was Mach? What did he actually accomplish in supersonic aerodynamics? Let us look at this man further.

Mach was born at Turas, Moravia, in Austria, on February 18, 1838. His father, Johann, was a student of classical literature who settled with his family on a farm in 1840. An extreme individualist, Johann raised his family in an atmosphere of seclusion, working on various improved methods of farming, including silkworm cultivation. Ernst's mother, on the other hand, came from a family of lawyers and doctors and brought with her a love of poetry and music. Ernst seemed to thrive in this family atmosphere. Until the age of 14, his education came exclusively from instruction by his father, who read extensively in the Greek and Latin classics. In 1853, Mach entered public school, where he became interested in the world of science. He went on to obtain a Ph.D. in physics in 1860 at the University of Vienna, writing his dissertation on electrical discharge and induction. In 1864, he became a full professor of mathematics at the University of Graz and was given the title of Professor of Physics in 1866. Mach's work during this period centered on optics—a subject which was to interest him for the rest of his life. The year 1867 was important for Mach—during that year he married, and he also became a professor of experimental physics at the University of Prague, a position he held for the next 28 years. While at Prague, Mach published over 100 technical papers—work that was to constitute the bulk of his technical contributions.

Mach's contribution to supersonic aerodynamics involves a series of experiments covering the period from 1873 to 1893. In collaboration with his son, Ludwig, Mach studied the flow over supersonic projectiles, as well as the propagation of sound waves and shock waves. His work included the flow fields associated with meteorites, explosions, and gas jets. The main experimental data were photographic results. Mach combined his interest in optics and supersonic motion by designing several photographic techniques for making shock waves in air visible. He was the first to use the schlieren system in aerodynamics; this system senses density gradients and allows shock waves to appear on screens or photographic negatives. He also devised an interferometric technique that senses directly the change in density in a flow. A pattern of alternate dark and light bands are set up on a screen by the superposition of light rays passing through regions of different density. Shock waves are visible as a shift in this pattern along the shock. Mach's optical device still perpetuates today in the form of the Mach-Zehnder interferometer, an instrument present in many aerodynamic laboratories. Mach's major contributions in supersonic aerodynamics are contained in a paper given to the Academy of Sciences in Vienna in 1887. Here, for the first time in history, Mach shows a photograph of a weak wave on a slender cone moving at supersonic speed, and he demonstrates that the angle μ between this wave and the direction of flight is given by $\sin \mu = a/V$. This angle was later denoted as the Mach angle by Prandtl and his colleagues after their work on shock and expansion waves in 1907 and 1908. Also, Mach was the first person to point out the discontinuous and marked changes in a flow field as the ratio V/a changes from below 1 to above 1.

It is interesting to note that the ratio V/a was not denoted as Mach number by Mach himself. Rather, the term "Mach number" was coined by the Swiss

engineer Jacob Ackeret in his inaugural lecture in 1929 as Privatdozent at the Eidgenossiche Technische Hochschule in Zurich. Hence, the term "Mach number" is of fairly recent usage, not being introduced into the English literature until the mid-1930s.

In 1895, the University of Vienna established the Ernst Mach chair in the philosophy of inductive sciences. Mach moved to Vienna to occupy this chair. In 1897 he suffered a stroke which paralyzed the right side of his body. Although he eventually partially recovered, he officially retired in 1901. From that time until his death on February 19, 1916 near Munich, Mach continued to be an active thinker, lecturer, and writer.

In our time, Mach is most remembered for his early experiments on supersonic flow and, of course, through the Mach number itself. However, Mach's contemporaries, as well as Mach himself, viewed him more as a philosopher and historian of science. Coming at the end of the nineteenth century, when most physicists felt comfortable with newtonian mechanics, and many believed that virtually all was known about physics, Mach's outlook on science is summarized by the following passage from his book *Die Mechanik:*

> The most important result of our reflections is that precisely the apparently simplest mechanical theorems are of a very complicated nature; that they are founded on incomplete experiences, even on experiences that never can be fully completed; that in view of the tolerable stability of our environment they are, in fact, practically safeguarded to serve as the foundation of mathematical deduction; but that they by no means themselves can be regarded as mathematically established truths, but only as theorems that not only admit of constant control by experience but actually require it.

In other words, Mach was a staunch experimentalist who believed that the established laws of nature were simply theories and that only observations that are apparent to the senses are the fundamental truth. In particular, Mach could not accept the elementary ideas of atomic theory or the basis of relativity, both of which were beginning to surface during Mach's later years and, of course, were to form the basis of twentieth-century modern physics. As a result, Mach's philosophy did not earn him favor with most of the important physicists of his day. Indeed, at the time of his death, Mach was planning to write a book pointing out the flaws of Einstein's theory of relativity.

Although Mach's philosophy was controversial, he was respected for being a thinker. In fact, in spite of Mach's critical outlook on the theory of relativity, Albert Einstein had the following to say in the year of Mach's death: "I even believe that those who consider themselves to be adversaries of Mach scarcely know how much of Mach's outlook they have, so to speak, adsorbed with their mother's milk."

Hopefully, this section has given you a new dimension to think about whenever you encounter the term "Mach number." Maybe you will pause now and then to reflect on the man himself and to appreciate that the term "Mach number" is in honor of a man who devoted his life to experimental physics, but who at the same time was bold enough to view the physical world through the eyes of a self-styled philosopher.

9.12 SUMMARY

The road map given in Figure 9.6 illustrates the flow of our discussion on oblique waves in supersonic flow. Review this road map, and make certain that you are familiar with all the ideas and results that are represented in Figure 9.6.

Some of the more important results are summarized as follows:

An infinitesimal disturbance in a multidimensional supersonic flow creates a Mach wave that makes an angle μ with respect to the upstream velocity. This angle is defined as the Mach angle and is given by

$$\mu = \sin^{-1} \frac{1}{M} \tag{9.1}$$

Changes across an oblique shock wave are determined by the normal component of velocity ahead of the wave. For a calorically perfect gas, the normal component of the upstream Mach number is the determining factor. Changes across an oblique shock can be determined from the normal shock relations derived in Chapter 8 by using $M_{n,1}$ in these relations, where

$$M_{n,1} = M_1 \sin \beta \tag{9.13}$$

Changes across an oblique shock depend on two parameters, for example, M_1 and β, or M_1 and θ. The relationship between M_1, β, and θ is given in Figure 9.9, which should be studied closely.

Oblique shock waves incident on a solid surface reflect from that surface in such a fashion to maintain flow tangency on the surface. Oblique shocks also intersect each other, with the results of the intersection depending on the arrangement of the shocks.

The governing factor in the analysis of a centered expansion wave is the Prandtl-Meyer function $\nu(M)$. The key equation which relates the downstream Mach number M_2, the upstream Mach number M_1, and the deflection angle θ is

$$\theta = \nu(M_2) - \nu(M_1) \tag{9.43}$$

The pressure distribution over a supersonic airfoil made up of straight-line segments can usually be calculated exactly from a combination of oblique and expansion waves—that is, from exact shock-expansion theory.

9.13 PROBLEMS

9.1 A slender missile is flying at Mach 1.5 at low altitude. Assume the wave generated by the nose of the missile is a Mach wave. This wave intersects the ground 559 ft behind the nose. At what altitude is the missile flying?

9.2 Consider an oblique shock wave with a wave angle of 30° in a Mach 4 flow. The upstream pressure and temperature are 2.65×10^4 N/m^2 and 223.3 K, respectively (corresponding to a standard altitude of 10,000 m). Calculate the pressure, temperature, Mach number, total pressure, and total temperature behind the wave and the entropy increase across the wave.

9.3 Equation (8.80) does *not* hold for an oblique shock wave, and hence the column in Appendix B labeled $p_{0,2}/p_1$ *cannot* be used, in conjunction with the normal component of the upstream Mach number, to obtain the total pressure behind an oblique shock wave. On the other hand, the column labeled $p_{0,2}/p_{0,1}$ can be used for an oblique shock wave, using $M_{n,1}$. Explain why all this is so.

9.4 Consider an oblique shock wave with a wave angle of 36.87°. The upstream flow is given by $M_1 = 3$ and $p_1 = 1$ atm. Calculate the total pressure behind the shock using

a. $p_{0,2}/p_{0,1}$ from Appendix B (the correct way)

b. $p_{0,2}/p_1$ from Appendix B (the incorrect way)

Compare the results.

9.5 Consider the flow over a 22.2° half-angle wedge. If $M_1 = 2.5$, $p_1 = 1$ atm, and $T_1 = 300$ K, calculate the wave angle and p_2, T_2, and M_2.

9.6 Consider a flat plate at an angle of attack α to a Mach 2.4 airflow at 1 atm pressure. What is the maximum pressure that can occur on the plate surface and still have an attached shock wave at the leading edge? At what value of α does this occur?

9.7 A 30.2° half-angle wedge is inserted into a freestream with $M_\infty = 3.5$ and $p_\infty = 0.5$ atm. A Pitot tube is located above the wedge surface and behind the shock wave. Calculate the magnitude of the pressure sensed by the Pitot tube.

9.8 Consider a Mach 4 airflow at a pressure of 1 atm. We wish to slow this flow to subsonic speed through a system of shock waves with as small a loss in total pressure as possible. Compare the loss in total pressure for the following three shock systems:

a. A single normal shock wave

b. An oblique shock with a deflection angle of 25.3°, followed by a normal shock

c. An oblique shock with a deflection angle of 25.3°, followed by a second oblique shock of deflection angle of 20°, followed by a normal shock

From the results of (*a*), (*b*), and (*c*), what can you induce about the efficiency of the various shock systems?

9.9 Consider an oblique shock generated at a compression corner with a deflection angle $\theta = 18.2^\circ$. A straight horizontal wall is present above the corner, as shown in Figure 9.19. If the upstream flow has the properties $M_1 = 3.2$, $p_1 = 1$ atm and $T_1 = 520^\circ$R, calculate M_3, p_3, and T_3 behind the reflected shock from the upper wall. Also, obtain the angle Φ which the reflected shock makes with the upper wall.

9.10 Consider the supersonic flow over an expansion corner, such as given in Figure 9.25. The deflection angle $\theta = 23.38^\circ$. If the flow upstream of the corner is given by $M_1 = 2$, $p_1 = 0.7$ atm, $T_1 = 630^\circ$R, calculate M_2, p_2, T_2, ρ_2, $p_{0,2}$, and $T_{0,2}$ downstream of the corner. Also, obtain the angles the forward and rearward Mach lines make with respect to the upstream direction.

9.11 A supersonic flow at $M_1 = 1.58$ and $p_1 = 1$ atm expands around a sharp corner. If the pressure downstream of the corner is 0.1306 atm, calculate the deflection angle of the corner.

9.12 A supersonic flow at $M_1 = 3$, $T_1 = 285$ K, and $p_1 = 1$ atm is deflected upward through a compression corner with $\theta = 30.6^\circ$ and then is subsequently expanded around a corner of the same angle such that the flow direction is the same as its original direction. Calculate M_3, p_3, and T_3 downstream of the expansion corner. Since the resulting flow is in the same direction as the original flow, would you expect $M_3 = M_1$, $p_3 = p_1$, and $T_3 = T_1$? Explain.

9.13 Consider an infinitely thin flat plate at an angle of attack α in a Mach 2.6 flow. Calculate the lift and wave-drag coefficients for
(*a*) $\alpha = 5^\circ$ (*b*) $\alpha = 15^\circ$ (*c*) $\alpha = 30^\circ$
(*Note:* Save the results of this problem for use in Chapter 12.)

9.14 Consider a diamond-wedge airfoil such as shown in Figure 9.36, with a half-angle $\varepsilon = 10^\circ$. The airfoil is at an angle of attack $\alpha = 15^\circ$ to a Mach 3 freestream. Calculate the lift and wave-drag coefficients for the airfoil.

9.15 Consider sonic flow. Calculate the maximum deflection angle through which this flow can be expanded via a centered expansion wave.

9.16 Consider a circular cylinder (oriented with its axis perpendicular to the flow) and a symmetric diamond-wedge airfoil with a half-angle of 5° at zero angle of attack; both bodies are in the same Mach 5 freestream. The thickness of the airfoil and the diameter of the cylinder are the same. The drag coefficient (based on projected frontal area) of the cylinder is 4/3. Calculate the *ratio* of the cylinder drag to the diamond airfoil drag. What does this say about the aerodynamic performance of a blunt body compared to a sharp-nosed slender body in supersonic flow?

9.17 Consider the supersonic flow over a flat plate at angle of attack, as sketched in Figure 9.35. As stated in Section 9.7, the flow direction downstream of

the trailing edge of the plate, behind the trailing edge shock and expansion waves, is not precisely in the free stream direction. Why? Outline a method to calculate the strengths of the trailing edge shock and expansion waves, and the direction of the flow downstream of the trailing edge.

9.18 (The purpose of this problem is to calculate a two-dimensional expanding supersonic flow and compare it with the analogous quasi-one-dimensional flow in Problem 10.15.) Consider a two-dimensional duct with a straight horizontal lower wall, and a straight upper wall inclined upward through the angle $\theta = 3^\circ$. The height of the duct entrance is 0.3 m. A uniform horizontal flow at Mach 2 enters the duct and goes through a Prandtl-Mayer expansion wave centered at the top corner of the entrance. The wave propagates to the bottom wall, where the leading edge (the forward Mach line) of the wave intersects the bottom wall at point A located at distance x_A from the duct entrance. Imagine a line drawn perpendicular to the lower wall at point A, and intersecting the upper wall at point B. The local height of the duct at point A is the length of this line AB. Calculate the *average* flow Mach number over AB, assuming that M varies linearly along that portion of AB inside the expansion wave.

9.19 Repeat Problem 9.18, except with $\theta = 30^\circ$. Again, we will use these results to compare with a quasi-one-dimensional calculation in Problem 10.16. The reason for repeating this calculation is to examine the effect of the much more highly two-dimensional flow generated in this case by a much larger expansion angle.

9.20 Consider a Mach 3 flow at 1 atm pressure initially moving over a flat horizontal surface. The flow then encounters a 20 degree expansion corner, followed by a 20 degree compression corner that turns the flow back to the horizontal. Calculate the pressure of the flow downstream of the compression corner. *Note:* You will find that the pressure downstream of the compression corner is different from the pressure upstream of the expansion corner, even though the upstream and downstream flows are in the same direction, namely horizontal. Why?

9.21 The purpose of this problem is to explain what causes the dramatic white cloud pattern generated in the flow field over the F-22 shown on the cover of this book. This problem is both a tutorial and a quantitative calculation involving the reader. We first discuss some necessary thermodynamic background, followed by an examination of the physical nature of the flowfield.

Necessary Thermodynamic Background

The white cloud surrounding the F-22 is condensed water vapor in the airflow over the airplane. Inside the cloud, water appears in two phases, vapor and liquid. Both inside and outside the cloud, the medium is a mixture of air and water; outside the cloud the water is completely in vapor form, and inside the cloud there is a mixture of air, water vapor, and

condensed liquid water. Both inside and outside the cloud, the gas pressure p is the sum of the partial pressure of air and the much smaller partial pressure of water vapor, p_{air} and p_{H2O}, respectively; i.e., $p = p_{air} + p_{H2O}$. On a graph of p_{H2O} versus temperature for this water-air mixture, we can construct a type of "phase diagram," where over part of this diagram the water is in the vapor phase only, and over another part the water is a liquid-vapor mixture. The curve dividing these two regions is called the saturation curve, which arches upward toward the right (on the saturation curve, p_{H2O} increases with increasing temperature, with the slope of this curve, dp_{H2O}/dT, increasing as T increases). To the right of this curve, the water is completely vapor, and to the left of this curve, the water is a mixture of vapor and liquid. What we see as the white cloud over the top of the F-22 is light reflecting from the condensed droplets of water; i.e., the white cloud is that region of the airflow that is in the liquid-vapor region of the phase diagram, namely that region to the left of the saturation curve.

Every point on the saturation curve corresponds to 100 percent relative humidity in the ambient air. The temperature at each point on the curve is the "dew-point" temperature. In the region to the right of this curve the air is at less than 100 percent humidity, the temperature is greater than the dew point, and there is no condensation of the water vapor. In the region to the left of the curve, the temperature falls below the dew point and condensation starts to occur.

The saturation curve is obtained experimentally. For this problem, we use the saturation curve for an ambient pressure of 1 atm found on page 4-86 of *Marks' Mechanical Engineer's Handbook*, 6th Ed., McGraw-Hill, 1958 (we use an older edition because it is on the author's personal book shelf and therefore handy). This is a curve of p_{H2O} versus T for air at 100 percent relative humidity with an ambient pressure of 1 atm.

For this problem, let us choose a particular point on this saturation curve, namely the point where $T = 520^\circ$ R and $p_{H2O} = 38.16$ lb/ft^2. Call this point A. At point A, air has 100 percent humidity and contains an amount of water vapor corresponding to a partial pressure of $p_{H2O} = 38.16$ lb/ft^2. The *slope* of the curve, dp_{H2O}/dT, at point A is important to this problem. The slope is

$$dp_{H2O}/dT = 1.08 \text{ lb}/(\text{ft}^2 \cdot {}^\circ\text{R}) \qquad \text{saturation curve}$$

Nature of the Flowfield

The F-22 on the cover of this book is flying supersonically at a small angle of attack; note that the trailing wing-tip vortices, also made visible by water condensation, are very thin and tightly wound, characteristic of tip vortices in a supersonic flow. Expansion waves occur over the top surface of the wings and fuselage. The expansion region is terminated by a trailing-edge shock wave. The objective of this problem is to prove that water condensation occurs within the expansion wave, but that the water

goes back to the purely vapor phase when it passes through the trailing-edge shock wave. We submit that the white cloud shown around the F-22 is caused by water condensation within the expansion wave, and that the rather sharp termination of the cloud is caused by water evaporation when the flow passes through the trailing-edge shock wave.

For a simplified model of this flow, consider the flat plate at an angle-of-attack in a supersonic flow as sketched in Figure 9.36. The supersonic flow over the top surface of the plate first goes through an expansion wave, then passes through the trailing-edge shock wave downstream. If the free stream upstream of the plate has sufficiently high humidity, a white cloud of condensed water vapor will form through and downstream of the expansion wave. However, the white cloud will disappear when the condensed water vapor is vaporized as the flow passes through the trailing-edge shock wave. So in Figure 9.36, the white cloud will occur in the region bounded upstream by the expansion wave, and downstream by the trailing-edge shock wave. This simulates the nature of the white cloud seen over the top of the F-22 on the cover of this book.

Statement of the Problem

Consider the flow field sketched in Figure 9.36. Assume the supersonic free stream is at a temperature of 520 °R and a pressure of 1 atm, and that it has 100 percent humidity. Therefore, the thermodynamic state of the free stream corresponds to point *A* on the saturation curve of the phase diagram discussed earlier.

(a) Prove that a fluid element from the free stream, when passing into the expansion wave, will experience water condensation.

(b) Prove that a fluid element already within and downstream of the expansion wave and hence with some condensed water, will experience water evaporation when passing through the trailing-edge shock wave.

Comment and Hint

Well, you might say this is obvious. When the flow expands through the expansion wave, the temperature decreases, and water will obviously condense. Hence the fluid element in the phase diagram will obviously penetrate through point *A* and enter the liquid-vapor region. But wait a minute! When the fluid element enters the expansion wave, *both* its temperature and pressure will decrease. The temperature decrease, by itself, will move the properties of the fluid element toward the left of point *A* into the liquid-vapor region, but the pressure decrease by itself will move the properties of the fluid element vertically downward from point *A*, farther away from the liquid-vapor region. The question is: Which change is more dominant, the temperature decrease or the pressure decease? This is for you to find out. As a hint, calculate the value of the

derivative dp_{H2O}/dT for the fluid element as it enters the expansion wave. If this derivative is smaller than the slope of the saturation curve at point A, given earlier as $dp_{H2O}/dT = 1.08$ lb/(ft$^2 \cdot {}^\circ$R), then the properties of the fluid element as it enters the expansion wave will also penetrate across point A into the liquid-vapor part of the phase diagram, and clearly water condensation will then occur.

Similarly, for a fluid element crossing the shock wave, you might also say that it is obvious that condensed water in the fluid element will evaporate because the temperature increases across the shock, which causes evaporation. But again, wait a minute! Consider that the properties of the fluid element just in front of the shock wave are in the liquid-vapor region just to the left of point A. *Both* the temperature and pressure will increase across the shock wave. The increase in temperature will move the properties of the fluid to the right of point A, away from the liquid-vapor region and into the purely vapor region, but the increase in pressure will move the properties of the fluid element vertically upward from point A, farther into the liquid-vapor region. Again, the question is: Which change is more dominant, the temperature increase or the pressure increase? This is for you to find out. As a hint, calculate the value $dp_{H2O}/dT \approx \Delta p_{H2O}/\Delta T$ across the shock wave. If this derivative is *smaller* than the slope of the saturation curve at point A, given earlier as $dp_{H2O}/dT = 1.08$ lb/(ft$^2 \cdot {}^\circ$R), then the properties of the fluid element as it crosses the shock wave will also exit the liquid-vapor phase at point A and enter the purely vapor region, and clearly evaporation will take place. To calculate $\Delta p_{H2O}/\Delta T$, you have to arbitrarily assume the strength of the shock wave. Assume the shock is a normal shock wave with an upstream Mach number of 1.2, and the upstream properties of the flow are those just minutely to the left of point A; i.e., assume ahead of the shock wave $M_1 = 1.2$, $p_1 = 2116$ lb/ft^2, and $p_{H2O} = 38.16$ lb/ft^2.

Answers

(a) When the fluid element enters the expansion wave, $dp_{H2O}/dT = 0.256$ lb/(ft$^2 \cdot {}^\circ$R). This is a smaller slope than that of the saturation curve at point A. The properties of the fluid will penetrate into the liquid-vapor region, and condensation will occur.

(b) When the fluid element in front of the shock wave with properties just inside the liquid-vapor region at point A crosses the shock wave, $\Delta p_{H2O}/\Delta T = 0.2923$ lb/(ft$^2 \cdot {}^\circ$R). This is a smaller slope than that of the saturation curve at point A. The properties of the fluid element will penetrate into the purely vapor region, and evaporation will occur.

CHAPTER 10

Compressible Flow through Nozzles, Diffusers, and Wind Tunnels

Having wondered from what source there is so much difficulty in successfully applying the principles of dynamics to fluids than to solids, finally, turning the matter over more carefully in my mind, I found the true origin of the difficulty; I discovered it to consist of the fact that a certain part of the pressing forces important in forming the throat *(so called by me, not considered by others) was neglected, and moreover regarded as if of no importance, for no other reason than the throat is composed of a very small, or even an infinitely small, quantity of fluid, such as occurs whenever fluid passes from a wider place to a narrower, or vice versa, from a narrower to a wider.*

Johann Bernoulli; from his Hydraulics, 1743

PREVIEW BOX

One of the best examples of the harnessing of nature by aerospace engineers is shown in Figure 10.1, which is a photograph of the main rocket engine for the space shuttle. This engine produces over 400,000 lb of thrust. The laws of nature have been used by aerospace engineers to design this engine. But what laws of nature? Take another look at Figure 10.1, and note the large bell-like *divergent* nozzle of the rocket engine. Why this shape instead of some convergent shape like the low-speed wind tunnel nozzles discussed in Section 3.3? In reality, in Figure 10.1, we are seeing only part of the rocket nozzle; hidden behind all the plumbing to the left of the divergent duct in Figure 10.1 is a combustion chamber that feeds the hot, high-pressure gas into a convergent duct that transitions to the divergent duct seen in

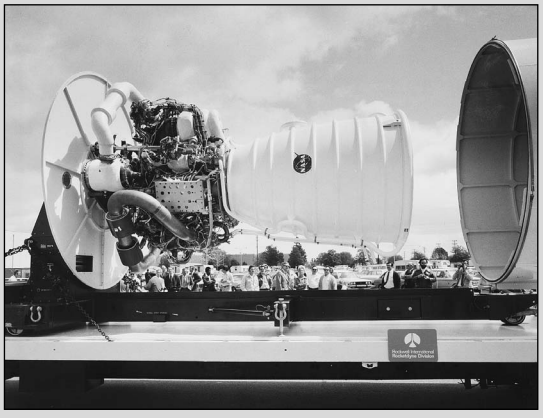


Figure 10.1 Space shuttle main rocket engine (NASA).

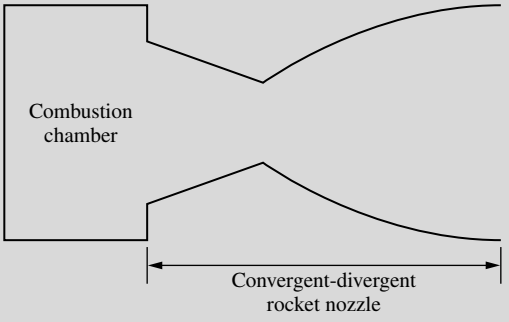


Figure 10.2 Schematic of rocket engine nozzle.

Figure 10.1. The whole rocket engine nozzle, from the combustion chamber to the exit, is a convergent–divergent shape such as sketched in Figure 10.2. Why?

The answers to these questions and more are given in the present chapter. Here we deal with the *internal* flow of a high-speed compressible gas through ducts of various shapes. The material in this chapter is essential to the design of rocket engines, jet engines, supersonic and hypersonic wind tunnels, and any flow device that involves the compressible flow of a gas internal to the device. We labeled the study of shock waves in the previous two chapters as the bread and butter of compressible flow. By comparison, the study of the flow through nozzles, diffusers, and wind tunnels in the present chapter is the rolls and jam of compressible flow. Applications of the material in this chapter are made daily in the modern world of aerospace engineering. Moreover, this material is packed with interesting physical phenomena, some of which you likely have not thought about before and which you will find almost amazing in some instances. Take this chapter very seriously. I predict that you will find it enjoyable and rewarding.

10.1 INTRODUCTION

Chapters 8 and 9 treated normal and oblique waves in supersonic flow. These waves are present on any aerodynamic vehicle in supersonic flight. Aeronautical engineers are concerned with observing the characteristics of such vehicles, especially the generation of lift and drag at supersonic speeds, as well as details of the flow field, including the shock- and expansion-wave patterns. To make such observations, we usually have two standard choices: (1) conduct flight tests using the actual vehicle, and (2) run wind-tunnel tests on a small-scale model of the vehicle. Flight tests, although providing the final answers in the full-scale environment, are costly and, not to say the least, dangerous if the vehicle is unproven. Hence, the vast bulk of supersonic aerodynamic data have been obtained in wind tunnels on the ground. What do such supersonic wind tunnels look like? How do

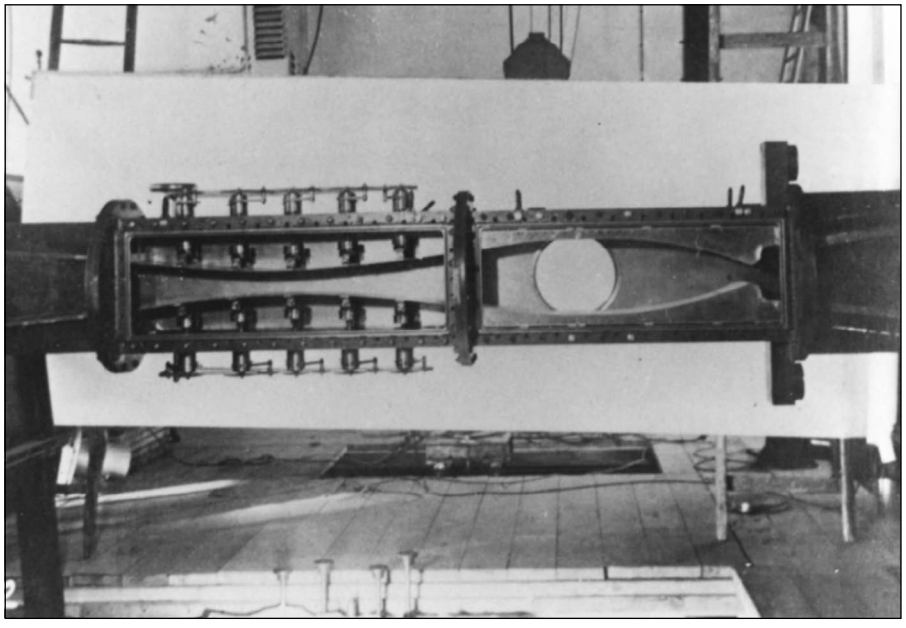


Figure 10.3 The first practical supersonic wind tunnel, built by A. Busemann in Germany in the mid-1930s. (*Courtesy of the John Anderson Collection.*)

we produce a uniform flow of supersonic gas in a laboratory environment? What are the characteristics of supersonic wind tunnels? The answers to these and other questions are addressed in this chapter.

The first practical supersonic wind tunnel was built and operated by Adolf Busemann in Germany in the mid-1930s, although Prandtl had a small supersonic facility operating as early as 1905 for the study of shock waves. A photograph of Busemann's tunnel is shown in Figure 10.3. Such facilities proliferated quickly during and after World War II. Today, all modern aerodynamic laboratories have one or more supersonic wind tunnels, and many are equipped with hypersonic tunnels as well. Such machines come in all sizes; an example of a moderately large hypersonic tunnel is shown in Figure 10.4.

In this chapter, we discuss the aerodynamic fundamentals of compressible flow through ducts. Such fundamentals are vital to the proper design of high-speed wind tunnels, rocket engines, high-energy gas-dynamic and chemical lasers, and jet engines, to list just a few. Indeed, the material developed in this chapter is used almost daily by practicing aerodynamicists and is indispensable toward a full understanding of compressible flow.

The road map for this chapter is given in Figure 10.5. After deriving the governing equations, we treat the cases of a nozzle and diffuser separately. Then we merge this information to examine the case of supersonic wind tunnels.



Figure 10.4 A large hypersonic wind tunnel at the U.S. Air Force Wright Aeronautical Laboratory, Dayton, Ohio. (*Courtesy of the U.S. Air Force and the John Anderson Collection.*)

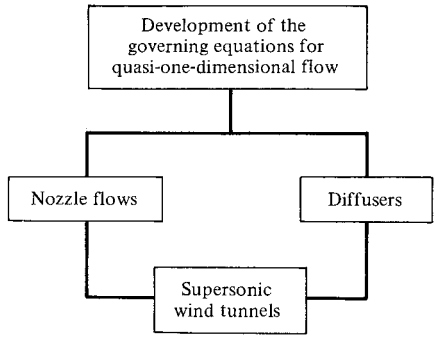
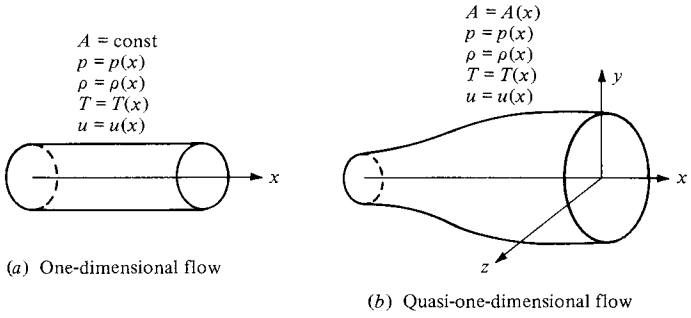


Figure 10.5 Road map for Chapter 10.

10.2 GOVERNING EQUATIONS FOR QUASI-ONE-DIMENSIONAL FLOW

Recall the one-dimensional flow treated in Chapter 8. There, we considered the flow-field variables to be a function of x only, that is, $p = p(x)$, $u = u(x)$, etc. Strictly speaking, a streamtube for such a flow must be of constant area; that is, the



(*a*) One-dimensional flow

(*b*) Quasi-one-dimensional flow

Figure 10.6 One-dimensional and quasi-one-dimensional flows.

one-dimensional flow discussed in Chapter 8 is constant-area flow, as sketched in Figure 10.6*a*.

In contrast, assume that the area of the streamtube changes as a function of x, that is, $A = A(x)$, as sketched in Figure 10.6*b*. Strictly speaking, this flow is three-dimensional; the flow-field variables are functions of x, y, and z, as can be seen simply by examining Figure 10.6*b*. In particular, the velocity at the boundary of the streamtube must be tangent to the boundary, and hence it has components in the y and z directions as well as the axial x direction. However, if the area variation is moderate, the components in the y and z directions are small in comparison with the component in the x direction. In such a case, the flow-field variables can be *assumed* to vary with x only (i.e., the flow can be assumed to be uniform across any cross section at a given x station). Such a flow, where $A = A(x)$, but $p = p(x)$, $\rho = \rho(x)$, $u = u(x)$, etc., is defined as *quasi-one-dimensional flow,* as sketched in Figure 10.6*b*. Such flow is the subject of this chapter. We have encountered quasi-one-dimensional flow earlier, in our discussion of incompressible flow through a duct in Section 3.3. Return to Section 3.3, and review the concepts presented there before progressing further.

Although the assumption of quasi-one-dimensional flow is an approximation to the actual flow in a variable-area duct, the integral forms of the conservation equations, namely, continuity [Equation (2.48)], momentum [Equation (2.64)], and energy [Equation (2.95)], can be used to obtain governing equations for quasi-one-dimensional flow which are physically consistent, as follows. Consider the control volume given in Figure 10.7. At station 1, the flow across area A_1 is assumed to be uniform with properties p_1, ρ_1, u_1, etc. Similarly, at station 2, the flow across area A_2 is assumed to be uniform with properties p_2, ρ_2, u_2, etc. The application of the integral form of the continuity equation was made to such a variable-area control volume in Section 3.3. The resulting continuity equation for steady, quasi-one-dimensional flow was obtained as Equation (3.21), which in terms of the nomenclature in Figure 10.7 yields

$$\boxed{\rho_1 u_1 A_1 = \rho_2 u_2 A_2} \tag{10.1}$$

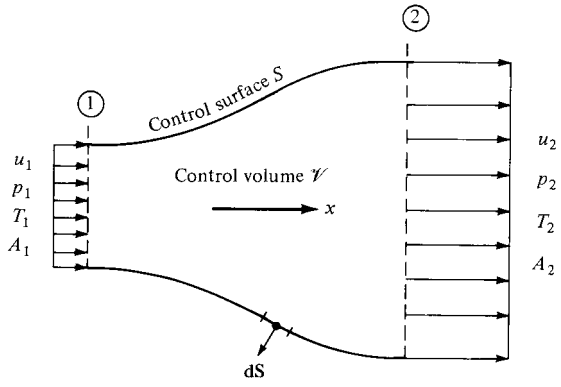


Figure 10.7 Finite control volume for quasi-one-dimensional flow.

Consider the integral form of the momentum equation, Equation (2.64). For a steady, inviscid flow with no body forces, this equation becomes

$$\oiint_S (\rho \mathbf{V} \cdot \mathbf{dS})\mathbf{V} = -\oiint_S p\, \mathbf{dS} \tag{10.2}$$

Since Equation (10.2) is a vector equation, let us examine its x component, given below:

$$\oiint_S (\rho \mathbf{V} \cdot \mathbf{dS})u = -\oiint_S (p\, dS)_x \tag{10.3}$$

where $(p\, dS)_x$ denotes the x component of the pressure force. Since Equation (10.3) is a scalar equation, we must be careful about the sign of the x components when evaluating the surface integrals. All components pointing to the right in Figure 10.7 are positive, and those pointing to the left are negative. The upper and lower surfaces of the control volume in Figure 10.7 are streamlines; hence, $\mathbf{V} \cdot \mathbf{dS} = 0$ along these surfaces. Also, recall that across A_1, $\mathbf{V}$ and $\mathbf{dS}$ are in opposite directions; hence, $\mathbf{V} \cdot \mathbf{dS}$ is negative. Therefore, the integral on the left of Equation (10.3) becomes $-\rho_1 u_1^2 A_1 + \rho_2 u_2^2 A_2$. The pressure integral on the right of Equation (10.2), evaluated over the faces A_1 and A_2 of the control volume, becomes $-(-p_1 A_1 + p_2 A_2)$. (The negative sign in front of $p_1 A$ is because $\mathbf{dS}$ over A_1 points to the left, which is the negative direction for the x components.) Evaluated over the upper and lower surface of the control volume, the pressure integral can be expressed as

$$-\int_{A_1}^{A_2} -p\, dA = \int_{A_1}^{A_2} p\, dA \tag{10.4}$$

where dA is simply the x component of the vector $\mathbf{dS}$, that is, the area dS projected on a plane perpendicular to the x axis. The negative sign *inside* the integral on the left of Equation (10.4) is due to the direction of $\mathbf{dS}$ along the upper and lower surfaces; note that $\mathbf{dS}$ points in the backward direction along these surfaces, as shown in Figure 10.7. Hence, the x component of $p\, \mathbf{dS}$ is to the left, and therefore

appears in our equations as a negative component. [Recall from Section 2.5 that the negative sign *outside* the pressure integral, that is, outside the integral on the left of Equation (10.4), is always present to account for the physical fact that the pressure force $p\,\mathbf{dS}$ exerted on a control surface always acts in the opposite direction of $\mathbf{dS}$. If you are unsure about this, review the derivation of the momentum equation in Section 2.5. Also, do not let the signs in the above results confuse you; they are all quite logical if you keep track of the direction of the x components.] With the above results, Equation (10.3) becomes

$$-\rho_1 u_1^2 A_1 + \rho_2 u_2^2 A_2 = -(-p_1 A_1 + p_2 A_2) + \int_{A_1}^{A_2} p\,dA$$

$$\boxed{p_1 A_1 + \rho_1 u_1^2 A_1 + \int_{A_1}^{A_2} p\,dA = p_2 A_2 + \rho_2 u_2^2 A_2} \tag{10.5}$$

Equation (10.5) is the momentum equation for steady, quasi-one-dimensional flow.

Consider the energy equation given by Equation (2.95). For inviscid, adiabatic, steady flow with no body forces, this equation becomes

$$\oiint_S \rho \left(e + \frac{V^2}{2}\right) \mathbf{V} \cdot \mathbf{dS} = -\oiint_S p\mathbf{V} \cdot \mathbf{dS} \tag{10.6}$$

Applied to the control volume in Figure 10.7, Equation (10.6) yields

$$\rho_1 \left(e_1 + \frac{u_1^2}{2}\right)(-u_1 A_1) + \rho_2 \left(e_2 + \frac{u_2^2}{2}\right)(u_2 A_2) = -(-p_1 u_1 A_1 + p_2 u_2 A_2)$$

or
$$p_1 u_1 A_1 + \rho_1 u_1 A_1 \left(e_1 + \frac{u_1^2}{2}\right) = p_2 u_2 A_2 + \rho_2 u_2 A_2 \left(e_2 + \frac{u_2^2}{2}\right) \tag{10.7}$$

Dividing Equation (10.7) by Equation (10.1), we have

$$\frac{p_1}{\rho_1} + e_1 + \frac{u_1^2}{2} = \frac{p_2}{\rho_2} + e_2 + \frac{u_2^2}{2} \tag{10.8}$$

Recall that $h = e + pv = e + p/\rho$. Hence, Equation (10.8) becomes

$$\boxed{h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2}} \tag{10.9}$$

which is the energy equation for steady, adiabatic, inviscid quasi-one-dimensional flow. Examine Equation (10.9) closely; it is a statement that the total enthalpy, $h_0 = h + u^2/2$, is a constant throughout the flow. Once again, this should come as no surprise; Equation (10.9) is simply another example of the general result for steady, inviscid, adiabatic flow discussed in Section 7.5. Hence, we can replace Equation (10.9) by

$$\boxed{h_0 = \text{const}} \tag{10.10}$$

Pause for a moment and examine our results given above. We have applied the integral forms of the conservation equations to the control volume in Figure 10.5. We have obtained, as a result, Equations (10.1), (10.5), and (10.9) or (10.10) as the governing continuity, momentum, and energy equations, respectively, for quasi-one-dimensional flow. Examine these equations—they are *algebraic* equations (with the exception of the single integral term in the momentum equation). In Figure 10.7, assume that the inflow conditions ρ_1, u_1, p_1, T_1, and h_1 are given and that the area distribution $A = A(x)$ is presented. Also, assume a calorically perfect gas, where

$$p_2 = \rho_2 R T_2 \tag{10.11}$$

and

$$h_2 = c_p T_2 \tag{10.12}$$

Equations (10.1), (10.5), (10.9) or (10.10), (10.11), and (10.12) constitute five equations for the five unknowns ρ_2, u_2, p_2, T_2, and h_2. We could, in principle, solve these equations directly for the unknown flow quantities at station 2 in Figure 10.7. However, such a direct solution would involve substantial algebraic manipulations. Instead, we take a simpler tack, as described in Section 10.3.

Before moving on to a solution of the governing equations, let us examine some physical characteristics of a quasi-one-dimensional flow. To help this examination, we first obtain some *differential* expressions for the governing equations, in contrast to the algebraic equations obtained above. For example, consider Equation (10.1), which states that

$$\rho u A = \text{const} \tag{10.13}$$

through a variable-area duct. Differentiating Equation (10.13), we have

$$\boxed{d(\rho u A) = 0} \tag{10.14}$$

which is the differential form of the continuity equation for quasi-one-dimensional flow.

To obtain a differential form of the momentum equation, apply Equation (10.5) to the infinitesimal control volume sketched in Figure 10.8. The flow going into the volume at station 1, where the area is A, has properties p, u, and ρ. In traversing the length dx, where the area changes by dA, the flow properties change by the corresponding amounts dp, $d\rho$, and du. Hence, the flow leaving at station 2 has the properties $p + dp$, $u + du$, and $\rho + d\rho$, as shown in Figure 10.8. For this case, Equation (10.5) becomes [recognizing that the integral in Equation (10.5) can be replaced by its integrand for the differential volume in Figure 10.8]

$$pA + \rho u^2 A + p\,dA = (p + dp)(A + dA) + (\rho + d\rho)(u + du)^2(A + dA) \tag{10.15}$$

In Equation (10.15), all products of differentials, such as $dp\,dA$, $d\rho(du)^2$, are very small and can be ignored. Hence, Equation (10.15) becomes

$$A\,dp + Au^2\,d\rho + \rho u^2\,dA + 2\rho u A\,du = 0 \tag{10.16}$$

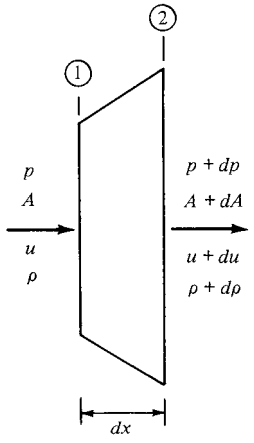


Figure 10.8 Incremental control volume.

Expanding the continuity equation, Equation (10.14), and multiplying by u, we have

$$\rho u^2\, dA + \rho u A\, du + A u^2\, d\rho = 0 \tag{10.17}$$

Subtracting Equation (10.17) from (10.16), we obtain

$$\boxed{dp = -\rho u\, du} \tag{10.18}$$

which is the differential form of the momentum equation for steady, inviscid, quasi-one-dimensional flow. Equation (10.18) is called *Euler's equation*. We have seen it before—as Equation (3.12). In Section 3.2, it was derived from the differential form of the general momentum equation in three dimensions. (Make certain to review that derivation before progressing further.) In Section 3.2, we demonstrated that Equation (3.12) holds along a streamline in a general three-dimensional flow. Now we see Euler's equation again, in Equation (10.18), which was derived from the governing equations for quasi-one-dimensional flow.

A differential form of the energy equation follows directly from Equation (10.9), which states that

$$h + \frac{u^2}{2} = \text{const}$$

Differentiating this equation, we have

$$\boxed{dh + u\, du = 0} \tag{10.19}$$

In summary, Equations (10.14), (10.18), and (10.19) are differential forms of the continuity, momentum, and energy equations, respectively, for a steady, inviscid, adiabatic, quasi-one-dimensional flow. We have obtained them from the algebraic forms of the equations derived earlier, applied essentially to the picture

shown in Figure 10.8. Now you might ask the question, Since we spent some effort obtaining partial differential equations for continuity, momentum, and energy in Chapter 2, applicable to a general three-dimensional flow, why would we not simply set $\partial/\partial y = 0$ and $\partial/\partial z = 0$ in those equations and obtain differential equations applicable to the one-dimensional flow treated in the present chapter? The answer is that we certainly could perform such a reduction, and we would obtain Equations (10.18) and (10.19) directly. [Return to the differential equations, Equations (2.113*a*) and (2.114), and prove this to yourself.] However, if we take the general continuity equation, Equation (2.52), and reduce it to one-dimensional flow, we obtain $d(\rho u) = 0$. Comparing this result with Equation (10.14) for quasi-one-dimensional flow, we see an inconsistency. This is another example of the physical inconsistency between the *assumption* of quasi-one-dimensional flow in a variable-area duct and the three-dimensional flow that actually occurs in such a duct. The result obtained from Equation (2.52), namely, $d(\rho u) = 0$, is a *truly* one-dimensional result, which applies to *constant-area* flows such as considered in Chapter 8. [Recall in Chapter 8 that the continuity equation was used in the form $\rho u =$ constant, which is compatible with Equation (2.52).] However, once we make the quasi-one-dimensional assumption, that is, that uniform properties hold across a given cross section in a variable-area duct, then Equation (10.14) is the only differential form of the continuity equation which insures mass conservation for such an assumed flow.

Let us now use the differential forms of the governing equations, obtained above, to study some physical characteristics of quasi-one-dimensional flow. Such physical information can be obtained from a particular combination of these equations, as follows. From Equation (10.14),

$$\frac{d\rho}{\rho} + \frac{du}{u} + \frac{dA}{A} = 0 \tag{10.20}$$

We wish to obtain an equation that relates the change in velocity du to the change in area dA. Hence, to eliminate $d\rho/\rho$ in Equation (10.20), consider Equation (10.18) written as

$$\frac{dp}{\rho} = \frac{dp}{d\rho}\frac{d\rho}{\rho} = -u\,du \tag{10.21}$$

Keep in mind that we are dealing with inviscid, adiabatic flow. Moreover, for the time being, we are assuming no shock waves in the flow. Hence, the flow is *isentropic*. In particular, any change in density $d\rho$ with respect to a change in pressure dp takes place isentropically; that is,

$$\frac{dp}{d\rho} \equiv \left(\frac{\partial p}{\partial \rho}\right)_s \tag{10.22}$$

From Equation (8.18) for the speed of sound, Equation (10.22) becomes

$$\frac{dp}{d\rho} = a^2 \tag{10.23}$$

Substituting Equation (10.23) into (10.21), we have

$$a^2 \frac{d\rho}{\rho} = -u\,du$$

or

$$\frac{d\rho}{\rho} = -\frac{u\,du}{a^2} = -\frac{u^2}{a^2}\frac{du}{u} = -M^2\frac{du}{u} \tag{10.24}$$

Substituting Equation (10.24) into (10.20), we have

$$-M^2\frac{du}{u} + \frac{du}{u} + \frac{dA}{A} = 0$$

or

$$\boxed{\frac{dA}{A} = (M^2 - 1)\frac{du}{u}} \tag{10.25}$$

Equation (10.25) is the desired equation which relates dA to du; it is called the *area-velocity relation*.

Equation (10.25) is very important; study it closely. In the process, recall the standard convention for differentials; for example, a positive value of du connotes an *increase* in velocity, a negative value of du connotes a *decrease* in velocity, etc. With this in mind, Equation (10.25) tells us the following information:

1. For $0 \leq M < 1$ (subsonic flow), the quantity in parentheses in Equation (10.25) is negative. Hence, an *increase* in velocity (positive du) is associated with a *decrease* in area (negative dA). Likewise, a decrease in velocity (negative du) is associated with an increase in area (positive dA). Clearly, for a subsonic compressible flow, to increase the velocity, we must have a convergent duct, and to decrease the velocity, we must have a divergent duct. These results are illustrated at the top of Figure 10.9. Also, these results are similar to the familiar trends for incompressible flow

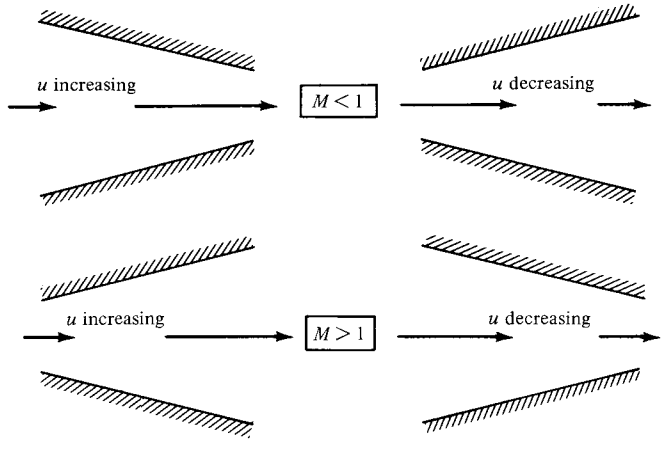


Figure 10.9 Compressible flow in converging and diverging ducts.

studied in Section 3.3. Once again we see that subsonic compressible flow is qualitatively (but not quantitatively) similar to incompressible flow.

2. For $M > 1$ (supersonic flow), the quantity in parentheses in Equation (10.25) is positive. Hence, an *increase* in velocity (positive du) is associated with an *increase* in area (positive dA). Likewise, a decrease in velocity (negative du) is associated with a decrease in area (negative dA). For a supersonic flow, to increase the velocity, we must have a divergent duct, and to decrease the velocity, we must have a convergent duct. These results are illustrated at the bottom of Figure 10.9; they are the *direct opposite* of the trends for subsonic flow.
3. For $M = 1$ (sonic flow), Equation (10.25) shows that $dA = 0$ even though a finite du exists. Mathematically, this corresponds to a local maximum or minimum in the area distribution. Physically, it corresponds to a minimum area, as discussed below.

Imagine that we want to take a gas at rest and isentropically expand it to supersonic speeds. The above results show that we must first accelerate the gas subsonically in a convergent duct. However, as soon as sonic conditions are achieved, we must further expand the gas to supersonic speeds by diverging the duct. Hence, a nozzle designed to achieve supersonic flow at its exit is a *convergent-divergent* duct, as sketched at the top of Figure 10.10. The minimum area of the duct is called the *throat*. Whenever an isentropic flow expands from

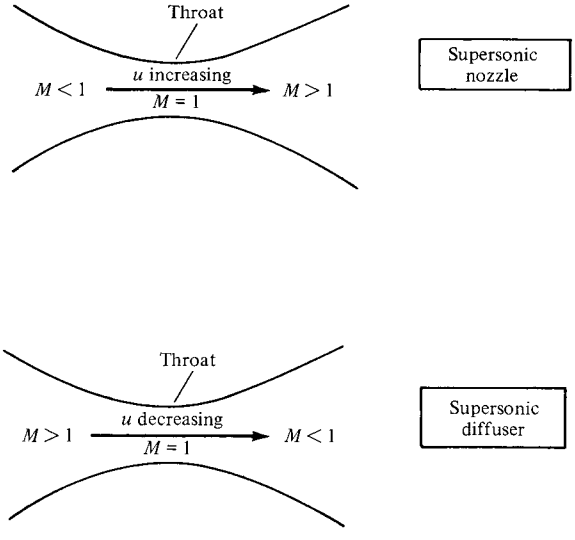


Figure 10.10 Illustration and comparison of a supersonic nozzle and a supersonic diffuser.

subsonic to supersonic speeds, the flow must pass through a throat; moreover, in such a case, $M = 1$ at the throat. The converse is also true; if we wish to take a supersonic flow and slow it down isentropically to subsonic speeds, we must first decelerate the gas in a convergent duct, and then as soon as sonic flow is obtained, we must further decelerate it to subsonic speeds in a divergent duct. Here, the convergent-divergent duct at the bottom of Figure 10.10 is operating as a diffuser. Note that whenever an isentropic flow is slowed from supersonic to subsonic speeds, the flow must pass through a throat; moreover, in such a case, $M = 1$ at the throat.

As a final note on Equation (10.25), consider the case when $M = 0$. Then we have $dA/A = -du/u$, which integrates to $Au =$ constant. This is the familiar continuity equation for incompressible flow in ducts as derived in Section 3.3 and as given by Equation (3.22).

10.3 NOZZLE FLOWS

In this section, we move to the left-hand branch of the road map given in Figure 10.5; that is, we study in detail the compressible flow through nozzles. To expedite this study, we first derive an important equation which relates Mach number to the ratio of duct area to sonic throat area.

Consider the duct shown in Figure 10.11. Assume that sonic flow exists at the throat, where the area is A^*. The Mach number and the velocity at the throat are denoted by M^* and u^*, respectively. Since the flow is sonic at the throat, $M^* = 1$ and $u^* = a^*$. (Note that the use of an asterisk to denote sonic conditions was introduced in Section 7.5; we continue this convention in our present discussion.) At any other section of this duct, the area, the Mach number, and the velocity are denoted by A, M, and u, respectively, as shown in Figure 10.11. Writing Equation (10.1) between A and A^*, we have

$$\rho^* u^* A^* = \rho u A \tag{10.26}$$

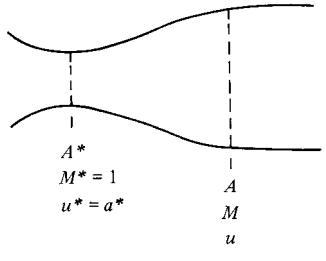


Figure 10.11 Geometry for the derivation of the area–Mach number relation.

Since $u^* = a^*$, Equation (10.26) becomes

$$\frac{A}{A^*} = \frac{\rho^*}{\rho}\frac{a^*}{u} = \frac{\rho^*}{\rho_0}\frac{\rho_0}{\rho}\frac{a^*}{u} \tag{10.27}$$

where ρ_0 is the stagnation density defined in Section 7.5 and is constant throughout an isentropic flow. From Equation (8.46), we have

$$\frac{\rho^*}{\rho_0} = \left(\frac{2}{\gamma+1}\right)^{1/(\gamma-1)} \tag{10.28}$$

Also, from Equation (8.43), we have

$$\frac{\rho_0}{\rho} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{1/(\gamma-1)} \tag{10.29}$$

Also, recalling the definition of M^* in Section 8.4, as well as Equation (8.48), we have

$$\left(\frac{u}{a^*}\right)^2 = M^{*2} = \frac{[(\gamma+1)/2]M^2}{1+[(\gamma-1)/2]M^2} \tag{10.30}$$

Squaring Equation (10.27) and substituting Equations (10.28) to (10.30), we obtain

$$\left(\frac{A}{A^*}\right)^2 = \left(\frac{\rho^*}{\rho_0}\right)^2\left(\frac{\rho_0}{\rho}\right)^2\left(\frac{a^*}{u}\right)^2$$

or

$$\left(\frac{A}{A^*}\right)^2 = \left(\frac{2}{\gamma+1}\right)^{2/(\gamma-1)}\left(1+\frac{\gamma-1}{2}M^2\right)^{2/(\gamma-1)}\frac{1+[(\gamma-1)/2]M^2}{[(\gamma+1)/2]M^2} \tag{10.31}$$

Algebraically simplifying Equation (10.31), we have

$$\boxed{\left(\frac{A}{A^*}\right)^2 = \frac{1}{M^2}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{(\gamma+1)/(\gamma-1)}} \tag{10.32}$$

Equation (10.32) is very important; it is called the *area–Mach number relation,* and it contains a striking result. "Turned inside out," Equation (10.32) tells us that $M = f(A/A^*)$; that is, the Mach number at any location in the duct is a function of the ratio of the local duct area to the sonic throat area. Recall from our discussion of Equation (10.25) that A must be greater than or at least equal to A^*; the case where $A < A^*$ is physically not possible in an isentropic flow. Thus, in Equation (10.32), $A/A^* \geq 1$. Also, Equation (10.32) yields *two* solutions for M at a given A/A^*—a subsonic value and a supersonic value. Which value of M that actually holds in a given case depends on the pressures at the inlet and exit of the duct, as explained later. The results for A/A^* as a function of M, obtained from Equation (10.32), are tabulated in Appendix A. Examining Appendix A, we note that for subsonic values of M, as M increases, A/A^* decreases (i.e., the duct converges). At $M = 1$, $A/A^* = 1$ in Appendix A. Finally, for supersonic values

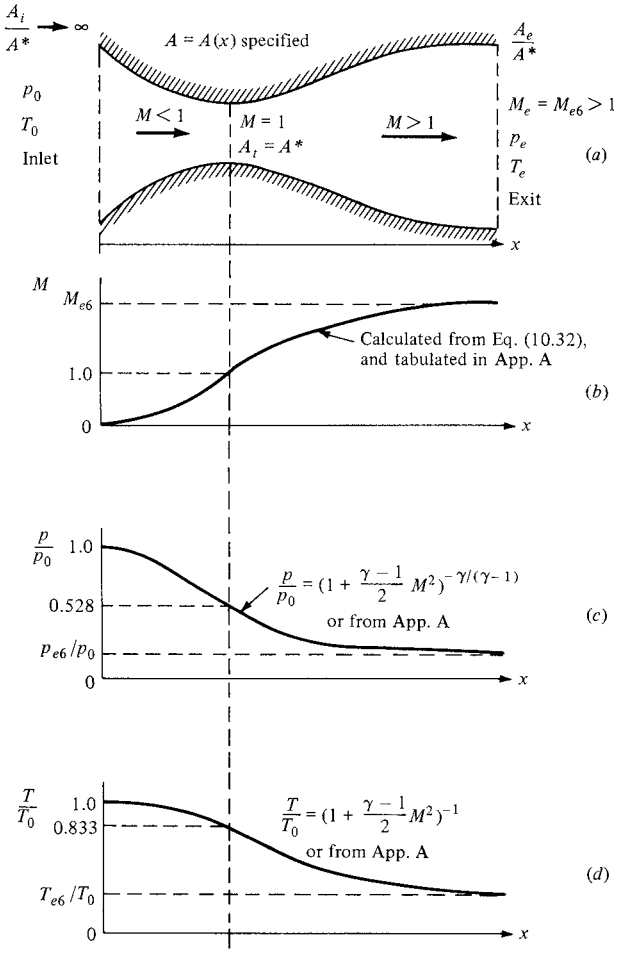


Figure 10.12 Isentropic supersonic nozzle flow.

of M, as M increases, A/A^* increases (i.e., the duct diverges). These trends in Appendix A are consistent with our physical discussion of convergent-divergent ducts at the end of Section 10.2. Moreover, Appendix A shows the double-valued nature of M as a function of A/A^*. For example, for $A/A^* = 2$, we have either $M = 0.31$ or $M = 2.2$.

Consider a given convergent-divergent nozzle, as sketched in Figure 10.12*a*. Assume that the area ratio at the inlet A_i/A^* is very large and that the flow at the inlet is fed from a large gas reservoir where the gas is essentially stationary. The reservoir pressure and temperature are p_0 and T_0, respectively. Since A_i/A^* is very large, the subsonic Mach number at the inlet is very small, $M \approx 0$. Thus, the pressure and temperature at the inlet are essentially p_0 and T_0, respectively. The area distribution of the nozzle $A = A(x)$ is specified, so that A/A^* is known

at every station along the nozzle. The area of the throat is denoted by A_t, and the exit area is denoted by A_e. The Mach number and static pressure at the exit are denoted by M_e and p_e, respectively. Assume that we have an isentropic expansion of the gas through this nozzle to a supersonic Mach number $M_e = M_{e,6}$ at the exit (the reason for the subscript 6 will be apparent later). The corresponding exit pressure is $p_{e,6}$. For this expansion, the flow is sonic at the throat; hence, $M = 1$ and $A_t = A^*$ at the throat. The flow properties through the nozzle are a function of the local area ratio A/A^* and are obtained as follows:

1. The local Mach number as a function of x is obtained from Equation (10.32), or more directly from the tabulated values in Appendix A. For the specified $A = A(x)$, we know the corresponding $A/A^* = f(x)$. Then read the related subsonic Mach numbers in the convergent portion of the nozzle from the first part of Appendix A (for $M < 1$) and the related supersonic Mach numbers in the divergent portion of the nozzle from the second part of Appendix A (for $M > 1$). The Mach number distribution through the complete nozzle is thus obtained and is sketched in Figure 10.12*b*.
2. Once the Mach number distribution is known, then the corresponding variation of temperature, pressure, and density can be found from Equations (8.40), (8.42), and (8.43), respectively, or more directly from Appendix A. The distributions of p/p_0 and T/T_0 are sketched in Figure 10.12*c* and *d*, respectively.

Examine the variations shown in Figure 10.12. For the isentropic expansion of a gas through a convergent-divergent nozzle, the Mach number monotonically increases from near 0 at the inlet to $M = 1$ at the throat, and to the supersonic value $M_{e,6}$ at the exit. The pressure monotonically decreases from p_0 at the inlet to $0.528p_0$ at the throat and to the lower value $p_{e,6}$ at the exit. Similarly, the temperature monotonically decreases from T_0 at the inlet to $0.833T_0$ at the throat and to the lower value $T_{e,6}$ at the exit. Again, for the isentropic flow shown in Figure 10.12, we emphasize that the distribution of M, and hence the resulting distributions of p and T, through the nozzle depends only on the local area ratio A/A^*. This is the key to the analysis of isentropic, supersonic, quasi-one-dimensional nozzle flows.

Imagine that you take a convergent-divergent nozzle, and simply place it on a table in front of you. What is going to happen? Is the air going to suddenly start flowing through the nozzle of its own accord? The answer is, of course not! Rather, by this stage in your study of aerodynamics, your intuition should tell you that we have to impose a force on the gas in order to produce any acceleration. Indeed, this is the essence of the momentum equation derived in Section 2.5. For the inviscid flows considered here, the only mechanism to produce an accelerating force on a gas is a pressure gradient. Thus, returning to the nozzle on the table, a pressure difference must be created between the inlet and exit; only then will the gas start to flow through the nozzle. The exit pressure must be less than the inlet pressure; that is, $p_e < p_0$. Moreover, if we wish to produce the isentropic

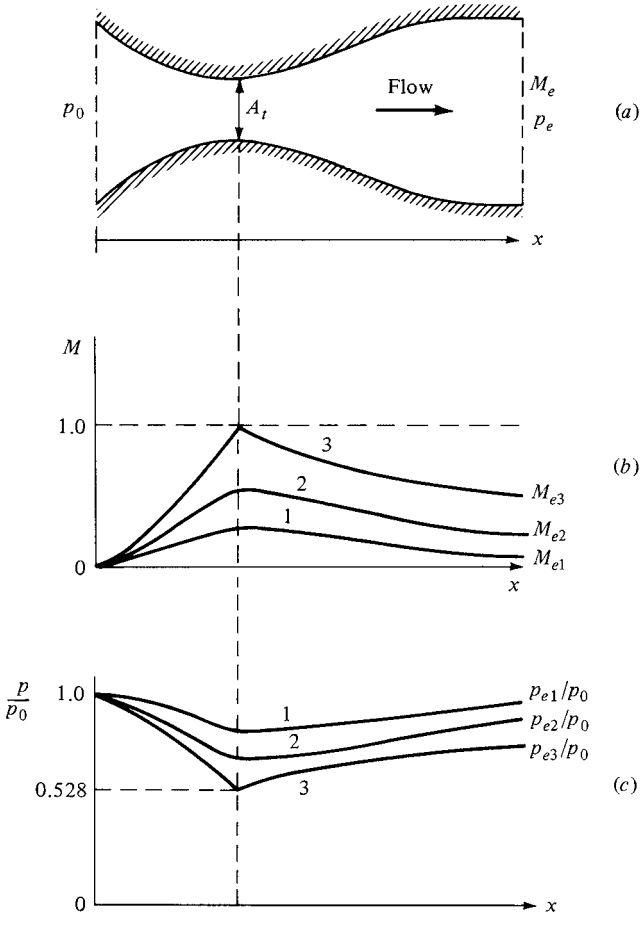


Figure 10.13 Isentropic subsonic nozzle flow.

supersonic flow sketched in Figure 10.12, the pressure p_e/p_0 must be *precisely* the value stipulated by Appendix A for the known exit Mach number $M_{e,6}$; that is, $p_e/p_0 = p_{e,6}/p_0$. If the pressure ratio is different from the above isentropic value, the flow either inside or outside the nozzle will be different from that shown in Figure 10.12.

Let us examine the type of nozzle flows that occur when p_e/p_0 is not equal to the precise isentropic value for $M_{e,6}$, that is, when $p_e/p_0 \neq p_{e,6}/p_0$. To begin with, consider the convergent-divergent nozzle sketched in Figure 10.13*a*. If $p_e = p_0$, no pressure difference exists, and no flow occurs inside the nozzle. Now assume that p_e is minutely reduced below p_0, say, $p_e = 0.999p_0$. This small pressure difference will produce a very low-speed subsonic flow inside the nozzle—essentially a gentle wind. The local Mach number will increase slightly through the convergent portion, reaching a maximum value at the throat, as shown

by curve 1 in Figure 10.13*b*. This Mach number at the throat will *not* be sonic; rather, it will be some small subsonic value. Downstream of the throat, the local Mach number will decrease in the divergent section, reaching a very small but finite value $M_{e,1}$ at the exit. Correspondingly, the pressure in the convergent section will gradually decrease from p_0 at the inlet to a minimum value at the throat, and then will gradually increase to the value $p_{e,1}$ at the exit. This variation is shown as curve 1 in Figure 10.13*c*. Please note that because the flow is not sonic at the throat in this case, A_t is not equal to A^*. Recall that A^*, which appears in Equation (10.32), is the *sonic* throat area. In the case of purely subsonic flow through a convergent-divergent nozzle, A^* takes on the character of a reference area; it is not the same as the actual geometric area of the nozzle throat A_t. Rather, A^* is the area the flow in Figure 10.13 would have *if* it were somehow accelerated to sonic velocity. If this did happen, the flow area would have to be decreased further than shown in Figure 10.13*a*. Hence, for a purely subsonic flow $A_t > A^*$.

Assume that we further decrease the exit pressure in Figure 10.13, say, to the value $p_e = p_{e,2}$. The flow is now illustrated by the curves labeled 2 in Figure 10.13. The flow moves faster through the nozzle, and the maximum Mach number at the throat increases but remains less than 1. Now, let us reduce p_e to the value $p_e = p_{e,3}$, such that the flow just reaches sonic conditions at the throat. This is shown by curve 3 in Figure 10.13. The throat Mach number is 1, and the throat pressure is $0.528p_0$. The flow downstream of the throat is subsonic.

Upon comparing Figures 10.12 and 10.13, we are struck by an important physical difference. For a given nozzle shape, there is only *one* allowable isentropic flow solution for the supersonic case shown in Figure 10.12. In contrast, there are an *infinite* number of possible isentropic subsonic solutions, each one corresponding to some value of p_e, where $p_0 \geq p_e \geq p_{e,3}$. Only three solutions of this infinite set of solutions are sketched in Figure 10.13. Hence, the key factors for the analysis of purely subsonic flow in a convergent-divergent nozzle are both A/A^* and p_e/p_0.

Consider the mass flow through the convergent-divergent nozzle in Figure 10.13. As the exit pressure is decreased, the flow velocity in the throat increases; hence, the mass flow increases. The mass flow can be calculated by evaluating Equation (10.1) at the throat; that is, $\dot{m} = \rho_t u_t A_t$. As p_e decreases, u_t increases and ρ_t decreases. However, the percentage increase in u_t is much greater than the decrease in ρ_t. As a result, $\dot{m}$ increases, as sketched in Figure 10.14. When $p_e = p_{e,3}$, sonic flow is achieved at the throat, and $\dot{m} = \rho^* u^* A^* = \rho^* u^* A_t$. Now, if p_e is further reduced below $p_{e,3}$, the conditions at the throat take on a new behavior; they remain *unchanged*. From our discussion in Section 10.2, the Mach number at the throat cannot exceed 1; hence, as p_e is further reduced, M will remain equal to 1 at the throat. Consequently, the mass flow will remain constant as p_e is reduced below $p_{e,3}$, as shown in Figure 10.14. In a sense, the flow at the throat, as well as upstream of the throat, becomes "frozen." Once the flow becomes sonic at the throat, disturbances cannot work their way upstream of the throat. Hence, the flow in the convergent section of the nozzle no longer communicates with the exit pressure and has no way of knowing that the exit pressure

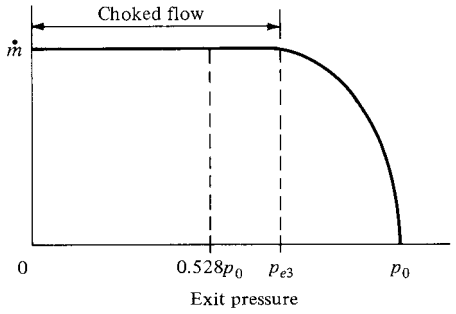


Figure 10.14 Variation of mass flow with exit pressure; illustration of choked flow.

is continuing to decrease. This situation—when the flow goes sonic at the throat, and the mass flow remains constant no matter how low p_e is reduced—is called *choked flow*. It is a vital aspect of the compressible flow through ducts, and we consider it further in our subsequent discussions.

Return to the subsonic nozzle flows sketched in Figure 10.13. *Question:* What happens in the duct when p_e is reduced below $p_{e,3}$? In the convergent portion, as described above, nothing happens. The flow properties remain fixed at the conditions shown by curve 3 in the convergent section of the duct (the left side of Figure 10.13*b* and *c*). However, a lot happens in the divergent section of the duct. As the exit pressure is reduced below $p_{e,3}$, a region of supersonic flow appears downstream of the throat. However, the exit pressure is too high to allow an isentropic supersonic flow throughout the entire divergent section. Instead, for p_e less than $p_{e,3}$ but substantially higher than the fully isentropic value $p_{e,6}$ (see Figure 10.12*c*), a normal shock wave is formed downstream of the throat. This situation is sketched in Figure 10.15.

In Figure 10.15, the exit pressure has been reduced to $p_{e,4}$, where $p_{e,4} < p_{e,3}$, but where $p_{e,4}$ is also substantially higher than $p_{e,6}$. Here we observe a normal shock wave standing inside the nozzle at a distance d downstream of the throat. Between the throat and the normal shock wave, the flow is given by the supersonic isentropic solution, as shown in Figure 10.15*b* and *c*. Behind the shock wave, the flow is subsonic. This subsonic flow sees the divergent duct and isentropically slows down further as it moves to the exit. Correspondingly, the pressure experiences a discontinuous increase across the shock wave and then is further increased as the flow slows down toward the exit. The flow on both the left and right sides of the shock wave is isentropic; however, the entropy increases across the shock wave. Hence, the flow on the left side of the shock wave is isentropic with one value of entropy s_1, and the flow on the right side of the shock wave is isentropic with another value of entropy s_2, where $s_2 > s_1$. The location of the shock wave inside the nozzle, given by d in Figure 10.15*a*, is determined by the requirement that the increase in static pressure across the wave plus that in the divergent portion of the subsonic flow behind the shock be just

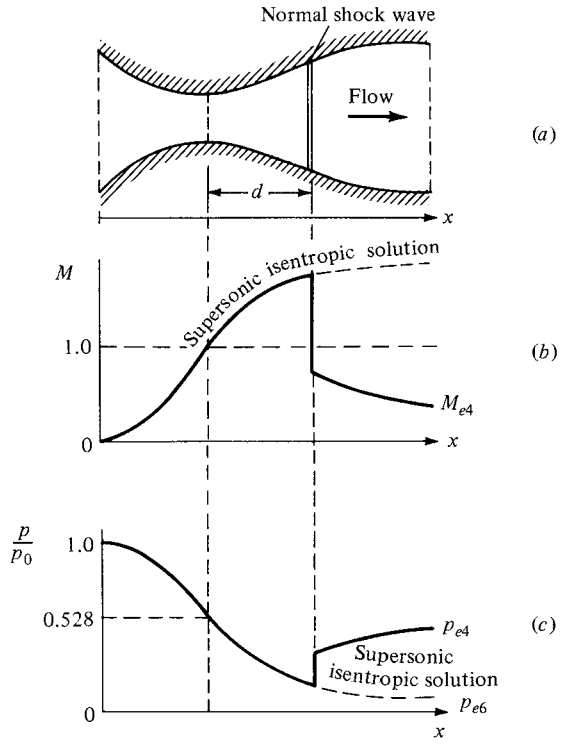


Figure 10.15 Supersonic nozzle flow with a normal shock inside the nozzle.

right to achieve $p_{e,4}$ at the exit. As p_e is further reduced, the normal shock wave moves downstream, closer to the nozzle exit. At a certain value of exit pressure, $p_e = p_{e,5}$, the normal shock stands precisely at the exit. This is sketched in Figure 10.16*a* to *c*. At this stage, when $p_e = p_{e,5}$, the flow through the entire nozzle, except precisely at the exit, is isentropic.

To this stage in our discussion, we have dealt with p_e, which is the pressure right at the nozzle exit. In Figures 10.12, 10.13, 10.15, and 10.16*a* to *c*, we have not been concerned with the flow downstream of the nozzle exit. Now imagine that the nozzle in Figure 10.16*a* exhausts directly into a region of surrounding gas downstream of the exit. These surroundings could be, for example, the atmosphere. In any case, the pressure of the surroundings downstream of the exit is defined as the *back pressure,* denoted by p_B. When the flow at the nozzle exit is subsonic, the exit pressure must equal the back pressure, $p_e = p_B$, because a pressure discontinuity cannot be maintained in a steady subsonic flow. That is, when the exit flow is subsonic, the surrounding back pressure is impressed on the exit flow. Hence, in Figure 10.13, $p_B = p_{e,1}$ for curve 1, $p_B = p_{e,2}$ for curve 2, and $p_B = p_{e,3}$ for curve 3. For the same reason, $p_B = p_{e,4}$ in Figure 10.15, and $p_B = p_{e,5}$ in Figure 10.16. Hence, in discussing these figures, instead of stating

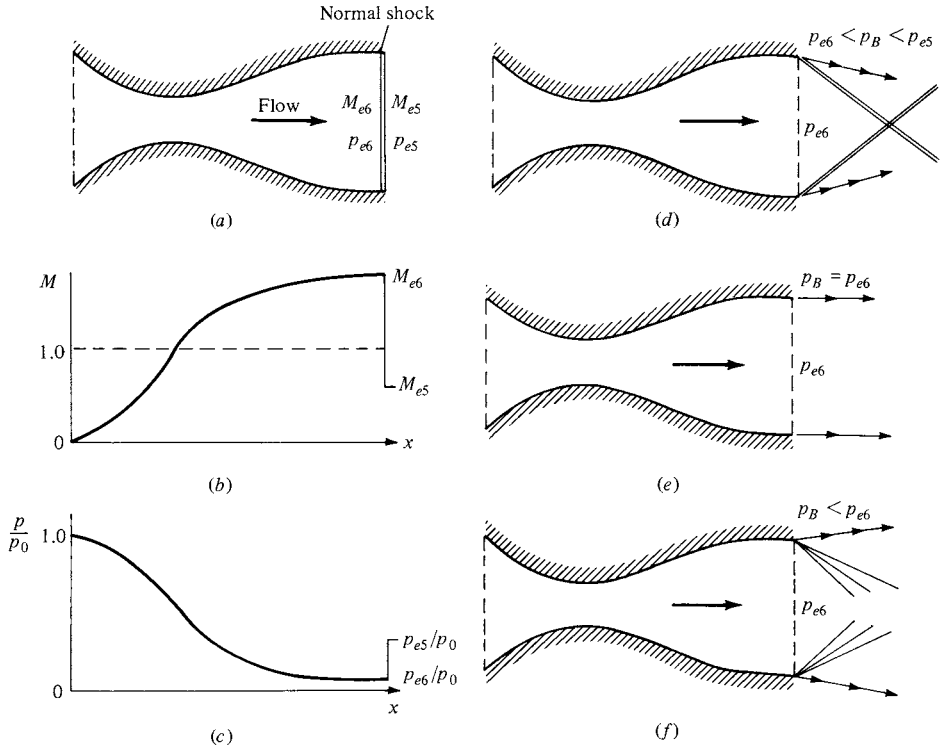


Figure 10.16 Supersonic nozzle flows with waves at the nozzle exit: (*a*), (*b*), and (*c*) pertain to a normal shock at the exit, (*d*) overexpanded nozzle, (*e*) isentropic expansion to the back pressure equal to the exit pressure, (*f*) underexpanded nozzle.

that we reduced the exit pressure p_e and observed the consequences, we could just as well have stated that we reduced the back pressure p_B. It would have amounted to the same thing.

For the remainder of our discussion in this section, let us now imagine that we have control over p_B and that we are going to continue to decrease p_B. Consider the case when the back pressure is reduced below $p_{e,5}$. When $p_{e,6} < p_B < p_{e,5}$, the back pressure is still above the isentropic pressure at the nozzle exit. Hence, in flowing out to the surroundings, the jet of gas from the nozzle must somehow be compressed such that its pressure is compatible with p_B. This compression takes place across oblique shock waves attached to the exit, as shown in Figure 10.16*d*. When p_B is reduced to the value such that $p_B = p_{e,6}$, there is no mismatch of the exit pressure and the back pressure; the nozzle jet exhausts smoothly into the surroundings without passing through any waves. This is shown in Figure 10.16*e*. Finally, as p_B is reduced below $p_{e,6}$ the jet of gas from the nozzle must expand further in order to match the lower back pressure. This expansion takes place across centered expansion waves attached to the exit, as shown in Figure 10.16*f*.

When the situation in Figure 10.16*d* exists, the nozzle is said to be *overexpanded,* because the pressure at the exit has expanded below the back pressure, $p_{e,6} < p_B$. That is, the nozzle expansion has gone too far, and the jet must pass through oblique shocks in order to come back up to the higher back pressure. Conversely, when the situation in Figure 10.16*f* exists, the nozzle is said to be *underexpanded*, because the exit pressure is higher than the back pressure, $p_{e,6} > p_B$, and hence the flow is capable of additional expansion after leaving the nozzle.

Surveying Figures 10.12 through 10.16, note that the purely isentropic supersonic flow originally illustrated in Figure 10.12 exists throughout the nozzle for all cases when $p_B \leq p_{e,5}$. For example, in Figure 10.16*a*, the isentropic supersonic flow solution holds throughout the nozzle except right at the exit, where a normal shock exists. In Figure 10.16*d* to *f*, the flow through the entire nozzle, including at the exit plane, is given by the isentropic supersonic flow solution.

Keep in mind that our entire discussion of nozzle flows in this section is predicated on having a duct of *given shape*. We assume that $A = A(x)$ is prescribed. When this is the case, the quasi-one-dimensional theory of this chapter gives a reasonable prediction of the flow inside the duct, where the results are interpreted as mean properties averaged over each cross section. This theory does *not* tell us how to design the *contour* of the nozzle. In reality, if the walls of the nozzle are not curved just right, then oblique shocks occur inside the nozzle. To obtain the proper contour for a supersonic nozzle so that it produces isentropic shock-free flow inside the nozzle, we must account for the three-dimensionality of the actual flow. This is one purpose of the method of characteristics, a technique for analyzing two- and three-dimensional supersonic flow. A brief introduction to the method of characteristics is given in Chapter 13.

EXAMPLE 10.1

Consider the isentropic supersonic flow through a convergent-divergent nozzle with an exit-to-throat area ratio of 10.25. The reservoir pressure and temperature are 5 atm and 600°R, respectively. Calculate M, p, and T at the nozzle exit.

■ Solution

From the supersonic portion of Appendix A, for $A_e/A^* = 10.25$,

$$M_e = \boxed{3.95}$$

Also,

$$\frac{p_e}{p_0} = \frac{1}{142} \quad \text{and} \quad \frac{T_e}{T_0} = \frac{1}{4.12}$$

Thus,

$$p_e = 0.007 p_0 = 0.007(5) = \boxed{0.035 \text{ atm}}$$

$$T_e = 0.2427 T_0 = 0.2427(600) = \boxed{145.6^\circ\text{R}}$$

EXAMPLE 10.2

Consider the isentropic flow through a convergent-divergent nozzle with an exit-to-throat area ratio of 2. The reservoir pressure and temperature are 1 atm and 288 K, respectively. Calculate the Mach number, pressure, and temperature at both the throat and the exit for the cases where (*a*) the flow is supersonic at the exit and (*b*) the flow is subsonic throughout the entire nozzle except at the throat, where $M = 1$.

■ Solution

(*a*) At the throat, the flow is sonic. Hence,

$$M_t = \boxed{1.0}$$

$$p_t = p^* = \frac{p^*}{p_0} p_0 = 0.528(1 \text{ atm}) = \boxed{0.528 \text{ atm}}$$

$$T_t = T^* = \frac{T^*}{T_0} = 0.833(288) = \boxed{240 \text{ K}}$$

At the exit, the flow is supersonic. Hence, from the supersonic portion of Appendix A, for $A_e/A^* = 2$,

$$M_e = \boxed{2.2}$$

$$p_e = \frac{p_e}{p_0} p_0 = \frac{1}{10.69}(1 \text{ atm}) = \boxed{0.0935 \text{ atm}}$$

$$T_e = \frac{T_e}{T_0} T_0 = \frac{1}{1.968}(288) = \boxed{146 \text{ K}}$$

(*b*) At the throat, the flow is still sonic. Hence, from above, $M_t = 1.0$, $p_t = 0.528$ atm, and $T_t = 240$ K. However, at all other locations in the nozzle, the flow is subsonic. At the exit, where $A_e/A^* = 2$, from the subsonic portion of Appendix A,

$$M_e = \boxed{0.3} \qquad \text{(rounded to the nearest entry in Appendix A)}$$

$$p_e = \frac{p_e}{p_0} p_0 = \frac{1}{1.064}(1 \text{ atm}) = \boxed{0.94 \text{ atm}}$$

$$T_e = \frac{T_e}{T_0} T_0 = \frac{1}{1.018}(288) = \boxed{282.9 \text{ K}}$$

EXAMPLE 10.3

For the nozzle in Example 10.2, assume the exit pressure is 0.973 atm. Calculate the Mach numbers at the throat and the exit.

■ Solution

In Example 10.2, we saw that if $p_e = 0.94$ atm, the flow is sonic at the throat, but subsonic elsewhere. Hence, $p_e = 0.94$ atm corresponds to $p_{e,3}$ in Figure 10.13. In the

present problem, $p_e = 0.973$ atm, which is higher than $p_{e,3}$. Hence, in this case, the flow is subsonic throughout the nozzle, including at the throat. For this case, A^* takes on a reference value, and the actual geometric throat area is denoted by A_t. At the exit,

$$\frac{p_0}{p_e} = \frac{1}{0.973} = 1.028$$

From the subsonic portion of Appendix A, for $p_0/p_e = 1.028$, we have

$$M_e = \boxed{0.2} \quad \text{and} \quad \frac{A_e}{A^*} = 2.964$$

$$\frac{A_t}{A^*} = \frac{A_t}{A_e}\frac{A_e}{A^*} = 0.5(2.964) = 1.482$$

From the subsonic portion of Appendix A, for $A_t/A^* = 1.482$, we have

$$M_t = \boxed{0.44} \qquad \text{(nearest entry)}$$

EXAMPLE 10.4

An equation for the thrust of a jet-propulsion device can be derived by applying the integral form of the momentum equation for a steady inviscid flow (Equation (2.71)) to a control volume wrapped around the jet engine. This derivation is carried out in great detail in Chapter 2 of Reference 21, and in a simpler form in Chapter 9 of Reference 2 where the result is also specialized to a rocket engine. You are encouraged to examine these derivations—they are an excellent example of the use of the control volume concept. The resulting thrust equation for a rocket engine (see Section 9.8 of Reference 2) is

$$T = \dot{m}u_e + (p_e - p_\infty)A_e \qquad \text{(E10.1)}$$

Where T is the thrust, $\dot{m}$ is the mass flow through the engine, u_e is the gas velocity at the nozzle exit, p_e is the gas pressure at the nozzle exit, p_∞ is the surrounding ambient atmospheric pressure, and A_e is the area of the exit.

Consider a rocket engine similar to that shown in Figure 10.1. Liquid hydrogen and oxygen are burned in the combustion chamber producing a combustion gas pressure and temperature of 30 atm and 3500 K, respectively. The area of the rocket nozzle throat is 0.4 m^2. The area of the exit is designed so that the exit pressure exactly equals the ambient atmospheric pressure at a standard altitude of 20 km. Assume an isentropic flow through the rocket engine nozzle with an effective value of the ratio of specific heats $\gamma = 1.22$, and a constant value of the specific gas constant $R = 520$ J/(kg)(K).

(a) Using Equation (E10.1), calculate the thrust of the rocket engine.

(b) Calculate the area of the nozzle exit.

■ Solution

(*a*) Examining Equation (E10.1), we first need to obtain the value of mass flow, $\dot{m}$, and exit velocity, u_e. The mass flow is constant through the nozzle and is equal to $\dot{m} = \rho u A$ evaluated at any location in the nozzle. A convenient location to evaluate $\dot{m}$ is at the

throat, where

$$\dot{m} = \rho^* u^* A^*$$

To obtain ρ^*, we need $\rho_0 = p_0/RT_0$. Noting that (1 atm) $= 1.01 \times 10^5$ N/m^2,

$$\rho_0 = \frac{(30)(1.01 \times 10^5)}{(520)(3500)} = 1.665 \text{ kg/m}^3$$

From Equation (8.46)

$$\frac{\rho*}{\rho_0} = \left(\frac{2}{\gamma+1}\right)^{\frac{1}{\gamma-1}} = \left(\frac{2}{1.22+1}\right)^{\frac{1}{1.22-1}} = \left(\frac{2}{2.22}\right)^{4.545} = 0.622$$

$$\rho^* = 0.622\rho_0 = 0.622(1.665) = 1.036 \text{ kg/m}^3$$

At the throat, the flow velocity is equal to the local speed of sound, $u^* = a^*$. From Equation (8.44),

$$\frac{T^*}{T_0} = \frac{2}{\gamma+1} = \frac{2}{2.22} = 0.901$$

$$T^* = 0.901T_0 = 0.901(3500) = 3154K$$

$$a^* = \sqrt{\gamma RT^*} = \sqrt{(1.22)(520)(3154)} = 1415 \text{ m/s}$$

$$\dot{m} = \rho^* u^* A^* = (1.036)(1415)(0.4) = 586.4 \text{ kg/s}$$

This is the value of $\dot{m}$ to be used in Equation (E10.1).

Next, we need to obtain the exit velocity u_e. We do this by first obtaining the exit Mach number from Equation (8.42).

$$\frac{p_0}{p_e} = \left(1 + \frac{\gamma-1}{2}M_e^2\right)^{\frac{\gamma}{\gamma-1}}$$

where, from the statement of the problem, p_e is equal to the ambient pressure at a standard altitude of 20 km. From Appendix D, at 20 km, $p_\infty = 5.5293 \times 10^3$ N/m^2. Hence,

$$p_e = p_\infty = 5529N/m^2$$

Thus, from Equation (8.42)

$$1 + \frac{\gamma-1}{2}M_e^2 = \left(\frac{p_0}{p_e}\right)^{\frac{\gamma-1}{\gamma}} = \left[\frac{(30)(1.01 \times 10^5)}{5529}\right]^{\frac{0.22}{1.22}} = (548)^{0.18} = 3.111$$

Note: For future use in this solution, we set aside the value:

$$1 + \frac{\gamma-1}{2}M_e^2 = 3.111 \qquad \text{(E10.2)}$$

Thus, from Equation (E10.2),

$$\frac{\gamma-1}{2}M_e^2 = 2.111$$

$$M_e^2 = (2.111)\left(\frac{2}{0.22}\right) = 19.19$$

$$M_e = 4.38$$

To obtain the speed of sound at the exit, from Equations (8.40) and (E10.2),

$$\frac{T_0}{T_e} = 1 + \frac{\gamma - 1}{2} M_e^2 = 3.111$$

$$T_e = \frac{T_0}{3.111} = \frac{3500}{3.111} = 1125 \text{ K}$$

$$a_e = \sqrt{\gamma R T_e} = \sqrt{(1.22)(520)(1125)} = 844.8 \text{ m/s}$$

Thus, $$u_e = M_e a_e = (4.38)(844.8) = 3700 \text{ m/s}$$

Intermediate check: We can check this value of 3700 m/s for u_e by directly using the energy equation, Equation (8.38)

$$c_p T_0 = c_p T_e + \frac{u_e^2}{2}$$

where, from Equation (7.9),

$$c_p = \frac{\gamma R}{\gamma - 1} = \frac{(1.22)(520)}{0.22} = 2883.6 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

Thus, from Equation (8.38)

$$u_e^2 = 2c_p(T_0 - T_e) = 2(2883.6)(3500 - 1125) = 1.3697 \times 10^7$$

or, $$u_e = 3700 \text{ m/s}$$

This checks with the result for u_e obtained earlier.

Finally, we are ready to calculate the thrust from Equation (E10.1). Since the statement of the problem gives $p_e = p_\infty$, the pressure term in Equation (E10.1) drops out, and we have

$$T = \dot{m} u_e = (586.4)(3700) = \boxed{2.17 \times 10^6 \text{ N}}$$

Since $$1\text{N} = 0.2247 \text{ lb}, \text{ we have}$$

$$T = (2.17 \times 10^6)(0.2247) = \boxed{487{,}600 \text{ lb}}$$

(b) From Equation (10.32),

$$\left(\frac{A_e}{A^*} \right)^2 = \frac{1}{M_e^2} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M_e^2 \right) \right]^{\frac{\gamma + 1}{\gamma - 1}}$$

In this equation, the numerical values of the various terms are

$$\frac{\gamma + 1}{\gamma - 1} = \frac{2.22}{0.22} = 10.1$$

$$\frac{2}{\gamma + 1} = \frac{2}{2.22} = 0.9$$

$$1 + \frac{\gamma - 1}{2} M_e^2 = 3.111 \qquad \text{(from Equation (E10.2))}$$

and $$M_e = 4.38$$

Inserting these values into Equation (10.32),

$$\left(\frac{A_e}{A^*}\right)^2 = \frac{1}{(4.38)^2}[(0.9)(3.111)]^{10.1} = 1710.8$$

$$\frac{A_e}{A^*} = 41.36$$

Thus,
$$A_e = (41.36)A^* = (41.36)(0.4) = \boxed{16.5 \text{ m}^2}$$

EXAMPLE 10.5

Calculate the mass flow through the rocket engine described in Example 10.4 using the closed-form analytical expression given in Problem 10.5 at the end of this chapter. Compare the result with that obtained in Example 10.4.

■ Solution

From Problem 10.5, the closed-form expression for the mass flow through a choked nozzle is

$$\dot{m} = \frac{p_0 A^*}{\sqrt{T_0}}\sqrt{\frac{\gamma}{R}\left(\frac{2}{\gamma+1}\right)^{(\gamma+1)/(\gamma-1)}} \qquad \text{(E10.3)}$$

From Example 10.4 we have $p_0 = 30$ atm, $T_0 = 3500$ K, $A^* = 0.4$ m^2, $R = 520$ J/(kg)(K), and $\gamma = 1.22$.

Noting that

$$p_0 = 30 \text{ atm} = (30)(1.01 \times 10^5) = 3.03 \times 10^6 \text{ N/m}^2$$

$$\gamma/R = 1.22/510 = 2.346 \times 10^{-3}$$

$$\frac{2}{\gamma+1} = \frac{2}{2.22} = 0.9$$

$$\frac{\gamma+1}{\gamma-1} = \frac{2.22}{0.22} = 10.09$$

from Equation (E10.3) we have

$$\dot{m} = \frac{(3.03 \times 10^6)(0.4)}{\sqrt{3500}}\sqrt{(2.346 \times 10^{-3})(0.9)^{10.09}} = \boxed{583.2 \text{ kg/s}}$$

This result, obtained from a single equation, compares well with the value of 586.4 kg/s obtained from a sequence of calculations that is subject to a larger cumulative roundoff error (the author is using a hand calculator and usually rounding off to the fourth significant figure). The result obtained here, using Equation (E10.3), should be considered more accurate.

10.3.1 More on Mass Flow

Equation (E10.3) in Example 10.5 has a distinct advantage over the piecemeal calculations of $\dot{m}$ in Example 10.4. Not only does it lead to a straightforward answer in one step, it also shows us exactly on what variables the mass flow depends,

and in what manner. From Equation (E10.3), we see that the mass flow depends primarily on p_0, T_0, and A^*, and that it varies directly with reservoir pressure and the area of the throat, and inversely as the square root of the reservoir temperature. If you double the reservoir pressure, you double the mass flow. If you double the throat area, you double the mass flow. If you quadruple the reservoir temperature, you cut the mass flow by a half. These variations are fundamental to the physics of choked flow in a nozzle. Make certain to fix in your mind this proportionality:

$$\dot{m} \propto \frac{p_0 A^*}{\sqrt{T_0}} \tag{10.33}$$

How is this discussion related to the variation of mass flow sketched in Figure 10.14? Recall that Figure 10.14 pertains to a nozzle flow with fixed reservoir conditions, including a fixed value of p_0. The mass flow is plotted versus exit pressure. If the exit pressure equals p_0, there is no pressure difference across the nozzle, hence no flow through the nozzle (i.e., in Figure 10.14 the point for $p_e = p_0$ corresponds to $\dot{m} = 0$). As the exit pressure decreases, $\dot{m}$ first increases, and then reaches a plateau when $p_e \leq p_{e,3}$. For $p_{e,3} \leq p_e \leq p_0$, the nozzle flow is *not* choked, and the value of the mass flow depends not only on p_0, A^*, and T_0, but also on p_e. When p_e falls below $p_{e,3}$, the flow is choked, and the mass flow becomes constant no matter how low p_e is decreased. The horizontal portion of the curve in Figure 10.14 pertains to choked flow, and the magnitude of this choked mass flow depends only on the values of p_0, A^*, and T_0 and *not* on p_e. For the case shown in Figure 10.14, the values of p_0, A^*, and T_0 are fixed, specific values. If, for whatever reason, the value of p_0, or A^*, or T_0 is changed, then the horizontal choked-flow line in Figure 10.14 would be raised or lowered appropriately, governed by the proportionality given by Equation (10.33).

10.4 DIFFUSERS

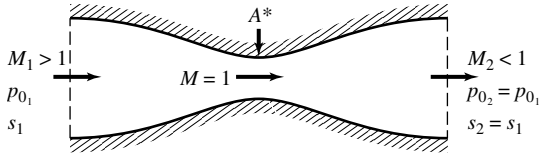
The role of a diffuser was first introduced in Section 3.3 in the context of a low-speed subsonic wind tunnel. There, a diffuser was a divergent duct downstream of the test section whose role was to slow the higher-velocity air from the test section down to a very low velocity at the diffuser exit (see Figure 3.8). Indeed, in general, we can define a diffuser as any duct designed to slow an incoming gas flow to lower velocity at the exit of the diffuser. The incoming flow can be subsonic, as discussed in Figure 3.8, or it can be supersonic, as discussed in the present section. However, the shape of the diffuser is drastically different, depending on whether the incoming flow is subsonic or supersonic.

Before pursuing this matter further, let us elaborate on the concept of total pressure p_0 as discussed in Section 7.5. In a semiqualitative sense, the total pressure of a flowing gas is a measure of the capacity of the flow to perform useful work. Let us consider two examples:

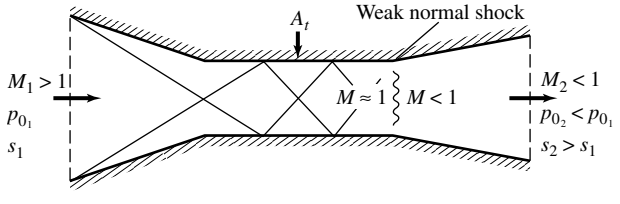
1. A pressure vessel containing stagnant air at 10 atm
2. A supersonic flow at $M = 2.16$ and $p = 1$ atm

In case 1, the air velocity is zero; hence, $p_0 = p = 10$ atm. Now, imagine that we want to use air to drive a piston in a piston-cylinder arrangement, where useful work is performed by the piston being displaced through a distance. The air is ducted into the cylinder from a large manifold, in the same vein as the reciprocating internal combustion engine in our automobile. In case, 1, the pressure vessel can act as the manifold; hence, the pressure on the piston is 10 atm, and a certain amount of useful work is performed, say, W_1. However, in case 2, the supersonic flow must be slowed to a low velocity before we can readily feed it into the manifold. If this slowing process can be achieved without loss of total pressure, then the pressure in the manifold in this case is also 10 atm (assuming $V \approx 0$), and the same amount of useful work W_1 is performed. On the other hand, assume that in slowing down the supersonic stream, a loss of 3 atm takes place in the total pressure. Then the pressure in the manifold is only 7 atm, with the consequent generation of useful work W_2, which is less than in the first case; that is, $W_2 < W_1$. The purpose of this simple example is to indicate that the total pressure of a flowing gas is indeed a measure of its capability to perform useful work. On this basis, a loss of total pressure is always an inefficiency—a loss of the capability to do a certain amount of useful work.

In light of the above, let us expand our definition of a diffuser. A diffuser is a duct designed to slow an incoming gas flow to lower velocity at the exit of the diffuser *with as small a loss in total pressure as possible*. Consequently, an ideal diffuser would be characterized by an *isentropic* compression to lower velocities; this is sketched in Figure 10.17*a*, where a supersonic flow enters the diffuser at M_1,



(*a*) Ideal (isentropic) supersonic diffuser



(*b*) Actual supersonic diffuser

Figure 10.17 The ideal (isentropic) diffuser compared with the actual situation.

is isentropically compressed in a convergent duct to Mach 1 at the throat, where the area is A^*, and then is further isentropically compressed in a divergent duct to a low subsonic Mach number at the exit. Because the flow is isentropic, $s_2 = s_1$, and from Equation (8.73), $p_{0,2} = p_{0,1}$. Indeed, p_0 is constant throughout the entire diffuser—a characteristic of isentropic flow. However, common sense should tell you that the ideal diffuser in Figure 10.17*a* can never be achieved. It is extremely difficult to slow a supersonic flow without generating shock waves in the process. For example, examine the convergent portion of the diffuser in Figure 10.17*a*. Note that the supersonic flow is turned into itself; hence, the converging flow will inherently generate oblique shock waves, which will destroy the isentropic nature of the flow. Moreover, in real life, the flow is viscous; there will be an entropy increase within the boundary layers on the walls of the diffuser. For these reasons, an ideal isentropic diffuser can never be constructed; an ideal diffuser is of the nature of a "perpetual motion machine"—only a utopian wish in the minds of engineers.

An actual supersonic diffuser is sketched in Figure 10.17*b*. Here, the incoming flow is slowed by a series of reflected oblique shocks, first in a convergent section usually consisting of straight walls, and then in a constant-area throat. Due to the interaction of the shock waves with the viscous flow near the wall, the reflected shock pattern eventually weakens and becomes quite diffuse, sometimes ending in a weak normal shock wave at the end of the constant-area throat. Finally, the subsonic flow downstream of the constant-area throat is further slowed by moving through a divergent section. At the exit, clearly $s_2 > s_1$; hence $p_{0,2} < p_{0,1}$. The art of diffuser design is to obtain as small a total pressure loss as possible, that is, to design the convergent, divergent, and constant-area throat sections so that $p_{0,2}/p_{0,1}$ is as close to unity as possible. Unfortunately, in most cases, we fall far short of that goal. For more details on supersonic diffusers, see Chapter 5 of Reference 21 and Chapter 12 of Reference 1.

Please note that due to the entropy increase across the shock waves and in the boundary layers, the real diffuser throat area A_t is larger than A^*, that is, in Figure 10.17, $A_t > A^*$.

10.5 SUPERSONIC WIND TUNNELS

Return to the road map given in Figure 10.5. The material for the left and right branches is covered in Sections 10.3 and 10.4, respectively. In turn, a mating of these two branches gives birth to the fundamental aspects of supersonic wind tunnels, to be discussed in this section.

Imagine that you want to create a Mach 2.5 uniform flow in a laboratory for the purpose of testing a model of a supersonic vehicle, say, a cone. How do you do it? Clearly, we need a convergent-divergent nozzle with an area ratio $A_e/A^* = 2.637$ (see Appendix A). Moreover, we need to establish a pressure ratio, $p_0/p_e = 17.09$, across the nozzle in order to obtain a shock-free expansion to $M_e = 2.5$ at the exit. Your first thought might be to exhaust the nozzle directly into the laboratory, as sketched in Figure 10.18. Here, the Mach 2.5 flow passes into the

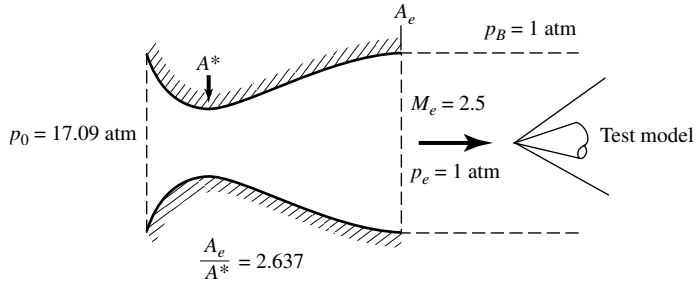


Figure 10.18 Nozzle exhausting directly to the atmosphere.

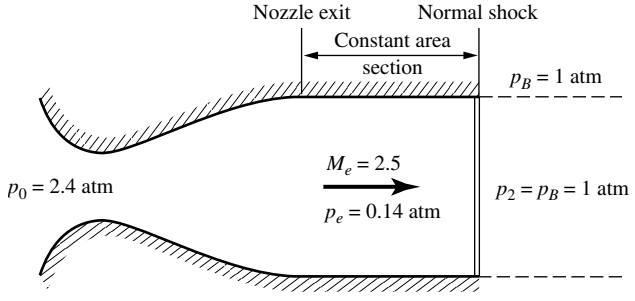


Figure 10.19 Nozzle exhausting into a constant-area duct, where a normal shock stands at the exit of the duct.

surroundings as a "free jet." The test model is placed in the flow downstream of the nozzle exit. In order to make certain that the free jet does not have shock or expansion waves, the nozzle exit pressure p_e must equal the back pressure p_B, as originally sketched in Figure 10.16*e*. Since the back pressure is simply that of the atmosphere surrounding the free jet, $p_B = p_e = 1$ atm. Consequently, to establish the proper isentropic expansion through the nozzle, you need a high-pressure reservoir with $p_0 = 17.09$ atm at the inlet to the nozzle. In this manner, you would be able to accomplish your objective, namely, to produce a uniform stream of air at Mach 2.5 in order to test a supersonic model, as sketched in Figure 10.18.

In the above example, you may have a problem obtaining the high-pressure air supply at 17.09 atm. You need an air compressor or a bank of high-pressure air bottles—both of which can be expensive. It requires work, hence money, to create reservoirs of high-pressure air—the higher the pressure, the more the cost. So, can you accomplish your objective in a more efficient way, at less cost? The answer is yes, as follows. Instead of the free jet as sketched in Figure 10.18, imagine that you have a long constant-area section downstream of the nozzle exit, with a normal shock wave standing at the end of the constant-area section; this is shown in Figure 10.19. The pressure downstream of the normal shock wave

is $p_2 = p_B = 1$ atm. At $M = 2.5$, the static pressure ratio across the normal shock is $p_2/p_e = 7.125$. Hence, the pressure upstream of the normal shock is 0.14 atm. Since the flow is uniform in the constant-area section, this pressure is also equal to the nozzle exit pressure; that is, $p_e = 0.14$ atm. Thus, in order to obtain the proper isentropic flow through the nozzle, which requires a pressure ratio of $p_0/p_e = 17.09$, we need a reservoir with a pressure of only 2.4 atm. This is considerably more efficient than the 17.09 atm required in Figure 10.18. Hence, we have created a uniform Mach 2.5 flow (in the constant-area duct) at a considerable reduction in cost compared with the scheme in Figure 10.18.

In Figure 10.19, the normal shock wave is acting as a diffuser, slowing the air originally at Mach 2.5 to the subsonic value of Mach 0.513 immediately behind the shock. Hence, by the addition of this "diffuser," we can more efficiently produce our uniform Mach 2.5 flow. This illustrates one of the functions of a diffuser. However, the "normal shock diffuser" sketched in Figure 10.19 has several problems:

1. A normal shock is the strongest possible shock, hence creating the largest total pressure loss. If we could replace the normal shock in Figure 10.19 with a weaker shock, the total pressure loss would be less, and the required reservoir pressure p_0 would be less than 2.4 atm.
2. It is extremely difficult to hold a normal shock wave stationary at the duct exit; in real life, flow unsteadiness and instabilities would cause the shock to move somewhere else and to fluctuate constantly in position. Thus, we could never be certain about the quality of the flow in the constant-area duct.
3. As soon as a test model is introduced into the constant-area section, the oblique waves from the model would propagate downstream, causing the flow to become two- or three-dimensional. The normal shock sketched in Figure 10.19 could not exist in such a flow.

Hence, let us replace the normal shock in Figure 10.19 with the oblique shock diffuser shown in Figure 10.17*b*. The resulting duct would appear as sketched in Figure 10.20. Examine this figure closely. We have a convergent-divergent nozzle feeding a uniform supersonic flow into the constant-area duct, which

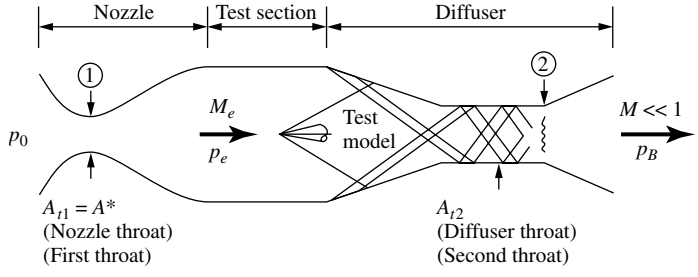


Figure 10.20 Sketch of a supersonic wind tunnel.

is called the *test section*. This flow is subsequently slowed to a low subsonic speed by means of a diffuser. This arrangement—namely, a convergent-divergent nozzle, a test section, and a convergent-divergent diffuser—is a *supersonic wind tunnel*. A test model, the cone in Figure 10.20, is placed in the test section, where aerodynamic measurements such as lift, drag, and pressure distribution are made. The wave system from the model propagates downstream and interacts with the multireflected shocks in the diffuser. The pressure ratio required to run the supersonic tunnel is p_0/p_B. This can be obtained by making p_0 large via a high-pressure reservoir at the inlet to the nozzle or by making p_B small via a vacuum source at the exit of the diffuser, or a combination of both.

The main source of total pressure loss in a supersonic wind tunnel is the diffuser. How does the oblique shock diffuser in Figure 10.20 compare with the hypothetical normal shock diffuser in Figure 10.19? Is the total pressure loss across all the reflected oblique shocks in Figure 10.20 greater or less than across the single normal shock wave in Figure 10.19? This is an important question, since the smaller the total pressure loss in the diffuser, the smaller is the pressure ratio p_0/p_B required to run the supersonic tunnel. There is no pat answer to this question. However, it is usually true that progressively reducing the velocity of a supersonic flow through a series of oblique shocks to a low supersonic value, and then further reducing the flow to subsonic speeds across a weak normal shock, results in a *smaller* total pressure loss than simply reducing the flow to subsonic speeds across a single, strong normal shock wave at the initially high supersonic Mach number. This trend is illustrated by Example 9.5. Therefore, the oblique shock diffuser shown in Figures 10.17*b* and 10.20 is usually more efficient than the simple normal shock diffuser shown in Figure 10.19. This is not always true, however, because in an actual real-life oblique shock diffuser, the shock waves interact with the boundary layers on the walls, causing local thickening and even possible separation of the boundary layers. This creates an additional total pressure loss. Moreover, the simple aspect of skin friction exerted on the surface generates a total pressure loss. Hence, actual oblique shock diffusers may have efficiencies greater or less than a hypothetical normal shock diffuser. Nevertheless, virtually all supersonic wind tunnels use oblique shock diffusers qualitatively similar to that shown in Figure 10.20.

Notice that the supersonic wind tunnel shown in Figure 10.20 has two throats: the nozzle throat with area $A_{t,1}$ is called the *first throat,* and the diffuser throat with area $A_{t,2}$ is called the *second throat*. The mass flow through the nozzle can be expressed as $\dot{m} = \rho u A$ evaluated at the first throat. This station is denoted as station 1 in Figure 10.20, and hence the mass flow through the nozzle is $\dot{m}_1 = \rho_1 u_1 A_{t,1} = \rho_1^* a_1^* A_{t,1}$. In turn, the mass flow through the diffuser can be expressed as $\dot{m} = \rho u A$ evaluated at station 2, namely, $\dot{m}_2 = \rho_2 u_2 A_{t,2}$. For steady flow through the wind tunnel, $\dot{m}_1 = \dot{m}_2$. Hence,

$$\rho_1^* a_1^* A_{t,1} = \rho_2 u_2 A_{t,2} \tag{10.34}$$

Since the thermodynamic state of the gas is irreversibly changed in going through the shock waves created by the test model and generated in the diffuser, clearly

ρ_2 and possibly u_2 are different from ρ_1^* and a_1^*, respectively. Hence, from Equation (10.34), *the second throat must have a different area from the first throat;* that is, $A_{t,2} \neq A_{t,1}$.

Question: How does $A_{t,2}$ differ from $A_{t,1}$? Let us assume that sonic flow occurs at *both* stations 1 and 2 in Figure 10.20. Thus, Equation (10.34) can be written as

$$\frac{A_{t,2}}{A_{t,1}} = \frac{\rho_1^* a_1^*}{\rho_2^* a_2^*} \tag{10.35}$$

Recall from Section 8.4 that a^* is constant for an adiabatic flow. Also, recall that the flow across shock waves is adiabatic (but not isentropic). Hence, the flow throughout the wind tunnel sketched in Figure 10.18 is adiabatic, and therefore $a_1^* = a_2^*$. In turn, Equation (10.35) becomes

$$\frac{A_{t,2}}{A_{t,1}} = \frac{\rho_1^*}{\rho_2^*} \tag{10.36}$$

Recall from Section 8.4 that T^* is also constant throughout the adiabatic flow of a calorically perfect gas. Hence, from the equation of state,

$$\frac{\rho_1^*}{\rho_2^*} = \frac{p_1^*/RT_1^*}{p_2^*/RT_2^*} = \frac{p_1^*}{p_2^*} \tag{10.37}$$

Substituting Equation (10.37) into (10.36), we have

$$\frac{A_{t,2}}{A_{t,1}} = \frac{p_1^*}{p_2^*} \tag{10.38}$$

From Equation (8.45), we have

$$p_1^* = p_{0,1}\left(\frac{2}{\gamma+1}\right)^{\gamma/(\gamma-1)}$$

and

$$p_2^* = p_{0,2}\left(\frac{2}{\gamma+1}\right)^{\gamma/(\gamma-1)}$$

Substituting the above into Equation (10.38), we obtain

$$\boxed{\frac{A_{t,2}}{A_{t,1}} = \frac{p_{0,1}}{p_{0,2}}} \tag{10.39}$$

Examining Figure 10.20, the total pressure always decreases across shock waves; therefore, $p_{0,2} < p_{0,1}$. In turn, from Equation (10.39), $A_{t,2} > A_{t,1}$. Thus, the second throat must always be *larger* than the first throat. Only in the case of an ideal isentropic diffuser, where $p_0 =$ constant, would $A_{t,2} = A_{t,1}$, and we have already discussed the impossibility of such an ideal diffuser.

Equation (10.39) is a useful relation to size the second throat relative to the first throat *if* we know the total pressure ratio across the tunnel. In the absence of such information, for the preliminary design of supersonic wind tunnels, the total pressure ratio across a normal shock is assumed.

For a given wind tunnel, if $A_{t,2}$ is less than the value given by Equation (10.39), the diffuser will "choke"; that is, the diffuser cannot pass the mass flow coming from the isentropic, supersonic expansion through the nozzle. In this case, nature adjusts the flow through the wind tunnel by creating shock waves in the nozzle, which in turn reduce the Mach number in the test section, producing weaker shocks in the diffuser with an attendant overall reduction in the total pressure loss; that is, nature adjusts the total pressure loss such that $p_{0,1}/p_{0,2} = p_{0,1}/p_B$ satisfies Equation (10.39). Sometimes this adjustment is so severe that a normal shock stands inside the nozzle, and the flow through the test section and diffuser is totally subsonic. Obviously, this choked situation is not desirable because we no longer have uniform flow at the desired Mach number in the test section. In such a case, the supersonic wind tunnel is said to be *unstarted*. The only way to rectify this situation is to make $A_{t,2}/A_{t,1}$ large enough so that the diffuser can pass the mass flow from the isentropic expansion in the nozzle, that is, so that Equation (10.39) is satisfied along with a shock-free isentropic nozzle expansion.

As a general concluding comment, the basic concepts and relations discussed in this chapter are not limited to nozzles, diffusers, and supersonic wind tunnels. Rather, we have been discussing quasi-one-dimensional flow, which can be applied in many applications involving flow in a duct. For example, inlets on jet engines, which diffuse the flow to lower speeds before entering the engine compressor, obey the same principles. Also, a rocket engine is basically a supersonic nozzle designed to optimize the thrust from the expanded jet. The applications of the ideas presented in this chapter are numerous, and you should make certain that you understand these ideas before progressing further.

In Section 1.2, we subdivided aerodynamics into external and internal flows. You are reminded that the material in this chapter deals exclusively with internal flows.

EXAMPLE 10.6

For the preliminary design of a Mach 2 supersonic wind tunnel, calculate the ratio of the diffuser throat area to the nozzle throat area.

■ Solution

Assuming a normal shock wave at the entrance of the diffuser (for starting), from Appendix B, $p_{0,2}/p_{0,1} = 0.7209$ for $M = 2.0$. Hence, from Equation (10.39),

$$\frac{A_{t,2}}{A_{t,1}} = \frac{p_{0,1}}{p_{0,2}} = \frac{1}{0.7209} = \boxed{1.387}$$

10.6 VISCOUS FLOW: SHOCK-WAVE/ BOUNDARY-LAYER INTERACTION INSIDE NOZZLES

Return to Figure 10.15. Here we see the case where the pressure ratio, $p_{e,4}/p_0$ is such that a normal shock wave stands inside the nozzle. This is a classic inviscid flow picture. In reality, there is a boundary layer growing along the nozzle wall, and the shock wave interacts with this boundary layer. One of the possible flow fields resulting from this interaction is sketched in Figure 10.21. The adverse pressure gradient across the shock causes the boundary layer to separate from the nozzle wall. A lambda-type shock pattern occurs at the two feet of the shock near the wall, and the core of the nozzle flow, now separated from the wall, flows downstream at almost constant area.

A series of schlieren photographs showing this type of flow is given in Figure 10.22, obtained from the recent work of Hunter (Craig A. Hunter, "Experimental Investigation of Separated Nozzle Flows," *Journal of Propulsion and Power,* vol. 20, no. 3, May-June 2004, pp. 527–532). For the nozzle shown in Figure 10.22, the exit-to-throat area ratio, A_e/A_t, is 1.797. In Figure 10.22*a*, the pressure ratio $p_{e,4}/p_0 = 0.5$; the normal shock stands inside the nozzle, and the lambda structure at both ends of the shock is clearly seen. The separated flow is seen trailing downstream from the lambda shock pattern. In Figures 10.22*b*, *c*, and *d* the shock pattern progressively moves closer to the nozzle exit as the pressure ratio is progressively reduced to 0.417, 0.333, and 0.294, respectively. A detailed schematic of the shock pattern for the pressure ratio of 0.417 is shown in Figure 10.23, corresponding to the flow in Figure 10.22*b*.

These results are an example of how the realities of a viscous flow can change the ideal picture obtained for an inviscid flow. The recent paper by Craig Hunter, referenced earlier, is an excellent discussion of the real flow in a supersonic nozzle under conditions where shock waves occur inside the nozzle. You are encouraged to study this reference for a revealing discussion of this interesting phenomena.

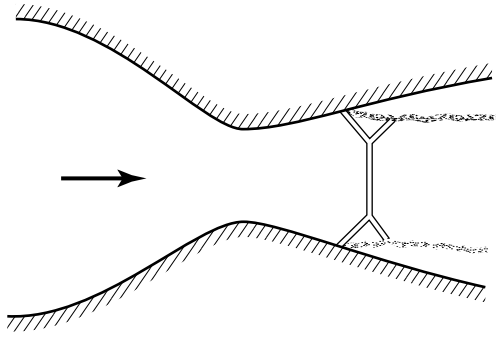


Figure 10.21 Sketch of an overexpanded nozzle flow with flow separation. (*Source: Craig Hunter, NASA.*).

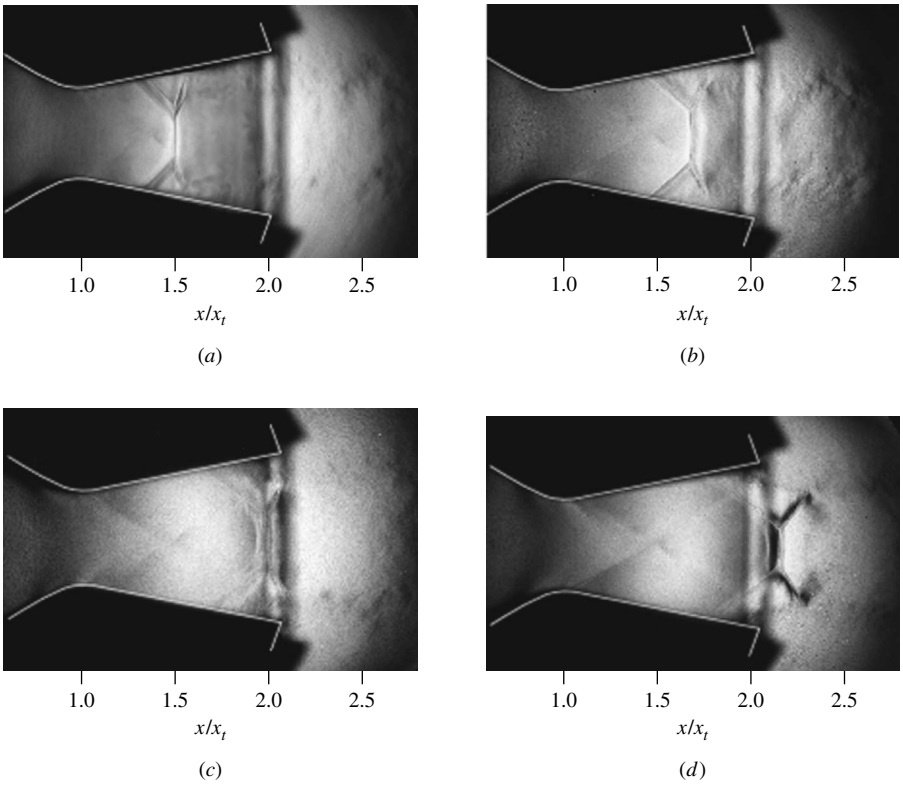


Figure 10.22 Schlieren photographs of the shock-wave/boundary-layer interaction inside an overexpanded nozzle flow. Exit-to-reservoir pressure ratio is (*a*) 0.5, (*b*) 0.417, (*c*) 0.333, (*d*) 0.294. (*Source: Craig Hunter, NASA.*)

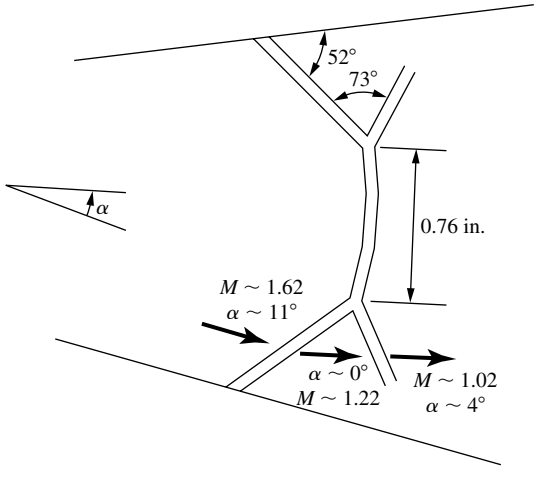


Figure 10.23 Detailed shock schematic for the case with a pressure ratio of 0.417. (*Source: Craig Hunter, NASA.*)

10.7 SUMMARY

The results of this chapter are highlighted below:

Quasi-one-dimensional flow is an approximation to the actual three-dimensional flow in a variable-area duct; this approximation assumes that $p = p(x)$, $u = u(x)$, $T = T(x)$, etc., although the area varies as $A = A(x)$. Thus, we can visualize the quasi-one-dimensional results as giving the mean properties at a given station, averaged over the cross section. The quasi-one-dimensional flow assumption gives reasonable results for many internal flow problems; it is a "workhorse" in the everyday application of compressible flow. The governing equations for this are

Continuity:
$$\rho_1 u_1 A_1 = \rho_2 u_2 A_2 \tag{10.1}$$

Momentum:
$$p_1 A_1 + \rho_1 u_1^2 A_1 + \int_{A_1}^{A_2} p\, dA = p_2 A_2 + \rho_2 u_2^2 A_2 \tag{10.5}$$

Energy:
$$h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2} \tag{10.9}$$

The area velocity relation

$$\frac{dA}{A} = (M^2 - 1)\frac{du}{u} \tag{10.25}$$

tells us that

1. To accelerate (decelerate) a subsonic flow, the area must decrease (increase).
2. To accelerate (decelerate) a supersonic flow, the area must increase (decrease).
3. Sonic flow can only occur at a throat or minimum area of the flow.

The isentropic flow of a calorically perfect gas through a nozzle is governed by the relation

$$\left(\frac{A}{A^*}\right)^2 = \frac{1}{M^2}\left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma-1}{2}M^2\right)\right]^{(\gamma+1)/(\gamma-1)} \tag{10.32}$$

This tells us that the Mach number in a duct is governed by the ratio of local duct area to the sonic throat area; moreover, for a given area ratio, there are two values of Mach number that satisfy Equation (10.32)—a subsonic value and a supersonic value.

For a given convergent-divergent duct, there is only one possible isentropic flow solution for supersonic flow; in contrast, there are an infinite number of subsonic isentropic solutions, each one associated with a different pressure ratio across the nozzle, $p_0/p_e = p_0/p_B$.

In a supersonic wind tunnel, the ratio of second throat area to first throat area should be approximately

$$\frac{A_{t,2}}{A_{t,1}} = \frac{p_{0,1}}{p_{0,2}} \tag{10.39}$$

If $A_{t,2}$ is reduced much below this value, the diffuser will choke and the tunnel will unstart.

10.8 PROBLEMS

10.1 The reservoir pressure and temperature for a convergent-divergent nozzle are 5 atm and 520°R, respectively. The flow is expanded isentropically to supersonic speed at the nozzle exit. If the exit-to-throat area ratio is 2.193, calculate the following properties at the exit: M_e, p_e, T_e, ρ_e, u_e, $p_{0,e}$, $T_{0,e}$.

10.2 A flow is isentropically expanded to supersonic speeds in a convergent-divergent nozzle. The reservoir and exit pressures are 1 and 0.3143 atm, respectively. What is the value of A_e/A^*?

10.3 A Pitot tube inserted at the exit of a supersonic nozzle reads 8.92×10^4 N/m^2. If the reservoir pressure is 2.02×10^5 N/m^2, calculate the area ratio A_e/A^* of the nozzle.

10.4 For the nozzle flow given in Problem 10.1, the throat area is 4 in^2. Calculate the mass flow through the nozzle.

10.5 A closed-form expression for the mass flow through a choked nozzle is

$$\dot{m} = \frac{p_0 A^*}{\sqrt{T_0}} \sqrt{\frac{\gamma}{R} \left(\frac{2}{\gamma + 1} \right)^{(\gamma+1)/(\gamma-1)}}$$

Derive this expression.

10.6 Repeat Problem 10.4, using the formula derived in Prob. 10.5, and check your answer from Prob. 10.4.

10.7 A convergent-divergent nozzle with an exit-to-throat area ratio of 1.616 has exit and reservoir pressures equal to 0.947 and 1.0 atm, respectively. Assuming isentropic flow through the nozzle, calculate the Mach number and pressure at the throat.

10.8 For the flow in Problem 10.7, calculate the mass flow through the nozzle, assuming that the reservoir temperature is 288 K and the throat area is 0.3 m^2.

10.9 Consider a convergent-divergent nozzle with an exit-to-throat area ratio of 1.53. The reservoir pressure is 1 atm. Assuming isentropic flow, except for the possibility of a normal shock wave inside the nozzle, calculate the exit Mach number when the exit pressure p_e is
(*a*) 0.94 atm (*b*) 0.886 atm (*c*) 0.75 atm (*d*) 0.154 atm

10.10 A 20° half-angle wedge is mounted at 0° angle of attack in the test section of a supersonic wind tunnel. When the tunnel is operating, the wave angle from the wedge leading edge is measured to be 41.8°. What is the exit-to-throat area ratio of the tunnel nozzle?

10.11 The nozzle of a supersonic wind tunnel has an exit-to-throat area ratio of 6.79. When the tunnel is running, a Pitot tube mounted in the test section measures 1.448 atm. What is the reservoir pressure for the tunnel?

10.12 We wish to design a supersonic wind tunnel that produces a Mach 2.8 flow at standard sea level conditions in the test section and has a mass flow of air equal to 1 slug/s. Calculate the necessary reservoir pressure and temperature, the nozzle throat and exit areas, and the diffuser throat area.

10.13 Consider a rocket engine burning hydrogen and oxygen. The total mass flow of the propellant plus oxidizer into the combustion chamber is 287.2 kg/s. The combustion chamber temperature is 3600 K. Assume that the combustion chamber is a low-velocity reservoir for the rocket engine. If the area of the rocket nozzle throat is 0.2 m^2, calculate the combustion chamber (reservoir) pressure. Assume that the gas that flows through the engine has a ratio of specific heats, $\gamma = 1.2$, and a molecular weight of 16.

10.14 For supersonic and hypersonic wind tunnels, a diffuser efficiency, η_D, can be defined as the ratio of the total pressures at the diffuser exit and nozzle reservoir, divided by the total pressure ratio across a normal shock at the test-section Mach number. This is a measure of the efficiency of the diffuser relative to normal shock pressure recovery. Consider a supersonic wind tunnel designed for a test-section Mach number of 3.0 which exhausts directly to the atmosphere. The diffuser efficiency is 1.2. Calculate the minimum reservoir pressure necessary for running the tunnel.

10.15 Return to Problem 9.18, where the average Mach number across the two-dimensional flow in a duct was calculated, and where θ for the upper wall was 3°. Assuming quasi-one-dimensional flow, calculate the Mach number at the location AB in the duct.

10.16 Return to Problem 9.19, where the average Mach number across the two-dimensional flow in a duct was calculated, and where θ for the upper wall was 30°. Assuming quasi-one-dimensional flow, calculate the Mach number at the location AB in the duct.

10.17 A horizontal flow initially at Mach 1 flows over a downward-sloping expansion corner, thus creating a centered Prandtl-Meyer expansion wave. The streamlines that enter the head of the expansion wave curve smoothly and continuously downward through the expansion fan, and emerge parallel to the downward sloping surface downstream of the tail of the wave, as shown in Figure 9.2*b*. Imagine a polar coordinate system r, Φ with its origin at the expansion corner (the vertex of the Prandtl-Meyer expansion wave), with r the usual radial distance along a ray from the origin and Φ the polar angle of r measured from the horizontal. Because the upstream flow is at Mach 1, the head of the expansion fan is a Mach wave perpendicular to the free stream. Consider a given streamline entering the expansion wave at the point $(r, \Phi) = (r^*, \pi/2)$. Construct a method for calculating the shape of this streamline as a function of r and Φ through the expansion fan. *Note:* To solve this problem, material from both Chapters 9 and 10 is required.

10.18 Consider a centered expansion wave where $M_1 = 1.0$ and $M_2 = 1.6$. Using the method developed in Problem 10.17, plot to scale a streamline that passes through the expansion wave.

CHAPTER 11

Subsonic Compressible Flow over Airfoils: Linear Theory

During the war a British engineer named Frank Whittle invented the jet engine, and deHavilland built the first production-type model. He produced a jet plane named Vampire, the first to exceed 500 mph. Then he built the experimental DH 108, and released it to young Geoffrey for test. In the first cautious trials the new plane behaved beautifully; but as Geoffrey stepped up the speed he unsuspectingly drew closer to an invisible wall in the sky then unknown to anyone, later named the sound barrier, which can destroy a plane not designed to pierce it. One evening he hit the speed of sound, and the plane disintegrated. Young Geoffrey's body was not found for ten days.

From the Royal Air Force Flying Review, as condensed in Reader's *Digest*, 1959

PREVIEW BOX

We learned all about airfoils in Chapter 4—but did we really? Chapter 4 dealt with airfoils in low-speed incompressible flow. What happens when we have a high-speed compressible flow over these airfoils? By this stage of our discussions about compressible flow, you might expect compressibility to make some differences, and you would be correct. But what differences, and by how much? This chapter and the next give some answers.

The present chapter deals with airfoils in a high-speed subsonic flow. What happens when an airfoil is flown at high subsonic Mach numbers, near the speed of sound? How does compressibility change the airfoil properties, and by how much? How do we analyze and calculate these compressibility effects? The answers are given in this chapter. In addition, we will see that as Mach 1 is approached, the drag of an airfoil suddenly skyrockets and looks like it is going out of sight (the so-called sound barrier, as sometimes referenced in the popular literature). What is going on here? How do we deal with it?

These are all very important and very practical questions, and they are all addressed in this chapter. Every time you fly in a high-speed jet transport, such as a Boeing 777 cruising at a Mach number of 0.85, you are encountering the phenomena that drive these questions. Without proper answers, we would never be able to design this type of airplane, and we would always be relegated to flying slower—something that the pace of modern life would not easily tolerate. So pay close attention to the material in this chapter.

But be aware. Our first four chapters dealing with the basics of compressible flow (Chapters 7–10) used mathematics at essentially the level of algebra. To go further in our study of compressible flow, especially to answer the questions posed here, we have to return to the world of partial differential equations. But this is no big deal; we have been there before in our discussions of incompressible flow in Part I of this book. So jump right in to the material of this chapter. I predict that you will experience some increase in technical maturity as you study this material, and that you will enjoy it. After all, here you will be dealing with some of the most important and exciting applications in modern aerodynamics.

11.1 INTRODUCTION

The above quotation refers to an accident that took place on September 27, 1946, when Geoffrey deHavilland, son of the famed British airplane designer Sir Geoffrey deHavilland, took the D. H. 108 Swallow up for an attack on the world's speed record. At that time, no airplane had flown at or beyond the speed of sound. The Swallow was an experimental jet-propelled aircraft with swept wings and no tail. During its first high-speed, low-level run, the Swallow encountered major compressibility problems and broke up in the air. deHavilland was killed instantly. This accident strengthened the opinion of many that Mach 1 stood as a barrier to manned flight and that no airplane would ever fly faster than the speed of sound. This myth of the "sound barrier" originated in the early 1930s. It was in full force by the time of the Volta Conference in 1935 (see Section 7.1). In light of the opening quotation, the idea of a sound barrier was still being discussed in the popular literature as late as 1959, 12 years after the first successful supersonic flight by Captain Charles Yeager on October 14, 1947.

Of course, we know today that the sound barrier is indeed a myth; the supersonic transport Concorde flew at Mach 2, and some military aircraft are capable of Mach 3 and slightly beyond. The X-15 hypersonic research airplane has flown at Mach 7, and the Apollo lunar return capsule successfully reentered the earth's atmosphere at Mach 36. Supersonic flight is now an everyday occurrence. So, what caused the early concern about a sound barrier? In the present chapter, we develop a theory applicable to high-speed subsonic flight, and we see how the theory predicts a monotonically increasing drag going to infinity as $M_\infty \rightarrow 1$. It was this type of result that led some people in the early 1930s to believe that flight beyond the speed of sound was impossible. However, we also show in this chapter that the approximations made in the theory break down near Mach 1 and that in reality, although the drag coefficient at Mach 1 is large, it is still a manageable finite number.

Specifically, the purpose of this chapter is to examine the properties of two-dimensional airfoils at Mach numbers above 0.3, where we can no longer assume incompressible flow, but below Mach 1. That is, this chapter is an extension of the airfoil discussions in Chapter 4 (which applied to incompressible flow) to the high-speed subsonic regime.

In the process, we climb to a new tier in our study of compressible flow. If you survey our discussions so far of compressible flow, you will observe that they treat one-dimensional cases such as normal shock waves and flows in ducts. Even oblique shock waves, which are two- and three-dimensional in nature, depend only on the component of Mach number normal to the wave. Therefore, we have not been explicitly concerned with a multidimensional flow. As a consequence, note that the types of equations which allow an analysis of these flows are *algebraic equations,* and hence are relatively easy to solve in comparison with partial differential equations. In Chapters 8 to 10, we have dealt primarily with such algebraic equations. These algebraic equations were obtained by applying the integral forms of the conservation equations [Equations (2.48), (2.64), and (2.95)] to appropriate control volumes where the flow properties were uniform over the inflow and outflow faces of the control volume. However, for general two- and three-dimensional flows, we are usually not afforded such a luxury. Instead, we must deal directly with the governing equations in their partial differential equation form (see Chapter 2). Such is the nature of the present chapter. Indeed, for the remainder of our aerodynamic discussions in this book, we appeal mainly to the differential forms of the continuity, momentum, and energy equations [such as Equations (2.52), (2.113*a* to *c*), and (2.114)].

The road map for this chapter is given in Figure 11.1. We are going to return to the concept of a velocity potential, first introduced in Section 2.15. We are going to combine our governing equations so as to obtain a single equation simply in

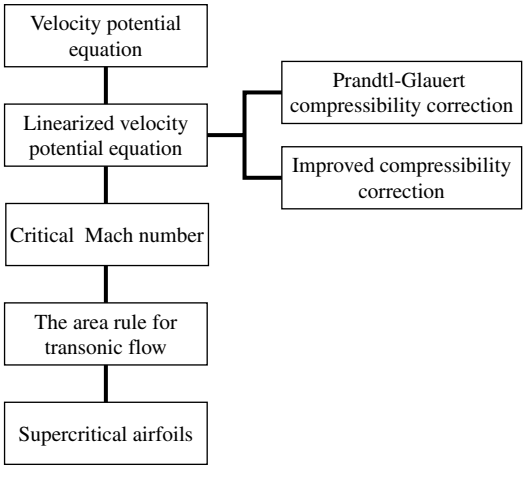


Figure 11.1 Road map for Chapter 11.

terms of the velocity potential; that is, we are going to obtain for compressible flow an equation analogous to Laplace's equation derived for incompressible flow in Section 3.7 [see Equation (3.40)]. However, unlike Laplace's equation, which is linear, the exact velocity potential equation for compressible flow is nonlinear. By making suitable approximations, we are able to linearize this equation and apply it to thin airfoils at small angles of attack. The results enable us to correct incompressible airfoil data for the effects of compressibility—so-called *compressibility corrections*. Finally, we conclude this chapter by discussing several practical aspects of airfoil and general wing-body aerodynamics at speeds near Mach 1.

11.2 THE VELOCITY POTENTIAL EQUATION

The inviscid, compressible, subsonic flow over a body immersed in a uniform stream is *irrotational;* there is no mechanism in such a flow to start rotating the fluid elements (see Section 2.12). Thus, a velocity potential (see Section 2.15) can be defined. Since we are dealing with irrotational flow and the velocity potential, review Sections 2.12 and 2.15 before progressing further.

Consider two-dimensional, steady, irrotational, isentropic flow. A velocity potential, $\phi = \phi(x, y)$, can be defined such that [from Equation (2.154)]

$$\mathbf{V} = \nabla\phi \tag{11.1}$$

or in terms of the cartesian velocity components,

$$u = \frac{\partial\phi}{\partial x} \tag{11.2a}$$

$$v = \frac{\partial\phi}{\partial y} \tag{11.2b}$$

Let us proceed to obtain an equation for ϕ which represents a combination of the continuity, momentum, and energy equations. Such an equation would be very useful, because it would be simply one governing equation in terms of one unknown, namely the velocity potential ϕ.

The continuity equation for steady, two-dimensional flow is obtained from Equation (2.52) as

$$\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0 \tag{11.3}$$

or

$$\rho\frac{\partial u}{\partial x} + u\frac{\partial\rho}{\partial x} + v\frac{\partial\rho}{\partial y} + \rho\frac{\partial v}{\partial y} = 0 \tag{11.4}$$

Substituting Equations (11.2*a* and *b*) into (11.4), we have

$$\rho\frac{\partial^2\phi}{\partial x^2} + \frac{\partial\phi}{\partial x}\frac{\partial\rho}{\partial x} + \frac{\partial\phi}{\partial y}\frac{\partial\rho}{\partial y} + \rho\frac{\partial^2\phi}{\partial y^2} = 0$$

or

$$\rho\left(\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2}\right) + \frac{\partial\phi}{\partial x}\frac{\partial\rho}{\partial x} + \frac{\partial\phi}{\partial y}\frac{\partial\rho}{\partial y} = 0 \tag{11.5}$$

We are attempting to obtain an equation completely in terms of ϕ; hence, we need to eliminate ρ from Equation (11.5). To do this, consider the momentum equation in terms of Euler's equation:

$$dp = -\rho V\, dV \tag{3.12}$$

This equation holds for a steady, compressible, inviscid flow and relates p and V along a streamline. It can readily be shown that Equation (3.12) holds in *any* direction throughout an irrotational flow, not just along a streamline (try it yourself). Therefore, from Equations (3.12) and (11.2*a* and *b*), we have

$$dp = -\rho V\, dV = -\frac{\rho}{2} d(V^2) = -\frac{\rho}{2} d(u^2 + v^2)$$

or

$$dp = -\frac{\rho}{2} d\left[\left(\frac{\partial \phi}{\partial x}\right)^2 + \left(\frac{\partial \phi}{\partial y}\right)^2\right] \tag{11.6}$$

Recall that we are also considering the flow to be isentropic. Hence, any change in pressure dp in the flow is automatically accompanied by a corresponding isentropic change in density $d\rho$. Thus, by definition

$$\frac{dp}{d\rho} = \left(\frac{\partial p}{\partial \rho}\right)_s \tag{11.7}$$

The right-hand side of Equation (11.7) is simply the square of the speed of sound. Thus, Equation (11.7) yields

$$dp = a^2\, d\rho \tag{11.8}$$

Substituting Equation (11.8) for the left side of Equation (11.6), we have

$$d\rho = -\frac{\rho}{2a^2} d\left[\left(\frac{\partial \phi}{\partial x}\right)^2 + \left(\frac{\partial \phi}{\partial y}\right)^2\right] \tag{11.9}$$

Considering changes in the x direction, Equation (11.9) directly yields

$$\frac{\partial \rho}{\partial x} = -\frac{\rho}{2a^2} \frac{\partial}{\partial x}\left[\left(\frac{\partial \phi}{\partial x}\right)^2 + \left(\frac{\partial \phi}{\partial y}\right)^2\right]$$

or

$$\frac{\partial \rho}{\partial x} = -\frac{\rho}{a^2}\left(\frac{\partial \phi}{\partial x}\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial \phi}{\partial y}\frac{\partial^2 \phi}{\partial x \partial y}\right) \tag{11.10}$$

Similarly, for changes in the y direction, Equation (11.9) gives

$$\frac{\partial \rho}{\partial y} = -\frac{\rho}{a^2}\left(\frac{\partial \phi}{\partial x}\frac{\partial^2 \phi}{\partial x\, \partial y} + \frac{\partial \phi}{\partial y}\frac{\partial^2 \phi}{\partial y^2}\right) \tag{11.11}$$

Substituting Equations (11.10) and (11.11) into (11.5), canceling the ρ which appears in each term, and factoring out the second derivatives of ϕ, we obtain

$$\boxed{\left[1-\frac{1}{a^2}\left(\frac{\partial\phi}{\partial x}\right)^2\right]\frac{\partial^2\phi}{\partial x^2}+\left[1-\frac{1}{a^2}\left(\frac{\partial\phi}{\partial y}\right)^2\right]\frac{\partial^2\phi}{\partial y^2} -\frac{2}{a^2}\left(\frac{\partial\phi}{\partial x}\right)\left(\frac{\partial\phi}{\partial y}\right)\frac{\partial^2\phi}{\partial x\,\partial y}=0} \tag{11.12}$$

which is called the *velocity potential equation*. It is almost completely in terms of ϕ; only the speed of sound appears in addition to ϕ. However, a can be readily expressed in terms of ϕ as follows. From Equation (8.33), we have

$$\begin{aligned} a^2 &= a_0^2-\frac{\gamma-1}{2}V^2 = a_0^2-\frac{\gamma-1}{2}(u^2+v^2) \\ &= a_0^2-\frac{\gamma-1}{2}\left[\left(\frac{\partial\phi}{\partial x}\right)^2+\left(\frac{\partial\phi}{\partial y}\right)^2\right] \end{aligned} \tag{11.13}$$

Since a_0 is a known constant of the flow, Equation (11.13) gives the speed of sound a as a function of ϕ. Hence, substitution of Equation (11.13) into (11.12) yields a single partial differential equation in terms of the unknown ϕ. This equation represents a combination of the continuity, momentum, and energy equations. In principle, it can be solved to obtain ϕ for the flow field around any two-dimensional shape, subject of course to the usual boundary conditions at infinity and along the body surface. These boundary conditions on ϕ are detailed in Section 3.7, and are given by Equations (3.47*a* and *b*) and (3.48*b*).

Because Equation (11.12) [along with Equation (11.13)] is a single equation in terms of one dependent variable ϕ, the analysis of isentropic, irrotational, steady, compressible flow is greatly simplified—we only have to solve one equation instead of three or more. Once ϕ is known, all the other flow variables are directly obtained as follows:

1. Calculate u and v from Equations (11.2*a* and *b*).
2. Calculate a from Equation (11.13).
3. Calculate $M = V/a = \sqrt{u^2+v^2}/a$.
4. Calculate T, p, and ρ from Equations (8.40), (8.42), and (8.43), respectively. In these equations, the total conditions T_0, p_0, and ρ_0 are known quantities; they are constant throughout the flow field and hence are obtained from the given freestream conditions.

Although Equation (11.12) has the advantage of being one equation with one unknown, it also has the distinct disadvantage of being a *nonlinear* partial differential equation. Such nonlinear equations are very difficult to solve analytically, and in modern aerodynamics, solutions of Equation (11.12) are usually sought by means of sophisticated finite-difference numerical techniques. Indeed, no general analytical solution of Equation (11.12) has been found to this day. Contrast

this situation with that for incompressible flow, which is governed by Laplace's equation—a *linear* partial differential equation for which numerous analytical solutions are well known.

Given this situation, aerodynamicists over the years have made assumptions regarding the physical nature of the flow field which are designed to simplify Equation (11.12). These assumptions limit our considerations to the flow over slender bodies at small angles of attack. For subsonic and supersonic flows, these assumptions lead to an *approximate* form of Equation (11.12) which is linear, and hence can be solved analytically. These matters are the subject of the next section.

Keep in mind that, within the framework of steady, irrotational, isentropic flow, Equation (11.12) is *exact* and holds for all Mach numbers, from subsonic to hypersonic, and for all two-dimensional body shapes, thin and thick.

11.3 THE LINEARIZED VELOCITY POTENTIAL EQUATION

Consider the two-dimensional, irrotational, isentropic flow over the body shown in Figure 11.2. The body is immersed in a uniform flow with velocity V_∞ oriented in the x direction. At an arbitrary point P in the flow field, the velocity is $\mathbf{V}$ with the x and y components given by u and v, respectively. Let us now visualize the velocity $\mathbf{V}$ as the sum of the uniform flow velocity plus some extra increments in velocity. For example, the x component of velocity u in Figure 11.2 can be visualized as V_∞ plus an increment in velocity (positive or negative). Similarly, the y component of velocity v can be visualized as a simple increment itself, because the uniform flow has a zero component in the y direction. These increments are called *perturbations,* and

$$u = V_\infty + \hat{u} \qquad v = \hat{v}$$

where $\hat{u}$ and $\hat{v}$ are called the *perturbation velocities*. These perturbation velocities are not necessarily small; indeed, they can be quite large in the stagnation region in front of the blunt nose of the body shown in Figure 11.2. In the same vein,

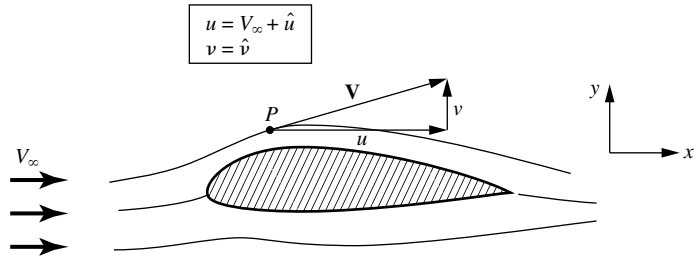


Figure 11.2 Uniform flow and perturbed flow.

because $\mathbf{V} = \nabla\phi$, we can define a perturbation velocity potential $\hat{\phi}$ such that

$$\phi = V_\infty x + \hat{\phi}$$

where

$$\frac{\partial\hat{\phi}}{\partial x} = \hat{u}$$

$$\frac{\partial\hat{\phi}}{\partial y} = \hat{v}$$

Hence,

$$\frac{\partial\phi}{\partial x} = V_\infty + \frac{\partial\hat{\phi}}{\partial x} \qquad \frac{\partial\phi}{\partial y} = \frac{\partial\hat{\phi}}{\partial y}$$

$$\frac{\partial^2\phi}{\partial x^2} = \frac{\partial^2\hat{\phi}}{\partial x^2} \qquad \frac{\partial^2\phi}{\partial y^2} = \frac{\partial^2\hat{\phi}}{\partial y^2} \qquad \frac{\partial^2\phi}{\partial x\,\partial y} = \frac{\partial^2\hat{\phi}}{\partial x\,\partial y}$$

Substituting the above definitions into Equation (11.12), and multiplying by a^2, we obtain

$$\left[a^2 - \left(V_\infty + \frac{\partial\hat{\phi}}{\partial x}\right)^2\right]\frac{\partial^2\hat{\phi}}{\partial x^2} + \left[a^2 - \left(\frac{\partial\hat{\phi}}{\partial y}\right)^2\right]\frac{\partial^2\hat{\phi}}{\partial y^2} - 2\left(V_\infty + \frac{\partial\hat{\phi}}{\partial x}\right)\left(\frac{\partial\hat{\phi}}{\partial y}\right)\frac{\partial^2\hat{\phi}}{\partial x\,\partial y} = 0 \tag{11.14}$$

Equation (11.14) is called the *perturbation velocity potential equation*. It is precisely the same equation as Equation (11.12) except that it is expressed in terms of $\hat{\phi}$ instead of ϕ. It is still a nonlinear equation.

To obtain better physical insight in some of our subsequent discussion, let us recast Equation (11.14) in terms of the perturbation velocities. From the definition of $\hat{\phi}$ given earlier, Equation (11.14) can be written as

$$[a^2 - (V_\infty + \hat{u})^2]\frac{\partial\hat{u}}{\partial x} + (a^2 - \hat{v}^2)\frac{\partial\hat{v}}{\partial y} - 2(V_\infty + \hat{u})\hat{v}\frac{\partial\hat{u}}{\partial y} = 0 \tag{11.14a}$$

From the energy equation in the form of Equation (8.32), we have

$$\frac{a_\infty^2}{\gamma - 1} + \frac{V_\infty^2}{2} = \frac{a^2}{\gamma - 1} + \frac{(V_\infty + \hat{u})^2 + \hat{v}^2}{2} \tag{11.15}$$

Substituting Equation (11.15) into (11.14a), and algebraically rearranging, we obtain

$$\begin{aligned}(1 - M_\infty^2)\frac{\partial\hat{u}}{\partial x} + \frac{\partial\hat{v}}{\partial y} = {} & M_\infty^2\left[(\gamma + 1)\frac{\hat{u}}{V_\infty} + \frac{\gamma + 1}{2}\frac{\hat{u}^2}{V_\infty^2} + \frac{\gamma - 1}{2}\frac{\hat{v}^2}{V_\infty^2}\right]\frac{\partial\hat{u}}{\partial x} \\ & + M_\infty^2\left[(\gamma - 1)\frac{\hat{u}}{V_\infty} + \frac{\gamma + 1}{2}\frac{\hat{v}^2}{V_\infty^2} + \frac{\gamma - 1}{2}\frac{\hat{u}^2}{V_\infty^2}\right]\frac{\partial\hat{v}}{\partial y} \\ & + M_\infty^2\left[\frac{\hat{v}}{V_\infty}\left(1 + \frac{\hat{u}}{V_\infty}\right)\left(\frac{\partial\hat{u}}{\partial y} + \frac{\partial\hat{v}}{\partial x}\right)\right]\end{aligned} \tag{11.16}$$

Equation (11.16) is still exact for irrotational, isentropic flow. Note that the left-hand side of Equation (11.16) is linear but the right-hand side is nonlinear. Also, keep in mind that the size of the perturbations $\hat{u}$ and $\hat{v}$ can be large or small; Equation (11.16) holds for both cases.

Let us now limit our considerations to *small* perturbations; that is, assume that the body in Figure 11.2 is a *slender* body at *small* angle of attack. In such a case, $\hat{u}$ and $\hat{v}$ will be small in comparison with V_∞. Therefore, we have

$$\frac{\hat{u}}{V_\infty}, \frac{\hat{v}}{V_\infty} \ll 1 \qquad \frac{\hat{u}^2}{V_\infty^2}, \frac{\hat{v}^2}{V_\infty^2} \lll 1$$

Keep in mind that products of $\hat{u}$ and $\hat{v}$ with their derivatives are also very small. With this in mind, examine Equation (11.16). Compare like terms (coefficients of like derivatives) on the left- and right-hand sides of Equation (11.16). We find

1. For $0 \leq M_\infty \leq 0.8$ or $M_\infty \geq 1.2$, the magnitude of

$$M_\infty^2 \left[(\gamma + 1) \frac{\hat{u}}{V_\infty} + \cdots \right] \frac{\partial \hat{u}}{\partial x}$$

is small in comparison with the magnitude of

$$\left(1 - M_\infty^2\right) \frac{\partial \hat{u}}{\partial x}$$

Thus, *ignore* the former term.

2. For $M_\infty < 5$ (approximately),

$$M_\infty^2 \left[(\gamma - 1) \frac{\hat{u}}{V_\infty} + \cdots \right] \frac{\partial \hat{v}}{\partial y}$$

is small in comparison with $\partial \hat{v}/\partial y$. So ignore the former term. Also,

$$M_\infty^2 \left[\frac{\hat{v}}{V_\infty} \left(1 + \frac{\hat{u}}{V_\infty} \right) \left(\frac{\partial \hat{u}}{\partial y} + \frac{\partial \hat{v}}{\partial x} \right) \right] \approx 0$$

With the above order-of-magnitude comparisons, Equation (11.16) reduces to

$$\left(1 - M_\infty^2\right) \frac{\partial \hat{u}}{\partial x} + \frac{\partial \hat{v}}{\partial y} = 0 \tag{11.17}$$

or in terms of the perturbation velocity potential,

$$\boxed{\left(1 - M_\infty^2\right) \frac{\partial^2 \hat{\phi}}{\partial x^2} + \frac{\partial^2 \hat{\phi}}{\partial y^2} = 0} \tag{11.18}$$

Examine Equation (11.18). It is a *linear* partial differential equation, and therefore is inherently simpler to solve than its parent equation, Equation (11.16). However, we have paid a price for this simplicity. Equation (11.18) is no longer exact. It is only an approximation to the physics of the flow. Due to the assumptions made in

obtaining Equation (11.18), it is reasonably valid (but not exact) for the following combined situations:

1. *Small* perturbation, that is, thin bodies at small angles of attack
2. *Subsonic* and *supersonic* Mach numbers

In contrast, Equation (11.18) is not valid for thick bodies and for large angles of attack. Moreover, it cannot be used for transonic flow, where $0.8 < M_\infty < 1.2$, or for hypersonic flow, where $M_\infty > 5$.

We are interested in solving Equation (11.18) in order to obtain the pressure distribution along the surface of a slender body. Since we are now dealing with approximate equations, it is consistent to obtain a linearized expression for the pressure coefficient—an expression that is approximate to the same degree as Equation (11.18), but which is extremely simple and convenient to use. The linearized pressure coefficient can be derived as follows.

First, recall the definition of the pressure coefficient C_p given in Section 1.5:

$$C_p \equiv \frac{p - p_\infty}{q_\infty} \tag{11.19}$$

where $q_\infty = \frac{1}{2}\rho_\infty V_\infty^2 =$ dynamic pressure. The dynamic pressure can be expressed in terms of M_∞ as follows:

$$q_\infty = \frac{1}{2}\rho_\infty V_\infty^2 = \frac{1}{2}\frac{\gamma p_\infty}{\gamma p_\infty}\rho_\infty V_\infty^2 = \frac{\gamma}{2}p_\infty\left(\frac{\rho_\infty}{\gamma p_\infty}\right)V_\infty^2 \tag{11.20}$$

From Equation (8.23), we have $a_\infty^2 = \gamma p_\infty/\rho_\infty$. Hence, Equation (11.20) becomes

$$q_\infty = \frac{\gamma}{2}p_\infty\frac{V_\infty^2}{a_\infty^2} = \frac{\gamma}{2}p_\infty M_\infty^2 \tag{11.21}$$

Substituting Equation (11.21) into (11.19), we have

$$\boxed{C_p = \frac{2}{\gamma M_\infty^2}\left(\frac{p}{p_\infty} - 1\right)} \tag{11.22}$$

Equation (11.22) is simply an alternate form of the pressure coefficient expressed in terms of M_∞. It is still an exact representation of the definition of C_p.

To obtain a linearized form of the pressure coefficient, recall that we are dealing with an adiabatic flow of a calorically perfect gas; hence, from Equation (8.39),

$$T + \frac{V^2}{2c_p} = T_\infty + \frac{V_\infty^2}{2c_p} \tag{11.23}$$

Recalling from Equation (7.9) that $c_p = \gamma R/(\gamma - 1)$, Equation (11.23) can be written as

$$T - T_\infty = \frac{V_\infty^2 - V^2}{2\gamma R/(\gamma - 1)} \tag{11.24}$$

Also, recalling that $a_\infty = \sqrt{\gamma RT_\infty}$, Equation (11.24) becomes

$$\frac{T}{T_\infty} - 1 = \frac{\gamma - 1}{2}\frac{V_\infty^2 - V^2}{\gamma RT_\infty} = \frac{\gamma - 1}{2}\frac{V_\infty^2 - V^2}{a_\infty^2} \tag{11.25}$$

In terms of the perturbation velocities

$$V^2 = (V_\infty + \hat{u})^2 + \hat{v}^2$$

Equation (11.25) can be written as

$$\frac{T}{T_\infty} = 1 - \frac{\gamma - 1}{2a_\infty^2}(2\hat{u}V_\infty + \hat{u}^2 + \hat{v}^2) \tag{11.26}$$

Since the flow is isentropic, $p/p_\infty = (T/T_\infty)^{\gamma/(\gamma-1)}$, and Equation (11.26) becomes

$$\frac{p}{p_\infty} = \left[1 - \frac{\gamma - 1}{2a_\infty^2}(2\hat{u}V_\infty + \hat{u}^2 + \hat{v}^2)\right]^{\gamma/(\gamma-1)}$$

or

$$\frac{p}{p_\infty} = \left[1 - \frac{\gamma - 1}{2}M_\infty^2\left(\frac{2\hat{u}}{V_\infty} + \frac{\hat{u}^2 + \hat{v}^2}{V_\infty^2}\right)\right]^{\gamma/(\gamma-1)} \tag{11.27}$$

Equation (11.27) is still an exact expression. However, let us now make the assumption that the perturbations are small, that is, $\hat{u}/V_\infty \ll 1$, $\hat{u}^2/V_\infty^2 \lll 1$, and $\hat{v}^2/V_\infty^2 \lll 1$. In this case, Equation (11.27) is of the form

$$\frac{p}{p_\infty} = (1 - \varepsilon)^{\gamma/(\gamma-1)} \tag{11.28}$$

where ε is small. From the binomial expansion, neglecting higher-order terms, Equation (11.28) becomes

$$\frac{p}{p_\infty} = 1 - \frac{\gamma}{\gamma - 1}\varepsilon + \cdots \tag{11.29}$$

Comparing Equation (11.27) to (11.29), we can express Equation (11.27) as

$$\frac{p}{p_\infty} = 1 - \frac{\gamma}{2}M_\infty^2\left(\frac{2\hat{u}}{V_\infty} + \frac{\hat{u}^2 + \hat{v}^2}{V_\infty^2}\right) + \cdots \tag{11.30}$$

Substituting Equation (11.30) into the expression for the pressure coefficient, Equation (11.22), we obtain

$$C_p = \frac{2}{\gamma M_\infty^2}\left[1 - \frac{\gamma}{2}M_\infty^2\left(\frac{2\hat{u}}{V_\infty} + \frac{\hat{u}^2 + \hat{v}^2}{V_\infty^2}\right) + \cdots - 1\right]$$

or

$$C_p = -\frac{2\hat{u}}{V_\infty} - \frac{\hat{u}^2 + \hat{v}^2}{V_\infty^2} \tag{11.31}$$

Since $\hat{u}^2/V_\infty^2$ and $\hat{v}^2/V_\infty^2 \lll 1$, Equation (11.31) becomes

$$\boxed{C_p = -\frac{2\hat{u}}{V_\infty}} \tag{11.32}$$

Equation (11.32) is the linearized form for the pressure coefficient; it is valid only for *small* perturbations. Equation (11.32) is consistent with the linearized perturbation velocity potential equation, Equation (11.18). Note the simplicity of Equation (11.32); it depends only on the x component of the velocity perturbation, namely, $\hat{u}$.

To round out our discussion on the basics of the linearized equations, we note that any solution to Equation (11.18) must satisfy the usual boundary conditions at infinity and at the body surface. At infinity, clearly $\hat{\phi}$ = constant; that is, $\hat{u} = \hat{v} = 0$. At the body, the flow-tangency condition holds. Let θ be the angle between the tangent to the surface and the freestream. Then, at the surface, the boundary condition is obtained from Equation (3.48*e*):

$$\tan\theta = \frac{v}{u} = \frac{\hat{v}}{V_\infty + \hat{u}} \tag{11.33}$$

which is an exact expression for the flow-tangency condition at the body surface. A simpler, approximate expression for Equation (11.33), consistent with linearized theory, can be obtained by noting that for small perturbations, $\hat{u} \ll V_\infty$. Hence, Equation (11.33) becomes

$$\hat{v} = V_\infty \tan\theta$$

or

$$\boxed{\frac{\partial \hat{\phi}}{\partial y} = V_\infty \tan\theta} \tag{11.34}$$

Equation (11.34) is an *approximate* expression for the flow-tangency condition at the body surface, with accuracy of the same order as Equations (11.18) and (11.32).

11.4 PRANDTL-GLAUERT COMPRESSIBILITY CORRECTION

The aerodynamic theory for incompressible flow over thin airfoils at small angles of attack was presented in Chapter 4. For aircraft of the period 1903–1940, such theory was adequate for predicting airfoil properties. However, with the rapid evolution of high-power reciprocating engines spurred by World War II, the velocities of military fighter planes began to push close to 450 mi/h. Then, with the advent of the first operational jet-propelled airplanes in 1944 (the German Me 262), flight velocities took a sudden spurt into the 550 mi/h range and faster. As a result, the incompressible flow theory of Chapter 4 was no longer applicable to such aircraft; rather, high-speed airfoil theory had to deal with compressible flow. Because a vast bulk of data and experience had been collected over the years in low-speed aerodynamics, and because there was no desire to totally discard such data, the natural approach to high-speed subsonic aerodynamics was to search for methods that would allow relatively simple *corrections* to existing incompressible flow results which would approximately take into account the effects of compressibility. Such

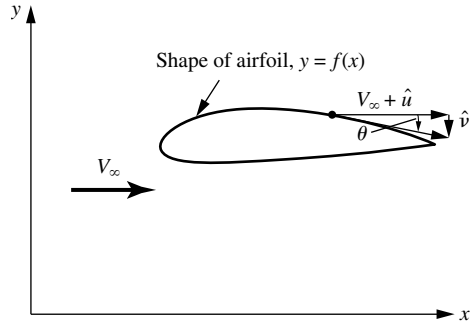


Figure 11.3 Airfoil in physical space.

methods are called *compressibility corrections*. The first, and most widely known of these corrections is the Prandtl-Glauert compressibility correction, to be derived in this section. The Prandtl-Glauert method is based on the linearized perturbation velocity potential equation given by Equation (11.18). Therefore, it is limited to thin airfoils at small angles of attack. Moreover, it is purely a subsonic theory and begins to give inappropriate results at values of $M_\infty = 0.7$ and above.

Consider the subsonic, compressible, inviscid flow over the airfoil sketched in Figure 11.3. The shape of the airfoil is given by $y = f(x)$. Assume that the airfoil is thin and that the angle of attack is small; in such a case, the flow is reasonably approximated by Equation (11.18). Define

$$\beta^2 \equiv 1 - M_\infty^2$$

so that Equation (11.18) can be written as

$$\beta^2 \frac{\partial^2 \hat{\phi}}{\partial x^2} + \frac{\partial^2 \hat{\phi}}{\partial y^2} = 0 \tag{11.35}$$

Let us transform the independent variables x and y into a new space, ξ and η, such that

$$\xi = x \tag{11.36a}$$

$$\eta = \beta y \tag{11.36b}$$

Moreover, in this transformed space, consider a new velocity potential $\bar{\phi}$ such that

$$\bar{\phi}(\xi, \eta) = \beta \hat{\phi}(x, y) \tag{11.36c}$$

To recast Equation (11.35) in terms of the transformed variables, recall the chain rule of partial differentiation; that is,

$$\frac{\partial \hat{\phi}}{\partial x} = \frac{\partial \hat{\phi}}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial \hat{\phi}}{\partial \eta}\frac{\partial \eta}{\partial x} \tag{11.37}$$

and
$$\frac{\partial \hat{\phi}}{\partial y} = \frac{\partial \hat{\phi}}{\partial \xi}\frac{\partial \xi}{\partial y} + \frac{\partial \hat{\phi}}{\partial \eta}\frac{\partial \eta}{\partial y} \tag{11.38}$$

From Equations (11.36*a* and *b*), we have

$$\frac{\partial \xi}{\partial x} = 1 \quad \frac{\partial \xi}{\partial y} = 0 \quad \frac{\partial \eta}{\partial x} = 0 \quad \frac{\partial \eta}{\partial y} = \beta$$

Hence, Equations (11.37) and (11.38) become

$$\frac{\partial \hat{\phi}}{\partial x} = \frac{\partial \hat{\phi}}{\partial \xi} \tag{11.39}$$

$$\frac{\partial \hat{\phi}}{\partial y} = \beta\frac{\partial \hat{\phi}}{\partial \eta} \tag{11.40}$$

Recalling Equation (11.36*c*), Equations (11.39) and (11.40) become

$$\frac{\partial \hat{\phi}}{\partial x} = \frac{1}{\beta}\frac{\partial \bar{\phi}}{\partial \xi} \tag{11.41}$$

and
$$\frac{\partial \hat{\phi}}{\partial y} = \frac{\partial \bar{\phi}}{\partial \eta} \tag{11.42}$$

Differentiating Equation (11.41) with respect to x (again using the chain rule), we obtain

$$\frac{\partial^2 \hat{\phi}}{\partial x^2} = \frac{1}{\beta}\frac{\partial^2 \bar{\phi}}{\partial \xi^2} \tag{11.43}$$

Differentiating Equation (11.42) with respect to y, we find that the result is

$$\frac{\partial^2 \hat{\phi}}{\partial y^2} = \beta\frac{\partial^2 \bar{\phi}}{\partial \eta^2} \tag{11.44}$$

Substitute Equations (11.43) and (11.44) into (11.35):

$$\beta^2\frac{1}{\beta}\frac{\partial^2 \bar{\phi}}{\partial \xi^2} + \beta\frac{\partial^2 \bar{\phi}}{\partial \eta^2} = 0$$

or
$$\frac{\partial^2 \bar{\phi}}{\partial \xi^2} + \frac{\partial^2 \bar{\phi}}{\partial \eta^2} = 0 \tag{11.45}$$

Examine Equation (11.45)—it should look familiar. Indeed, Equation (11.45) is Laplace's equation. Recall from Chapter 3 that Laplace's equation is the governing relation for *incompressible flow*. Hence, starting with a subsonic compressible flow in physical (x, y) space where the flow is represented by $\hat{\phi}(x, y)$ obtained from Equation (11.35), we have related this flow to an incompressible

flow in transformed (ξ, η) space, where the flow is represented by $\bar{\phi}(\xi, \eta)$ obtained from Equation (11.45). The relation between $\bar{\phi}$ and $\hat{\phi}$ is given by Equation (11.36*c*).

Consider again the shape of the airfoil given in physical space by $y = f(x)$. The shape of the airfoil in the transformed space is expressed as $\eta = q(\xi)$. Let us compare the two shapes. First, apply the approximate boundary condition, Equation (11.34), in physical space, noting that $df/dx = \tan\theta$. We obtain

$$V_\infty \frac{df}{dx} = \frac{\partial \hat{\phi}}{\partial y} = \frac{1}{\beta}\frac{\partial \bar{\phi}}{\partial y} = \frac{\partial \bar{\phi}}{\partial \eta} \tag{11.46}$$

Similarly, apply the flow-tangency condition in transformed space, which from Equation (11.34) is

$$V_\infty \frac{dq}{d\xi} = \frac{\partial \bar{\phi}}{\partial \eta} \tag{11.47}$$

Examine Equations (11.46) and (11.47) closely. Note that the right-hand sides of these two equations are identical. Thus, from the left-hand sides, we obtain

$$\frac{df}{dx} = \frac{dq}{d\xi} \tag{11.48}$$

Equation (11.48) implies that the shape of the airfoil in the transformed space is the same as in the physical space. Hence, the above transformation relates the compressible flow over an airfoil in (x, y) space to the incompressible flow in (ξ, η) space over the *same* airfoil.

The above theory leads to an immensely practical result, as follows. Recall Equation (11.32) for the linearized pressure coefficient. Inserting the above transformation into Equation (11.32), we obtain

$$C_p = \frac{-2\hat{u}}{V_\infty} = -\frac{2}{V_\infty}\frac{\partial \hat{\phi}}{\partial x} = -\frac{2}{V_\infty}\frac{1}{\beta}\frac{\partial \bar{\phi}}{\partial x} = -\frac{2}{V_\infty}\frac{1}{\beta}\frac{\partial \bar{\phi}}{\partial \xi} \tag{11.49}$$

Question: What is the significance of $\partial\bar{\phi}/\partial\xi$ in Equation (11.49)? Recall that $\bar{\phi}$ is the perturbation velocity potential for an incompressible flow in transformed space. Hence, from the definition of velocity potential, $\partial\bar{\phi}/\partial\xi = \bar{u}$, where $\bar{u}$ is a perturbation velocity for the incompressible flow. Hence, Equation (11.49) can be written as

$$C_p = \frac{1}{\beta}\left(-\frac{2\bar{u}}{V_\infty}\right) \tag{11.50}$$

From Equation (11.32), the expression $(-2\bar{u}/V_\infty)$ is simply the linearized pressure coefficient for the incompressible flow. Denote this incompressible pressure coefficient by $C_{p,0}$. Hence, Equation (11.50) gives

$$C_p = \frac{C_{p,0}}{\beta}$$

or recalling that $\beta \equiv \sqrt{1 - M_\infty^2}$, we have

$$\boxed{C_p = \frac{C_{p,0}}{\sqrt{1 - M_\infty^2}}} \tag{11.51}$$

Equation (11.51) is called the *Prandtl-Glauert rule;* it states that, if we know the incompressible pressure distribution over an airfoil, then the compressible pressure distribution over the same airfoil can be obtained from Equation (11.51). Therefore, Equation (11.51) is truly a *compressibility correction* to incompressible data.

Consider the lift and moment coefficients for the airfoil. For an inviscid flow, the aerodynamic lift and moment on a body are simply integrals of the pressure distribution over the body, as described in Section 1.5. (If this is somewhat foggy in your mind, review Section 1.5 before progressing further.) In turn, the lift and moment *coefficients* are obtained from the integral of the pressure coefficient via Equations (1.15) to (1.19). Since Equation (11.51) relates the compressible and incompressible pressure coefficients, the same relation must therefore hold for lift and moment coefficients:

$$\boxed{c_l = \frac{c_{l,0}}{\sqrt{1 - M_\infty^2}}} \tag{11.52}$$

$$\boxed{c_m = \frac{c_{m,0}}{\sqrt{1 - M_\infty^2}}} \tag{11.53}$$

The Prandtl-Glauert rule, embodied in Equations (11.51) to (11.53), was historically the first compressibility correction to be obtained. As early as 1922, Prandtl was using this result in his lectures at Göttingen, although without written proof. The derivation of Equations (11.51) to (11.53) was first formally published by the British aerodynamicist, Hermann Glauert, in 1928. Hence, the rule is named after both men. The Prandtl-Glauert rule was used exclusively until 1939, when an improved compressibility correction was developed. Because of their simplicity, Equations (11.51) to (11.53) are still used today for initial estimates of compressibility effects.

Recall that the results of Chapters 3 and 4 proved that inviscid, incompressible flow over a closed, two-dimensional body theoretically produces zero drag—the well-known d'Alembert's paradox. Does the same paradox hold for inviscid, subsonic, compressible flow? The answer can be obtained by again noting that the only source of drag is the integral of the pressure distribution. If this integral is zero for an incompressible flow, and since the compressible pressure coefficient differs from the incompressible pressure coefficient by only a constant scale factor, β, then the integral must also be zero for a compressible flow. Hence, d'Alembert's paradox also prevails for inviscid, subsonic, compressible flow. However, as soon as the freestream Mach number is high enough to produce locally supersonic flow

on the body surface with attendant shock waves, as shown in Figure 1.43*b*, then a positive wave drag is produced, and d'Alembert's paradox no longer prevails.

EXAMPLE 11.1

At a given point on the surface of an airfoil, the pressure coefficient is −0.3 at very low speeds. If the freestream Mach number is 0.6, calculate C_p at this point.

■ Solution

From Equation (11.51),

$$C_p = \frac{C_{p,0}}{\sqrt{1 - M^2}} = \frac{-0.3}{\sqrt{1 - (0.6)^2}} = \boxed{-0.375}$$

EXAMPLE 11.2

From Chapter 4, the theoretical lift coefficient for a thin, symmetric airfoil in an incompressible flow is $c_l = 2\pi\alpha$. Calculate the lift coefficient for $M_\infty = 0.7$.

■ Solution

From Equation (11.52),

$$c_l = \frac{c_{l,0}}{\sqrt{1 - M_\infty^2}} = \frac{2\pi\alpha}{\sqrt{1 - (0.7)^2}} = \boxed{8.8\alpha}$$

Note: The effect of compressibility at Mach 0.7 is to increase the lift slope by the ratio $8.8/2\pi = 1.4$, or by 40 percent.

11.5 IMPROVED COMPRESSIBILITY CORRECTIONS

The importance of accurate compressibility corrections reached new highs during the rapid increase in airplane speeds spurred by World War II. Efforts were made to improve upon the Prandtl-Glauert rule discussed in Section 11.4. Several of the more popular formulas are given below.

The Karman-Tsien rule states

$$C_p = \frac{C_{p,0}}{\sqrt{1 - M_\infty^2} + \left[M_\infty^2/\left(1 + \sqrt{1 - M_\infty^2}\right)\right] C_{p,0}/2} \tag{11.54}$$

This formula, derived in References 27 and 28, has been widely adopted by the aeronautical industry since World War II.

Laitone's rule states

$$C_p = \frac{C_{p,0}}{\sqrt{1 - M_\infty^2} + \left(M_\infty^2\left\{1 + [(\gamma - 1)/2]M_\infty^2\right\}/2\sqrt{1 - M_\infty^2}\right) C_{p,0}} \tag{11.55}$$

This formula is more recent than either the Prandtl-Glauert or the Karman-Tsien rule; it is derived in Reference 29.

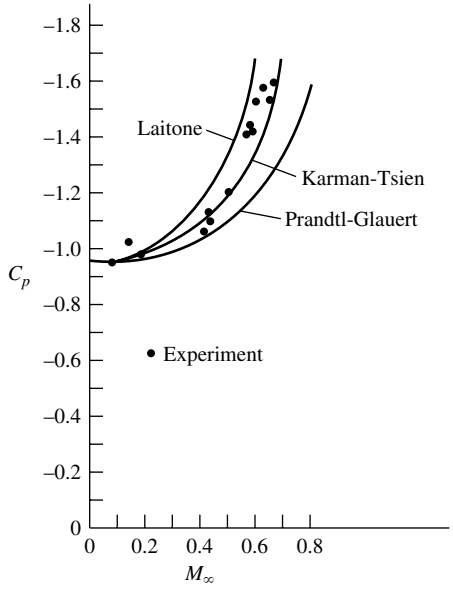


Figure 11.4 Several compressibility corrections compared with experimental results for an NACA 4412 airfoil at an angle of attack $\alpha = 1^\circ 53'$. The experimental data are chosen for their historical significance; they are from NACA report no. 646, published in 1938 *(Reference 30)*. This was the first major NACA publication to address the compressibility problem in a systematic fashion; it covered work performed in the 2-ft high-speed tunnel at the Langley Aeronautical Laboratory and was carried out during 1935–1936.

These compressibility corrections are compared in Figure 11.4, which also shows experimental data for the C_p variation with M_∞ at the 0.3-chord location on an NACA 4412 airfoil. Note that the Prandtl-Glauert rule, although the simplest to apply, underpredicts the experimental data, whereas the improved compressibility corrections are clearly more accurate. Recall that the Prandtl-Glauert rule is based on linear theory. In contrast, both the Laitone and Karman-Tsien rules attempt to account for some of the nonlinear aspects of the flow.

11.6 CRITICAL MACH NUMBER

Return to the road map given in Figure 11.1. We have now finished our discussion of linearized flow and the associated compressibility corrections. Keep in mind that such linearized theory does *not* apply to the transonic flow regime,

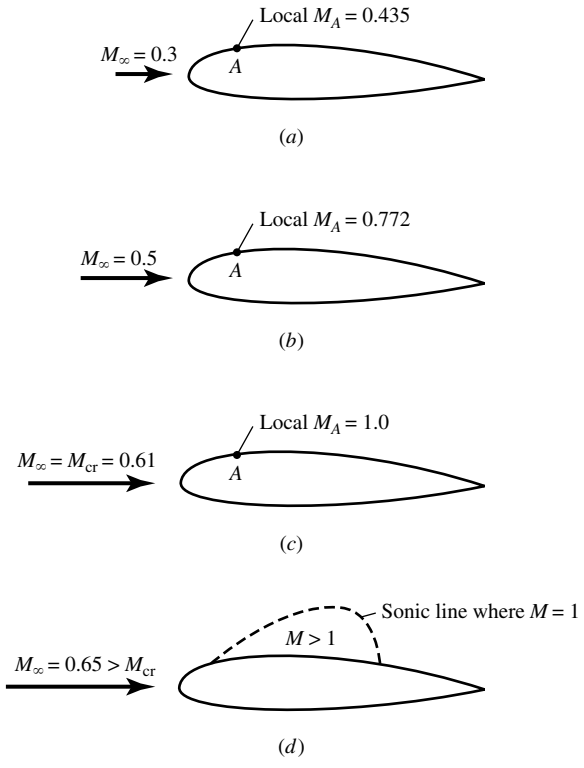


Figure 11.5 Definition of critical Mach number. Point A is the location of minimum pressure on the top surface of the airfoil. (See end-of-chapter Problem 11.7 for the calculation of the numbers in this figure.)

$0.8 \leq M_\infty \leq 1.2$. Transonic flow is highly nonlinear, and theoretical transonic aerodynamics is a challenging and sophisticated subject. For the remainder of this chapter, we deal with several aspects of transonic flow from a qualitative point of view. The theory of transonic aerodynamics is beyond the scope of this book.

Consider an airfoil in a low-speed flow, say, with $M_\infty = 0.3$, as sketched in Figure 11.5*a*. In the expansion over the top surface of the airfoil, the local flow Mach number M increases. Let point A represent the location on the airfoil surface where the pressure is a minimum, hence where M is a maximum. In Figure 11.5*a*, let us say this maximum is $M_A = 0.435$. Now assume that we gradually increase the freestream Mach number. As M_∞ increases, M_A also increases. For example, if M_∞ is increased to $M = 0.5$, the maximum local value of M will be 0.772, as shown in Figure 11.5*b*. Let us continue to increase M_∞ until we achieve just the right value such that the local Mach number at the minimum pressure point equals 1, that is, such that $M_A = 1.0$, as shown in Figure 11.5*c*. When this happens, the freestream Mach number M_∞ is called the *critical Mach number,*

denoted by M_{cr}. By definition, the critical Mach number is that *freestream* Mach number at which sonic flow is first achieved on the airfoil surface. In Figure 11.5*c*, $M_{cr} = 0.61$.

One of the most important problems in high-speed aerodynamics is the determination of the critical Mach number of a given airfoil, because at values of M_∞ slightly above M_{cr}, the airfoil experiences a dramatic increase in drag coefficient (discussed in Section 11.7). The purpose of the present section is to give a rather straightforward method for estimating M_{cr}.

Let p_∞ and p_A represent the static pressures in the freestream and at point A, respectively, in Figure 11.5. For isentropic flow, where the total pressure p_0 is constant, these static pressures are related through Equation (8.42) as follows:

$$\frac{p_A}{p_\infty} = \frac{p_A/p_0}{p_\infty/p_0} = \left(\frac{1 + [(\gamma - 1)/2]M_\infty^2}{1 + [(\gamma - 1)/2]M_A^2} \right)^{\gamma/(\gamma-1)} \tag{11.56}$$

The pressure coefficient at point A is given by Equation (11.22) as

$$C_{p,A} = \frac{2}{\gamma M_\infty^2} \left(\frac{p_A}{p_\infty} - 1 \right) \tag{11.57}$$

Combining Equations (11.56) and (11.57), we have

$$C_{p,A} = \frac{2}{\gamma M_\infty^2} \left[\left(\frac{1 + [(\gamma - 1)/2]M_\infty^2}{1 + [(\gamma - 1)/2]M_A^2} \right)^{\gamma/(\gamma-1)} - 1 \right] \tag{11.58}$$

Equation (11.58) is useful in its own right; for a given freestream Mach number, it relates the local value of C_p to the local Mach number. [Note that Equation (11.58) is the compressible flow analogue of Bernoulli's equation, Equation (3.13), which for incompressible flow with a given freestream velocity and pressure relates the local pressure at a point in the flow to the local velocity at that point.] However, for our purposes here, we ask the question, What is the value of the local C_p when the local Mach number is unity? By definition, this value of the pressure coefficient is called the *critical pressure coefficient,* denoted by $C_{p,cr}$. For a given freestream Mach number M_∞, the value of $C_{p,cr}$ can be obtained by inserting $M_A = 1$ into Equation (11.58):

$$C_{p,cr} = \frac{2}{\gamma M_\infty^2} \left[\left(\frac{1 + [(\gamma - 1)/2]M_\infty^2}{1 + (\gamma - 1)/2} \right)^{\gamma/(\gamma-1)} - 1 \right] \tag{11.59}$$

Equation (11.59) allows us to calculate the pressure coefficient at any point in the flow where the local Mach number is 1, for a given freestream Mach number M_∞. For example, if M_∞ is slightly greater than M_{cr}, say, $M_\infty = 0.65$ as shown in Figure 11.5*d*, then a finite region of supersonic flow will exist above the airfoil; Equation (11.59) allows us to calculate the pressure coefficient at only those points where $M = 1$, that is, at only those points that fall on the sonic line in Figure 11.5*d*. Now, returning to Figure 11.5*c*, when the freestream Mach number is precisely equal to the critical Mach number, there is only one point where $M = 1$, namely,

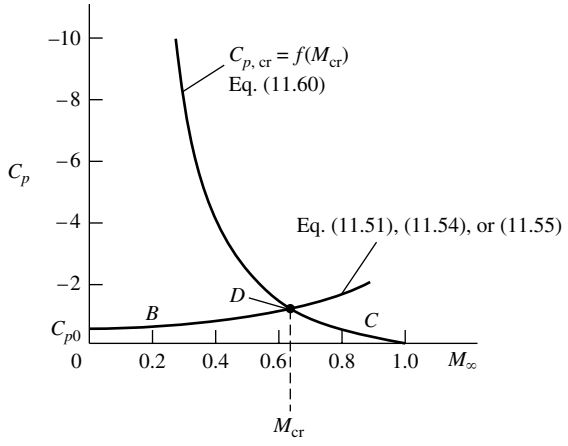


Figure 11.6 Estimation of critical Mach number.

point A. The pressure coefficient at point A will be $C_{p,\text{cr}}$, which is obtained from Equation (11.59). In this case, M_∞ in Equation (11.59) is precisely M_{cr}. Hence,

$$C_{p,\text{cr}} = \frac{2}{\gamma M_{\text{cr}}^2}\left[\left(\frac{1+[(\gamma-1)/2]M_{\text{cr}}^2}{1+(\gamma-1)/2}\right)^{\gamma/(\gamma-1)} - 1\right] \tag{11.60}$$

Equation (11.60) shows that $C_{p,\text{cr}}$ is a unique function of M_{cr}; this variation is plotted as curve C in Figure 11.6. Note that Equation (11.60) is simply an aerodynamic relation for isentropic flow—it has no connection with the shape of a given airfoil. In this sense, Equation (11.60), and hence curve C in Figure 11.6, is a type of "universal relation" which can be used for all airfoils.

Equation (11.60), in conjunction with any one of the compressibility corrections given by Equations (11.51), (11.54), or (11.55), allows us to estimate the critical Mach number for a given airfoil as follows:

1. By some means, either experimental or theoretical, obtain the low-speed incompressible value of the pressure coefficient $C_{p,0}$ at the minimum pressure point on the given airfoil.
2. Using any of the compressibility corrections, Equation (11.51), (11.54), or (11.55), plot the variation of C_p with M_∞. This is represented by curve B in Figure 11.6.
3. Somewhere on curve B, there will be a single point where the pressure coefficient corresponds to locally sonic flow. Indeed, this point must coincide with Equation (11.60), represented by curve C in Figure 11.6. Hence, the *intersection* of curves B and C represents the point corresponding to sonic flow at the minimum pressure location on the airfoil. In turn, the value of M_∞ at this intersection is, by definition, the critical Mach number, as shown in Figure 11.6.

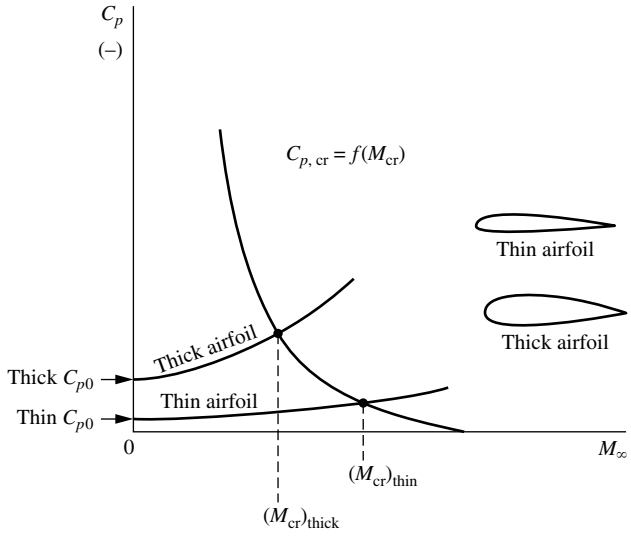


Figure 11.7 Effect of airfoil thickness on critical Mach number.

The graphical construction in Figure 11.6 is not an exact determination of M_{cr}. Although curve C is exact, curve B is approximate because it represents the approximate compressibility correction. Hence, Figure 11.6 gives only an estimation of M_{cr}. However, such an estimation is quite useful for preliminary design, and the results from Figure 11.6 are accurate enough for most applications.

Consider two airfoils, one thin and the other thick, as sketched in Figure 11.7. First consider the low-speed incompressible flow over these airfoils. The flow over the thin airfoil is only slightly perturbed from the freestream. Hence, the expansion over the top surface is mild, and $C_{p,0}$ at the minimum pressure point is a negative number of only small absolute magnitude, as shown in Figure 11.7. [Recall from Equation (11.32) that $C_p \propto \hat{u}$; hence, the smaller the perturbation, the smaller is the absolute magnitude of C_p.] In contrast, the flow over the thick airfoil experiences a large perturbation from the freestream. The expansion over the top surface is strong, and $C_{p,0}$ at the minimum pressure point is a negative number of large magnitude, as shown in Figure 11.7. If we now perform for each airfoil the same construction as given in Figure 11.6, we see that the thick airfoil will have a lower critical Mach number than the thin airfoil. This is clearly illustrated in Figure 11.7. For high-speed airplanes, it is desirable to have M_{cr} as high as possible. Hence, modern high-speed subsonic airplanes are usually designed with relatively thin airfoils. (The development of the supercritical airfoil has somewhat loosened this criterion, as discussed in Section 11.8.) For example, the Gates Lear jet high-speed jet executive transport utilizes a 9 percent thick airfoil; contrast this with the low-speed Piper Aztec, a twin-engine propeller-driven general aviation aircraft designed with a 14 percent thick airfoil.

EXAMPLE 11.3

In this example, we illustrate the estimation of the critical Mach number for an airfoil using (*a*) the graphical solution discussed in this section, and (*b*) an analytical solution using a closed-form equation obtained from a combination of Equations (11.51) and (11.60). Consider the NACA 0012 airfoil at zero angle of attack shown at the top of Figure 11.8. The pressure coefficient distribution over this airfoil, measured in a wind tunnel at low speed, is given at the bottom of Figure 11.8. From this information, estimate the critical Mach number of the NACA 0012 airfoil at zero angle of attack.

■ Solution

(*a*) *Graphical Solution.* First, let us accurately plot the curve of $C_{p,\text{cr}}$ versus M_{cr} from Equation (11.60),

$$C_{p,\text{cr}} = \frac{2}{\gamma M_{\text{cr}}^2}\left[\left(\frac{1+[(\gamma-1)/2]M_{\text{cr}}^2}{1+(\gamma-1)/2}\right)^{\gamma/(\gamma-1)} - 1\right] \tag{11.60}$$

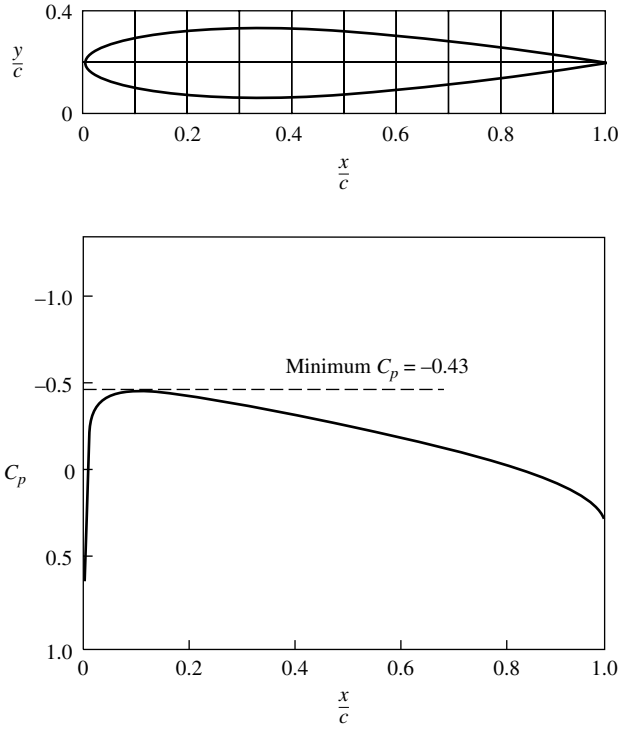


Figure 11.8 Low-speed pressure coefficient distribution over the surface of an NACA 0012 airfoil at zero angle of attack. $\text{Re} = 3.65 \times 10^6$. (*Source:* R. J. Freuler and G. M. Gregorek, "An Evaluation of Four Single Element Airfoil Analytical Methods," in *Advanced Technology Airfoil Research,* NASA CP 2045, 1978, pp. 133–162.)

For $\gamma = 1.4$, from Equation (11.60) we can tabulate

M_∞	0.4	0.5	0.6	0.7	0.8	0.9	1.0
$C_{p,\text{cr}}$	−3.66	−2.13	−1.29	−0.779	−0.435	−0.188	0

These numbers are plotted as curve C in Figure 11.9. Note that $C_{p,\text{cr}} = 0$ when $M_{\text{cr}} = 1.0$. This makes physical sense; if the free stream Mach number is already 1, then no change in the pressure is required to achieve Mach 1 at a local point in the flow, and hence the pressure difference $(p_{\text{cr}} - p_\infty)$ is zero and $C_{p,\text{cr}} = 0$.

Following the three-step procedure mentioned earlier, in step one we obtain the low-speed incompressible value of the minimum pressure coefficient $(C_{p,0})_{\text{min}}$ from the pressure coefficient distribution given in Figure 11.8. The *minimum* value of C_p on the surface is −0.43. Following step two, in Equation (11.51), $(C_{p,0})_{\text{min}} = -0.43$, and we have from Equation (11.51),

$$(C_p)_{\text{min}} = \frac{(C_{p,0})_{\text{min}}}{\sqrt{1 - M_\infty^2}} = \frac{-0.43}{\sqrt{1 - M_\infty^2}} \tag{11.61}$$

Some values of $(C_p)_{\text{min}}$ are tabulated below

M_∞	0	0.2	0.4	0.6	0.8
$(C_p)_{\text{min}}$	−0.43	−0.439	−0.469	−0.538	−0.717

Following step three, these values are plotted as Curve B in Figure 11.9. The intersection of curves B and C is at point D. The freestream Mach number associated with point D is the critical Mach number for the NACA 0012 airfoil. From Figure 11.9, we have

$$\boxed{M_{\text{cr}} = 0.74}$$

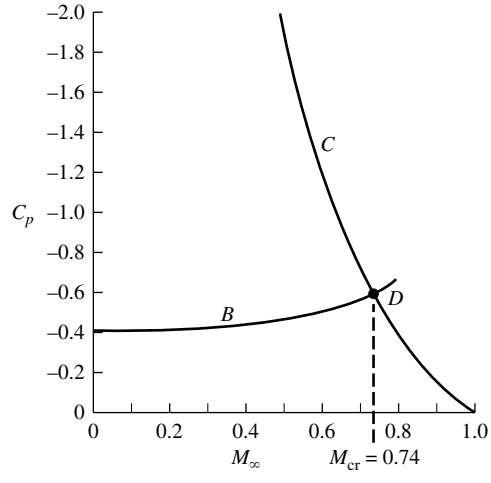


Figure 11.9 Graphical solution for the critical Mach number.

(*b*) *Analytical Solution.* In Figure 11.9, curve B is given by Equation (11.61)

$$(C_p)_{\min} = \frac{-0.43}{\sqrt{1 - M_\infty^2}} \tag{11.61}$$

At the intersection point D, $(C_p)_{\min}$ in Equation (11.61) is the critical pressure coefficient and M_∞ is the critical Mach number

$$C_{p,\text{cr}} = \frac{-0.43}{\sqrt{1 - M_{\text{cr}}^2}} \quad \text{(at point } D\text{)} \tag{11.62}$$

Also, at point D the value of $C_{p,\text{cr}}$ is given by Equation (11.60). Hence, *at point D* we can equate the right-hand sides of Equations (11.62) and (11.60),

$$\frac{-0.43}{\sqrt{1 - M_{\text{cr}}^2}} = \frac{2}{\gamma M_{\text{cr}}^2}\left[\left(\frac{1 + [(\gamma - 1)/2]M_{\text{cr}}^2}{1 + (\gamma - 1)/2}\right)^{\gamma/\gamma - 1} - 1\right] \tag{11.63}$$

Equation (11.63) is one equation with one unknown, namely, M_{cr}. The solution of Equation (11.63) gives the value of M_{cr} associated with the intersection point D in Figure 11.9, that is, the solution of Equation (11.63) is the critical Mach number for the NACA 0012 airfoil. Since M_{cr} appears in a complicated fashion on both sides of Equation (11.63), we solve the equation by trial-and-error by assuming different values of M_{cr}, calculating the values of both sides of Equation (11.63), and iterating until we find a value of M_{cr} that results in both the right and left sides being the same value.

M_{cr}	$\dfrac{-0.43}{\sqrt{1 - M_{\text{cr}}^2}}$	$\dfrac{2}{\gamma M_{\text{cr}}^2}\left[\left(\dfrac{1 + [(\gamma - 1)/2]M_{\text{cr}}^2}{1 + (\gamma - 1)/2}\right)^{\gamma/\gamma - 1} - 1\right]$
0.72	−0.6196	−0.6996
0.73	−0.6292	−0.6621
0.74	−0.6393	−0.6260
0.738	−0.6372	−0.6331
0.737	−0.6362	−0.6367
0.7371	−0.6363	−0.6363

To four-place accuracy, when $M_{\text{cr}} = 0.7371$, both the left and right sides of Equation (11.63) have the same value. Therefore, the analytical solution yields

$$\boxed{M_{\text{cr}} = 0.7371}$$

Note: Within the two-place accuracy of the graphical solution in part (a), both the graphical and analytical solutions give the same value of M_{cr}.

Question: How accurate is the estimate of the critical Mach number in this example? To answer this question, we examine some experimental pressure coefficient distributions for the NACA 0012 airfoil obtained at higher freestream Mach numbers. Wind tunnel measurements of the surface pressure distributions for this airfoil at zero angle of attack in a high-speed flow are shown in Figure 11.10; for Figure 11.10*a*, $M_\infty = 0.575$, and for Figure 11.10*b*, $M_\infty = 0.725$. In Figure 11.10*a*, the value of $C_{p,\text{cr}} = -1.465$ at $M_\infty = 0.575$ is shown as the dashed horizontal line. From the definition of critical pressure coefficient, any local value of C_p above this horizontal line corresponds to locally

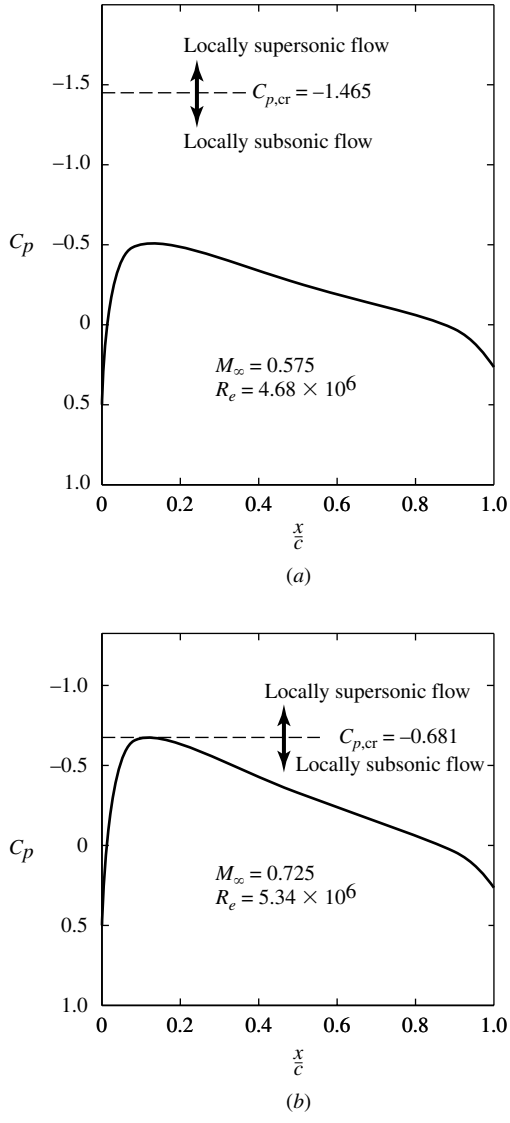


Figure 11.10 Wind tunnel measurements of surface pressure coefficient distribution for the NACA 0012 airfoil at zero angle of attack. Experimental data of Frueler and Gregorek, NASA CP 2045. (*a*) $M_\infty = 0.575$, (*b*) $M_\infty = 0.725$.

supersonic flow, and any local value below the horizontal line corresponds to locally subsonic flow. Clearly from the measured surface pressure coefficient distribution at $M_\infty = 0.575$ shown in Figure 11.10*a*, the flow is locally subsonic at every point on the surface. Hence, $M_\infty = 0.575$ is *below* the critical Mach number. In Figure 11.10*b*, which is for a higher Mach number, the value of $C_{p,\text{cr}} = -0.681$ at $M_\infty = 0.725$ is

shown as the dashed horizontal line. Here, the local pressure coefficient on the airfoil is higher than $C_{p,\text{cr}}$ at every point on the surface *except* at the point of minimum pressure, where $(C_p)_{\text{min}}$ is essentially equal to $C_{p,\text{cr}}$. This means that for $M_\infty = 0.725$, the flow is locally subsonic at every point on the surface *except* the point of minimum pressure, where the flow is essentially sonic. Hence, these experimental measurements indicate that the critical Mach number of the NACA 0012 airfoil at zero angle of attack is approximately 0.73. Comparing this experimental result with the calculated value of $M_{\text{cr}} = 0.74$ in this example, we see that our calculations are amazingly accurate, to within about one percent.

11.6.1 A Comment on the Location of Minimum Pressure (Maximum Velocity)

Examining the shape of the NACA 0012 airfoil shown at the top of Figure 11.8, note that the maximum thickness occurs at $x/c = 0.3$. However, examining the pressure coefficient distribution shown at the bottom of Figure 11.8, note that the point of minimum pressure occurs on the surface at $x/c = 0.11$, considerably ahead of the point of maximum thickness. This is a graphic illustration of the general fact that the point of minimum pressure (hence maximum velocity) does *not* correspond to the location of maximum thickness of the airfoil. Intuition might at first suggest that, at least for a symmetric airfoil at zero degrees angle of attack, the location of minimum pressure (maximum velocity) on the surface might be at the maximum thickness of the airfoil, but our intuition would be completely wrong. Nature places the maximum velocity at a point which satisfies the physics of the *whole flow field,* not just what is happening in a local region of the flow. The point of maximum velocity is dictated by the *complete* shape of the airfoil, not just by the shape in a local region.

We also note that it is implicit in the approximate compressibility corrections discussed in Sections 11.4 and 11.5, and their use for the estimation of the critical Mach number as discussed in Section 11.6, that the point of minimum pressure remains at a fixed location on the body surface as M_∞ is increased from a very low to a high subsonic value. This is indeed approximately the case. Examine the experimental pressure distributions in Figures 11.8 and 11.10, which are for three different Mach numbers ranging from a low, incompressible value (Figure 11.8) to $M_\infty = 0.725$ (Figure 11.10*b*). Note that in each case the minimum pressure point is at the same approximate location, that is, at $x/c = 0.11$.

11.7 DRAG-DIVERGENCE MACH NUMBER: THE SOUND BARRIER

Imagine that we have a given airfoil at a fixed angle of attack in a wind tunnel, and we wish to measure its drag coefficient c_d as a function of M_∞. To begin with, we measure the drag coefficient at low subsonic speed to be $c_{d,0}$, shown in Figure 11.11. Now, as we gradually increase the freestream Mach number, we observe that c_d remains relatively constant all the way to the critical Mach number, as illustrated in Figure 11.11. The flow fields associated with points a, b, and c in Figure 11.11 are represented by Figure 11.5*a*, *b*, and *c*, respectively. As we very

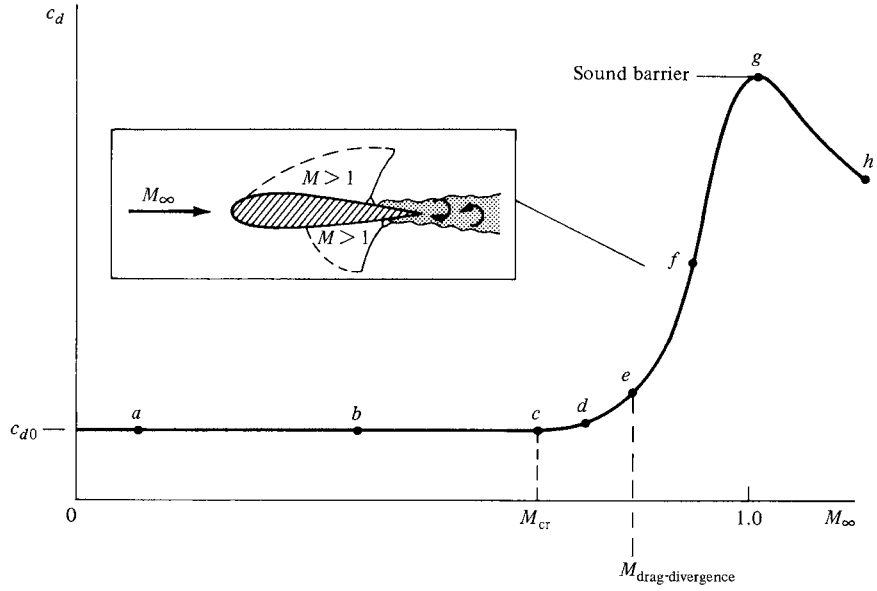


Figure 11.11 Sketch of the variation of profile drag coefficient with freestream Mach number, illustrating the critical and drag-divergence Mach numbers and showing the large drag rise near Mach 1.

carefully increase M_∞ slightly above M_{cr}, say, to point d in Figure 11.11, a finite region of supersonic flow appears on the airfoil, as shown in Figure 11.5d. The Mach number in this bubble of supersonic flow is only slightly above Mach 1, typically 1.02 to 1.05. However, as we continue to nudge M_∞ higher, we encounter a point where the drag coefficient suddenly starts to increase. This is given as point e in Figure 11.11. The value of M_∞ at which this sudden increase in drag starts is defined as the *drag-divergence Mach number*. Beyond the drag-divergence Mach number, the drag coefficient can become very large, typically increasing by a factor of 10 or more. This large increase in drag is associated with an extensive region of supersonic flow over the airfoil, terminating in a shock wave, as sketched in the insert in Figure 11.11. Corresponding to point f on the drag curve, this insert shows that as M_∞ approaches unity, the flow on both the top and bottom surfaces can be supersonic, both terminated by shock waves. For example, consider the case of a reasonably thick airfoil, designed originally for low-speed applications, when M_∞ is beyond drag-divergence; in such a case, the local Mach number can reach 1.2 or higher. As a result, the terminating shock waves can be relatively strong. These shocks generally cause severe flow separation downstream of the shocks, with an attendant large increase in drag.

Now, put yourself in the place of an aeronautical engineer in 1936. You are familiar with the Prandtl-Glauert rule, given by Equation (11.51). You recognize that as $M_\infty \rightarrow 1$, this equation shows the absolute magnitude of C_p approaching infinity. This hints at some real problems near Mach 1. Furthermore, you know



Figure 11.12 The Bell XS-1—the first manned airplane to fly faster than sound, October 14, 1947. (*Courtesy of the National Air and Space Museum.*)

of some initial high-speed subsonic wind-tunnel tests that have generated drag curves that resemble the portion of Figure 11.11 from points a to f. How far will the drag coefficient increase as we get closer to $M_\infty = 1$? Will c_d go to infinity? At this stage, you might be pessimistic. You might visualize the drag increase to be so large that no airplane with the power plants existing in 1936, or even envisaged for the future, could ever overcome this "barrier." It was this type of thought that led to the popular concept of a sound barrier and that prompted many people to claim that humans would never fly faster than the speed of sound.

Of course, today we know the sound barrier was a myth. We cannot use the Prandtl-Glauert rule to argue that c_d will become infinite at $M_\infty = 1$, because the Prandtl-Glauert rule is invalid at $M_\infty = 1$ (see Sections 11.3 and 11.4). Moreover, early transonic wind-tunnel tests carried out in the late 1940s clearly indicated that c_d peaks at or around Mach 1 and then actually decreases as we enter the supersonic regime, as shown by points g and h in Figure 11.11. All we need is an aircraft with an engine powerful enough to overcome the large drag rise at Mach 1. The myth of the sound barrier was finally put to rest on October 14, 1947, when Captain Charles (Chuck) Yeager became the first human being to fly faster than sound in the sleek, bullet-shaped Bell XS-1. This rocket-propelled research aircraft is shown in Figure 11.12. Of course, today supersonic flight is

a common reality; we have jet engines powerful enough to accelerate military fighters through Mach 1 flying straight up! Such airplanes can fly at Mach 3 and beyond. Indeed, we are limited only by aerodynamic heating at high speeds (and the consequent structural problems). Right now, NASA is conducting research on supersonic combustion ramjet engines for flight at Mach 5 and higher (see the Design Box at the end of Section 9.6). Keep in mind, however, that because of the large power requirements for very high-speed flight, the fuel consumption becomes large. In today's energy-conscious world, this constraint can be as much a barrier to high-speed flight as the sound barrier was once envisaged.

Since 1945, research in transonic aerodynamics has focused on reducing the large drag rise shown in Figure 11.11. Instead of living with a factor of 10 increase in drag at Mach 1, can we reduce it to a factor of 2 or 3? This is the subject of the remaining sections of this chapter.

DESIGN BOX

In order to cope with the large drag rise near Mach 1, as seen in Figure 11.11, the designers of high-speed airplanes after World War II utilized two aerodynamic design features to increase the critical Mach number, and hence the drag-divergence Mach number, for such aircraft. These two features are now classic and are discussed here.

The first design ploy was the use of *thin airfoil sections* on the airplane wing. We have already discussed that, everything else being equal, the thinner is the airfoil the higher is the critical Mach number; this is shown in Figure 11.7. This phenomena was observed as early as 1918 by two research engineers, F. W. Caldwell and E. N. Fales, at the Army's McCook Field in Dayton, Ohio, and was solidly confirmed by various experiments during the 1920s and 1930s. More historical details are given in Section 11.11, and the detailed story of the early research in the twentieth century on compressibility effects is told in Reference 62. Indeed, the Bell X-1 (Figure 11.12) was designed with the full knowledge of the importance of thin airfoils. As a result, the X-1 was designed with two sets of wings, one with a 10 percent thick airfoil for more routine flights and another with an 8 percent thick airfoil for flights intended to penetrate through Mach 1. The airfoil sections for the two wings were NACA 65-110 and NACA 65-108, respectively. These wings were much thinner than conventional airplanes at that time, which had wing thickness typically of 15 percent or higher. Moreover, the horizontal tail was even thinner, utilizing a 6-percent thick NACA 65-006 airfoil section. This was done to ensure that when the wing encountered major compressibility effects, the horizontal tail and elevator would still be free of such problems and would be functional for stability and control. However, hedging their bets, the Bell engineers also made the tail all-moving, that is, the incidence angle of the complete horizontal tail surface could be changed in flight in case elevator effectiveness was lost, and to help trim the airplane as it flew through Mach 1. Chuck Yeager made good use of this all-moving tail feature during his history-making flights in the X-1.

The use of thin airfoils on high-speed aircraft is almost a standard design practice today. The design trend to thin airfoils is illustrated in Figure 11.13, which shows the variation of airfoil thickness for different airplanes as a function of their design Mach number. Clearly, as M_∞ increases, the trend is towards thinner wings.

The second design ploy was the use of swept wings. The story surrounding the history of the swept wing concept is told in Reference 62; the invention of the concept is shared by two people, the German engineer Adolf Busemann in 1935 and the American aerodynamicist R. T. Jones who independently conceived

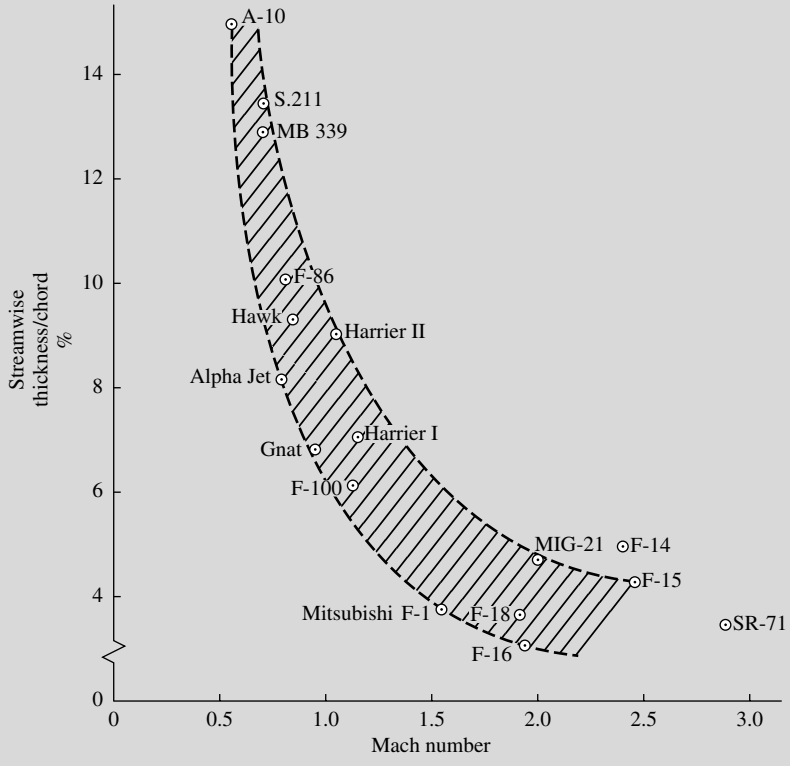


Figure 11.13 Variation of thickness-to-chord ratio with Mach number for a representative sampling of different airplanes. (*Data Source:* Ray Whitford, *Design for Air Combat,* Jane's Information Group, Surry, England, 1989.)

the idea in 1945. There are several ways of explaining how a swept wing works (e.g., see Reference 2). However, for our purposes here we will explain the benefit of a swept wing using the ideas set forth about thin airfoils, as discussed above. Consider a straight wing, a portion of which is sketched in Figure 11.14*a*. We define a straight wing as one for which the mid-chord line is perpendicular to the freestream; this is certainly the case for the rectangular planform shown in Figure 11.14*a*. Assume the straight wing has an airfoil section with a thickness-to-chord ratio of 0.15 as shown at the left of Figure 11.14*a*. Streamline AB flowing over this wing sees the airfoil with $t_1/c_1 = 0.15$. Now consider the same wing swept back through the angle $\Lambda = 45°$, as shown in Figure 11.14*b*. Streamline CD that flows over this wing (ignoring any three-dimensional curvature effects) sees an effective airfoil shape with the same thickness as before ($t_2 = t_1$), but the effective chord length c_2 is longer by a factor of 1.41 (i.e., $c_2 = 1.41c_1$). This makes the effective thickness-to-chord ratio seen by streamline CD equal to $t_2/c_2 = 0.106$—thinner by almost one-third compared to the straight-wing case. Hence, by sweeping the wing, the flow behaves as if the airfoil section is thinner, with a consequent increase in the critical Mach number of the wing. Everything else being equal, a swept wing has a larger critical Mach number, hence a larger drag-divergence Mach number than a straightwing. For this reason, most high-speed airplanes designed since the

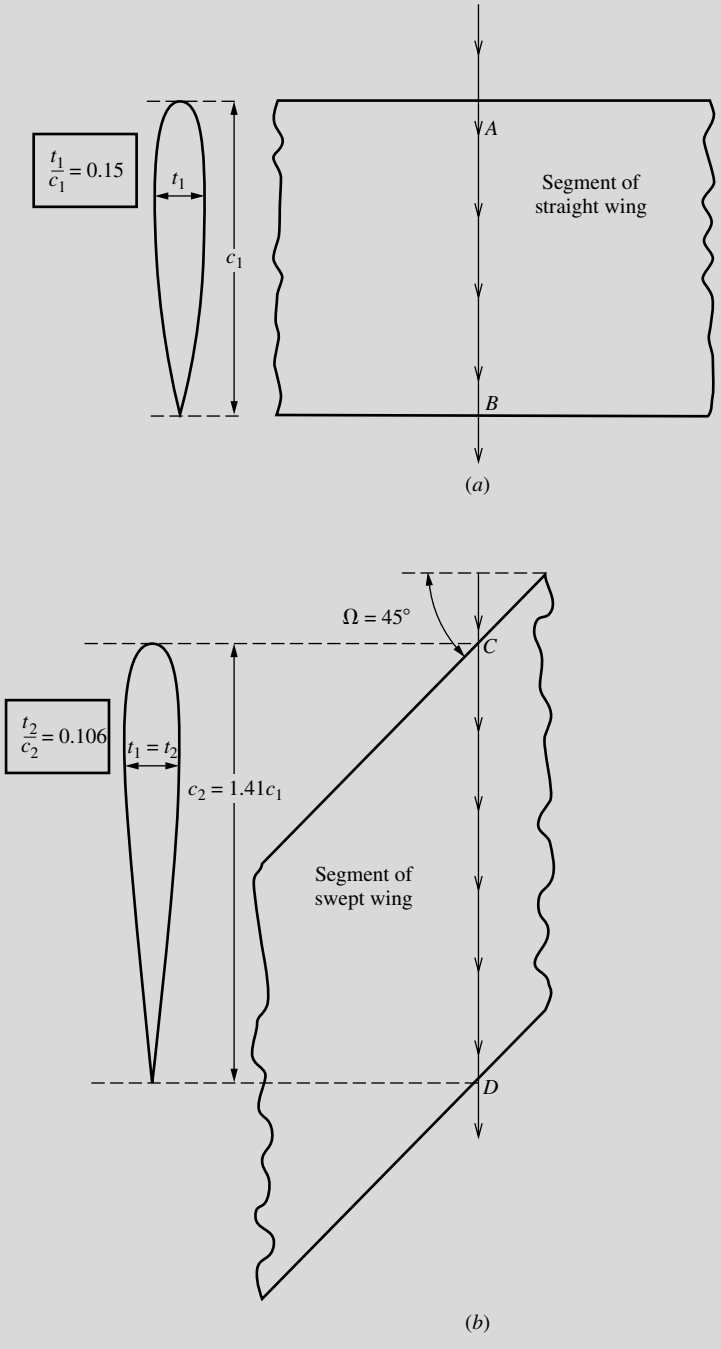


Figure 11.14 By sweeping the wing, a streamline effectively sees a thinner airfoil.

middle 1940s have swept wings. (The only reason why the Bell X-1, shown in Figure 11.12, had straight wings is because its design commenced in 1944 before any knowledge or data about swept wings was available in the United States. Later, when such swept-wing data flooded into the United States from Germany in mid-1945, the Bell designers were conservative, and stuck with the straight wing.) A wonderful example of an early swept-wing fighter is the North American F-86 of Korean War vintage, shown in Figure 11.15.

Finally, we note some compressibility corrections that apply to Equations (5.69), (5.81), and (5.82) given earlier for the estimation of the lift slope for high aspect ratio straight wings, low aspect ratio straight wings, and swept wings respectively. These equations apply to low-speed, incompressible flow. The Prandtl-Glauert rule for an airfoil section expressed in terms of pressure coefficient as given by Equation (11.51) also holds for the lift slope for the airfoil. Letting a_0 be the lift slope for an infinite wing for incompressible flow, and $a_{0,\text{comp}}$ be the lift slope for an infinite wing in a subsonic compressible flow, from the Prandtl-Glauert rule we have

$$a_{0,\text{comp}} = \frac{a_0}{\sqrt{1 - M_\infty^2}} \tag{11.64}$$

We assume that Equation (5.70) relating the lift slope of a finite wing to that for an infinite wing, as obtained from Prandtl's lifting line theory, holds for subsonic compressible flow as well. Letting the term $(1+\tau)^{-1}$ in Equation (5.70) be replaced by a span efficiency factor for lift slope denoted by e_1, where $e_1 = (1+\tau)^{-1}$, and denoting the compressible lift slope for the finite wing by a_{comp}, the compressible analogue of Equation (5.70) is

$$a_{\text{comp}} = \frac{a_{0,\text{comp}}}{1 + a_{0,\text{comp}}/(\pi e_1 \text{AR})} \tag{11.65}$$

Substituting Equation (11.64) into Equation (11.65), and simplifying, we obtain

$$\boxed{a_{\text{comp}} = \frac{a_0}{\sqrt{1 - M_\infty^2} + a_0/(\pi e_1 \text{AR})}} \tag{11.66}$$

Equation (11.66) is an equation for estimating the lift slope for a *high-aspect-ratio straight wing* in a compressible flow with a subsonic M_∞ from the known lift slope for an infinite wing in an incompressible flow a_0.

Helmbold's equation for low-aspect-ratio straight wings for incompressible flow, Equation (5.81), can be modified for compressibility effects by replacing a_0 in that equation by $a_0/\sqrt{1 - M_\infty^2}$, yielding

$$\boxed{a_{\text{comp}} = \frac{a_0}{\sqrt{1 - M_\infty^2 + [a_0/(\pi e_1 \text{AR})]^2} + a_0/(\pi \text{AR})}} \tag{11.67}$$

Equation (11.67) is an equation for estimating the lift slope for a *low-aspect-ratio straight wing* in a compressible flow with a subsonic M_∞ from the known lift slope for an infinite wing in an incompressible flow a_0.

Finally, Equation (5.82) for a swept wing can be modified for compressibility effects by replacing a_0 in that equation by $a_0/\sqrt{1 - (M_{\infty,n})^2}$ where $M_{\infty,n}$ is the component of the freestream Mach number perpendicular to the half-chord line of the swept wing. If the half-chord line is swept by the angle Λ, then $M_{\infty,n} = M_\infty \cos\Lambda$. Hence, replace a_0 in Equation (5.82) with $a_0/\sqrt{1 - M_\infty^2 \cos^2\Lambda}$. The result is in Equation (11.68) below. Equation (11.68) is an equation for estimating the lift slope for a *swept wing* in a compressible flow with a subsonic M_∞ from the known lift slope for an infinite wing in an incompressible flow a_0.

$$\boxed{a_{\text{comp}} = \frac{a_0 \cos\Lambda}{\sqrt{1 - M_\infty^2 \cos^2\Lambda + [(a_0 \cos\Lambda)/(\pi \text{AR})]^2} + (a_0 \cos\Lambda)/(\pi \text{AR})}} \tag{11.68}$$

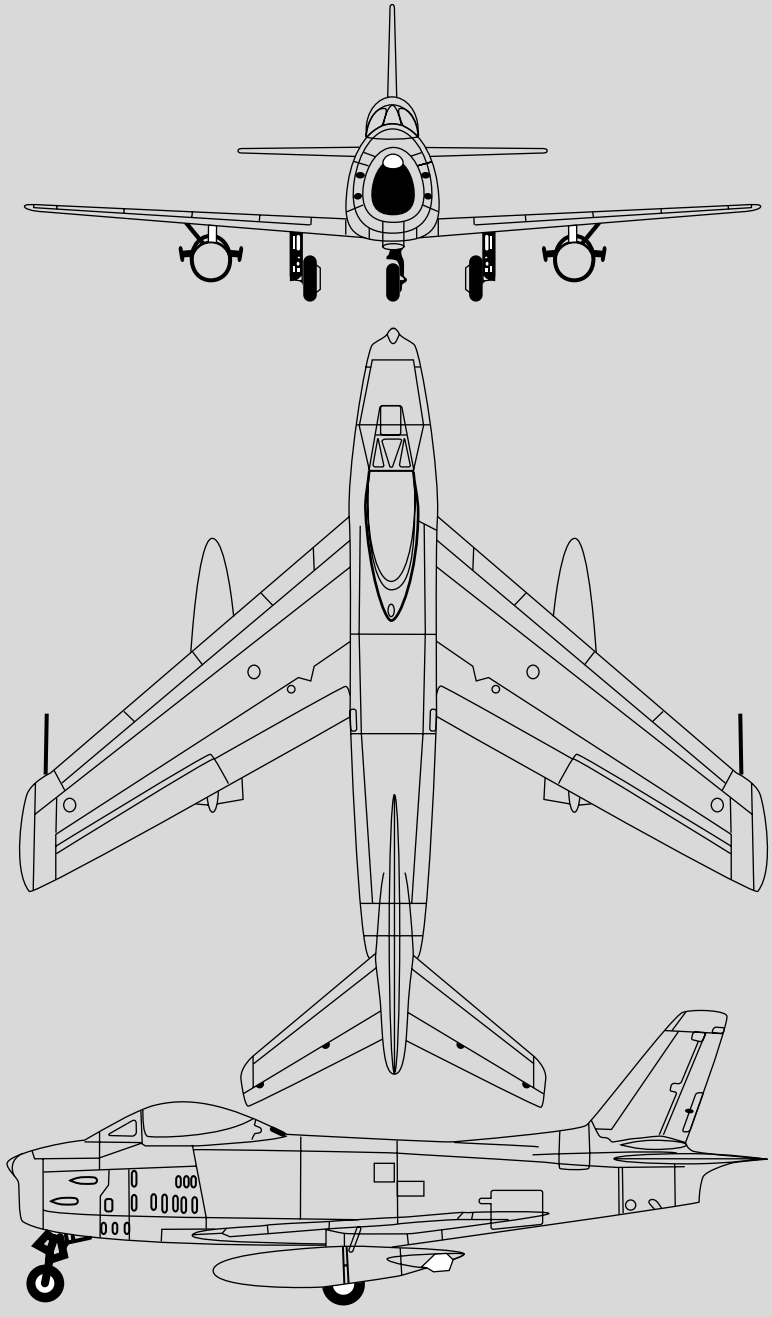


Figure 11.15 A typical example of a swept-wing aircraft. The North American F-86 Sabre of Korean War fame.

11.8 THE AREA RULE

In addition to the classic approaches of using thin airfoils and swept wings to cope with the large drag rise near Mach 1, in recent years two rather revolutionary concepts have helped greatly to break down the "sound barrier" near and beyond the speed of sound. One of these—the area rule—is discussed in this section; the other—the supercritical airfoil—is the subject of Section 11.9.

For a moment, let us expand our discussion from two-dimensional airfoils to a consideration of a complete airplane. In this section, we introduce a design concept that has effectively reduced the drag rise near Mach 1 for a complete airplane.

As stated before, the first practical jet-powered aircraft appeared at the end of World War II in the form of the German Me 262. This was a subsonic fighter plane with a top speed near 550 mi/h. The next decade saw the design and production of many types of jet aircraft—all limited to subsonic flight by the large drag near Mach 1. Even the "century" series of fighter aircraft designed to give the U.S. Air Force supersonic capability in the early 1950s, such as the Convair F-102 delta-wing airplane, ran into difficulty and could not at first readily penetrate the sound barrier in level flight. The thrust of jet engines at that time simply could not overcome the large peak drag near Mach 1.

A planview, cross section, and area distribution (cross-sectional area versus distance along the axis of the airplane) for a typical airplane of that decade are sketched in Figure 11.16. Let A denote the total cross-sectional area at any given station. Note that the cross-sectional area distribution experiences some abrupt

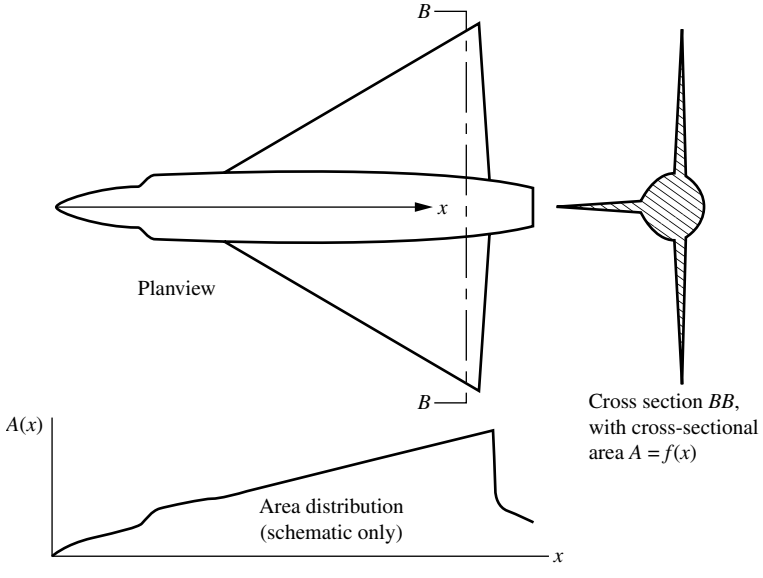


Figure 11.16 A schematic of a non-area-ruled aircraft.

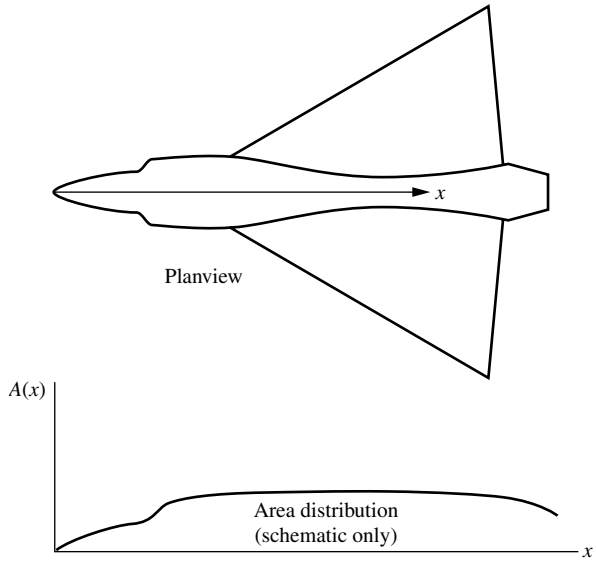


Figure 11.17 A schematic of an area-ruled aircraft.

changes along the axis, with discontinuities in both A and dA/dx in the regions of the wing.

In contrast, for almost a century, it was well known by ballisticians that the speed of a supersonic bullet or artillery shell with a *smooth* variation of cross-sectional area was higher than projectiles with abrupt or discontinuous area distributions. In the mid-1950s, an aeronautical engineer at the NACA Langley Aeronautical Laboratory, Richard T. Whitcomb, put this knowledge to work on the problem of transonic flight of airplanes. Whitcomb reasoned that the variation of cross-sectional area for an airplane should be smooth, with no discontinuities. This meant that, in the region of the wings and tail, the fuselage cross-sectional area should decrease to compensate for the addition of the wing and tail cross-sectional area. This led to a "coke bottle" fuselage shape, as shown in Figure 11.17. Here, the planview and area distribution are shown for an aircraft with a relatively smooth variation of $A(x)$. This design philosophy is called the *area rule,* and it successfully reduced the peak drag near Mach 1 such that practical airplanes could fly supersonically by the mid-1950s. The variations of drag coefficient with M_∞ for an area-ruled and non-area-ruled airplane are schematically compared in Figure 11.18; typically, the area rule leads to a factor-of-2 reduction in the peak drag near Mach 1.

The development of the area rule was a dramatic breakthrough in high-speed flight, and it earned a substantial reputation for Richard Whitcomb—a reputation that was to be later garnished by a similar breakthrough in transonic airfoil design, to be discussed in Section 11.9. The original work on the area rule was presented by Whitcomb in Reference 31, which should be consulted for more details.

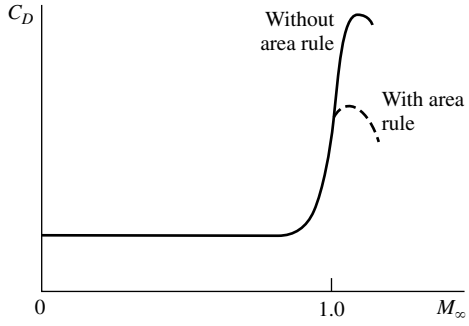


Figure 11.18 The drag-rise properties of area-ruled and non-area-ruled aircraft (schematic only).

11.9 THE SUPERCRITICAL AIRFOIL

Let us return to a consideration of two-dimensional airfoils. A natural conclusion from the material in Section 11.6, and especially from Figure 11.11, is that an airfoil with a high critical Mach number is very desirable, indeed necessary, for high-speed subsonic aircraft. If we can increase M_{cr}, then we can increase $M_{\text{drag-divergence}}$, which follows closely after M_{cr}. This was the philosophy employed in aircraft design from 1945 to approximately 1965. Almost by accident, the NACA 64-series airfoils (see Section 4.2), although originally designed to encourage laminar flow, turned out to have relative high values of M_{cr} in comparison with other NACA shapes. Hence, the NACA 64 series has seen wide application on high-speed airplanes. Also, we know that thinner airfoils have higher values of M_{cr} (see Figure 11.7); hence, aircraft designers have used relatively thin airfoils on high-speed airplanes.

However, there is a limit to how thin a practical airfoil can be. For example, considerations other than aerodynamic influence the airfoil thickness; the airfoil requires a certain thickness for structural strength, and there must be room for the storage of fuel. This prompts the following question: For an airfoil of given thickness, how can we delay the large drag rise to higher Mach numbers? To increase M_{cr} is one obvious tack, as described above, but there is another approach. Rather than increasing M_{cr}, let us strive to increase the Mach number *increment* between M_{cr} and $M_{\text{drag-divergence}}$. That is, referring to Figure 11.11, let us increase the distance between points e and c. This philosophy has been pursued since 1965, leading to the design of a new family of airfoils called *supercritical airfoils,* which are the subject of this section.

The purpose of a supercritical airfoil is to increase the value of $M_{\text{drag-divergence}}$, although M_{cr} may change very little. The shape of a supercritical airfoil is compared with an NACA 64-series airfoil in Figure 11.19. Here, an NACA 64_2-A215 airfoil is sketched in Figure 11.19*a*, and a 13-percent thick supercritical airfoil is shown in Figure 11.19*c*. (Note the similarity between the supercritical profile

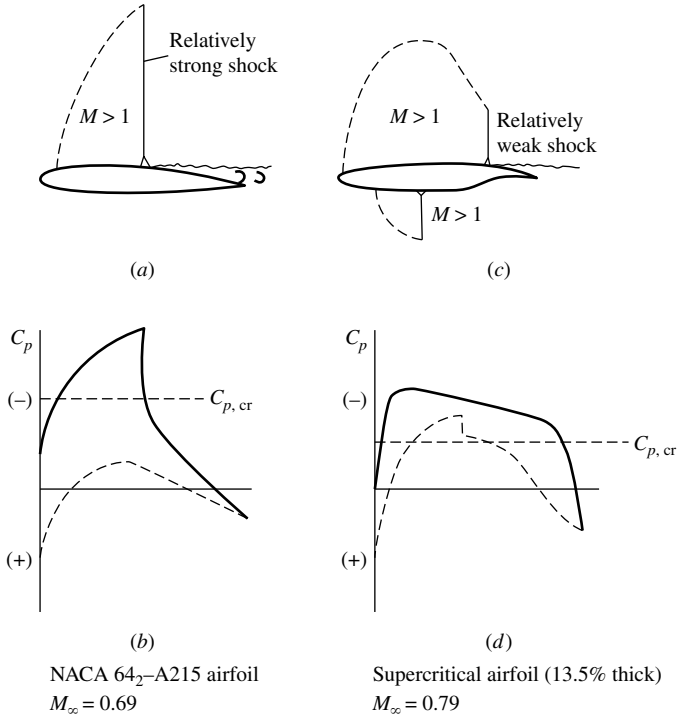


Figure 11.19 Standard NACA 64-series airfoil compared with a supercritical airfoil at cruise lift conditions. *(From Reference 32.)*

and the modern low-speed airfoils discussed in Section 4.11.) The supercritical airfoil has a relatively flat top, thus encouraging a region of supersonic flow with lower local values of M than the NACA 64 series. In turn, the terminating shock is weaker, thus creating less drag. Similar trends can be seen by comparing the C_p distributions for the NACA 64 series (Figure 11.19*b*) and the supercritical airfoil (Figure 11.19*d*). Indeed, Figure 11.19*a* and *b* for the NACA 64-series airfoil pertain to a lower freestream Mach number, $M_\infty = 0.69$, than Figure 11.19*c* and *d*, which pertain to the supercritical airfoil at a higher freestream Mach number, $M_\infty = 0.79$. In spite of the fact that the 64-series airfoil is at a lower M_∞, the extent of the supersonic flow reaches farther above the airfoil, the local supersonic Mach numbers are higher, and the terminating shock wave is stronger. Clearly, the supercritical airfoil shows more desirable flow-field characteristics; namely, the extent of the supersonic flow is closer to the surface, the local supersonic Mach numbers are lower, and the terminating shock wave is weaker. As a result, the value of $M_{\text{drag-divergence}}$ will be higher for the supercritical airfoil. This is verified by the experimental data given in Figure 11.20, taken from Reference 32. Here, the value of $M_{\text{drag-divergence}}$ is 0.79 for the supercritical airfoil in comparison with 0.67 for the NACA 64 series.

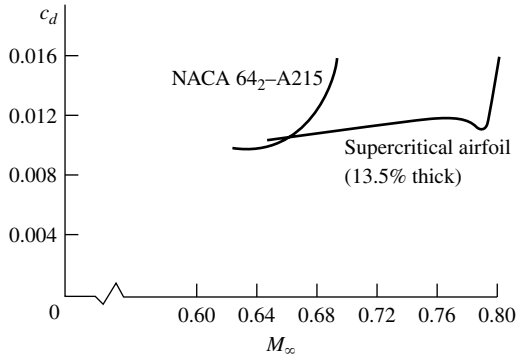


Figure 11.20 The drag-divergence properties of a standard NACA 64-series airfoil and a supercritical airfoil.

Because the top of the supercritical airfoil is relatively flat, the forward 60 percent of the airfoil has negative camber, which lowers the lift. To compensate, the lift is increased by having extreme positive camber on the rearward 30 percent of the airfoil. This is the reason for the cusplike shape of the bottom surface near the trailing edge.

The supercritical airfoil was developed by Richard Whitcomb in 1965 at the NASA Langley Research Center. A detailed description of the rationale as well as some early experimental data for supercritical airfoils are given by Whitcomb in Reference 32, which should be consulted for more details. The supercritical airfoil, and many variations of such, are now used by the aircraft industry on modern high-speed airplane designs. Examples are the Boeing 757 and 767, and the latest model Lear jets. The supercritical airfoil is one of two major breakthroughs made in transonic airplane aerodynamics since 1945, the other being the area rule discussed in Section 11.8. It is a testimonial to the man that Richard Whitcomb was mainly responsible for both.

11.10 CFD APPLICATIONS: TRANSONIC AIRFOILS AND WINGS

The analysis of subsonic compressible flow over airfoils discussed in this chapter, resulting in classic compressibility corrections such as the Prandtl-Glauert rule (Section 11.4), fits into the category of "closed-form" theory as discussed in Section 2.17.1. Although this theory is elegant and useful, it is restricted to:

1. Thin airfoils at small angles of attack
2. Subsonic numbers that do not approach too close to one, that is, Mach numbers typically below 0.7
3. Inviscid, irrotational flow

However, modern subsonic transports (Boeing 747, 777, etc.) cruise at freestream Mach numbers on the order of 0.85, and high-performance military combat airplanes spend time at high subsonic speeds near Mach one. These airplanes are in the *transonic* flight regime, as discussed in Section 1.10.4 and noted in Figure 1.44. The closed-form theory discussed in this chapter does not apply in this flight regime. The only approach that allows the accurate calculation of airfoil and wing characteristics at transonic speeds is to use computational fluid dynamics; the basic philosophy of CFD is discussed in Section 2.17.2, which should be reviewed before you progress further.

The need to calculate accurately the transonic flow over airfoils and wings was one of the two areas that drove advances in CFD in the early days of its development, the other area being hypersonic flow. The growing importance of high-speed jet civil transports during the 1960s and 1970s made the accurate calculation of transonic flow imperative, and CFD was (and still is) the only way of making such calculations. In this section we will give only the flavor of such calculations; see Chapter 14 of Reference 21 for more details, as well as the modern aerodynamic literature for the latest developments.

Beginning in the 1960s, transonic CFD calculations historically evolved through four distinct steps, as follows:

1. The earliest calculations numerically solved the nonlinear small-perturbation potential equation for transonic flow, obtained from Equation (11.6) by dropping all terms on the right-hand side except for the leading term, which is not small near $M_\infty = 1$. This yields

$$(1 - M_\infty^2)\frac{\partial \hat{u}}{\partial x} + \frac{\partial \hat{v}}{\partial y} = M_\infty^2 \left[(\gamma + 1)\frac{\hat{u}}{V_\infty}\right]\frac{\partial \hat{u}}{\partial x}$$

which in terms of the perturbation velocity potential is

$$(1 - M_\infty^2)\frac{\partial^2 \hat{\phi}}{\partial x^2} + \frac{\partial^2 \hat{\phi}}{\partial y^2} = M_\infty^2 \left[(\gamma + 1)\frac{\partial \hat{\phi}}{\partial x}\frac{1}{V_\infty}\right]\frac{\partial^2 \hat{\phi}}{\partial x^2} \tag{11.69}$$

Equation (11.69) is the transonic small perturbation potential equation; it is non-linear due to the term on the right-hand side, which involves a product of derivatives of the dependent variable $\hat{\phi}$. This necessitated a numerical CFD solution. However, the results were limited to the assumptions embodied in this equation, namely, small perturbations and hence thin airfoils at small angles of attack.

2. The next step was numerical solutions of the full potential equation, Equation (11.12). This allowed applications to airfoils of any shape at any angle of attack. However, the flow was still assumed to be isentropic, and even though shock waves appeared in the results, the properties of these shocks were not always accurately predicted.

3. As CFD algorithms became more sophisticated, numerical solutions of the Euler equations [the full continuity, momentum, and energy equations for inviscid flow, such as Equations (7.40), (7.42), and (7.44)] were obtained.

The advantage of these Euler solutions was that shock waves were properly treated. However, none of the approaches discussed in steps 1–3 accounted for the effects of viscous flow, the importance of which in transonic flows soon became more appreciated because of the interaction of the shock wave with the boundary layer. This interaction, with the attendant flow separation is dominant in the prediction of drag.

4. This led to the CFD solution of the viscous flow equations (the Navier–Stokes equations, such as Equations (2.43), (2.61), and (2.87) with the viscous terms included) for transonic flow. The Navier–Stokes equations are developed in detail in Chapter 15. Such CFD solutions of the Navier–Stokes equations are currently the state of the art in transonic flow calculations. These solutions contain all of the realistic physics of such flows, with the exception that some type of turbulence model must be included to deal with turbulent boundary layers, and such turbulent models are frequently the Achilles heel of these calculations.

An example of a CFD calculation for the transonic flow over an NACA 0012 airfoil at 2° angle of attack with $M_\infty = 0.8$ is shown in Figure 11.21. The contour lines shown here are lines of constant Mach number, and the bunching of these lines together clearly shows the nearly normal shock wave occurring on the top surface. In reference to our calculation in Example 11.3 showing that the critical Mach number for the NACA 0012 airfoil at zero angle of attack is 0.74, and the experimental confirmation of this shown in Figure 11.10*b*, clearly the flow over the same airfoil shown in Figure 11.21 is well beyond the critical Mach number. Indeed, the boundary layer downstream of the shock wave in Figure 11.21 is separated, and the airfoil is squarely in the drag-divergence region. The CFD calculations predict this separated flow because a version of the Navier–Stokes equations (called the thin shear layer approximation) is being numerically solved, taking into account the viscous flow effects. The results shown in Figure 11.21 are from the work of Nakahashi and Deiwert at the NASA Ames Research Center (Reference 74); these results are a graphic illustration of the power of CFD applied to transonic flow. For details on these types of CFD calculations, see the definitive books by Hirsch (Reference 75).

Today, CFD is an integral part of modern transonic airfoil and wing design. A recent example of how CFD is combined with modern optimization design techniques for the design of complete wings for transonic aircraft is shown in Figures 11.22 and 11.23, taken from the survey paper by Jameson (Reference 76). On the left side of Figure 11.22*a* the airfoil shape distribution along the semispan of a baseline, initial wing shape at $M_\infty = 0.83$ is given, with the computed pressure coefficient distributions shown at the right. The abrupt drop in C_p in these distributions is due to a relatively strong shock wave along the wing. After repeated iterations, the optimized design at the same $M_\infty = 0.83$ is shown in Figure 11.22*b*. Again, the new airfoil shape distribution is shown on the left, and the C_p distribution is given on the right. The new, optimized wing design shown in Figure 11.22*b* is virtually shock free, as indicated by the smooth C_p distributions,

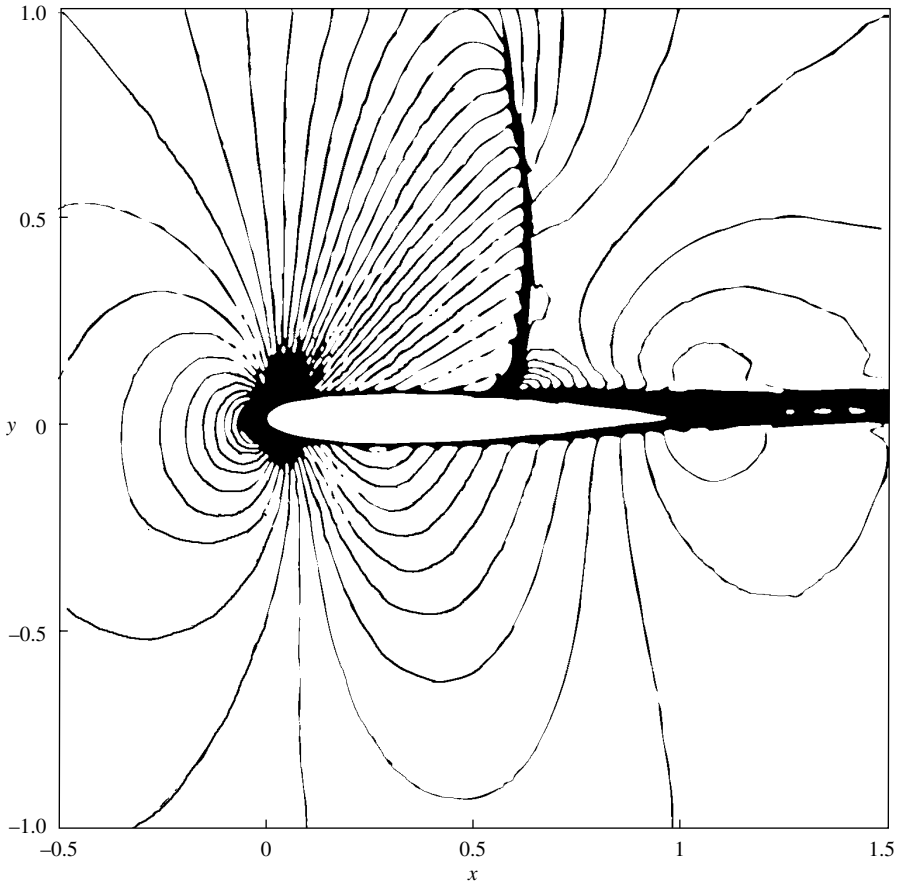


Figure 11.21 Mach number contours in the transonic flow over an NACA 0012 airfoil at $M_\infty = 0.8$ and at 2° angle of attack. (*Source:* Nakahashi and Deiwert, Reference 74.)

with a consequent reduction in drag of 7.6 percent. The optimization shown in Figure 11.22 was subject to the constraint of keeping the wing thickness the same. Another but similar case of wing design optimization is shown in Figure 11.23. Here, the final optimized wing planform shape is shown for $M_\infty = 0.86$, with the final computed pressure contour lines shown on the planform. Straddling the wing planform on both the left and right of Figure 11.23 are the pressure coefficient plots at six spanwise stations. The dashed curves show the C_p variations for the initial baseline wing, with the tell-tale oscillations indicating a shock wave, whereas the solid curves are the final C_p variations for the optimized wing, showing smoother variations that are almost shock-free. The results shown in Figures 11.22 and 11.23 are reflective of multidisciplinary design optimization using CFD for transonic wings. For more details on this and other design applications, see the special issue of the *Journal of Aircraft,* vol. 36, no. 1, Jan./Feb. 1999, devoted to aspects of multidisciplinary design optimization.

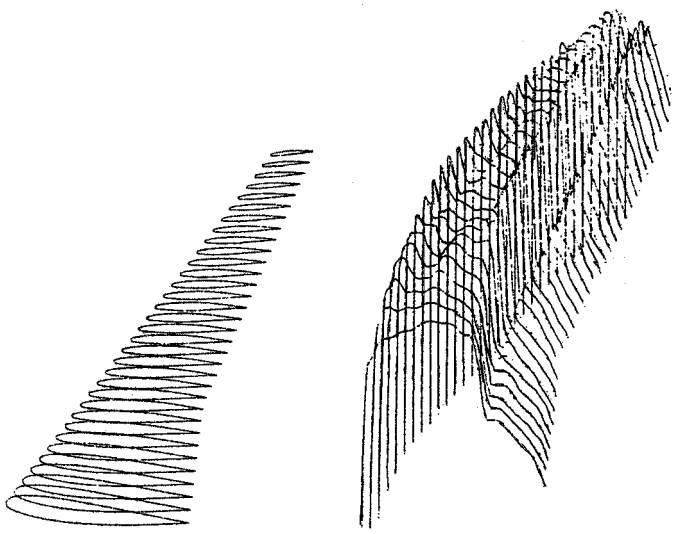


Figure 11.22 The use of CFD for optimized transonic wing design. $M_\infty = 0.83$. (*a*) Baseline wing with a shock wave. (*b*) Optimized wing, virtually shock free. *Source:* Jameson, Reference 76.

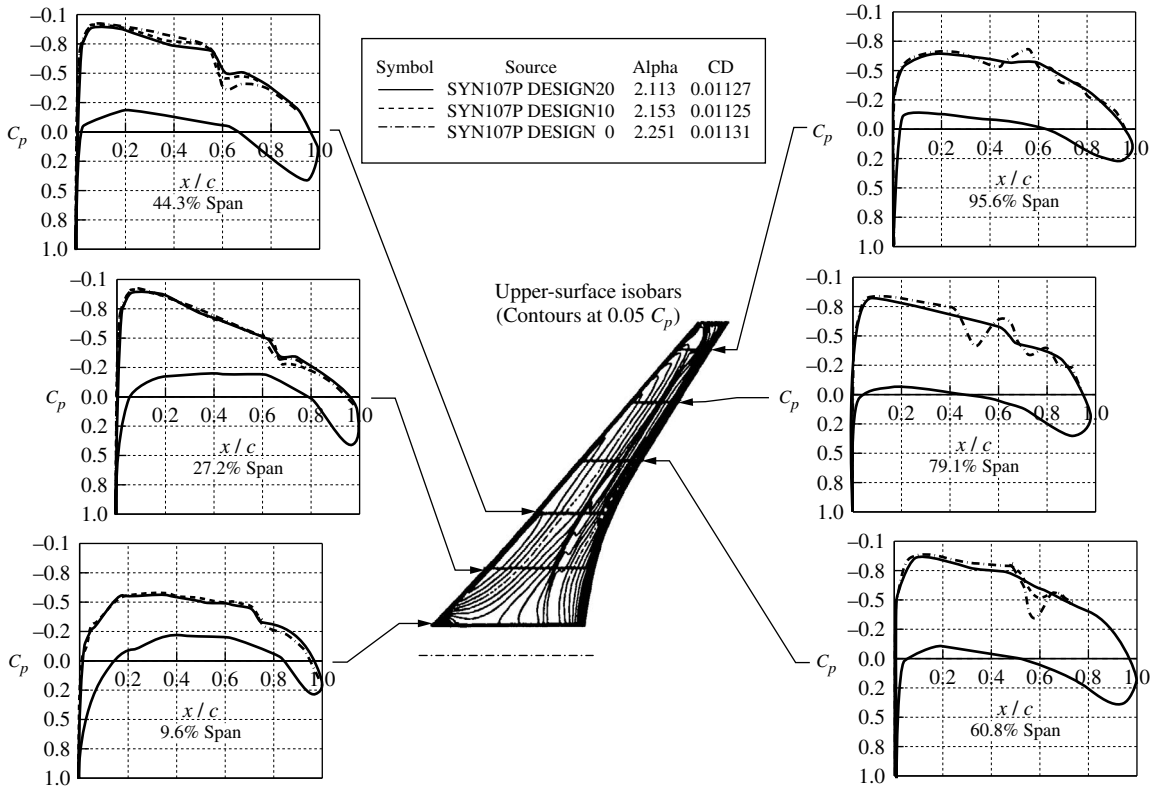


Figure 11.23 Another example of optimized transonic wing design using CFD. $M_\infty = 0.86$. (*Source:* Jameson, Reference 76.)

11.11 APPLIED AERODYNAMICS: THE BLENDED WING BODY

This book began with images of six flight vehicles that illustrate six good reasons to learn about aerodynamics. In particular, Figure 1.6 shows an artist's conception of the blended wing body, a new vehicle concept that promises to create a renaissance in long distance air transport. The aerodynamics of the blended wing body (BWB) is deeply rooted in the aerodynamic fundamentals discussed in this book, and so we take this opportunity to examine some of the applied aerodynamics associated with the BWB.

As background, the BWB was the outgrowth of a challenge issued by Dennis Bushnell to the aircraft industry in 1988. Bushnell, the chief scientist of the NASA Langley Research Center, asked if new, innovative thinking could result in a commercial jet transport that would provide a quantum leap in efficiency and performance in comparison to the standard tube-fuselage, swept wing airplane with jet engines pod-mounted under the wings, such as pioneered by the

historic Boeing 707 (Figure 1.2). After 50 years, this configuration remains the same for virtually all transport aircraft, as reinforced by Boeing's latest design, the 787 Dreamliner shown in Figure 2.2. Responding to this challenge, a small group of aerodynamicists at McDonnell Douglas led by Dr. Robert Liebeck conceived the blended wing body, one version of which is shown in Figure 11.24.

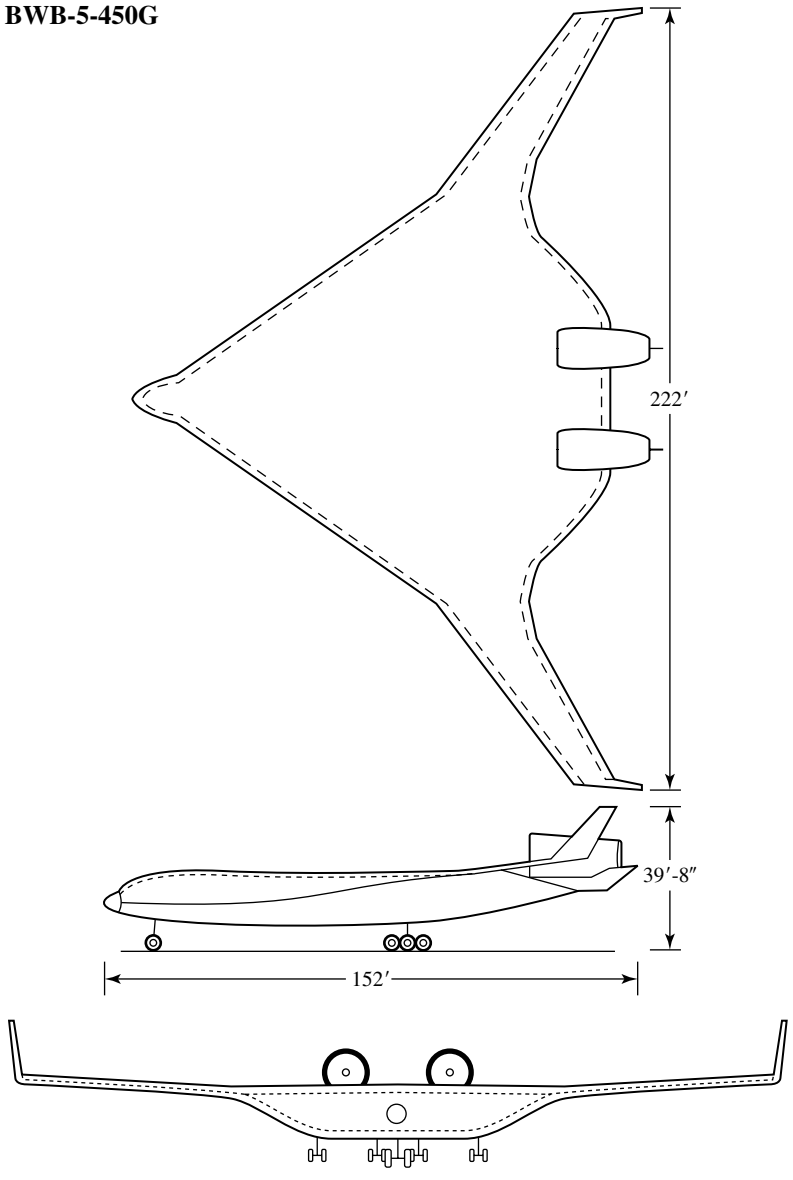


Figure 11.24 Three-view of a blended wing body.

When McDonnell Douglas was absorbed by Boeing, Liebeck continued his work on the BWB funded by NASA. Figure 11.24 shows a configuration obtained from a baseline study of the BWB carried out by Boeing circa 2002. Now a Boeing senior technical fellow, Liebeck continues to spearhead the BWB concept at Boeing.

The aerodynamics of the blended wing body is a graphic illustration of the application of many of the fundamentals highlighted in this book. For this reason, the blended wing body is chosen for attention in this applied aerodynamics section. We will see that the BWB is an advanced futuristic flight vehicle applying the very fundamental aerodynamics that is the subject of this book, underscoring the fact that such fundamentals are timeless.

To begin with, examining Figure 11.24 shows that the BWB is clearly a flying wing merged with a center body that is also an airfoil shape with a bullet nose. By replacing the conventional tube fuselage with a center body that itself is an efficient lifting surface, the spanwise lift distribution from one wing tip to the other is closer to the ideal elliptical distribution. Our study of the aerodynamics of finite wings in Chapter 5 underscored that an elliptical lift distribution yields the minimum induced drag. The BWB is designed to preserve such an elliptical lift distribution, as illustrated in Figure 11.25. Here we see the BWB lift distribution along with plots of the spanwise airfoil lift coefficient c_l and the airfoil thickness-to-chord ratio t/c. There is a direct connection between the variations of c_l and t/c. The center body must be large enough and thick enough to accommodate the passenger and cargo load, which drives the increase of both the airfoil chord length and t/c for the center body section. In order to preserve the elliptical lift

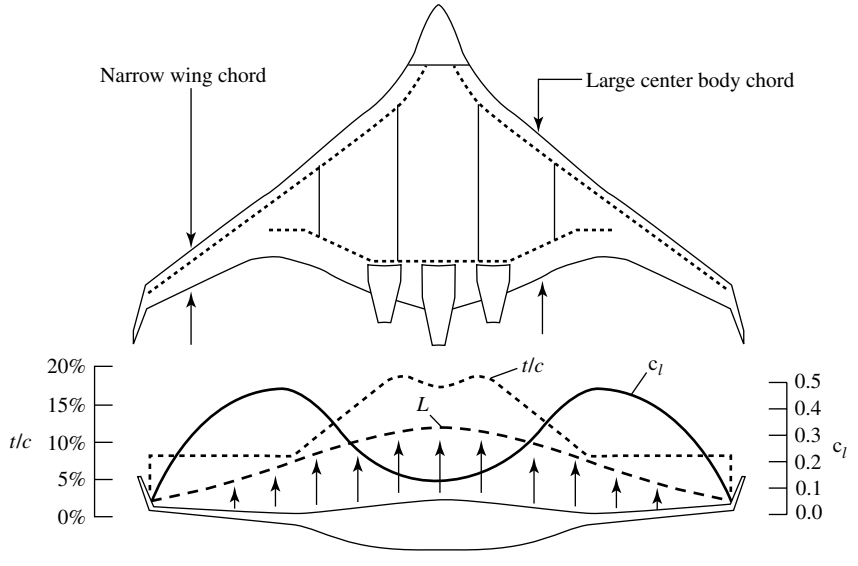


Figure 11.25 Typical spanwise distribution of lift L, lift coefficient c_l, and thickness-to-chord ratio for the blended wing body.

distribution over the center body, the airfoil section of the center body is different from that used for the outer wing panels. It is chosen to have a lower lift coefficient to counterbalance the larger chord length and thus preserve the smooth spanwise elliptical lift distribution. (Recall that the lift distribution is the variation of the lift *force* per unit span, and this lift force is proportional to both the local value of c_l and the chord length.)

The BWB is a high-speed subsonic airplane intended to fly at the lower end of the transonic flight regime. Hence, major efforts are made to obtain as high a drag-divergence Mach number as possible. Toward that end, the BWB incorporates two design features, both of which deal with aerodynamic fundamentals discussed in this chapter.

1. *Supercritical airfoils.* The function of a supercritical airfoil is discussed in Section 11.9. The outer portions of the BWB wing incorporate a modern supercritical airfoil section with aft camber, similar to that shown in Figure 11.19*c*. The center body profile is also an airfoil section. In the first generation of the BWB development, the airfoil shape chosen was a Liebeck LW102A airfoil (Ref. 93) point designed for $c_l = 0.25$ at a Mach number of 0.7. A side view of the resulting center body profile is shown in Figure 11.26*a*. The new-generation BWB uses an advanced customized transonic airfoil design for the center body profile. Taking into account the constraints in cross-sectional area required to effectively hold passengers, baggage, and cargo, the new transonic airfoil design dealt with a careful three-dimensional contouring of the center body smoothly blending into the outer wing panels. The resulting center body profile is shown in

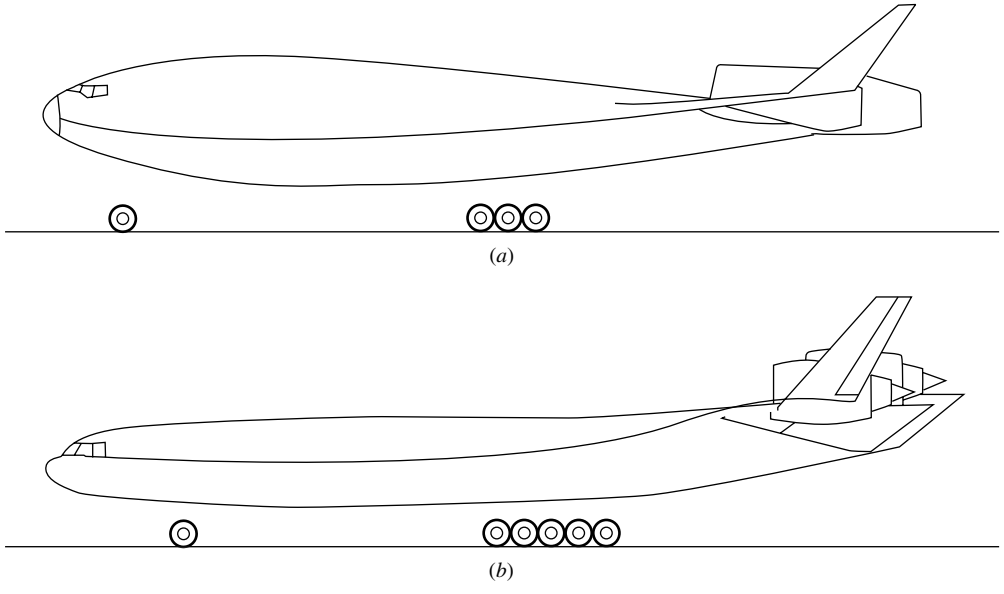


Figure 11.26 Center body profiles: (*a*) first generation; (*b*) recent generation.

Figure 11.26*b*, giving a cleaner, more streamlined appearance than the original profile in Figure 11.26*a*, and providing a higher critical Mach number. Indeed, the new centerbody profile in Figure 11.2*b* increased the BWB lift-to-drag ratio by 4 percent.

2. *Area rule.* The notion of the area rule is discussed in Section 11.8. The blended wing body, with its smooth contours and smoothly varying cross section, is almost naturally area-ruled. Figure 11.27 compares the cross-sectional area distributions as a function of longitudinal coordinate for the BWB (solid curve) and a conventional subsonic transport, the MD-11 (dashed curve). Clearly, the BWB area distribution is much smoother than that of the MD-11, thus exhibiting good area-rule qualities. Liebeck (Ref. 94) states that for the BWB "there appears to be no explicit boundary for increasing the cruise Mach number beyond 0.88." Indeed, a set of blended wing bodies have been designed for Mach numbers of 0.85, 0.9., 0.93, and 0.95.

The role of computational fluid dynamics (CFD) in the calculation of transonic inviscid flows is discussed in Section 11.10. This discussion is extended in Chapter 20 to CFD solutions for viscous flows via numerical solutions of the Navier-Stokes equations. Computational fluid dynamics is an essential tool in modern aircraft design, and this is particularly important for the blended wing body as discussed by Roman et al. (Ref. 95) and Liebeck (Ref. 96). For example, Figure 11.28 shows the static pressure contours over the top surface of the BWB obtained with a Navier-Stokes CFD solution. Shock waves are indicated by regions where the pressure contours cluster together. These results show that the typical transonic shock wave, which is well defined on the outboard wing, becomes smeared into a weaker compression wave on the center body. The CFD solutions showed that the flow pattern on the center body was relatively insensitive to the angle of attack. Also, the results indicated the initiation of flow separation in the kink region between the outboard wing and the center body. More CFD results are compared in Figure 11.29 with experimental data obtained on a blended wing body model tested in the National Transonic Facility (NFT) at the NASA Langley Research Laboratory at almost full-scale Reynolds number. Figure 11.29*a* gives the variation of drag coefficient C_D with lift coefficient C_L (a portion of the drag polar). Figure 11.29*b* is a plot of C_L versus angle of attack, and Figure 11.29*c* gives lift coefficient versus moment coefficient. Even though, for proprietary reasons, numbers are not given on the axes of these graphs, the most important conclusion from these comparisons is that CFD results for the BWB agree within 1% with experimental data. In the words of Robert Liebeck (Reference 96), "The remarkable agreement indicated that CFD could be reliably utilized for the aerodynamic design and analysis."

In summary, we offer this applied aerodynamics section as a clear example of how the understanding of the fundamental aerodynamics presented in this book is so essential to the design of the flight vehicles of the future. For more information on the blended wing body, see References 93–96.

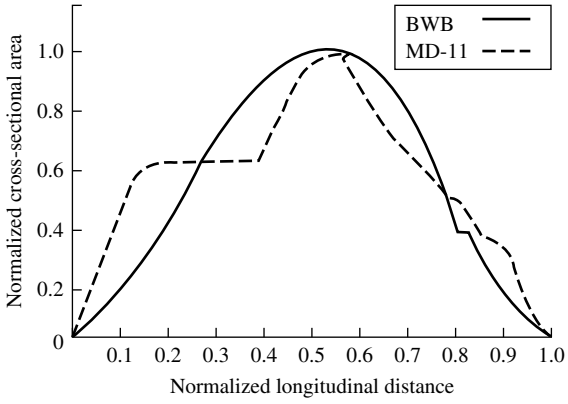


Figure 11.27 Longitudinal distributions of cross-sectional area comparing the blended wing body with a conventional wide-body civil transport, the McDonnell-Douglas MD-11.

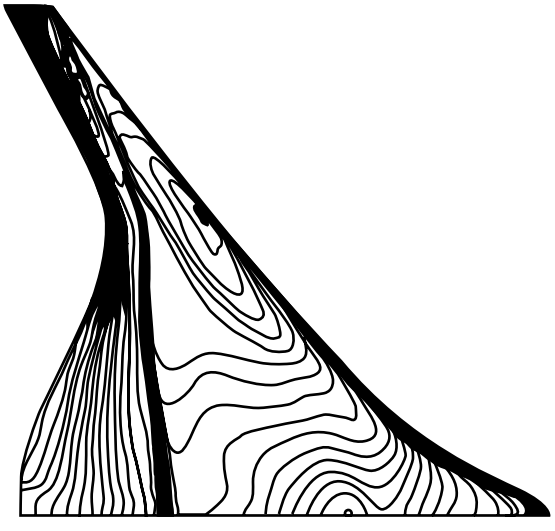


Figure 11.28 Representative CFD calculations of the static pressure contours over the top surface of the BWB at midcruise condition.

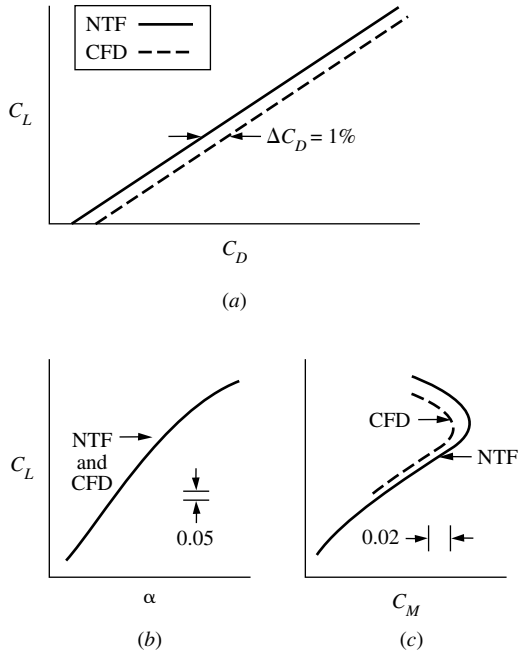


Figure 11.29 Comparison between CFD calculations and experimental data for the blended wing body. The CFD calculations are made with the CFL3D code, and the experimental data are from the National Transonic Facility at the NASA Langley Research Laboratoty. (*a*) C_L versus CD (drag polar); (*b*) C_L versus angle of attack; (*c*) C_L versus moment coefficient.

11.12 HISTORICAL NOTE: HIGH-SPEED AIRFOILS—EARLY RESEARCH AND DEVELOPMENT

Twentieth-century aerodynamics does not have the exclusive rights to the observation of the large drag rise on bodies flying at near the speed of sound; rather, in the eighteenth century the Englishman Benjamin Robins, inventor of the ballistic pendulum, reported that "the velocity at which the body shifts its resistance (from a V^2 to a V^3 relation) is nearly the same with which sound is propagated through air." His statement was based on a large number of experiments during which projectiles were fired into his ballistic pendulum. However, these results had little relevance to the early aerodynamicists of this century, who were struggling to push aircraft speeds to 150 mi/h during and just after World War I. To these people, flight near the speed of sound was just fantasy.

With one exception! World War I airplanes such as the Spad and Nieuport had propeller blades where the tips were moving at near the speed of sound. By

1919, British researchers had already observed the loss in thrust and large increase in blade drag for a propeller with tip speeds up to 1180 ft/s—slightly above the speed of sound. To examine this effect further, F. W. Caldwell and E. N. Fales, both engineers at the U.S. Army's Engineering Division at McCook Field near Dayton, Ohio (the forerunner of the massive Air Force research and development facilities at Wright-Patterson Air Force Base today), conducted a series of high-speed airfoil tests. They designed and built the first high-speed wind tunnel—a facility with a 14-in-diameter test section capable of velocities up to 675 ft/s. In 1918, they conducted the first wind-tunnel tests involving the high-speed flow over a stationary airfoil. Their results showed large decreases in lift coefficient and major increases in drag coefficient for the thicker airfoils at angle of attack. These were the first measured "compressibility effects" on an airfoil in history. Caldwell and Fales noted that such changes occurred at a certain air velocity, which they denoted as the "critical speed"—a term that was to evolve into the critical Mach number at a later date. It is interesting to note that Orville Wright was a consultant to the Army at this time (Wilbur had died prematurely in 1912 of typhoid fever) and observed some of the Caldwell and Fales tests. However, a fundamental understanding and explanation of this critical-speed phenomenon was completely lacking. Nobody at that time had even the remotest idea of what was really happening in this high-speed flow over the airfoil.

Members of the National Advisory Committee for Aeronautics were well aware of the Caldwell-Fales results. Rather than let the matter die, in 1922 the NACA contracted with the National Bureau of Standards (NBS) for a study of high-speed flows over airfoils, with an eye toward improved propeller sections. The work at NBS included the building of a high-speed wind tunnel with a 12-in-diameter test section, capable of producing a Mach number of 0.95. The aerodynamic testing was performed by Lyman J. Briggs (soon to become director of NBS) and Hugh Dryden (soon to become one of the leading aerodynamicists of the twentieth century). In addition to the usual force data, Briggs and Dryden also measured pressure distributions over the airfoil surface. These pressure distributions allowed more insight into the nature of the flow and definitely indicated flow separation on the top surface of the airfoil. We now know that such flow separation is induced by a shock wave, but these early researchers did not at that time know about the presence of such shocks.

During the same period, the only meaningful theoretical work on high-speed airfoil properties was carried out by Ludwig Prandtl in Germany and Hermann Glauert in England—work which led to the Prandtl-Glauert compressibility correction, given by Equation (11.51). As early as 1922, Prandtl is quoted as stating that the lift coefficient increased according to $(1 - M_\infty^2)^{-1/2}$; he mentioned this conclusion in his lectures at Göttingen, but without written proof. This result was mentioned again 6 years later by Jacob Ackeret, a colleague of Prandtl, in the famous German series *Handbuch der Physik,* again without proof. Subsequently, in 1928 the concept was formally established by Hermann Glauert, a British aerodynamicist working for the Royal Aircraft Establishment. (See Chapter 9 of Reference 21 for a biographical sketch of Glauert.) Using only six pages in the *Proceedings of the Royal Society,* vol. 118, p. 113, Glauert presented a derivation

based on linearized small-perturbation theory (similar to that described in Section 11.4) which confirmed the $(1 - M_\infty^2)^{-1/2}$ variation. In this paper, entitled "The Effect of Compressibility on the Lift of an Airfoil," Glauert derived the famous Prandtl-Glauert compressibility correction, given here as Equations (11.51) to (11.53). This result was to stand alone, unaltered, for the next 10 years.

Hence, in 1930 the state of the art of high-speed subsonic airfoil research was characterized by experimental proof of the existence of the drag-divergence phenomenon, some idea that it was caused by flow separation, but no fundamental understanding of the basic flow field. In turn, there was virtually no theoretical background outside of the Prandtl-Glauert rule. Also, keep in mind that all the above work was paced by the need to understand propeller performance, because in that day the only component of airplanes to encounter compressibility effects was the propeller tips.

All this changed in the 1930s. In 1928, the NACA had constructed its first rudimentary high-speed subsonic wind tunnel at the Langley Aeronautical Laboratory, utilizing a 1-ft-diameter test section. With Eastman Jacobs as tunnel director and John Stack as the chief researcher, a series of tests was run on various standard airfoil shapes. Frustrated by their continual lack of understanding about the flow field, they turned to optical techniques, following in the footsteps of Ernst Mach (see Section 9.10). In 1933, they assembled a crude schlieren optical system consisting of 3-in-diameter reading-glass-quality lenses and a short-duration-spark light source. In their first test using the schlieren system, dealing with flow over a cylinder, the results were spectacular. Shock waves were seen, along with the resulting flow separation. Visitors flocked to the wind tunnel to observe the results, including Theodore Theodorsen, one of the ranking theoretical aerodynamicists of that period. An indicator of the psychology at that time is given by Theodorsen's comment that since the freestream flow was subsonic, what appeared as shock waves must be an "optical illusion." However, Eastman Jacobs and John Stack knew differently. They proceeded with a major series of airfoil testing, using standard NACA sections. Their schlieren pictures revealed the secrets of flow over the airfoils above the critical Mach number. (See Figure 1.38*b* and its attendant discussion of such supercritical flow.) In 1935, Jacobs traveled to Italy, where he presented results of the NACA high-speed airfoil research at the fifth Volta Conference (see Section 7.1). This is the first time in history that photographs of the transonic flow field over standard-shaped airfoils were presented in a large public forum.

During the course of such work in the 1930s, the incentive for high-speed aerodynamic research shifted from propeller applications to concern about the airframe of the airplane itself. By the mid-1930s, the possibility of the 550 mi/h airplane was more than a dream—reciprocating engines were becoming powerful enough to consider such a speed regime for propeller-driven aircraft. In turn, the entire airplane itself (wings, cowling, tail, etc.) would encounter compressibility effects. This led to the design of a large 8-ft high-speed tunnel at Langley, capable of test-section velocities above 500 mi/h. This tunnel, along with the earlier 1-ft tunnel, established the NACA's dominance in high-speed subsonic research in the late 1930s.

In the decade following 1930, the picture had changed completely. By 1940, the high-speed flow over airfoils was relatively well understood. During this period, Stack and Jacobs had not only highlighted the experimental aspects of such high-speed flow, but they also derived the expression for $C_{p,\text{cr}}$ as a function of M_{cr} given by Equation (11.60), and had shown how to estimate the critical Mach number for a given airfoil as discussed in Section 11.6. Figure 11.30 shows some representative schlieren photographs taken by the NACA of the flow over

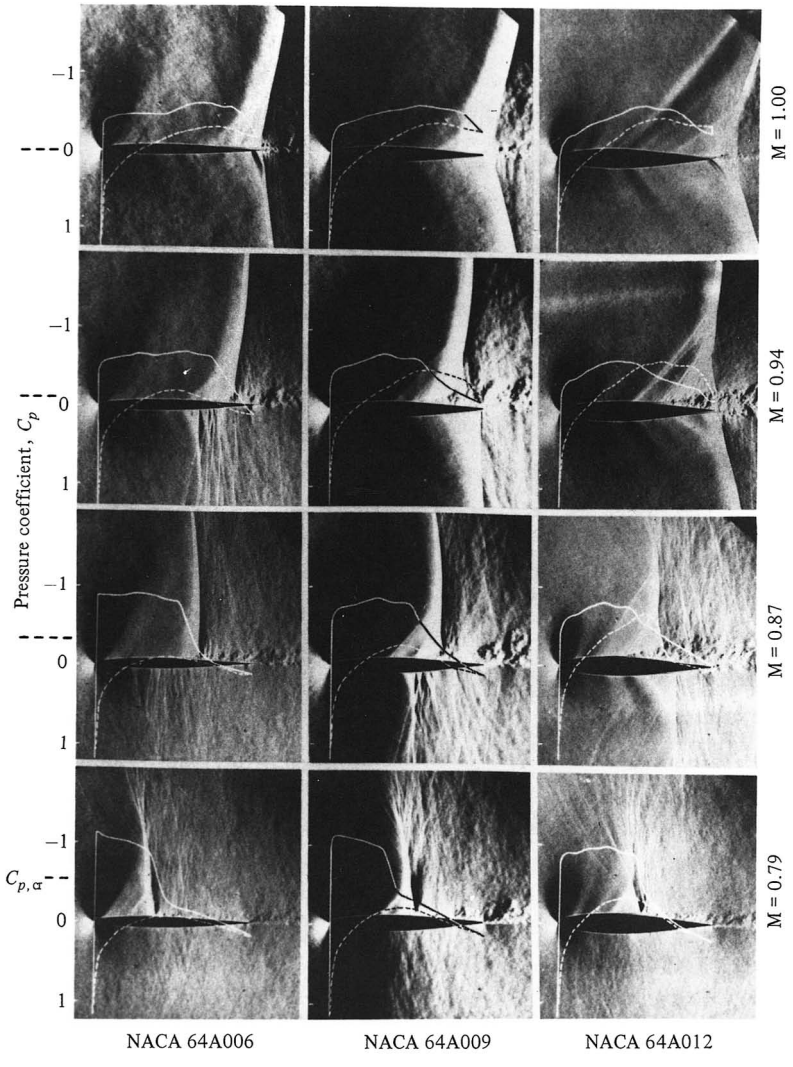


Figure 11.30 Schlieren pictures and pressure distributions for transonic flows over several NACA airfoils. These pictures were taken by the NACA in 1949. (*Source:* John V. Becker, "The High-Speed Frontier," *NASA SP-445, 1980.*)

standard NACA airfoils. Although these photographs were taken in 1949, they are similar to the results obtained by Stack and Jacobs in the 1930s. Superimposed on these photographs are the measured pressure distributions over the top (solid curve) and bottom (dashed curve) surfaces of the airfoil. Study these pictures carefully. Moving from bottom to top, you can see the influence of increasing freestream Mach number, and going from left to right, you can observe the effect of increasing airfoil thickness. Note how the shock wave moves downstream as M_∞ is increased, finally reaching the trailing edge at $M_\infty = 1.0$. For this case, the top row of pictures shows almost completely supersonic flow over the airfoil. Note also the large regions of separated flow downstream of the shock waves for the Mach numbers of 0.79, 0.87, and 0.94—this separated flow is the primary reason for the large increase in drag near Mach 1. By 1940, it was well understood that the almost discontinuous pressure increase across the shock wave creates a strong adverse pressure gradient on the airfoil surface, and this adverse pressure gradient is responsible for separating the flow.

The high-speed airfoil research program continues today within NASA. It led to the supercritical airfoils in the 1960s (see Sections 11.9 and 11.14). It has produced a massive effort in modern times to use computational techniques for theoretically solving the transonic flow over airfoils. Such efforts are beginning to be successful, and in many respects, today we have the capability to design transonic airfoils on the computer. However, such abilities today have roots which reach all the way back to Caldwell and Fales in 1918.

For a more detailed account of the history of high-speed airfoil research, you are encouraged to read the entertaining story portrayed by John V. Becker in *The High-Speed Frontier,* NASA SP-445, 1980.

11.13 HISTORICAL NOTE: THE ORIGIN OF THE SWEPT-WING CONCEPT

The concept of swept wings for high-speed flight was first introduced at the fifth Volta Conference in Rome in 1935 by the German aerodynamicist Dr. Adolf Busemann. The importance of this conference to the advancement of high-speed flight in general is noted in Section 7.1; please review that section before reading further. One of the most farsighted and important papers given at this conference was presented by Busemann (see Figure 11.31). Entitled "Aerodynamischer Auftrieb bei Überschallgeschwindigkert" ("Aerodynamic Forces at Supersonic Speeds"), this paper introduced for the first time in history the concept of the swept wing as a mechanism for reducing the large drag increase encountered at supersonic speeds. Busemann reasoned that the flow over a wing is governed mainly by the component of velocity perpendicular to the leading edge. If the wing is swept, this component will decrease, as illustrated in Figure 11.32, which is taken directly from Busemann's original paper. Consequently the supersonic wave drag will decrease. If the sweep angle is large enough, the normal component of velocity will be subsonic (the supersonic wing is then said to have a "subsonic leading

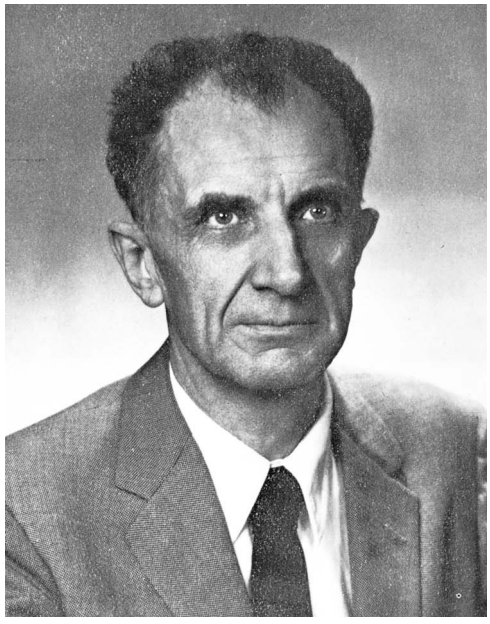


Figure 11.31 Adolf Busemann (1901–1986)
Source: (NASA.)

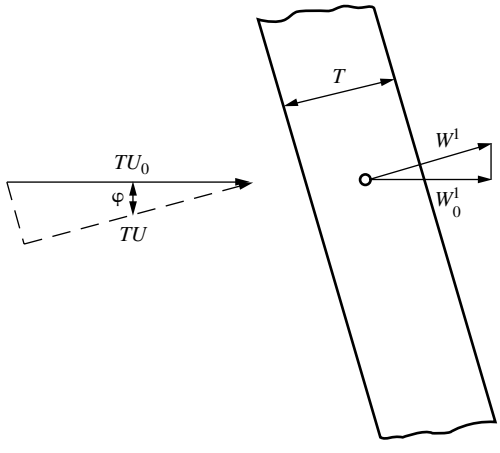


Figure 11.32 The swept-wing concept as it appeared in Busemann's original paper in 1935.

edge") with a dramatic reduction of wave drag. Figure 11.33, showing a complete swept-wing planform, is also from Busemann's paper.

At the time of the Volta conference, Adolf Busemann was a relatively young (age 34) but accomplished aerodynamicist. Born in Lübeck, Germany, in 1901, he completed high school in his home town and received his engineering diploma

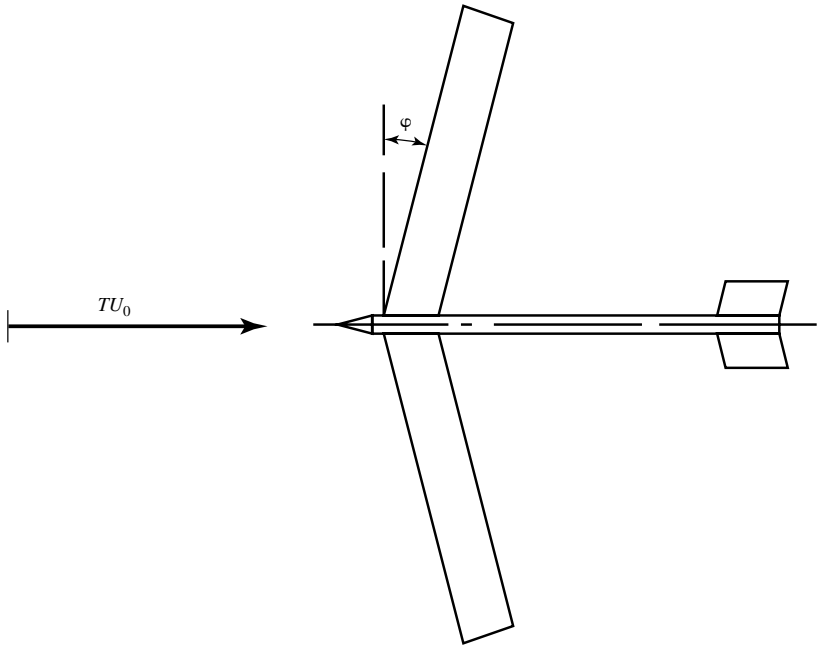


Figure 11.33 A swept-wing airplane planform, from Busemann's original paper in 1935.

and doctorate in engineering in 1924 and 1925, respectively, from the Technische Hochschule in Braunschweig. Busemann was one of the few important German aerodynamicists of that era who did not begin as one of Ludwig Prandtl's students (see Section 5.8), but in 1925 Busemann began his professional career at the Kaiser Wilhelm Institute in Göttingen and soon entered Prandtl's sphere. From 1931 to 1935, Busemann broke away from that sphere to teach in the Engine Laboratory of the Technische Hochschule in Dresden. He was still at Dresden when he gave his seminal paper at the Volta Conference. Shortly thereafter, he went to Braunschweig as chief of the Gas Dynamics Division of the Aeronautical Research Laboratory (DFL). When the Allied technical teams moved into German laboratories at the end of World War II, they not only scooped up masses of technical aerodynamic data but also effectively scooped up Busemann, who first went to the Royal Aircraft Establishment in Farnborough, England, but then accepted an invitation to join the NACA Langley Memorial Laboratory under Operation Paperclip in 1947. Busemann continued his research on high-speed aerodynamics for the NACA after joining Langley. He subsequently became chairman of the advanced-study committee at Langley and among other responsibilities supervised the preparation of science lectures used for training the early group of astronauts in the manned space program. In 1963, Busemann became a professor in the Department of Aerospace Engineering Sciences at the University of Colorado in Boulder. After retirement, he remained in Boulder, leading an active life until his death in 1986.

The swept-wing concept in Busemann's 1935 Volta Conference paper was, for everybody outside of Germany, an idea before its time. It is difficult to understand how the premier aerodynamicist in the United States, Theodor von Kármán, and other attendees failed to appreciate the significance of Busemann's idea, even forgetting it entirely, for that very evening Busemann went to dinner with von Kármán, Hugh Dryden (see Section 11.11), and General Arturo Crocco, the organizer of the conference. During dinner, Crocco sketched on the back of a menu card an airplane with swept wings, swept tail, and swept propeller, calling it, facetiously, "Busemann's airplane of the future."

There was no such facetiousness in Germany. The German Luftwaffe recognized the military significance of the swept wing, and classified the concept in 1936 – one year after the conference. From that time until the end of World War II, the Germans produced a large bulk of swept-wing research, a secret known only to themselves. Moreover, they expanded the horizons of the swept wing to high-speed subsonic and transonic airplanes, recognizing that the same aerodynamic mechanism described in Busemann's Volta Conference paper would serve to increase the critical Mach number for such aircraft. Experimental results for swept wings at high subsonic Mach numbers were first reported by Hubert Ludweig in 1939, as shown in Figure 11.34. Here the drag polars (defined in Section 5.3.3 as lift coefficient C_A versus drag coefficient C_W, in the German notation) for a straight wing and a swept wing are compared. These are the original figures from the German report AVA-Bericht 39/H/18, 1939. This was top-secret data obtained in a new high-speed wind tunnel at the Aerodynamische Versuchsanstalt (AVA) in Göttingen, and represented just the tip of the iceberg of German swept-wing research to follow. In his recent article, Peter Hamel ("Birth of Sweepback: Related Research at Luftfahrtforschungsanstalt–Germany," *Journal of Aircraft*, vol. 42, no. 4, July–August 2005, pp 801–813) labels this data as the first from what was to be "established systematic wind-tunnel tests to generate a world-first database for future transonic aircraft configurations with wing sweep."

Meanwhile, Robert T. Jones (Figure 11.35), a leading aerodynamicist at the NACA Langley Memorial Laboratory, independently discovered the advantages of a swept wing. Jones was a self-made person. Born in Macon, Missouri, in 1910, Jones was totally captivated by aeronautics at an early age. He later wrote in 1977 ("Recollections from an Earlier Period in American Aeronautics," in *Annual Review of Fluid Mechanics*, vol. 9, M. Van Dyke et al (eds), pp. 1–11):

> All during the late twenties the weekly magazine *Aviation* appeared on the local newsstand in my hometown, Macon, Missouri. *Aviation* carried technical articles by eminent aeronautical engineers such as B. V. Korvin-Krovkovsky, Alexander Klemin, and others. Included in both *Aero Digest* and *Aviation* were notices of forthcoming *NACA Technical Reports* and *Notes*. These could be procured from the Government Printing Office usually for ten cents and sometimes even free simply for writing NACA Headquarters in Washington. The contents of these reports seemed much more interesting to me than the regular high school and college curricula, and I suspect that my English teachers may have been quite perplexed by the essays I wrote for them on aeronautical subjects.

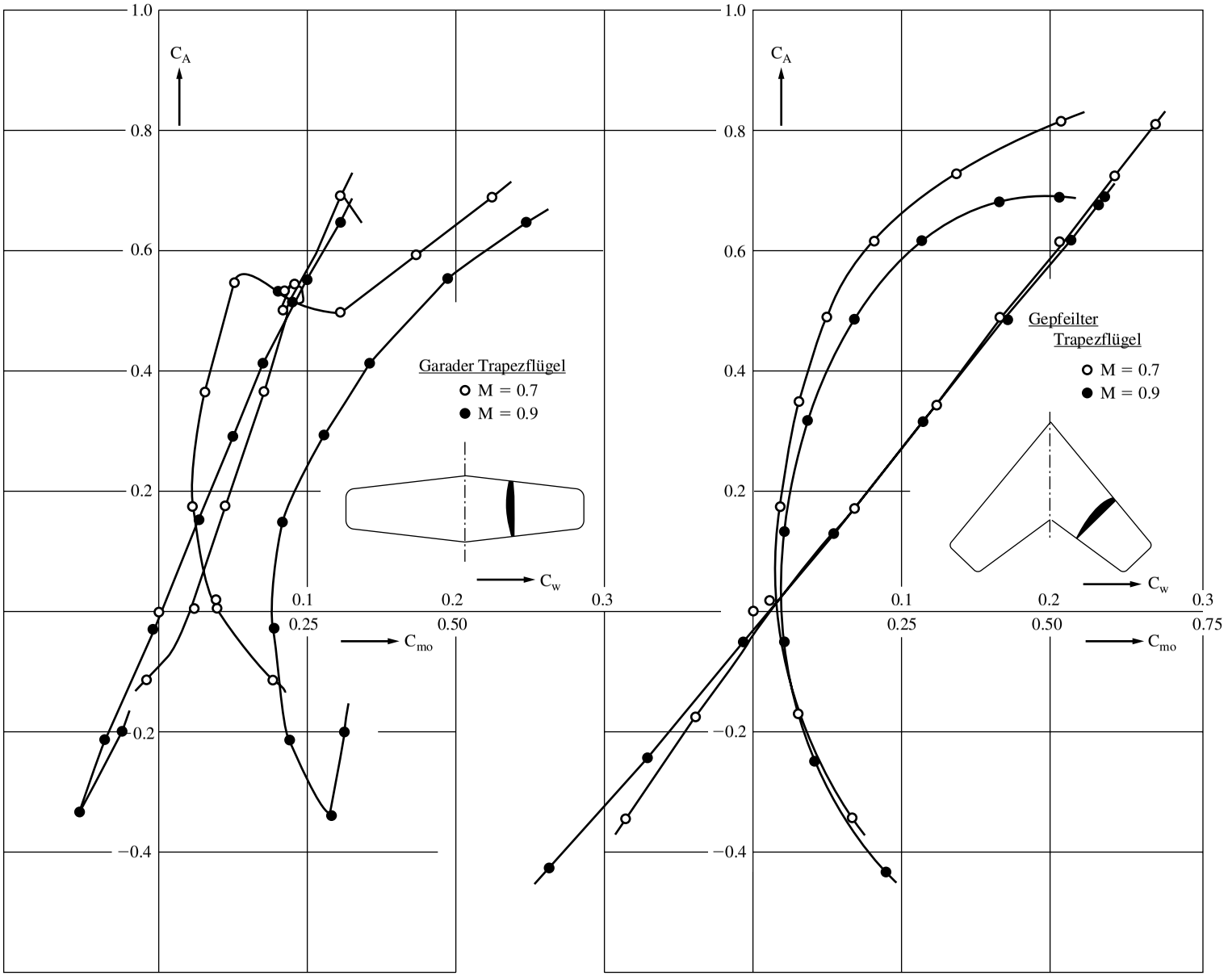


Figure 11.34 Drag polars at Mach 0.7 and 0.9 for a straight wing (left) and swept wing (right), as reported by Ludwieg in 1939.

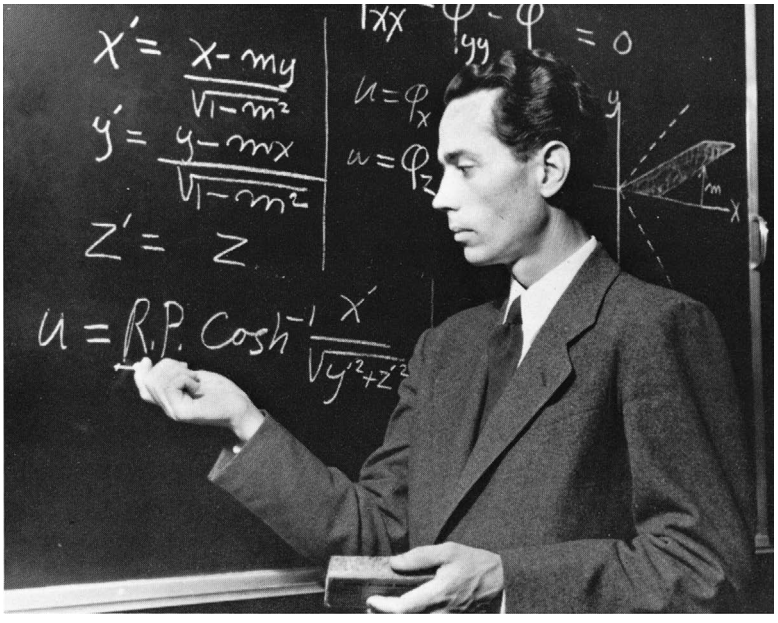


Figure 11.35 Robert T. Jones (1910–1999). (NASA)

Jones attended the University of Missouri for one year, but left to take a series of aeronautics-related jobs, first as a crew member with Marie Meyer Flying Circus, and then with the Nicholas-Beazley Airplane Company in Marshall, Missouri, which was just starting to produce a single-engine, low-wing monoplane designed by the noted British aeronautical engineer Walter H. Barling. At one time, Nicholas-Beazley was producing and selling one of those aircraft each day. However, the company became a victim of the Depression, and in 1933 Jones found himself working as an elevator operator in Washington, D.C., and taking night classes in aeronautics at Catholic University, taught by Max Monk. That contact began a lifelong friendship between Jones and Monk. In 1934 the Public Works Administration created a number of temporary scientific positions in the federal government. On the recommendation of Congressman David J. Lewis, from Jones's hometown, Jones received a nine-month appointment at the NACA Langley Memorial Laboratory. That was the beginning of a lifetime career for Jones at the NACA/NASA. Through a passionate interest and self-study in aeronautics, Jones had become exceptionally knowledgeable in aerodynamic theory. His talents were recognized at Langley, and he was kept on at the laboratory through a series of temporary and emergency reappointments for the next two years. Unable to promote him into the lowest professional engineering grade because of civil-service regulations that required a college degree, in 1936 the laboratory management was finally able to hire Jones permanently via a loophole: It hired Jones at the next grade above the lowest, for which the requirement of a college degree was not specifically stated (although presumed).

By 1944, Jones was one of the most respected aerodynamicists in the NACA. At that time he was working on the design of an experimental air-to-air missile for the Army Air Force and he was also studying the aerodynamics of a proposed glide bomb having a low-aspect-ratio delta wing. The Ludington-Griswald Company of Saybrook, Connecticut, had carried out wind-tunnel tests on a dart-shaped missile of their design, and Roger Griswold, president of the company, showed the data to Jones in 1944. Griswold had compared the lift data for the missile's low-aspect-ratio delta wing with calculations made from Prandtl's tried-and-proved lifting-line theory (Section 5.3). Jones realized that Prandtl's lifting-line theory was not valid for low-aspect-ratio wings, and he began to construct a more appropriate theory for the delta-wing planform. Jones obtained rather simple analytical equations for the low-speed, incompressible flow over delta wings, but considered the theory to be "so crude" that "nobody would be interested in it." He placed his analysis in his desk and went on with other matters.

In early 1945 Jones began to look at the mathematical theory of supersonic potential (irrotational) flows. When applied to delta wings, Jones found that he was obtaining equations similar to those he had found for incompressible flow using the crude theory that was now buried in his desk. Searching for an explanation, he recalled the statement by Max Munk in 1924 that the aerodynamic characteristics of a wing were governed mainly by the component of the freestream velocity perpendicular to the leading edge. The answer suddenly was quite simple: For the delta wing, the reason his supersonic findings were the same as his earlier low-speed findings was that the leading edge of the delta wing was swept far enough that the component of the supersonic free stream Mach number perpendicular to the leading edge was subsonic, and hence the supersonic swept wing acted as if it were in a subsonic flow. With that revelation, Jones had independently discovered the high-speed aerodynamic advantage of swept wings, albeit 10 years after Busemann's paper at the Volta conference.

Jones began to discuss his swept-wing theory with colleagues at NACA Langley. In mid-February 1945 he outlined his thoughts to Jean Roche and Ezra Kotcher of the Army Air Force at Wright Field. On March 5, 1945, he sent a memo to Gus Crowley, chief of research at Langley, stated that he had "recently made a theoretical analysis which indicates that a V-shaped wing traveling point foremost would be less affected by compressibility than other planforms. In fact, if the angle of the V is kept small relative to the Mach angle, the lift and center of pressure remain the same at speeds both above and below the speed of sound." In the same memo, Jones asked Crowley to approve experimental work on swept wings. Such work was quickly initiated by the Flight Research Section of Langley, under the direction of Robert Gilruth, beginning with a series of free-flight tests using bodies with swept wings dropped from high altitude.

Jones finished a formal report on his low-aspect-ratio wing theory in late April 1945, including the effects of compressibility and the concept of a swept wing. However, during the in-house editorial review of that report, Theodore Theodorsen raised some serious objections. Theodorsen did not like the heavily intuitive nature of Jones's theory, and he asked Jones to clarify the "hocus-pocus" with some "real mathematics." Furthermore, because supersonic flow was so

different physically and mathematically from subsonic flow, Theodorsen could not accept the "subsonic" behavior of Jones's highly swept wings at supersonic speeds. Criticizing Jones's entire swept-wing concept, calling it "a snare and a delusion," Theodorsen insisted that Jones take out the part about swept wings.

Theodorsen's insistence prevailed, and publication of Jones's report was delayed. However, at the end of May 1945, Gilruth's free-flight tests dramatically verified Jones's predictions, showing a factor of 4 reduction in drag due to sweeping the wings. Quickly following those data, wind-tunnel tests carried out in a small supersonic wind tunnel at Langley showed a large reduction in drag on a section of wire in the test section when the wire was placed at a substantial angle of sweep relative to the flow in the test section. With that experimental proof of the validity of the swept-wing concept, Langley forwarded Jones's report to NACA Headquarters in Washington for publication. But Theodorsen would not give up: The transmittal letter to NACA Headquarters contained the statement that "Dr. Theodore Theodorsen (still) does not agree with the arguments presented and the conclusions reached and accordingly declined to participate in editing the paper." Such recalcitrance on the part of Theodorsen is reminiscent of his refusal to believe that the shock waves seen in John Stack's schlieren photographs of the transonic flow over an airfoil 11 years earlier were real (see Section 11.11). Theodorsen certainly made important contributions to airfoil theory in the 1930s, but he was also capable of errors in judgment (i.e., he was human).

On June 21, 1945, the NACA issued Jones's report as a confidential memorandum, chiefly for the Army and Navy. Three weeks later, the report was reissued as an advance confidential report, sent by registered mail to those people in industry with a "need to know." Entitled "Wing Plan Forms for High-Speed Flight," Jones's report quickly spread the idea of the swept wing to selected members of the aeronautical community in the United States, but by that time, information about the German swept-wing research was beginning to reach the same aeronautical community. Jones's work appeared in the open literature about a year later, as NACA TR 863, a technical report only five pages long, but a classic explanation of how a swept wing works aerodynamically.

Credit for the idea of the swept wing for high-speed flight is shared between Busemann and Jones. Separated by a time interval of 10 years, and the closed shops of military security in both Germany and the United States, each independently developed the concept, not knowing of the other's work. The full impact of the swept-wing concept on the aeronautical industry came directly after the end of World War II, with almost simultaneous release of similar information from both sides of the ocean, thus promoting confidence in the validity of the concept.

The speed at which this information was used for airplane design is nothing short of amazing. Boeing's chief aerodynamicist, George Schairer, was a member of one of the Allied technical intelligence teams sent to Germany in April 1945. On the flight over, Schairer, who was aware of Jones's swept-wing concept and that Jones's report was being held up in the editorial process, reported that the concept was the main topic of conversation. Concluding that sweepback was a valid concept, Schairer needed no convincing of its value when the team saw the German swept-wing data at Baunschweig on May 7. On May 10 he sent a

seminal and historically important letter to Boeing directing that a new design for a straight-wing jet bomber immediately be changed to a swept-wing configuration, and that other aircraft manufacturers be informed about the German swept-wing research. At Boeing, the result of this letter was the Boeing B-47, shown in Figure 11.36, and at North American the result was the famous F-86 (Figure 11.15). The B-47 sowed the seeds for the Boeing 707 swept-wing jet transport (Figure 1.2) and for all subsequent large commercial jet transports to the present day.

Indeed, by 1948 the swept wing had become an accepted airplane design feature. It had done for the high-speed jet airplane what streamlining had done in the 1930s for the advanced propeller-driven airplane, namely, provided the aerodynamic means for efficient flight in the desired flight regime. Virtually all

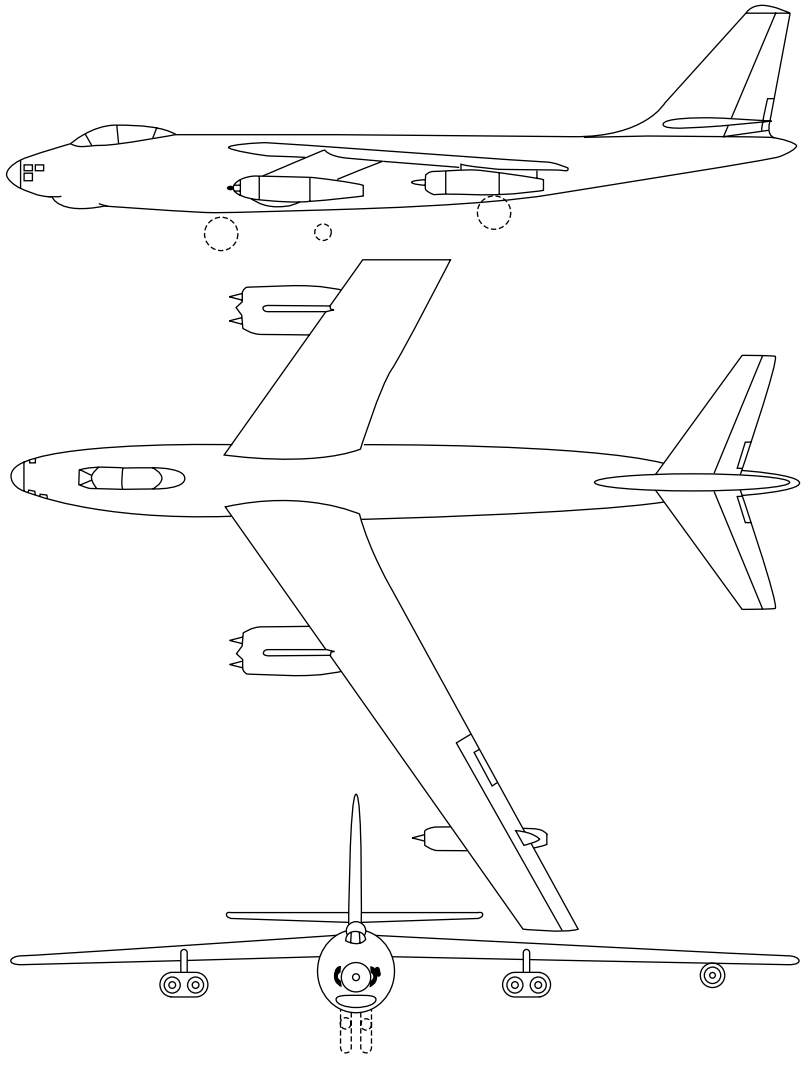


Figure 11.36 Three-view of the Boeing B-47 swept-wing bomber, circa 1948.

high-speed, jet-propelled airplanes today have wings with highly swept leading edges, and these contemporary aircraft can trace their ancestry directly back to the B-47 and the F-86, and to the innovative ideas and genius of Adolf Busemann and Robert Jones.

For a more in-depth discussion of the history of the swept wing, see References 62 and 115.

11.14 HISTORICAL NOTE: RICHARD T. WHITCOMB—ARCHITECT OF THE AREA RULE AND THE SUPERCRITICAL WING

The developments of the area rule (Section 11.8) and the supercritical airfoil (Section 11.9) are two of the most important advancements in aerodynamics since 1950. That both developments were made by the same man—Richard T. Whitcomb—is remarkable. Who is this man? What qualities lead to such accomplishments? Let us pursue these matters further.

Richard Whitcomb was born on February 21, 1921, in Evanston, Illinois. At an early age, he was influenced by his grandfather, who had known Thomas A. Edison. In an interview with *The Washington Post* on August 31, 1969, Whitcomb is quoted as saying: "I used to sit around and hear stories about Edison. He sort of developed into my idol." Whitcomb entered the Worcester Polytechnic Institute in 1939. (This is the same school from which the rocket pioneer, Robert H. Goddard, had graduated 31 years earlier.) Whitcomb distinguished himself in college and graduated with a mechanical engineering degree with honors in 1943. Informed by a *Fortune* magazine article on the research facilities at the NACA Langley Memorial Laboratory, Whitcomb immediately joined the NACA. He became a wind-tunnel engineer, and as an early assignment he worked on design problems associated with the Boeing B-29 Superfortress. He remained with the NACA and later its successor, NASA, until his retirement in 1980—spending his entire career with the wind tunnels at the Langley Research Center. In the process, he rose to become head of the Eight-foot Tunnel Branch at Langley. He died of pneumonia at the age of 88 on October 13, 2009 in Newport News, Virginia.

Whitcomb conceived the idea of the area rule as early as 1951. He tested his idea in the transonic wind tunnel at Langley. The results were so promising that the aeronautical industry changed designs in midstream. For example, the Convair F-102 delta-wing fighter had been designed for supersonic flight, but was having major difficulty even exceeding the speed of sound—the increase in drag near Mach 1 was simply too large. The F-102 was redesigned to incorporate Whitcomb's area rule and afterward was able to achieve its originally intended supersonic Mach number. The area rule was such an important aerodynamic breakthrough that it was classified "secret" from 1952 to 1954, when airplanes incorporating the area rule began to roll off the production line. In 1954, Whitcomb was given the Collier Trophy—an annual award for the "greatest achievement in aviation in America."

In the early 1960s, Whitcomb turned his attention to airfoil design, with the objective again of decreasing the large drag rise near Mach 1. Using the existing knowledge about airfoil properties, a great deal of wind-tunnel testing, and

intuition honed by years of experience, Whitcomb produced the supercritical airfoil. Again, this development had a major impact on the aeronautical industry, and today virtually all new commercial transport and executive aircraft designs are incorporating a supercritical wing. Because of his development of the supercritical airfoil, in 1974 NASA gave Whitcomb a cash award of $25,000—the largest cash award ever given by NASA to a single individual.

There are certain parallels between the personalities of the Wright brothers and Richard Whitcomb: (1) they all had powerful intuitive abilities which they brought to bear on the problem of flight, (2) they were totally dedicated to their work (none of them ever married), and (3) they did a great deal of their work themselves, trusting only their own results. For example, here is a quote from Whitcomb which appears in the same *Washington Post* interview mentioned above. Concerning the detailed work on the development of the supercritical airfoil, Whitcomb says:

> I modified the shape of the wing myself as we tested it. It's just plain easier this way. In fact my reputation for filing the wing's shape has become so notorious that the people at North American have threatened to provide me with a 10-foot file to work on the real airplane, also.

Perhaps the real ingredient for Whitcomb's success is his personal philosophy, as well as his long hours at work daily. In his own words:

> There's been a continual drive in me ever since I was a teenager to find a better way to do everything. A lot of very intelligent people are willing to adapt, but only to a certain extent. If a human mind can figure out a better way to do something, let's do it. I can't just sit around. I have to think.

Students take note!

11.15 SUMMARY

Review the road map in Figure 11.1, and make certain that you have all the concepts listed on this map well in mind. Some of the highlights of this chapter are as follows:

For two-dimensional, irrotational, isentropic, steady flow of a compressible fluid, the exact velocity potential equation is

$$\left[1-\frac{1}{a^2}\left(\frac{\partial\phi}{\partial x}\right)^2\right]\frac{\partial^2\phi}{\partial x^2}+\left[1-\frac{1}{a^2}\left(\frac{\partial\phi}{\partial y}\right)^2\right]\frac{\partial^2\phi}{\partial y^2} -\frac{2}{a^2}\left(\frac{\partial\phi}{\partial x}\right)\left(\frac{\partial\phi}{\partial y}\right)\frac{\partial^2\phi}{\partial x\,\partial y}=0 \tag{11.12}$$

where

$$a^2=a_0^2-\frac{\gamma-1}{2}\left[\left(\frac{\partial\phi}{\partial x}\right)^2+\left(\frac{\partial\phi}{\partial y}\right)^2\right] \tag{11.13}$$

This equation is exact, but it is nonlinear and hence difficult to solve. At present, no general analytical solution to this equation exists.

For the case of small perturbations (slender bodies at low angles of attack), the exact velocity potential equation can be approximated by

$$\left(1 - M_\infty^2\right)\frac{\partial^2 \hat{\phi}}{\partial x^2} + \frac{\partial^2 \hat{\phi}}{\partial y^2} = 0 \tag{11.18}$$

This equation is approximate, but linear, and hence more readily solved. This equation holds for subsonic ($0 \leq M_\infty \leq 0.8$) and supersonic ($1.2 \leq M_\infty \leq 5$) flows; it does not hold for transonic ($0.8 \leq M_\infty \leq 1.2$) or hypersonic ($M_\infty > 5$) flows.

The Prandtl-Glauert rule is a compressibility correction that allows the modification of existing incompressible flow data to take into account compressibility effects:

$$C_p = \frac{C_{p,0}}{\sqrt{1 - M_\infty^2}} \tag{11.51}$$

Also,

$$c_l = \frac{c_{l,0}}{\sqrt{1 - M_\infty^2}} \tag{11.52}$$

and

$$c_m = \frac{c_{m,0}}{\sqrt{1 - M_\infty^2}} \tag{11.53}$$

The critical Mach number is that freestream Mach number at which sonic flow is first obtained at some point on the surface of a body. For thin airfoils, the critical Mach number can be estimated as shown in Figure 11.6.

The drag-divergence Mach number is that freestream Mach number at which a large rise in the drag coefficient occurs, as shown in Figure 11.11.

The area rule for transonic flow states that the cross-sectional area distribution of an airplane, including fuselage, wing, and tail, should have a smooth distribution along the axis of the airplane.

Supercritical airfoils are specially designed profiles to increase the drag-divergence Mach number.